FPT UNIVERSITY

Capstone Project Report

Report 4 – Software Design Document

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Definition and Acronyms

Acronym	Definition
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
CDN	Content Delivery Network
CI/CD	Continuous Integration / Continuous Deployment
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
CV	Curriculum Vitae
DTO	Data Transfer Object
eKYC	Electronic Know Your Customer
ERD	Entity–Relationship Diagram
HTML	HyperText Markup Language
HTTP / HTTPS	HyperText Transfer Protocol / HTTP Secure
JWT	JSON Web Token
ORM	Object-Relational Mapping
OTP	One-Time Password
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDK	Software Development Kit
SCM	Source Code Management
SQL	Structured Query Language
SRS	Software Requirements Specification
SSL / TLS	Secure Sockets Layer / Transport Layer Security
UI	User Interface
UML	Unified Modeling Language

Acronym	Definition
URI / URL	Uniform Resource Identifier / Uniform Resource Locator
UX	User Experience
WBS	Work Breakdown Structure
WebSocket	Full-Duplex Bidirectional Communication Protocol

I. Record of Changes

Date	A* M, D	In charge	Change Description

**A - Added M - Modified D - Deleted*

II. Software Design Document

1. System Design

1.1 System Architecture

The Frevia system follows an N-layer architecture, separating the client applications, presentation layer, business logic layer, data access layer, shared library layer, and external integration systems (Figure 1). The system also natively integrates with PostgreSQL, Cloudinary, Stripe, Google OAuth 2.0, Nodemailer, Infisical, and local Python FastAPI AI microservices backed by Ollama LLM.

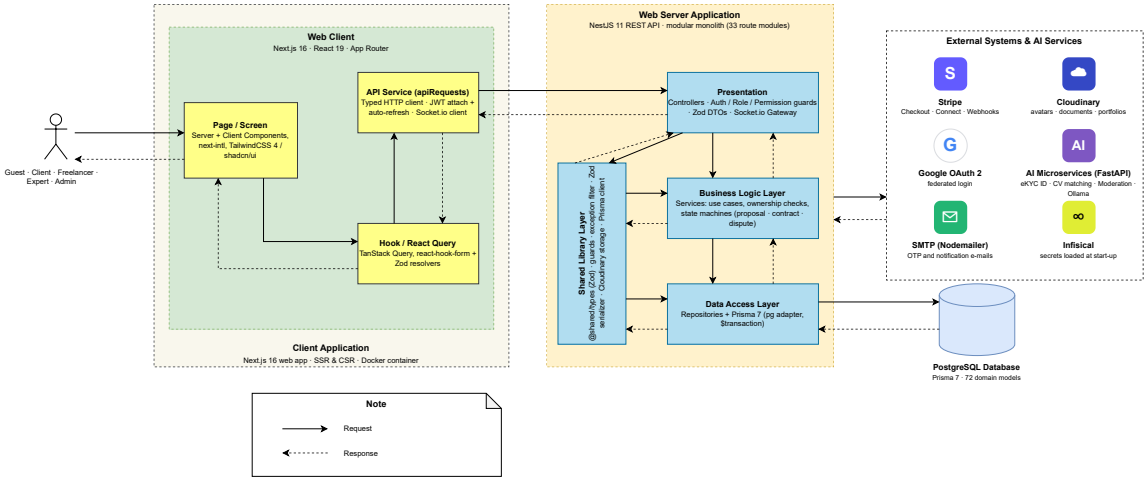

Figure 1: System Architecture Diagram

Client Application

Frevia delivers a unified, responsive web application catering to five actor roles: Guest, Client, Freelancer, Expert, and Administrator.

- **Web Client:** Developed with Next.js 16 (React 19) and TypeScript utilizing the App Router architecture. It consists of:
  - *Pages and Screens:* Built with React Server Components (RSC) for optimal search engine indexing and rapid initial page loads, complemented by Client Components for interactive user interfaces. Styled using TailwindCSS 4 and the `@repo/ui` component library (based on shadcn/ui), and localized into English and Vietnamese via `next-intl`.
  - *Hooks and State Management:* Driven by TanStack Query (React Query) for server state caching, background revalidation, and optimistic mutations. Client-side form validation is managed via `react-hook-form` paired with Zod schemas.
  - *API Services (`apiRequests/`):* A typed HTTP client layer encapsulating `lib/http.ts`. It automatically binds JWT access tokens to request headers, transparently handles token

refreshes upon expiration, and interfaces with the Socket.io client for real-time messaging and notifications.

The client application transmits requests exclusively over HTTPS REST APIs and displays returned responses. It contains no direct database access logic and enforces no authoritative business rules.

Presentation Layer

The Presentation Layer is implemented within the NestJS 11 Web Server Application (`apps/api`). It serves as the entry boundary for incoming client requests and real-time events:

- **Controllers:** Expose structured RESTful API endpoints (versioned under `/api/v1`), receive client requests, bind validated request payloads, and delegate processing to domain services.
- **Authentication and Authorization Guards:** Intercept requests to verify JWT tokens and enforce role-based access control (`RoleGuard`) and fine-grained permission checks (`PermissionGuard`).
- **Validation and Serialization:** Validates incoming request payloads using shared Zod DTOs and sanitizes outgoing responses using the global `ZodSerializerInterceptor`.
- **Socket.io Gateway:** Manages real-time bidirectional WebSocket connections for peer-to-peer conversations and immediate notification pushes.

Business Logic Layer

The Business Logic Layer represents the core of the Frevia platform, encapsulating business rules, ownership verifications, and deterministic domain state machines across 33 route modules:

- **Authentication and Access Control Module:** Manages user registration, OTP verification, credential login, Google OAuth federated sign-in, token refresh, device sessions, roles, and granular permissions.
- **User and Profile Management Module:** Handles administrator user accounts, account profiles, freelancer professional portfolios, CV uploads, and expert consultation profiles.
- **Job and Proposal Management Module:** Governs client job postings, faceted job browsing, bookmarking, freelancer proposal submissions, interviews, and proposal acceptance/rejection workflows.
- **Contract and Milestone Module:** Enforces contract lifecycle transitions (draft, agreement, digital signature, completion, cancellation) and milestone delivery reviews.
- **Payment and Escrow Module:** Integrates Stripe Checkout for milestone escrow funding, platform fee collections, dispute fee handling, and automated Stripe Connect payouts to freelancer bank accounts.

- **Communication and Community Module:** Facilitates real-time direct messaging, multi-category discussion forums, post interactions, and automated user notification feeds.
- **Dispute Arbitration Module:** Provides formal dispute filing, evidence attachment, and administrator arbitration workflows.

Data Access Layer

The Data Access Layer encapsulates all persistence logic using the Repository Pattern combined with Prisma 7 ORM and the native PostgreSQL driver adapter (`@prisma/adapter-pg`):

- **Repositories:** Implement dedicated query methods, data filtering, pagination helpers, and soft-delete semantics (`isDeleted`). Controllers and services are strictly decoupled from raw database queries.
- **Transaction Management:** Wraps complex multi-table workflows (such as escrow fund transfers, contract creation, and user onboarding cascades) inside atomic `$transaction` blocks to maintain absolute relational integrity.

Shared Library Layer

The Shared Library Layer provides shared definitions and cross-cutting utilities utilized across the application tiers:

- **Contract Single Source of Truth (`@shared/types`):** Monorepo package defining shared Zod validation schemas, TypeScript interfaces, system constants, database enumerations, and standardized error codes.
- **Backend Cross-Cutting Infrastructure (`src/shared`):** Contains global exception filters, authentication decorators, Cloudinary storage wrappers, Nodemailer integration, and the shared Prisma client service.

Data Storage

- **PostgreSQL Database:** Stores all persistent platform data across 72 relational domain models and 43 enumerations, maintaining relational integrity with foreign keys, compound indexes, and audit timestamps.
- **Cloudinary Object Storage:** Dedicated cloud repository storing binary multimedia files, including user profile avatars, citizen identity document photographs (CCCD), candidate resumes, and contract deliverable archives.

External Systems and Internal AI Services

- **Google OAuth 2.0:** Facilitates federated single sign-on. The frontend retrieves a Google authorization credential, which the backend verifies to establish an authenticated Frevia session.

- **Stripe:** Processes online financial transactions, managing escrow funding via Stripe Checkout, platform fee collection, dispute deposits, and automated freelancer payouts via Stripe Connect. Status updates are synchronized securely via cryptographic webhooks.
- **Internal AI Microservices (FastAPI & Python):**
  - *eKYC ID Verification (`id-verification`, port 8001):* Employs RapidOCR for card text extraction, InsightFace for biometric facial embeddings, and MediaPipe for liveness detection to detect fraudulent identity documents.
  - *Section-Aware CV and Job Matching (`cv-matching`, port 8000):* BGE-M3 embeddings, ChromaDB vector indexing, and local Ollama LLM inference to parse candidate CVs and compute semantic candidate-to-job matching scores.
  - *AI Content Moderation (`ai-moderation`, port 8001):* Automatically scores community forum posts and comments, quarantining toxic or abusive content into the administrative review queue.
- **SMTP Mail Service (Nodemailer):** Delivers transactional notifications, one-time passwords (OTP), and contract status alerts.
- **Infisical:** Secure secrets manager that injects environment variables and credentials into application containers at startup.

Request-Response Data Flow

A typical user request originates on a Next.js Page or Screen, triggers a TanStack Query hook, and invokes a typed API method in `apiRequests/`. The service binds the JWT access token and issues an HTTPS REST call to the NestJS backend. The request passes through the Authentication Guard, Role/Permission Guards, and Zod DTO validation before reaching the appropriate Controller. The Controller delegates execution to the Service Layer, which validates business rules and state machines, then commands the Repository to execute queries through Prisma 7 to PostgreSQL. The resulting entity follows the reverse path, is sanitized by the Zod serializer interceptor, and is returned as a uniform JSON response to update the client-side cache and re-render the user interface. Real-time events follow an asynchronous parallel path via the Socket.io Gateway.

1.2 Package Diagram

The Frevia repository is organized as a high-cohesion monorepo managed with **pnpm workspaces** and **Turborepo** (Figure 2). This structure cleanly separates the presentation tier (`apps/web`), the application tier (`apps/api`), the shared workspace packages (`packages/*`), and the independent AI microservices (`ai-services/`).

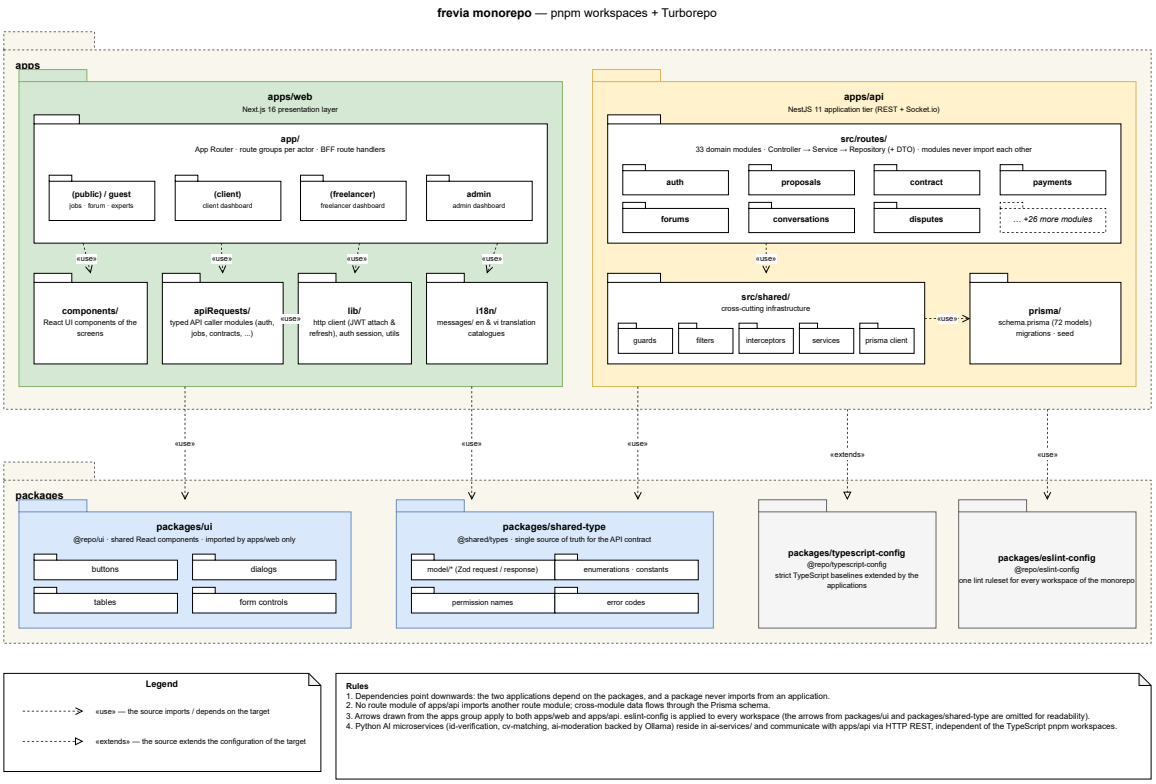

Figure 2: Package diagram

Monorepo Architecture Rules

The monorepo enforces four key architectural rules illustrated in Figure 2:

1. **Strict Downward Dependencies:** Applications (`apps/web` and `apps/api`) depend on shared packages (`packages/*`); a shared package never imports code or types from an application.

2. **Module Isolation:** Domain route modules inside `apps/api/src/routes/` are strictly decoupled and never import controllers, services, or repositories from sibling modules. Cross-module data flows exclusively through database relationships at the Prisma ORM schema level.

3. **Universal Configuration:** Common configurations for linting (`@repo/eslint-config`) and TypeScript compilation (`@repo/typescript-config`) are inherited by all workspaces to guarantee uniform code quality.

4. **Independent AI Service Boundaries:** Python-based AI microservices (`id-verification`, `cv-matching`, `ai-moderation` backed by Ollama) reside in `ai-services/` and communicate with the backend via standard HTTP REST endpoints, maintaining complete independence from the TypeScript pnpm workspaces.

Package Descriptions

Tables 1, 2, 3, and 4 detail the responsibilities of every package in the Frevia project.

Table 1: Frontend package description (`apps/web`)

No	Package	Description
01	`app`	Next.js App Router entry points, layout definitions, error boundaries, and route groups (`(auth)`, `(client)`, `(freelancer)`, `admin/`, and public pages).
02	`components`	Reusable UI components, page section blocks, modals, and screen layouts specific to the web application.
03	`apiRequests`	Typed HTTP client callers for all domain modules (auth, jobs, proposals, contracts, payments, etc.) that attach the JWT token and coordinate API communication.
04	`lib`	Core HTTP fetch wrapper (`http.ts`), authentication session token management, form validation resolvers, string utilities, and formatters.
05	`i18n`	Internationalization configuration and localization message catalogues (`messages/en.json`, `messages/vi.json`) powered by `next-intl`.
06	`hooks`	Custom React hooks for server state caching (TanStack Query), form management, debouncing, and authentication monitoring.
07	`configs`	Client application environment configuration variables, API endpoints, and system runtime constants.
08	`providers`	Application-wide React context providers, including QueryClientProvider, NextIntlClientProvider, and theme state providers.

Table 2: Backend package description (`apps/api`)

No	Package	Description
01	`src/routes`	33 domain modules structured into Controller, Service, Repository, and DTO layers.
02	`src/routes/auth`	User registration, OTP delivery, login, Google OAuth, token refresh, and role switching.
03	`src/routes/sessions`	Device session telemetry tracking, session listing, and remote revocation.
04	`src/routes/account-profile`, `identity-verifications`	Account profile management, avatar uploads, citizen identity document submission (CCCD), and administrator verification reviews.
05	`src/routes/users`	Administrator user management, account status control, and role assignments.
06	`src/routes/roles`, `permissions`	Role-based access control (RBAC) administration and granular permission mapping.

No	Package	Description
07	src/routes/profiles, portfolios, profile-revisions	Freelancer professional resumes, CV analysis integration, portfolio showcases, skill tags, and profile revision audit history.
08	src/routes/expert-profile	Expert directory listings, consultation profiles, and domain certifications.
09	src/routes/browse-job	Public job search, keyword filtering, category navigation, and slug-based job details.
10	src/routes/manage-job	Client job posting, draft editing, applicant review, and bookmark management.
11	src/routes/job-categories, job-categories-admin, skills-admin	Job category taxonomy, category hierarchy management, and administrative skill tags.
12	src/routes/proposals	Freelancer bid submissions, cover letters, proposed terms, and proposal lifecycle management.
13	src/routes/contract, milestone	Contract generation, digital signing, milestone deliverable submissions, and progress reviews.
14	src/routes/milestone-file, milestone-submission, shared-file	Milestone file attachments, deliverable review submissions, and shared project documents.
15	src/routes/payments	Stripe Connect onboarding, milestone escrow deposits, fund releases, platform fee deductions, and webhook handlers.
16	src/routes/disputes	Contract dispute arbitration, claim filing, evidence submission, and administrator resolution.
17	src/routes/reviews	Bilateral post-contract rating reviews, score aggregations, and public responses.
18	src/routes/conversations	Direct real-time messaging, conversation threads, and message history persistence.
19	src/routes/notifications	User notification feed generation, unread status tracking, and push alerts.
20	src/routes/forums, forums-admin	Public discussion forums, posts, comments, likes, reports, AI moderation queues, and soft-delete trash recovery.
21	src/routes/saved-search, job-alert	Saved job filter criteria and automated alert notifications for matching postings.
22	src/routes/recommendations	AI-assisted candidate and job recommendations based on semantic CV matching.
23	src/routes/banners, banners-admin	Promotional banner delivery, placement targeting, and administrative banner management.
24	src/shared/guards	JWT authentication guards, role-based authorization guards, and permission guards.
25	src/shared/filters	Global exception filter converting internal exceptions into standard JSON error bodies.

No	Package	Description
26	src/shared/interceptors	Zod response serialization and transformation interceptor.
27	src/shared/services	Cross-cutting infrastructure services including Cloudinary media storage, Nodemailer email dispatch, and AI moderation client.
28	src/shared/repositories	Base repository definitions and database query execution helpers.
29	prisma	Prisma 7 ORM persistence configuration, schema.prisma (72 models, 43 enums), database migrations, and seed scripts.

Table 3: Shared and tooling packages description (packages/)

No	Package	Description
01	packages/ui	Shared React component library (@repo/ui) styled with TailwindCSS 4 and shadcn/ui (buttons, dialogs, form controls, tables) imported exclusively by the web application.
02	packages/shared-type	API contract single source of truth (@shared/types) providing Zod validation models, TypeScript interfaces, database enumerations, permission names, and error codes to both applications.
03	packages/typescript-config	Shared TypeScript configuration baselines (@repo/typescript-config) that the applications extend for strict type checking.
04	packages/eslint-config	Centralized ESLint configuration ruleset (@repo/eslint-config) enforced across every monorepo package.

Table 4: AI microservices package description (ai-services/)

No	Package	Description
01	id-verification	FastAPI microservice (port 8001) performing Vietnamese citizen identity card (CCCD) OCR via RapidOCR, facial biometric embedding via InsightFace, and anti-spoofing liveness verification via MediaPipe.
02	cv-matching	FastAPI microservice (port 8000) performing section-aware candidate CV parsing, dense semantic vector embedding with BGE-M3, ChromaDB vector indexing, and candidate-job matching supported by Ollama LLM.
03	ollama	Containerized local Large Language Model runtime (port 11434) running instruction-tuned models (e.g., Qwen) for structured CV extraction and semantic matching.
04	ai-moderation	FastAPI microservice (port 8001) providing automated toxic content classification, sensitive keyword detection, and moderation scoring for forum posts and user submissions.

2. Database Design

Frevia persists every domain object in a single PostgreSQL database whose schema is declared with Prisma 7 in `apps/api/prisma/schema.prisma`. The schema contains 72 models together with 43 enumerations, every table has a surrogate integer primary key generated by the database (nine membership tables use a composite key instead), and deletion is either restricted by a foreign-key rule (identity and money-related rows are never removed) or implemented as a soft delete through a `deletedAt` column (content that an administrator must be able to restore: jobs, forum categories and posts, skills, banners, contract term templates). Binary files are not stored in the database: only their Cloudinary URL is kept in the owning row.

Figure 3 shows the entity–relationship diagram of the whole schema. Every table is drawn as a frameless box: the header carries the table name, each of the following rows is one column, and the key columns are flagged with `PK` (primary key) or `FK` (foreign key). A relationship is a line that leaves the many-side table with a crow's foot and ends on the one-side table with a single bar, so it reads from the row that holds the foreign key to the row it references.

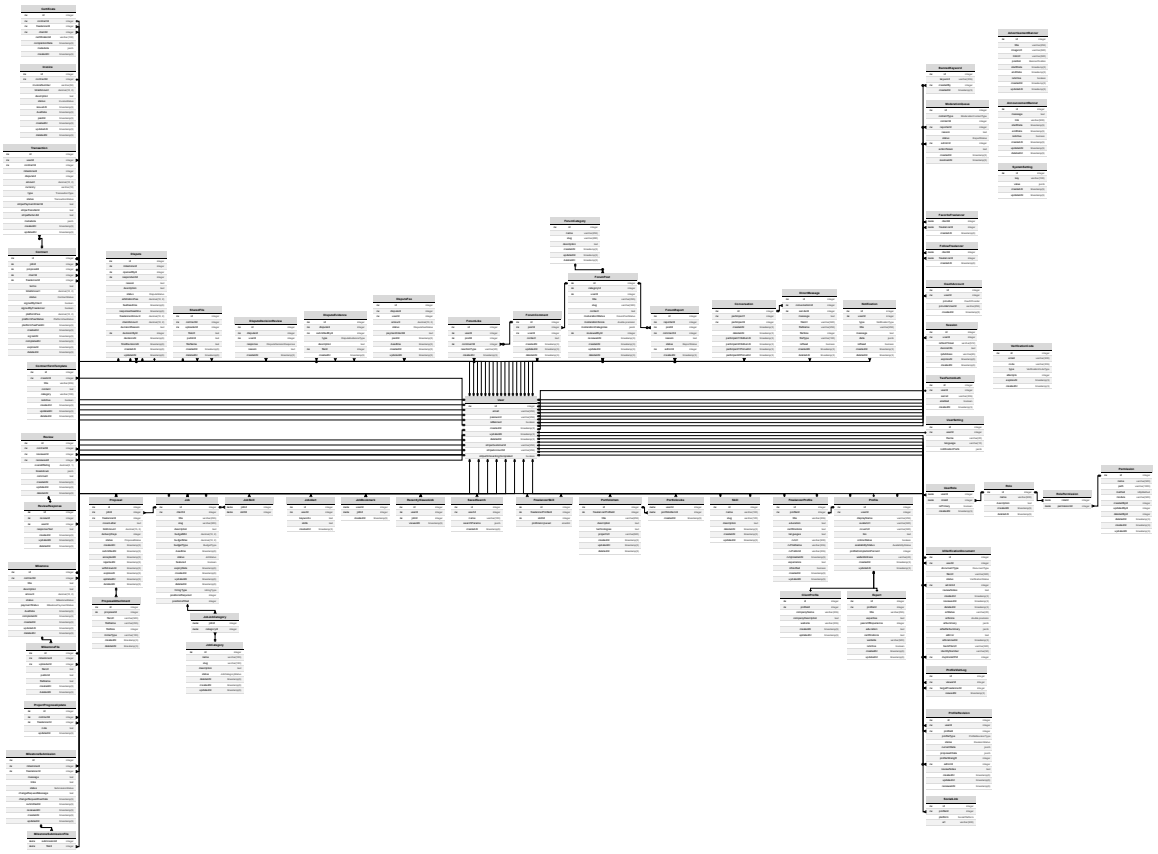

Figure 3: Entity–relationship diagram of the Frevia database

2.1 Table Relationships

The 64 tables are grouped into nine families; inside a family the relationships are one-to-many, and membership tables carry the many-to-many links:

- **Identity and access** — `User` is the root of the schema. `UserRole` links an account to a `Role`, and `RolePermission` links a role to a `Permission`, which yields the RBAC model enforced by the guards. Authentication adds `Session` (one row per device), `VerificationCode` (OTP for registration and password reset), `OauthAccount` (Google login), `TwoFactorAuth` and `UserSetting`.

- **Profiles** — `Profile` is shared by every account; `FreelancerProfile` and `ClientProfile` extend it with role-specific columns, `Expert` publishes an expert entry in the public directory, and `FreelancerSkill` joins a freelancer to the `Skill` catalogue. `PortfolioItem`, `SocialLink`, `Certificate`, `IdVerificationDocument` and `ProfileRevision` hang from the profile.

- **Jobs** — `Job` belongs to a client and is classified through `JobJobCategory` (`JobCategory`) and `JobSkill` (`Skill`). `JobBookmark`, `RecentlyViewedJob`, `SavedSearch` and `JobAlert` record the browsing behaviour of a freelancer.

- **Proposals and contracts** — `Proposal` references one `Job` and one freelancer; accepting it creates exactly one `Contract`, which carries `ContractTermTemplate` snapshots and a list of `Milestone` rows. `ProjectProgressUpdate`, `SharedFile` and `MilestoneFile` document the work in progress, while `MilestoneSubmission` (with `MilestoneSubmissionFile`) records the deliverable and its review decision.

- **Payments** — `Transaction` stores every Stripe movement (funding, release, refund, platform fee, dispute fee) with its payment-intent identifier, and `Invoice` holds the invoice issued per contract. Money is never derived: the escrow balance of a milestone is the sum of its transactions.

- **Disputes** — `Dispute` is opened on a contract or a milestone, `DisputeEvidence` collects the attachments of both parties, `DisputeFee` records the fee charged to open it, and `DisputeDecisionReview` stores the request of a party that asks for the decision to be reconsidered.

- **Reviews** — one `Review` per completed contract, answered by at most one `ReviewResponse`.

- **Messaging and notifications** — `Conversation` groups the `DirectMessage` rows exchanged by two accounts (pin and read state are kept on the message and on the conversation), and `Notification` feeds the in-app bell.

- **Community forum and configuration** — `ForumCategory` contains `ForumPost`, a post contains `ForumComment`, and `ForumLike` / `ForumReport` attach reactions and reports to posts or comments. Administrators work through `ModerationQueue` and `BannedKeyword`; platform-wide configuration lives in `SystemSetting`, `AnnouncementBanner` and `AdvertisementBanner`.

Table Descriptions

Table 5 describes the purpose of every table of the schema, in the order in which the models are declared in `schema.prisma`, together with its primary key and its foreign keys.

Table 5: Table descriptions of the Frevia database

No	Table	Description
01	`Role`	Role catalogue of the RBAC model (admin, client, freelancer, expert); carries the role code, name and description. • **Primary keys:** `id` • **Foreign keys:** —
02	`UserRole`	Assignment of a role to an account, so one account can hold several roles and switch between them. • **Primary keys:** `userId`, `roleId` • **Foreign keys:** `userId` → `User`, `roleId` → `Role`
03	`Permission`	Catalogue of route-level permissions (`module.action` names) used by the permission guard. • **Primary keys:** `id` • **Foreign keys:** —
04	`RolePermission`	Permissions granted to a role; drives the effective permission list of every signed-in account. • **Primary keys:** `roleId`, `permissionId` • **Foreign keys:** `roleId` → `Role`, `permissionId` → `Permission`
05	`User`	Account identity: e-mail, username, bcrypt password hash, status, verification and ban flags, preferred language and timestamps. • **Primary keys:** `id` • **Foreign keys:** —
06	`VerificationCode`	One-time code (OTP) issued for e-mail verification or password reset, with its purpose, expiry and consumed flag. • **Primary keys:** `id` • **Foreign keys:** —
07	`Session`	Server-side session per device: refresh-token hash, device/user-agent information, last activity and revocation time. • **Primary keys:** `id` • **Foreign keys:** `userId` → `User`

No	Table	Description
08	OauthAccount	Link between an account and an external identity provider (Google) for federated login. • **Primary keys:** id • **Foreign keys:** userId → User
09	TwoFactorAuth	Two-factor authentication secret and enabled flag of an account (one row per account). • **Primary keys:** id • **Foreign keys:** userId → User
10	UserSetting	Per-account preferences: interface theme, language and the JSON notification preferences. • **Primary keys:** id • **Foreign keys:** userId → User
11	Profile	Base profile shared by all accounts: display name, avatar, cover, headline and biography. • **Primary keys:** id • **Foreign keys:** userId → User
12	FreelancerProfile	Professional data of a freelancer: professional title, overview, hourly rate, availability and total earnings. • **Primary keys:** id • **Foreign keys:** profileId → Profile
13	FreelancerSkill	Many-to-many link between a freelancer profile and a skill, with the self-declared proficiency level. • **Primary keys:** id • **Foreign keys:** freelancerProfileId → FreelancerProfile, skillId → Skill
14	ClientProfile	Client-side data: company name, company size, industry and billing information. • **Primary keys:** id • **Foreign keys:** profileId → Profile
15	Expert	Expert entry published in the public expert directory, with specialisation and verification state. • **Primary keys:** id • **Foreign keys:** profileId → Profile
16	ProfileRevision	Pending change request on a profile, reviewed and approved or rejected by an administrator. • **Primary keys:** id • **Foreign keys:** userId → User, adminId → User, profileId → Profile

No	Table	Description
17	PortfolioItem	Portfolio entry of a profile: title, description, media URL, project link and ordering position. • **Primary keys:** id • **Foreign keys:** freelancerProfileId → FreelancerProfile
18	SocialLink	External link of a profile (website, LinkedIn, GitHub, Behance) shown on the public profile. • **Primary keys:** id • **Foreign keys:** profileId → Profile
19	IdVerificationDocument	Identity document uploaded for eKYC, with the AI check result and the administrator decision. • **Primary keys:** id • **Foreign keys:** userId → User, adminId → User, duplicateOfId → IdVerificationDocument
20	ProfileVisitLog	View log of a freelancer profile (viewerId, targetFreelancerId, viewedAt); source of the profile statistics. • **Primary keys:** id • **Foreign keys:** viewerId → User, targetFreelancerId → User
21	JobCategory	Hierarchical job category (parent reference, slug, name, icon) with soft-delete support. • **Primary keys:** id • **Foreign keys:** —
22	Job	Job posting: slug, title, description, budget range, duration, deadline, visibility, status and the owning client. • **Primary keys:** id • **Foreign keys:** clientId → User
23	JobJobCategory	Many-to-many link between a job and the job categories it is published in. • **Primary keys:** jobId, categoryId • **Foreign keys:** jobId → Job, categoryId → JobCategory
24	Skill	Skill catalogue shared by freelancer profiles and jobs, with soft-delete support. • **Primary keys:** id • **Foreign keys:** —
25	JobSkill	Skills required by a job, with the required proficiency level. • **Primary keys:** jobId, skillId • **Foreign keys:** jobId → Job, skillId → Skill

No	Table	Description
26	JobBookmark	Jobs saved by a freelancer. • **Primary keys:** userId, jobId • **Foreign keys:** userId → User, jobId → Job
27	RecentlyViewedJob	Recently opened job postings of an account, with the viewing timestamp. • **Primary keys:** id • **Foreign keys:** userId → User, jobId → Job
28	SavedSearch	Stored job filter criteria (keyword, categories, budget, skills) that the freelancer can reapply. • **Primary keys:** id • **Foreign keys:** userId → User
29	JobAlert	Alert configuration attached to a saved search (frequency, channel, active flag). • **Primary keys:** id • **Foreign keys:** userId → User
30	Proposal	Proposal submitted by a freelancer for a job: cover letter, bid amount, estimated duration and lifecycle status. • **Primary keys:** id • **Foreign keys:** jobId → Job, freelancerId → User
31	ProposalAttachment	File attached to a proposal (portfolio sample, reference document) stored in Cloudinary. • **Primary keys:** id • **Foreign keys:** proposalId → Proposal
32	Contract	Contract created when a proposal is accepted: client, freelancer, total amount, terms, signature state, start/end dates and status. • **Primary keys:** id • **Foreign keys:** jobId → Job, proposalId → Proposal, clientId → User, freelancerId → User
33	ContractTermTemplate	Reusable contract term template owned by a user, with category, content and soft-delete support. • **Primary keys:** id • **Foreign keys:** creatorId → User
34	Milestone	Milestone inside a contract: title, description, amount, due date, escrow status, progress percentage and approval state. • **Primary keys:** id • **Foreign keys:** contractId → Contract

No	Table	Description
35	MilestoneFile	Reference file attached to a milestone and shared with both parties. • **Primary keys:** id • **Foreign keys:** milestoneId → Milestone, uploaderId → User
36	MilestoneSubmission	Work submitted for a milestone, with the submitted note, timestamp and the client review decision. • **Primary keys:** id • **Foreign keys:** milestoneId → Milestone, freelancerId → User
37	MilestoneSubmissionFile	Join table linking a submission to the milestone files it delivers. • **Primary keys:** submissionId, fileId • **Foreign keys:** submissionId → MilestoneSubmission, fileId → MilestoneFile
38	Invoice	Invoice of a contract (one per contract): invoice number, total amount, status, issued, due and paid dates. • **Primary keys:** id • **Foreign keys:** contractId → Contract
39	Certificate	Completion certificate generated when a contract is finished: unique certificate UID, completion date and JSON metadata. • **Primary keys:** id • **Foreign keys:** contractId → Contract, freelancerId → User, clientId → User
40	Dispute	Dispute opened on a contract or a milestone: reason, description, opened by, status, resolution and decision timestamps. • **Primary keys:** id • **Foreign keys:** milestoneId → Milestone, openedById → User, respondentId → User, decisionById → User
41	DisputeFee	Fee charged to open a dispute, with its amount, payment state and the transaction that settled it. • **Primary keys:** id • **Foreign keys:** disputeId → Dispute, userId → User
42	DisputeEvidence	Evidence attached by either party to a dispute (file, description, uploader). • **Primary keys:** id • **Foreign keys:** disputeId → Dispute, submittedById → User, fileId → SharedFile

No	Table	Description
43	DisputeDecisionReview	Request of a party to reconsider a dispute decision, with the response, the reason and the timestamp. • **Primary keys:** id • **Foreign keys:** disputeId → Dispute, userId → User
44	Transaction	Payment movement reconciled with Stripe (funding, release, refund, platform fee, dispute fee): type, amount, status, payment-intent id and timestamps. • **Primary keys:** id • **Foreign keys:** userId → User, contractId → Contract
45	Review	Review written after a contract is completed: rating, comment, author, recipient and the reviewed contract. • **Primary keys:** id • **Foreign keys:** contractId → Contract, reviewerId → User, revieweeId → User
46	ReviewResponse	Reply of the reviewed party to a review (at most one per review). • **Primary keys:** id • **Foreign keys:** reviewId → Review, userId → User
47	ForumCategory	Forum category: slug, name, description, display order, post counter and soft-delete support. • **Primary keys:** id • **Foreign keys:** —
48	ForumPost	Forum post: title, content, author, category, moderation status, AI score, reviewer and soft-delete support. • **Primary keys:** id • **Foreign keys:** categoryId → ForumCategory, userId → User, reviewedById → User
49	ForumComment	Comment on a forum post; comments are one level deep and can be soft-deleted. • **Primary keys:** id • **Foreign keys:** postId → ForumPost, userId → User
50	ForumLike	Reaction of a member on a post or a comment; a member can like a target only once. • **Primary keys:** id • **Foreign keys:** userId → User, postId → ForumPost, commentId → ForumComment

No	Table	Description
51	ForumReport	Content report filed by a member: reason, description, status and the administrator who handled it. • **Primary keys:** id • **Foreign keys:** reporterId → User, postId → ForumPost, commentId → ForumComment, adminId → User
52	Notification	In-app notification of an account: type, title, body, target link and read state. • **Primary keys:** id • **Foreign keys:** userId → User
53	AnnouncementBanner	Platform announcement shown in the application: message, link, display window, active flag and soft delete. • **Primary keys:** id • **Foreign keys:** —
54	ModerationQueue	Queue of AI-flagged or user-reported content waiting for an administrator decision, with the item type, the score and the assignee. • **Primary keys:** id • **Foreign keys:** reporterId → User, adminId → User
55	SystemSetting	Platform-wide configuration values stored as key/value rows. • **Primary keys:** id • **Foreign keys:** —
56	BannedKeyword	Keyword list used by the moderation filter before the AI service is called. • **Primary keys:** id • **Foreign keys:** createdBy → User
57	AdvertisementBanner	Advertising banner with image, target URL, position, schedule and click counter. • **Primary keys:** id • **Foreign keys:** —
58	FavoriteFreelancer	Freelancers marked as favourite by a client account. • **Primary keys:** clientId, freelancerId • **Foreign keys:** clientId → User, freelancerId → User
59	FollowFreelancer	Follow relationship between two accounts, used by the activity feed. • **Primary keys:** clientId, freelancerId • **Foreign keys:** clientId → User, freelancerId → User

No	Table	Description
60	PortfolioLike	Like given on a portfolio item. • **Primary keys:** userId, portfolioItemId • **Foreign keys:** userId → User, portfolioItemId → PortfolioItem
61	ProjectProgressUpdate	Progress note posted by the freelancer on a contract, with the update timestamp. • **Primary keys:** id • **Foreign keys:** contractId → Contract, freelancerId → User
62	SharedFile	File shared in the workspace of a contract (both parties can upload and read). • **Primary keys:** id • **Foreign keys:** contractId → Contract, uploaderId → User
63	Conversation	Direct conversation between two accounts, with last-message cache and the pin state of each participant. • **Primary keys:** id • **Foreign keys:** participant1 → User, participant2 → User
64	DirectMessage	Message of a conversation: sender, content, attachments, read state, recall (soft delete) flag and timestamp. • **Primary keys:** id • **Foreign keys:** conversationId → Conversation, senderId → User

2.2 Module Entity–Relationship Diagrams

The whole schema is easier to navigate when it is split along the functional modules used in Report 3. Each figure below shows one module together with the tables that the module references, so every foreign key of the module is visible in the same picture. The notation is the one of Figure 3: the crow's foot marks the many side of a relationship and the single bar marks the one side.

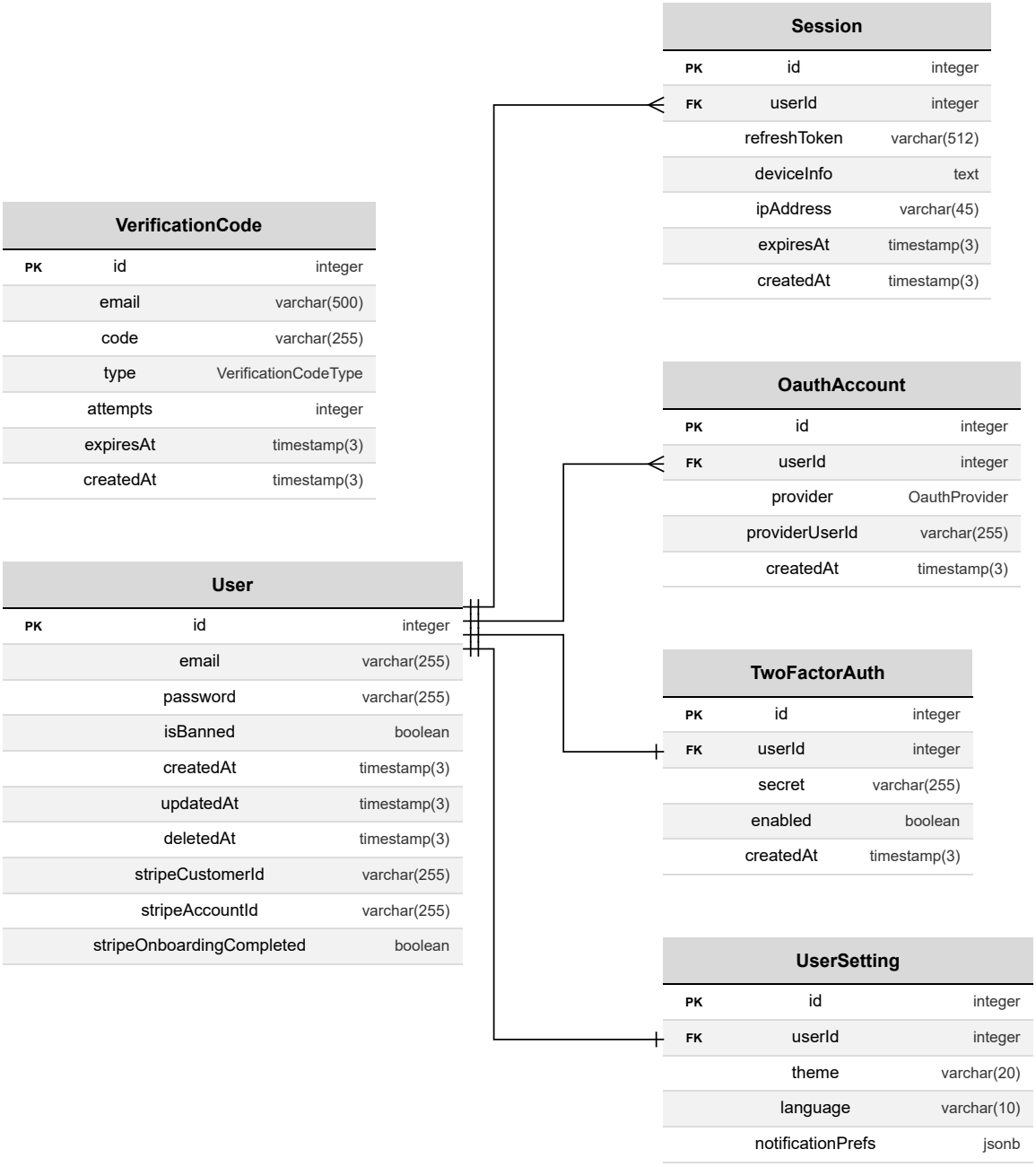

Figure 4: ERD1 — Authentication & Sessions

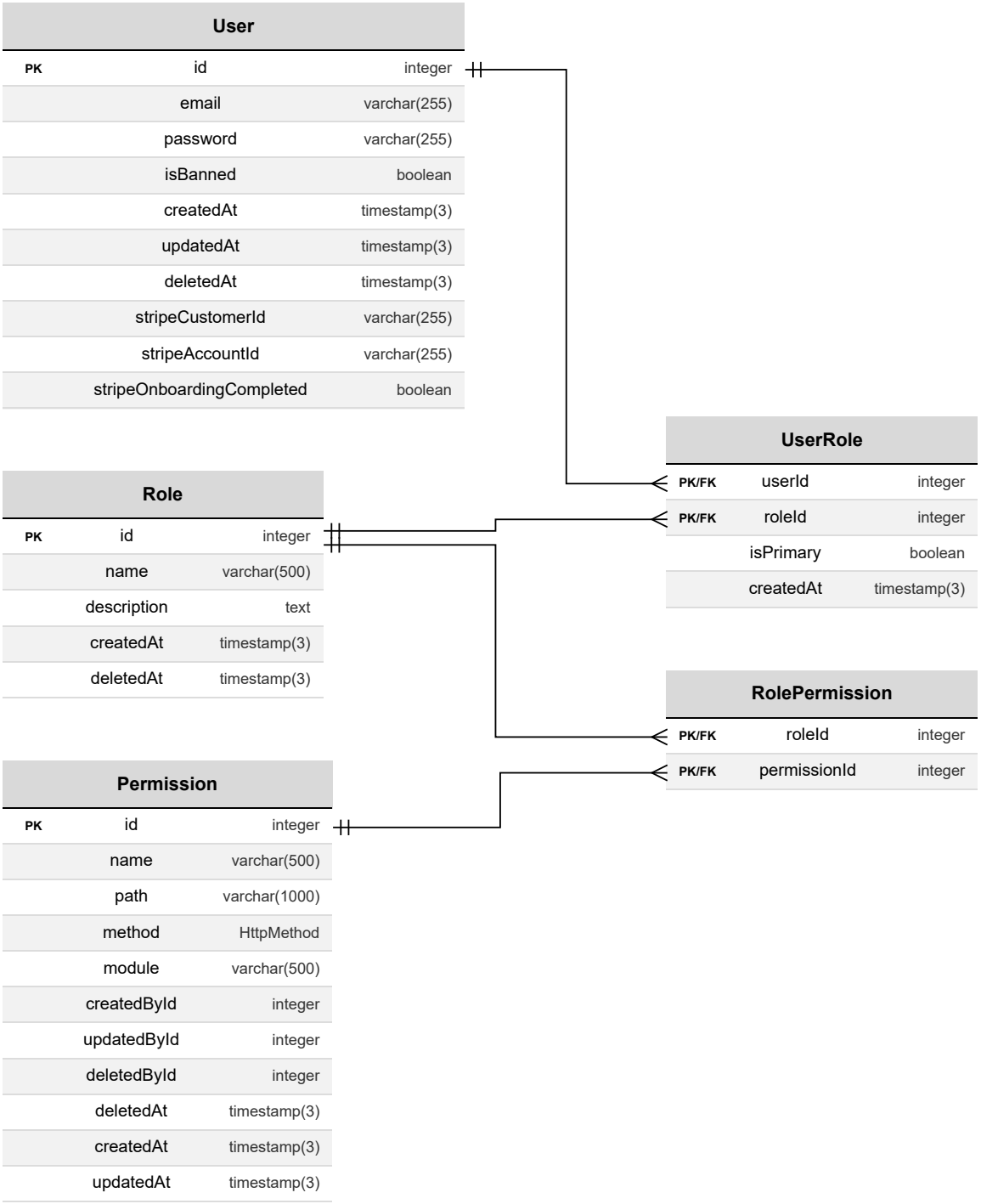

Figure 5: ERD2 — Role-Based Access Control

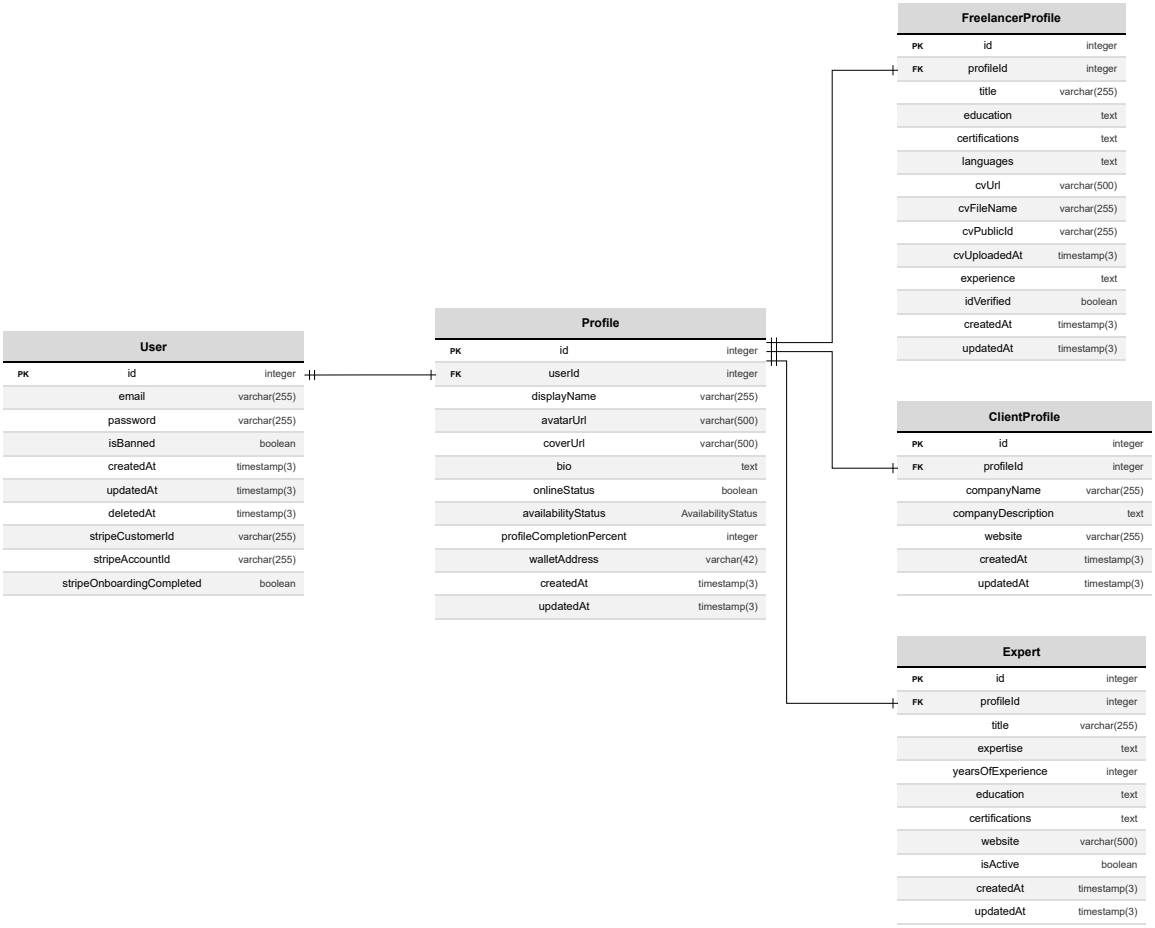

Figure 6: ERD3 — Profile Core

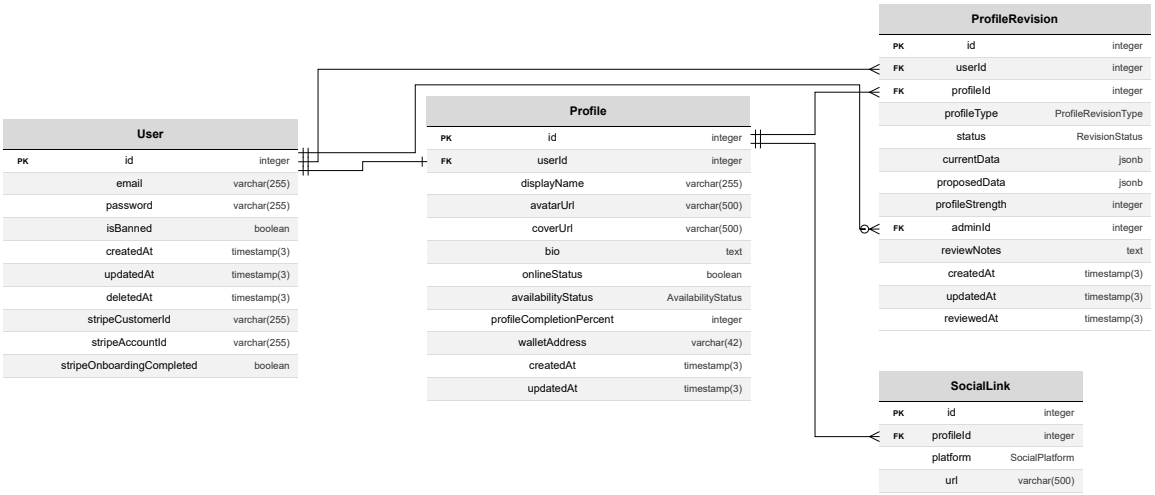

Figure 7: ERD4 — Profile History & Social Links

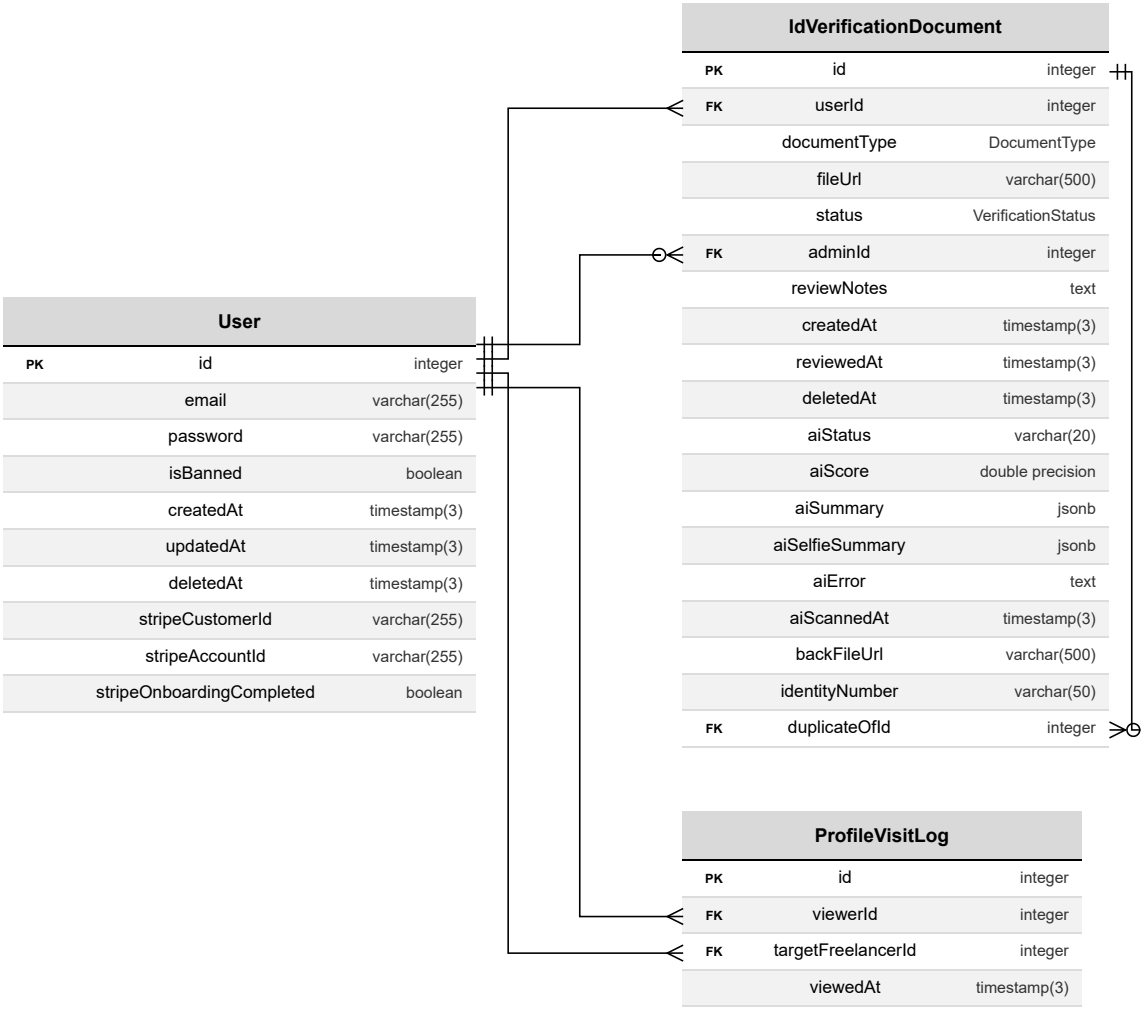

Figure 8: ERD5 — Identity Verification & Profile Visits

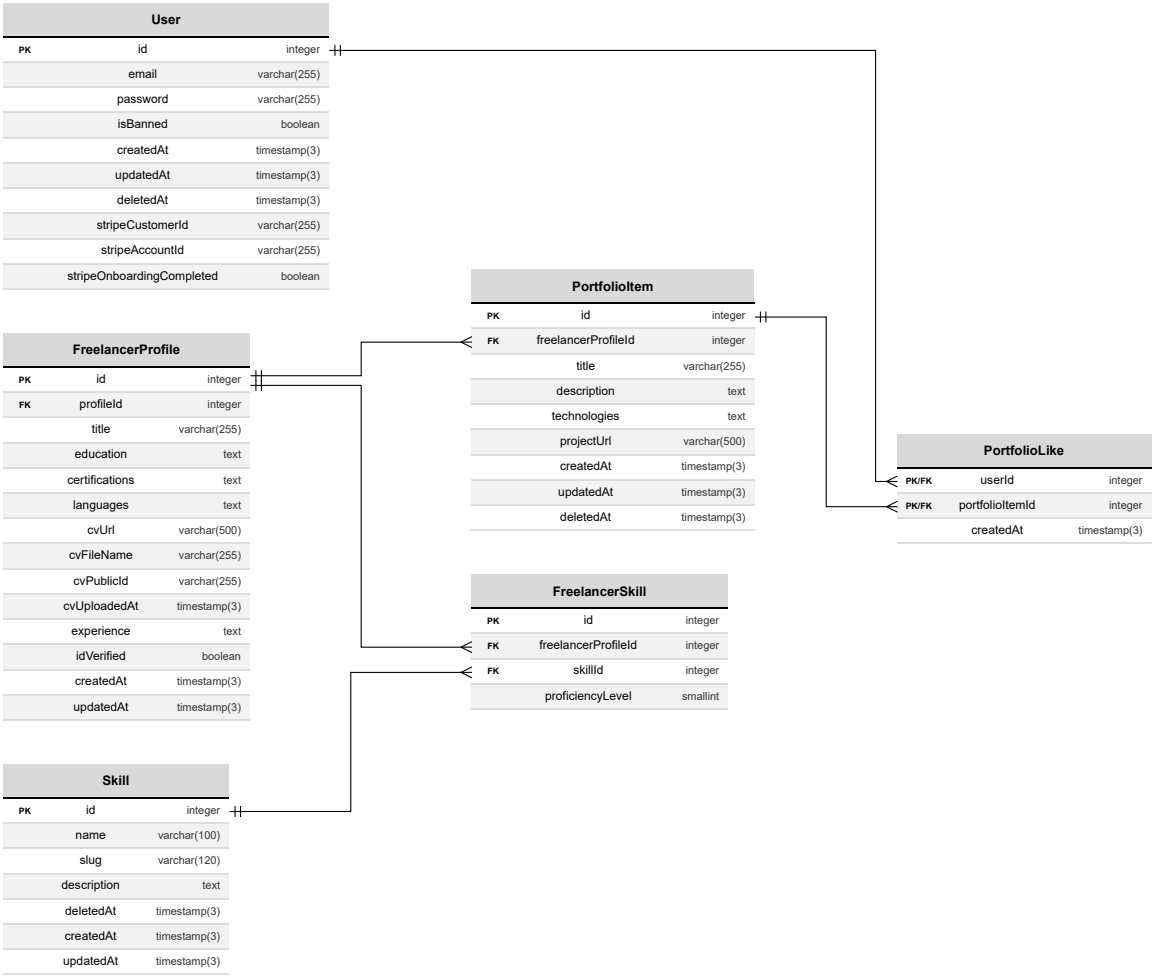

Figure 9: ERD6 — Portfolio & Skills

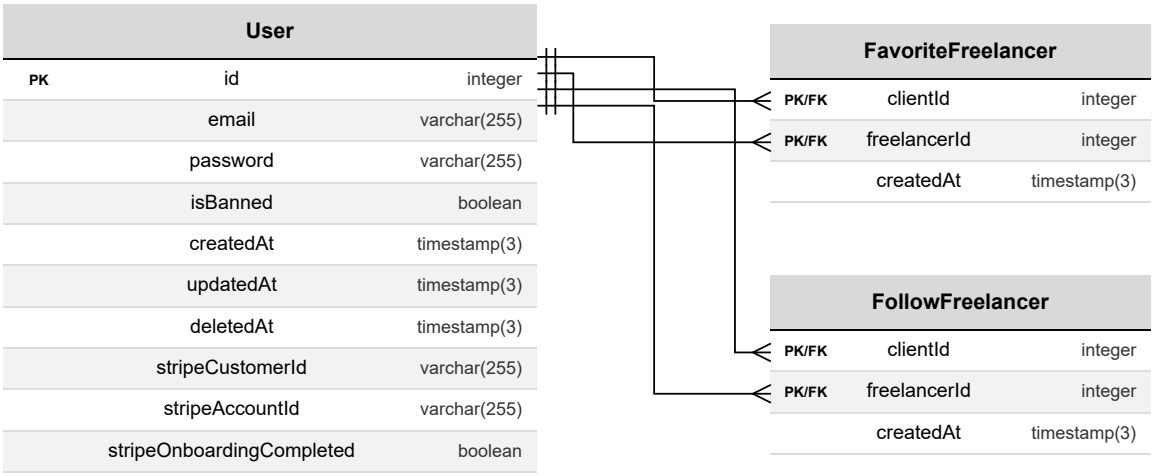

Figure 10: ERD7 — Favorites & Follows

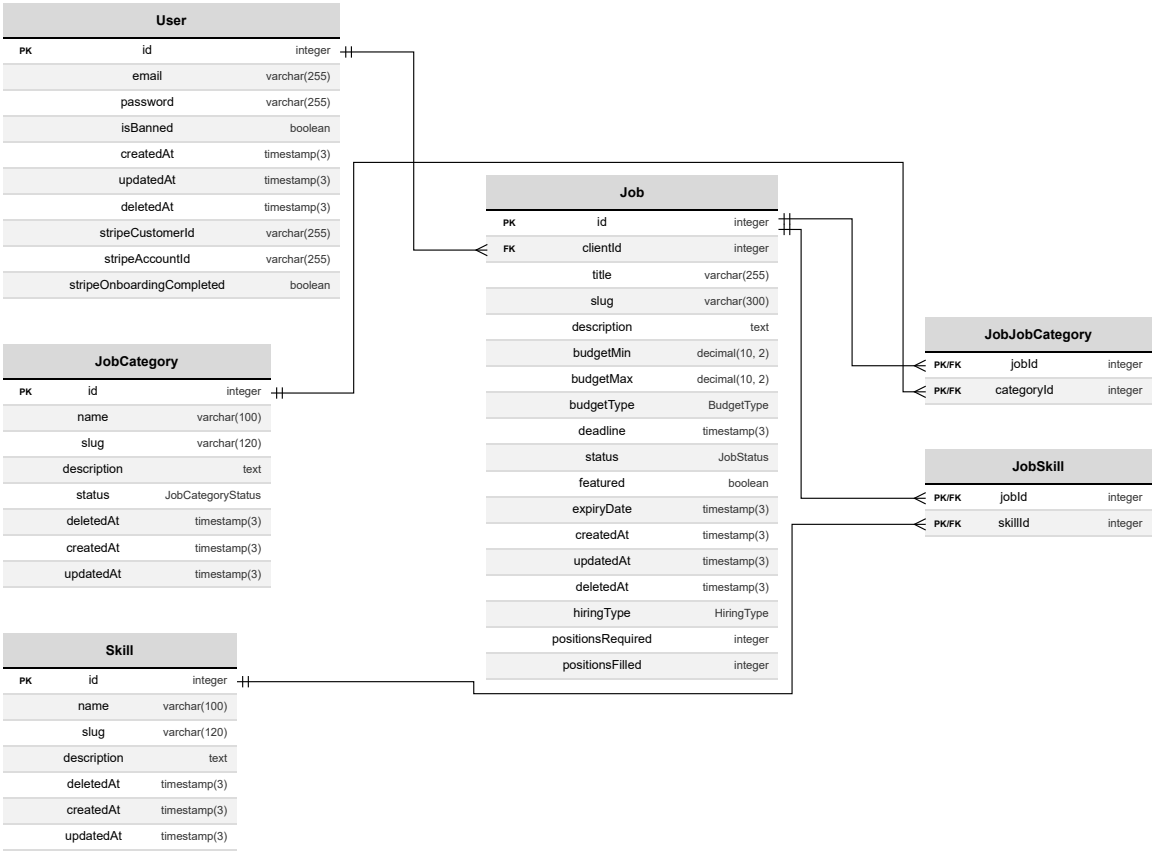

Figure 11: ERD8 — Job Posting

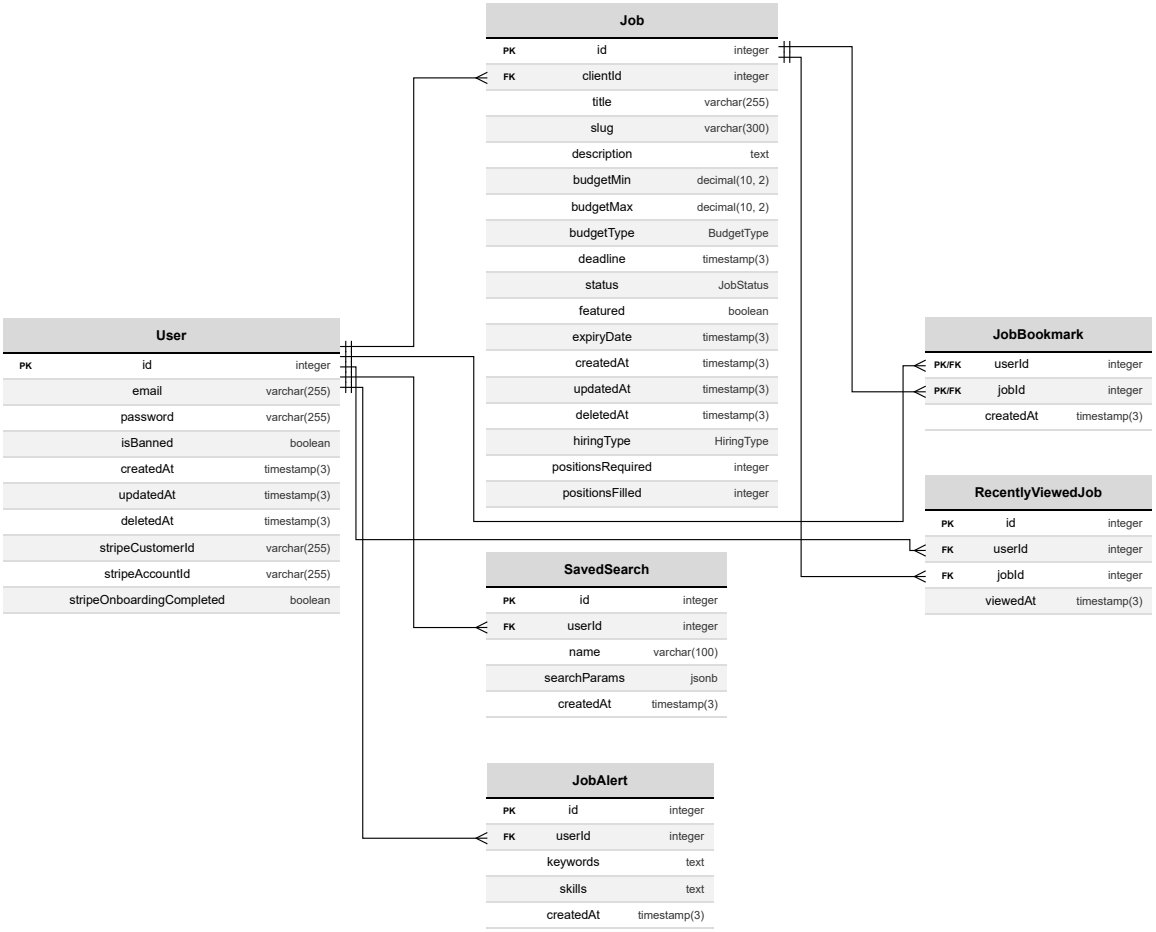

Figure 12: ERD9 — Job Discovery & Alerts

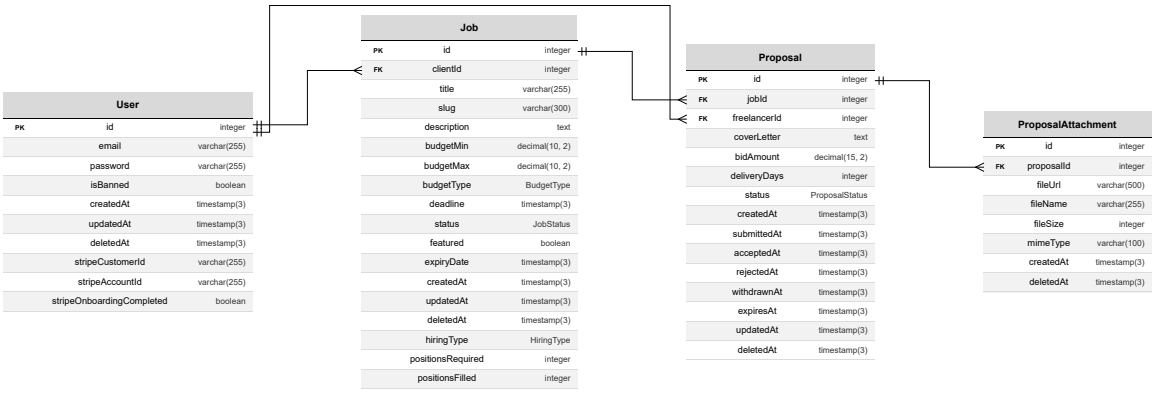

Figure 13: ERD10 — Proposal

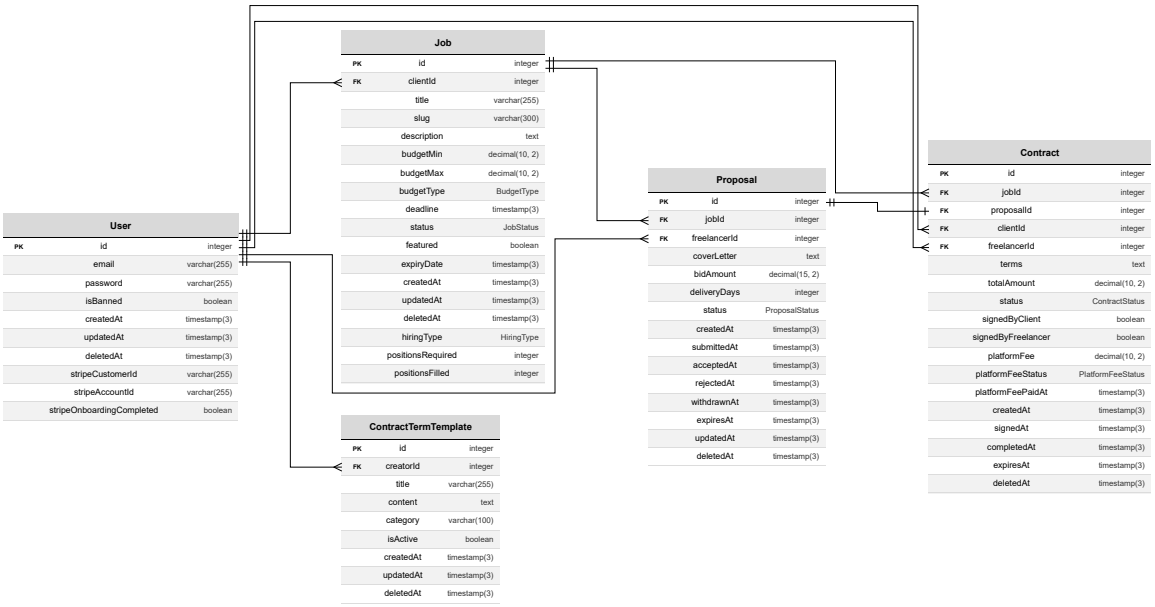

Figure 14: ERD11 — Contract

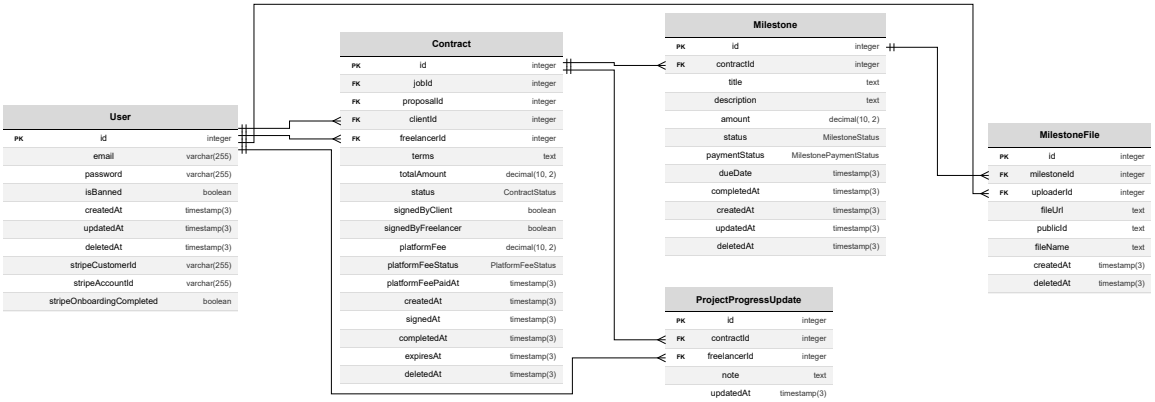

Figure 15: ERD12 — Milestone & Project Progress

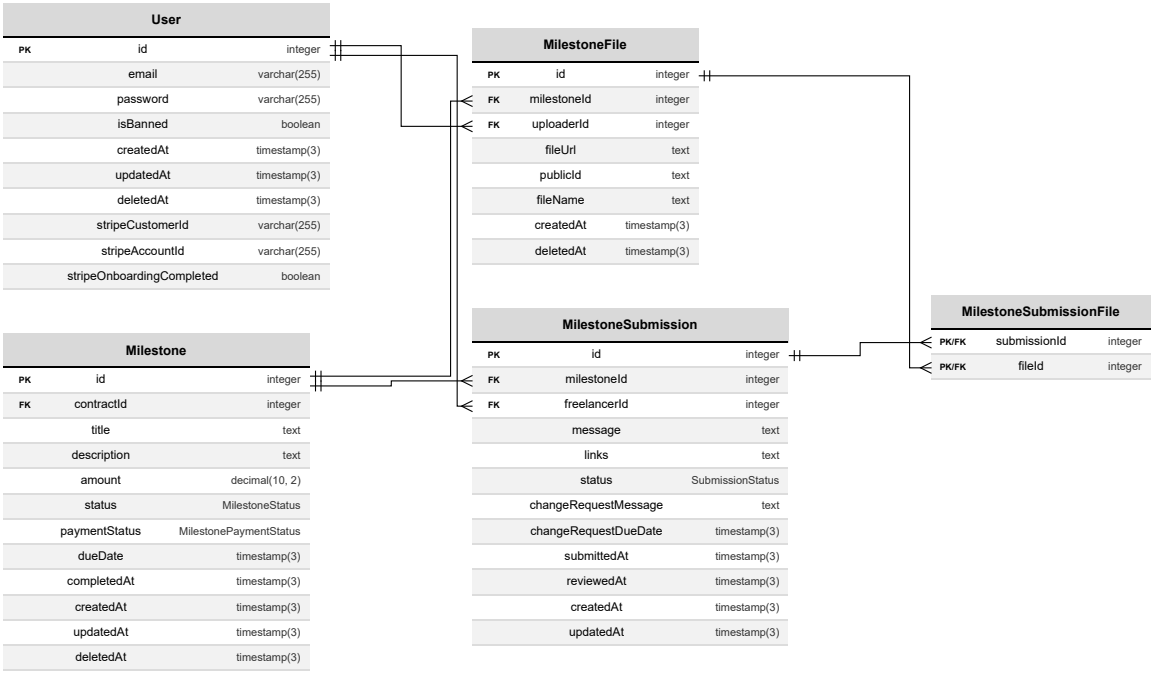

Figure 16: ERD13 — Milestone Submission

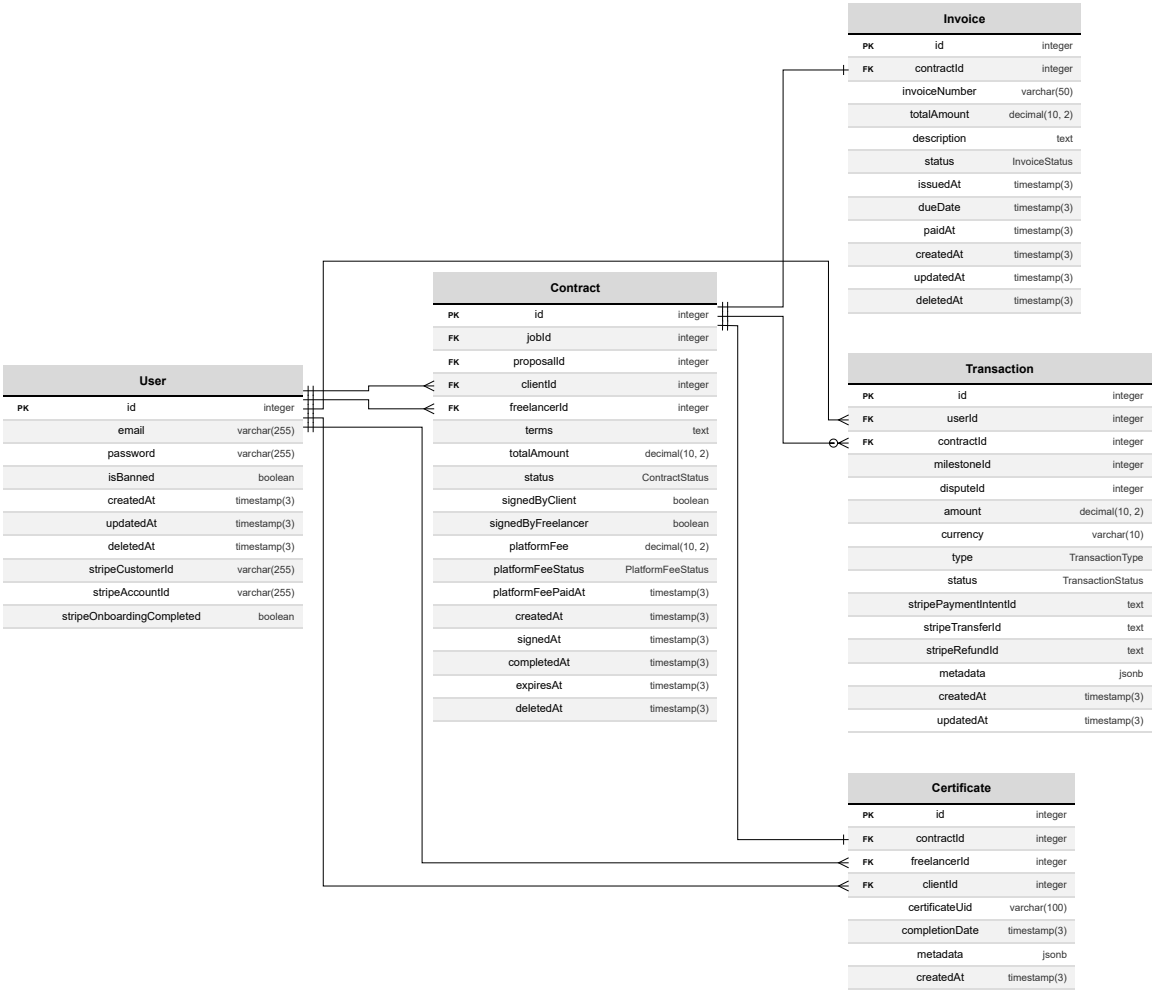

Figure 17: ERD14 — Escrow, Payments & Certificates

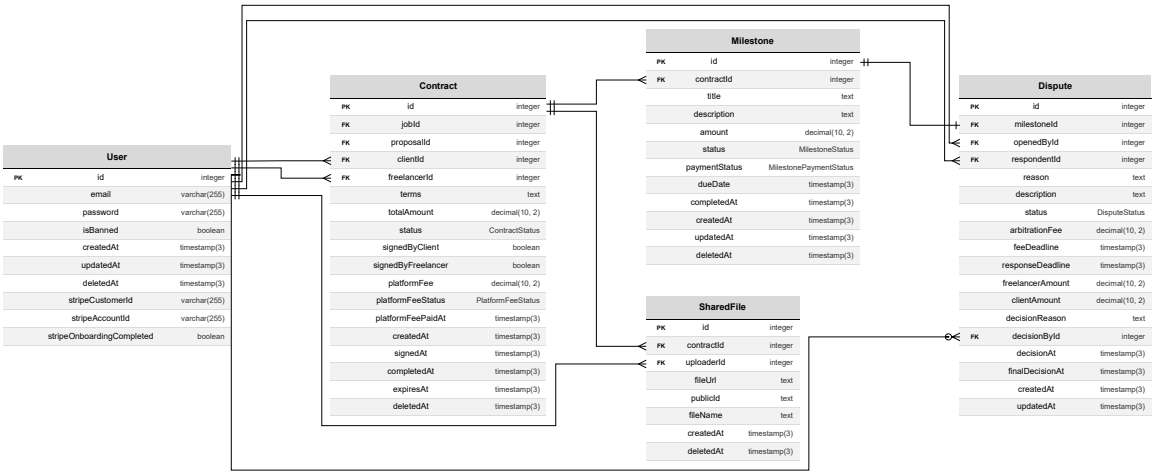

Figure 18: ERD15 — Dispute Core

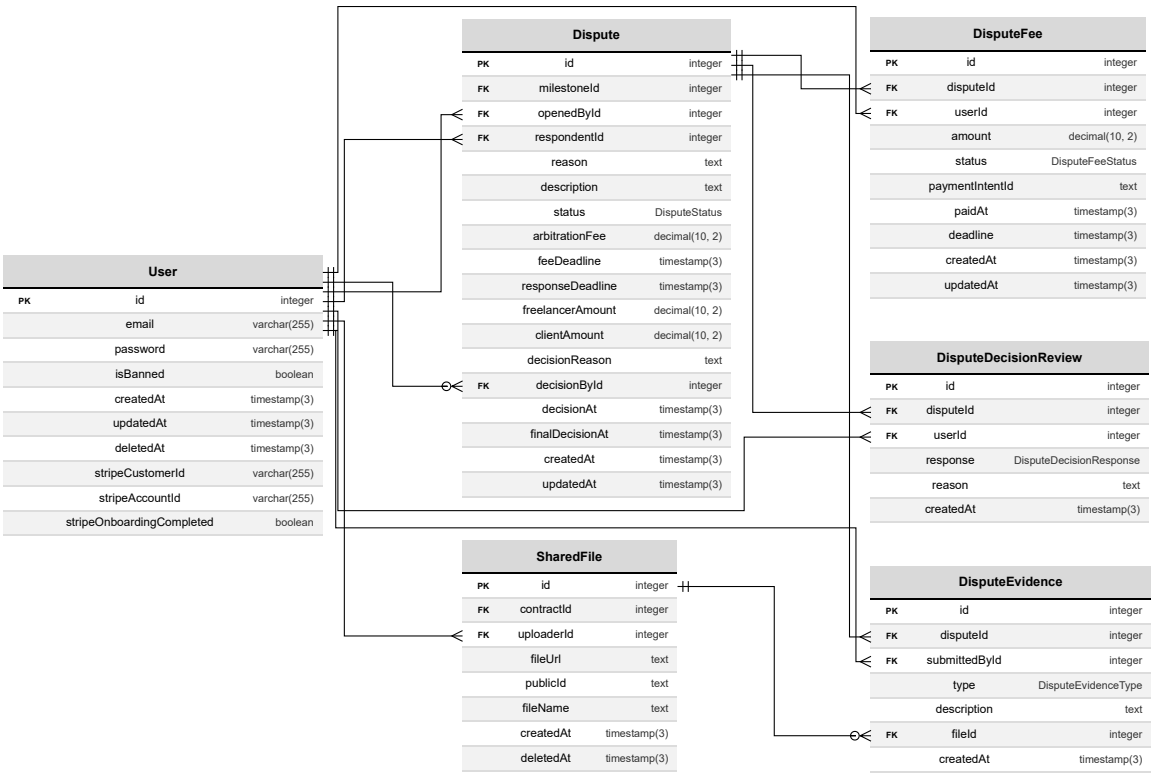

Figure 19: ERD16 — Dispute Resolution

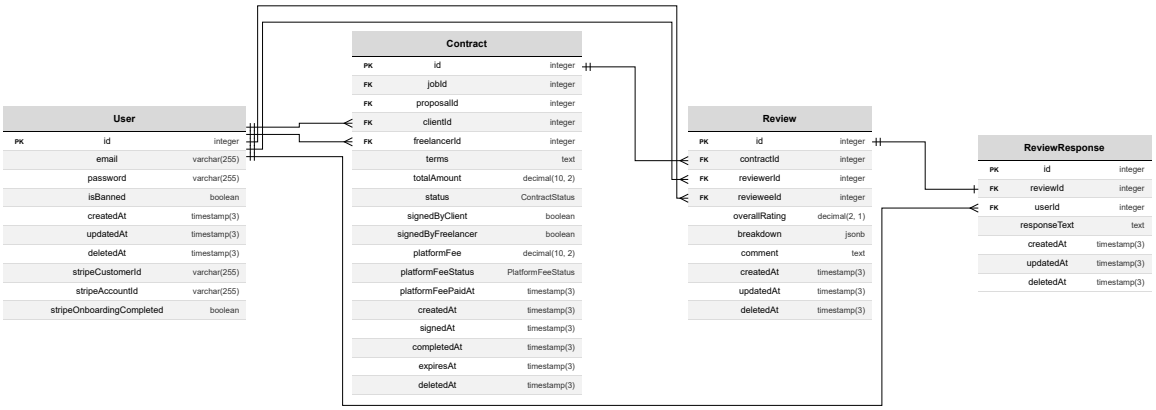

Figure 20: ERD17 — Reviews & Reputation

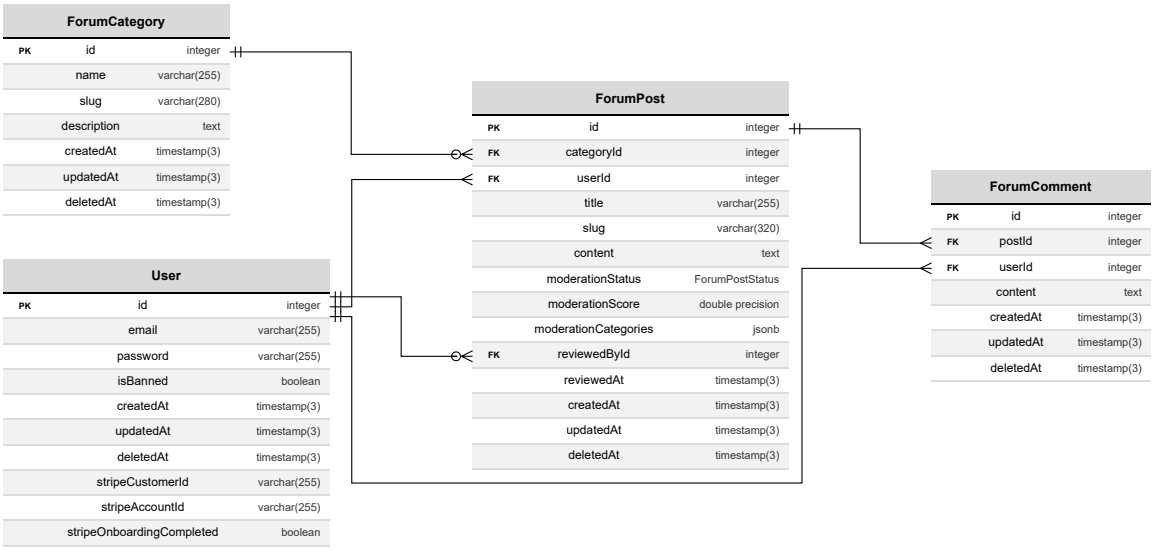

Figure 21: ERD18 — Forum Posts & Comments

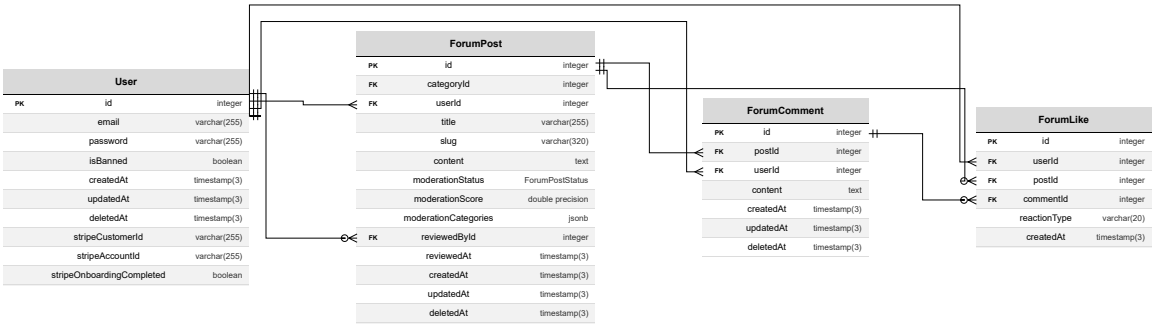

Figure 22: ERD19 — Forum Likes

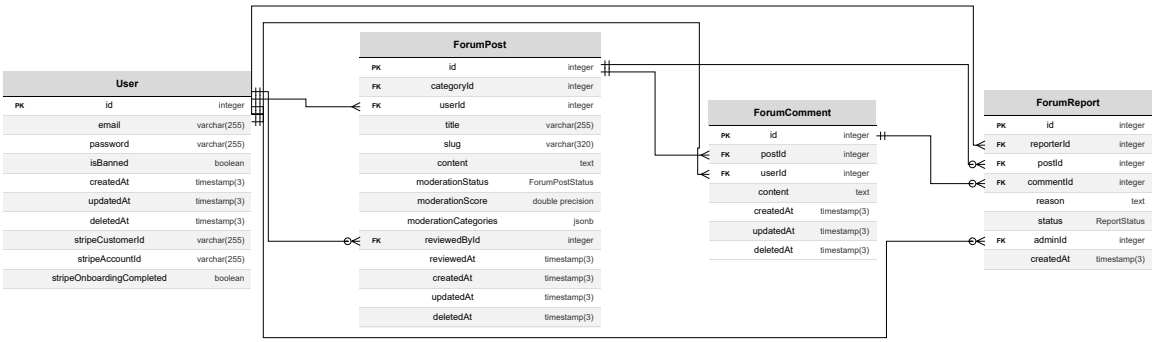

Figure 23: ERD20 — Forum Reports & Moderation

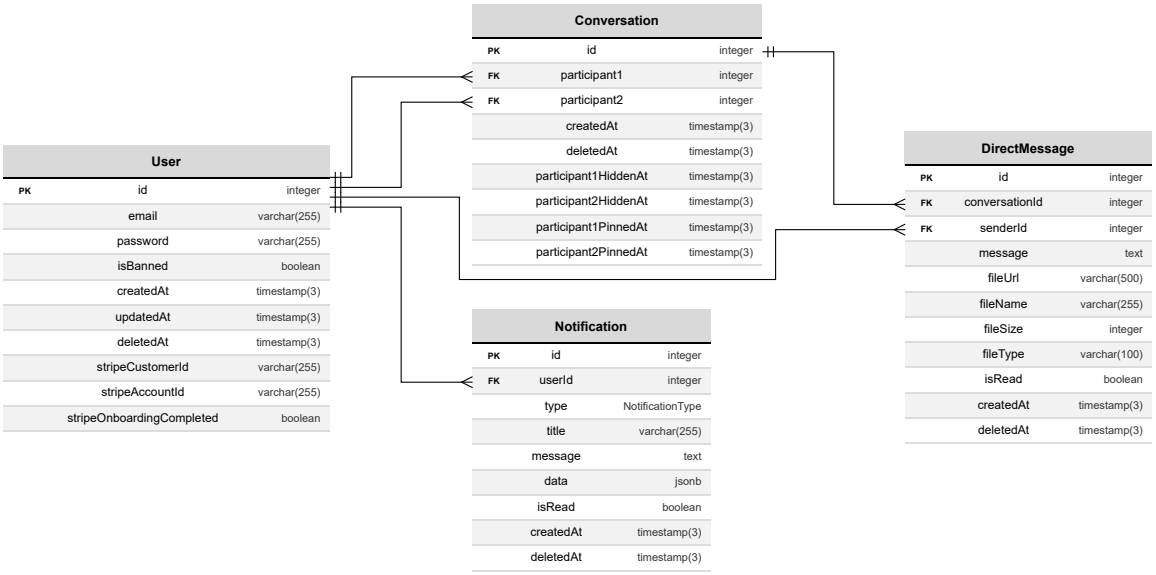

Figure 24: ERD21 — Messaging & Notifications

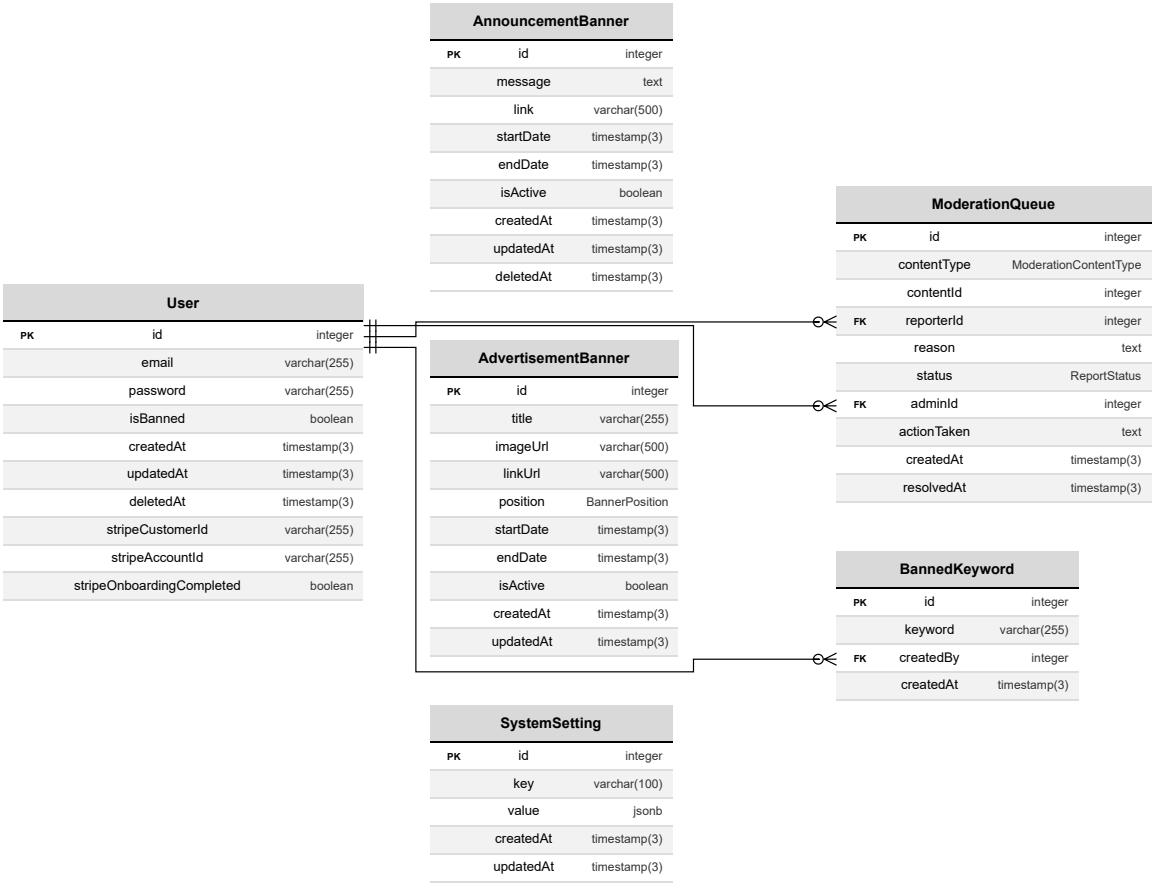

Figure 25: ERD22 — Administration & System

3. Detailed Design

This chapter presents the detailed design of every feature of the platform. For each use case, the architectural structure is specified through its dedicated Class Diagram followed by its behavioral Sequence Diagram, illustrating both static dependencies and runtime interactions.

3.1 Verify JWT

Verify JWT is the shared security guard protecting API endpoints across all modules. Every incoming HTTP request with a Bearer token is intercepted, validated for cryptographic signature and expiration time, and verified against the database to guarantee the associated session remains active and has not been revoked.

3.1.1 UC-0 – Verify JWT

a. Class diagram

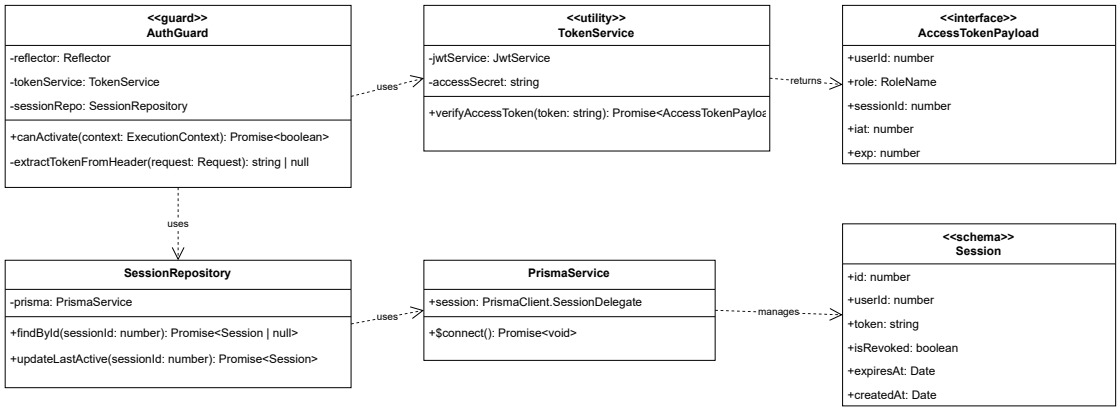

Figure 26: Class diagram of UC-0 Verify JWT

b. Sequence diagram

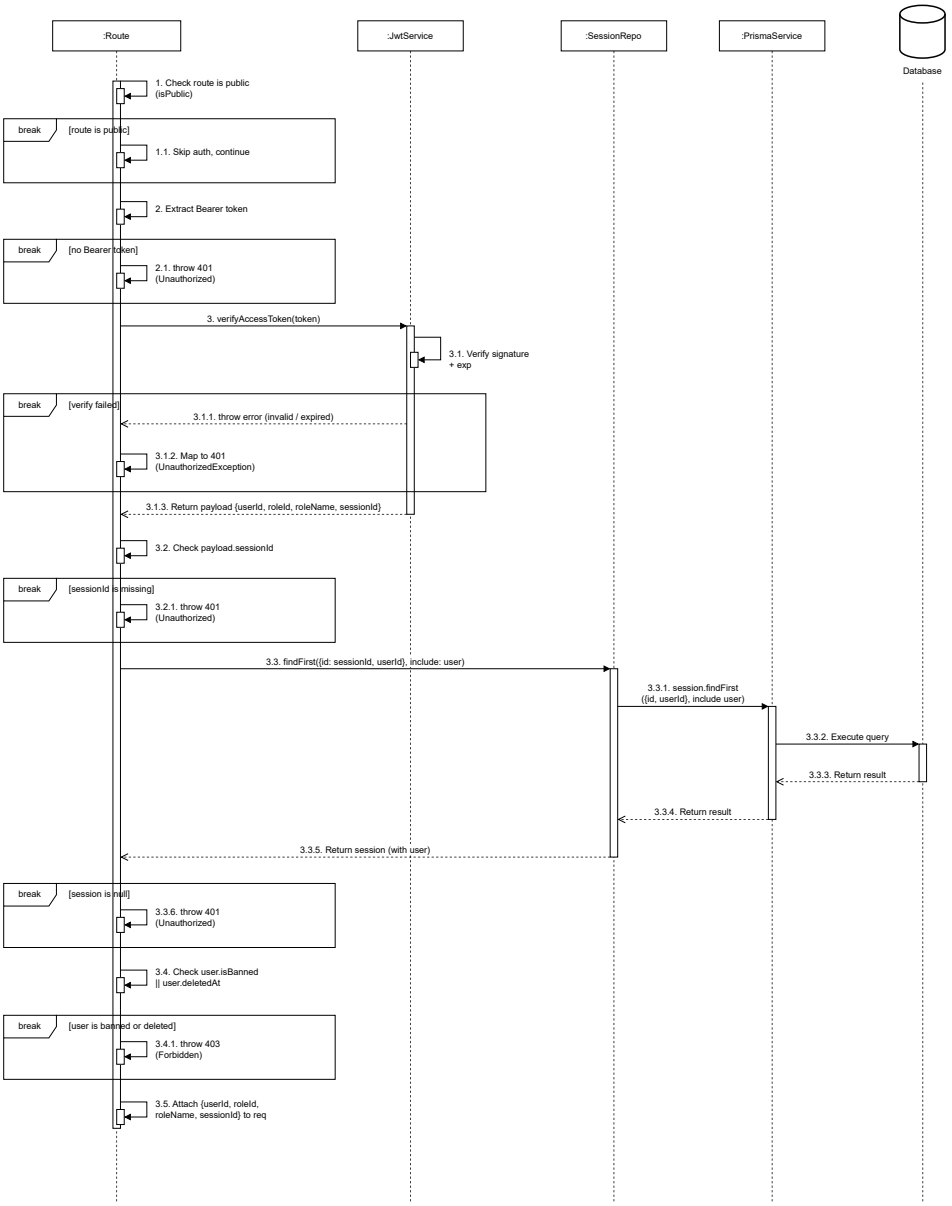

Figure 27: Sequence diagram of UC-0 Verify JWT

3.2 Authentication

The authentication module covers account registration, email/password and Google OAuth login, logout, password recovery, and the JWT/session flow shared by all protected endpoints.

3.2.1 UC-01 – Login with Google

a. Class diagram

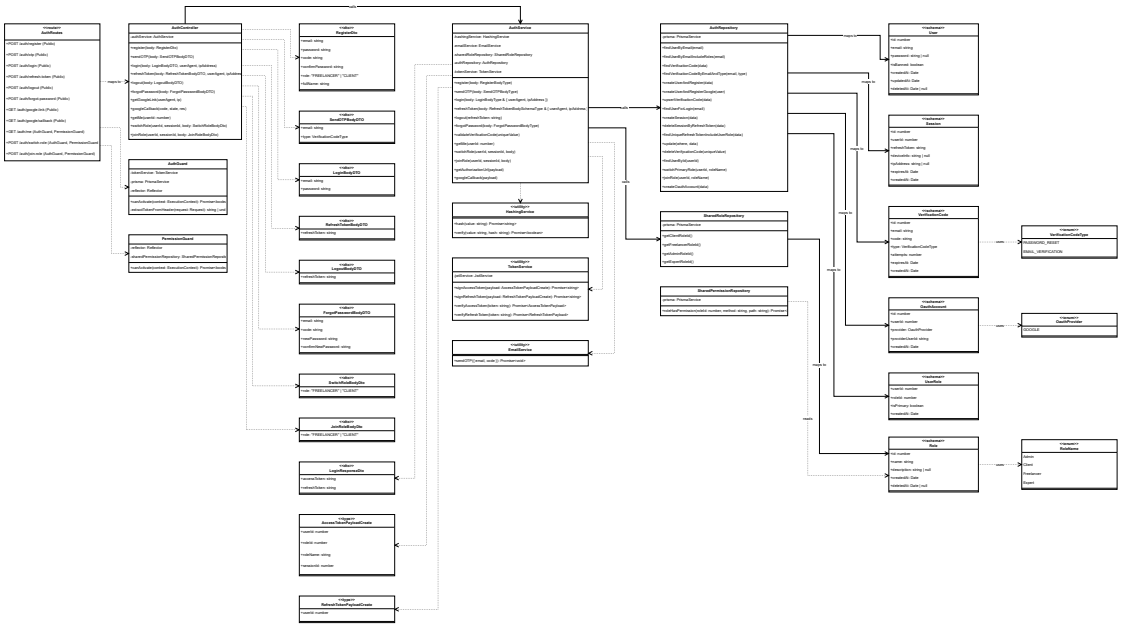

Figure 28: Class diagram of UC-01 Login with Google

b. Sequence diagram

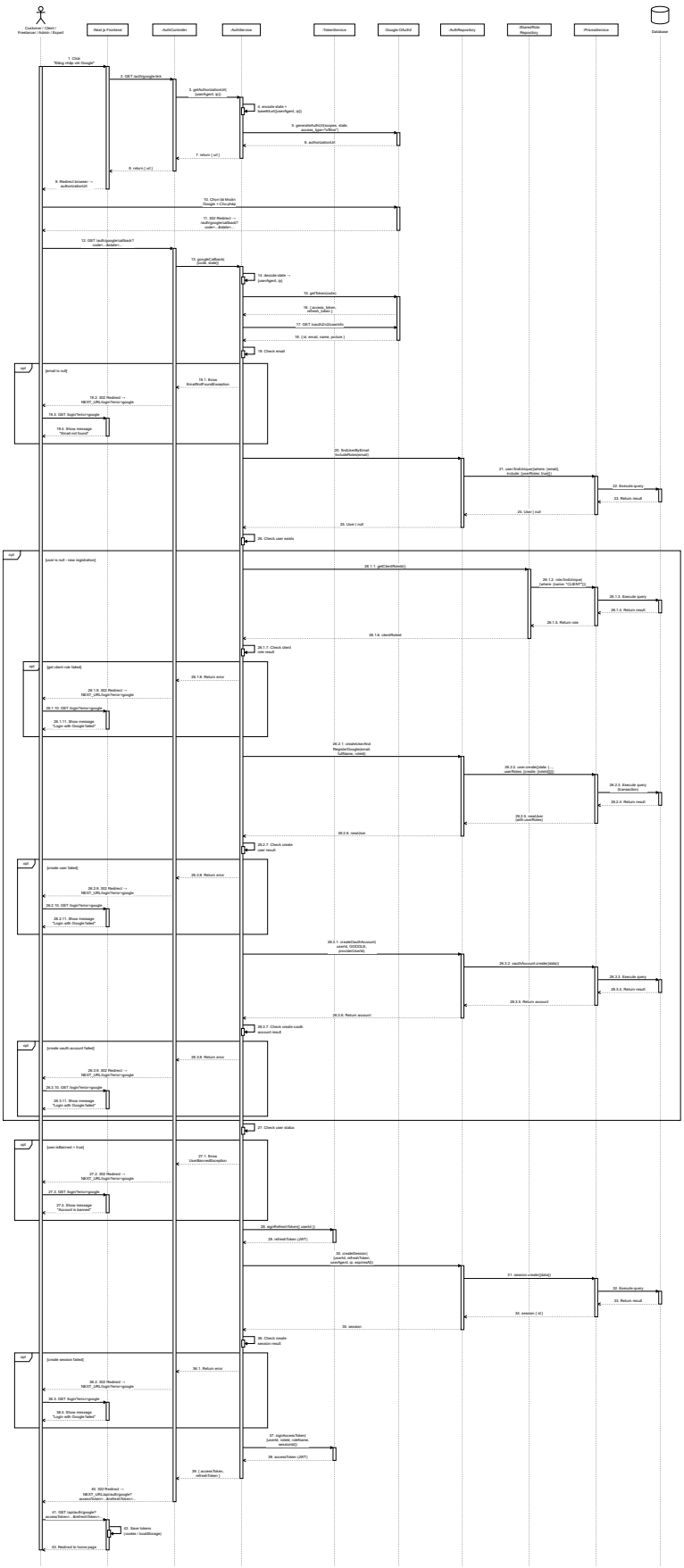

Figure 29: Sequence diagram of UC-01 Login with Google

3.2.2 UC-02 – Login

a. Class diagram

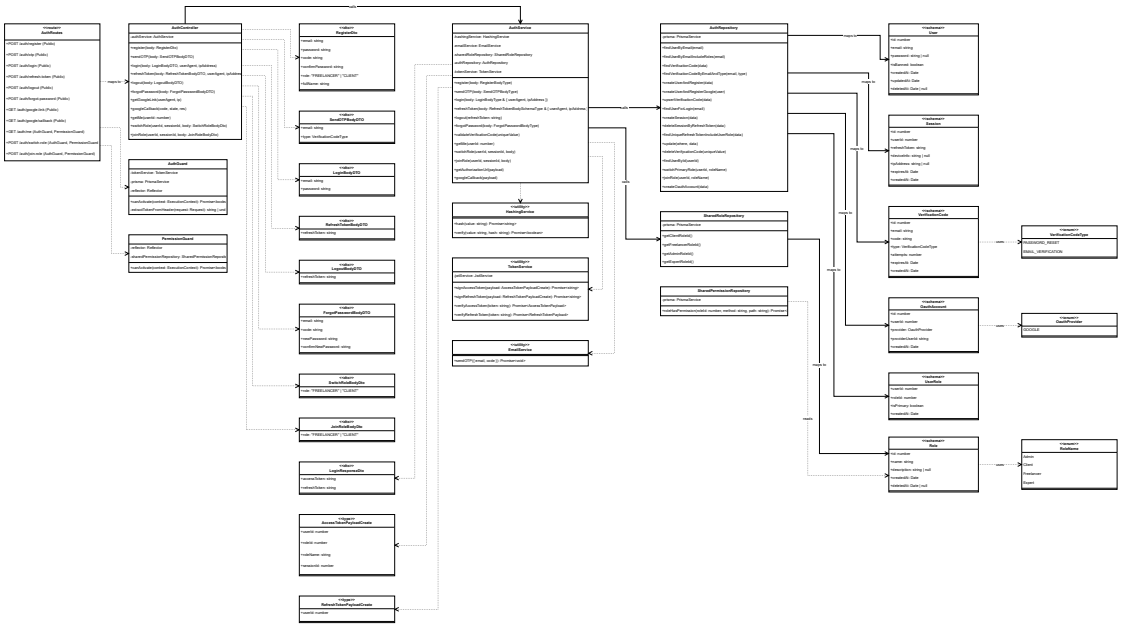

Figure 30: Class diagram of UC-02 Login

b. Sequence diagram

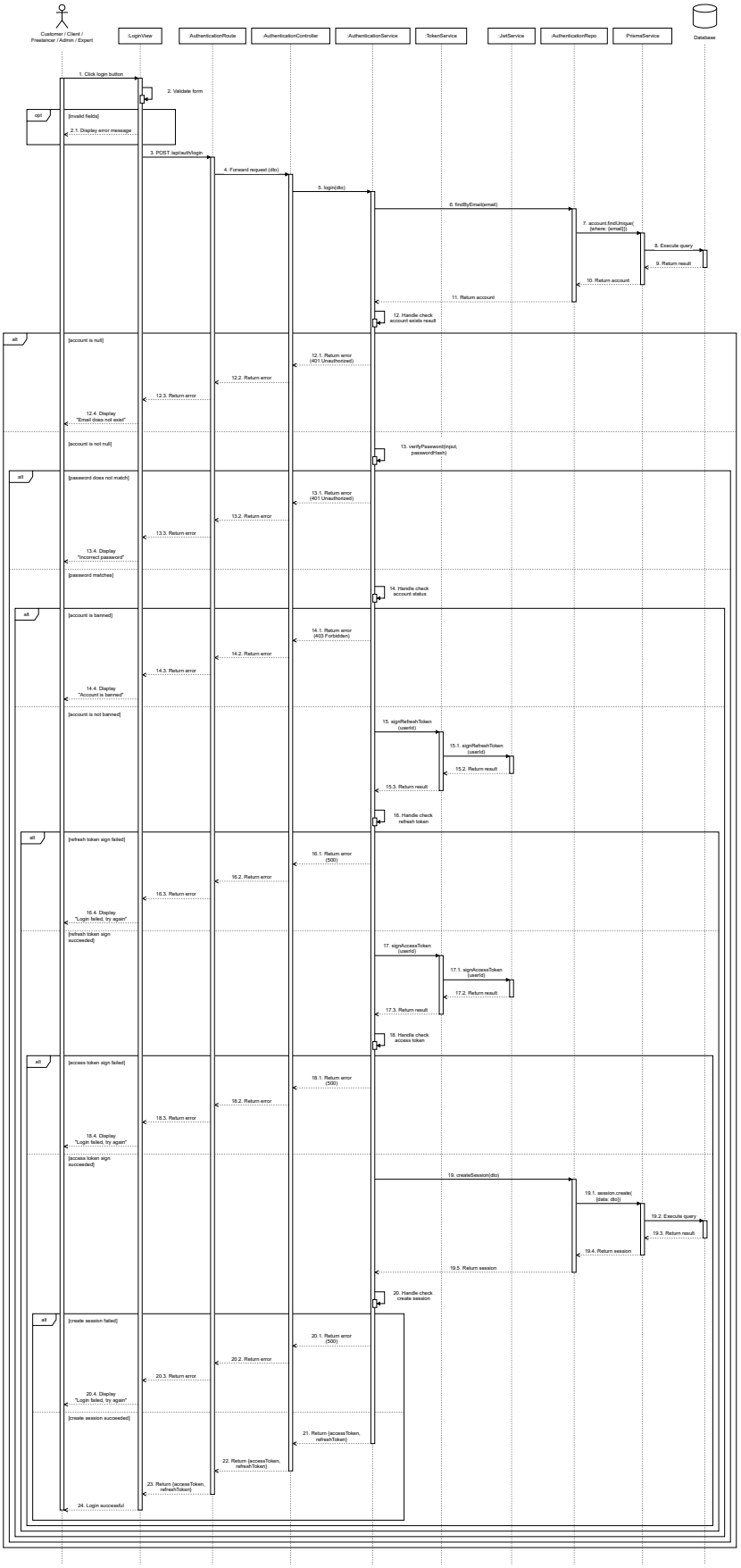

Figure 31: Sequence diagram of UC-02 Login

3.2.3 UC-03 – Logout

a. Class diagram

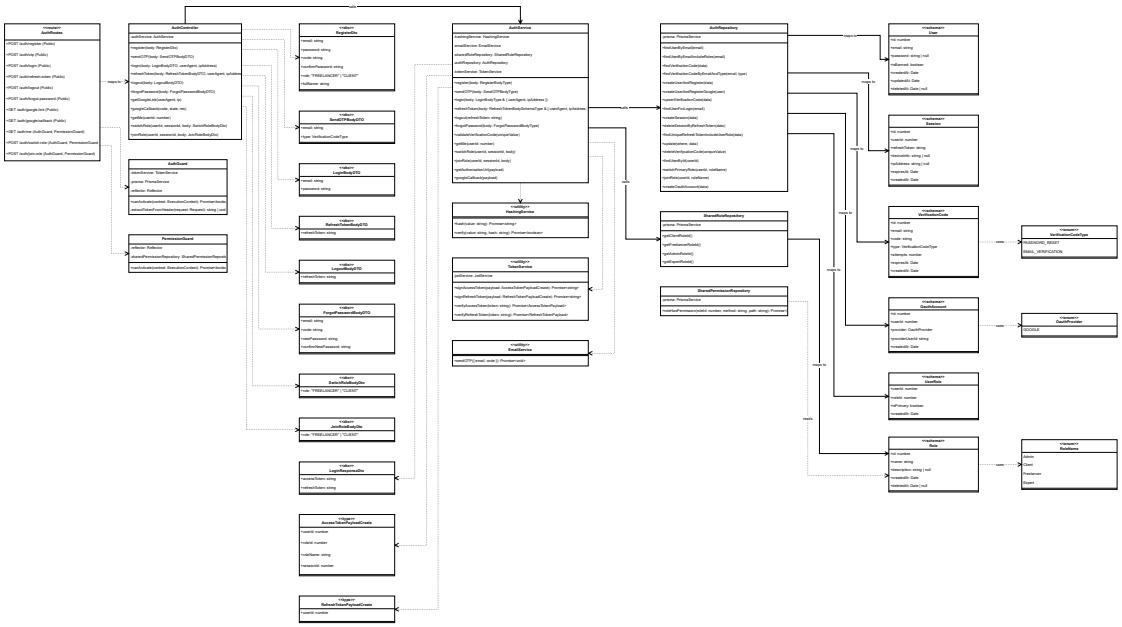

Figure 32: Class diagram of UC-03 Logout

b. Sequence diagram

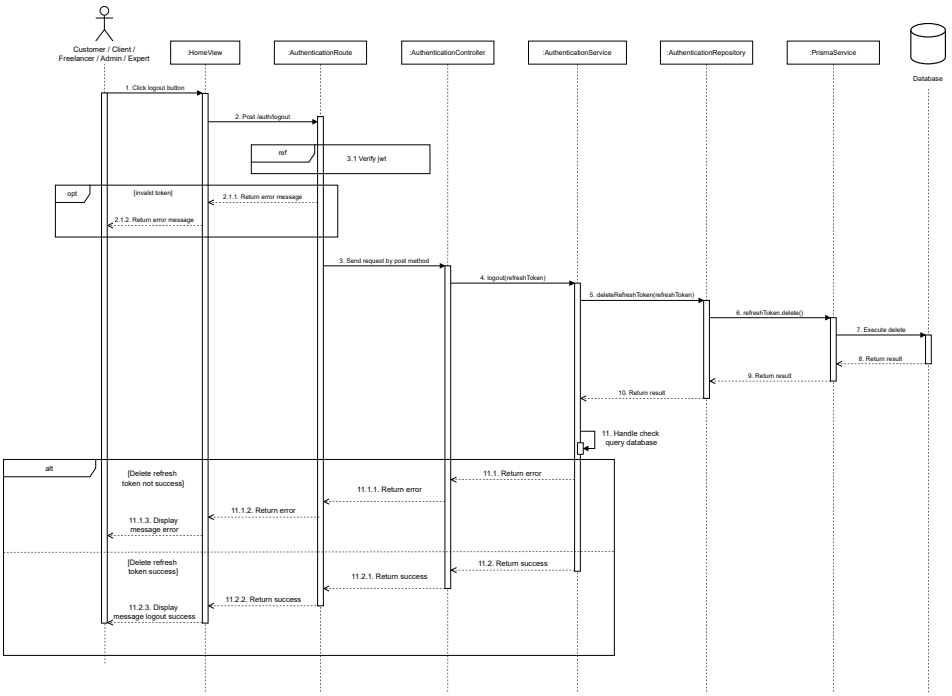

Figure 33: Sequence diagram of UC-03 Logout

3.2.4 UC-04 – Register

a. Class diagram

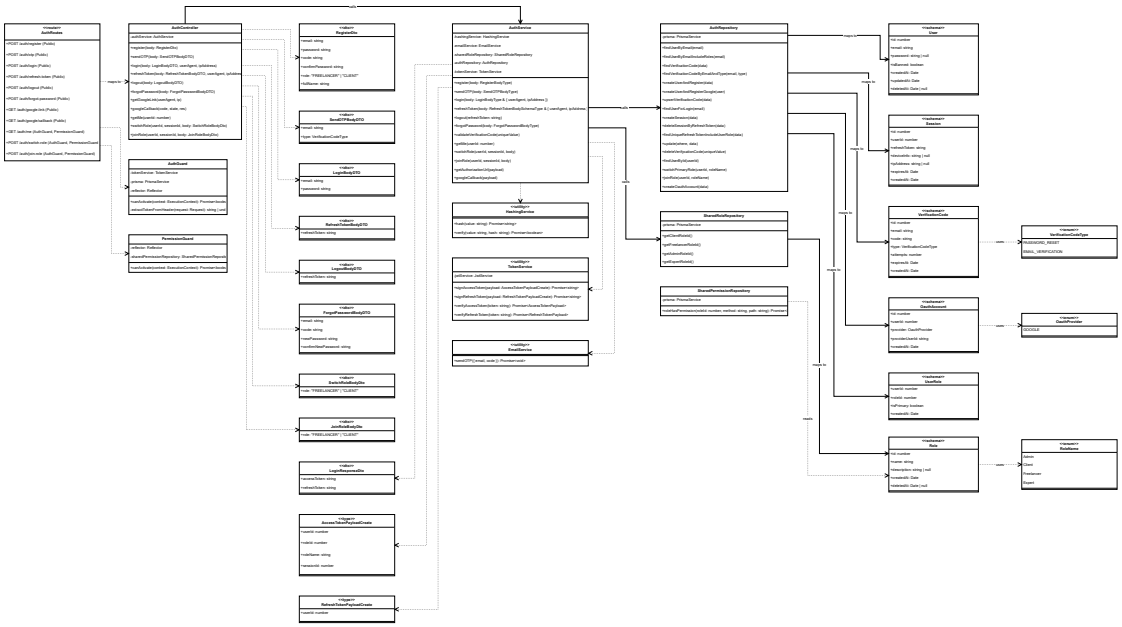

Figure 34: Class diagram of UC-04 Register

b. Sequence diagram

Figure 35: Sequence diagram of UC-04 Register

3.2.5 UC-05 – Forgot Password

a. Class diagram

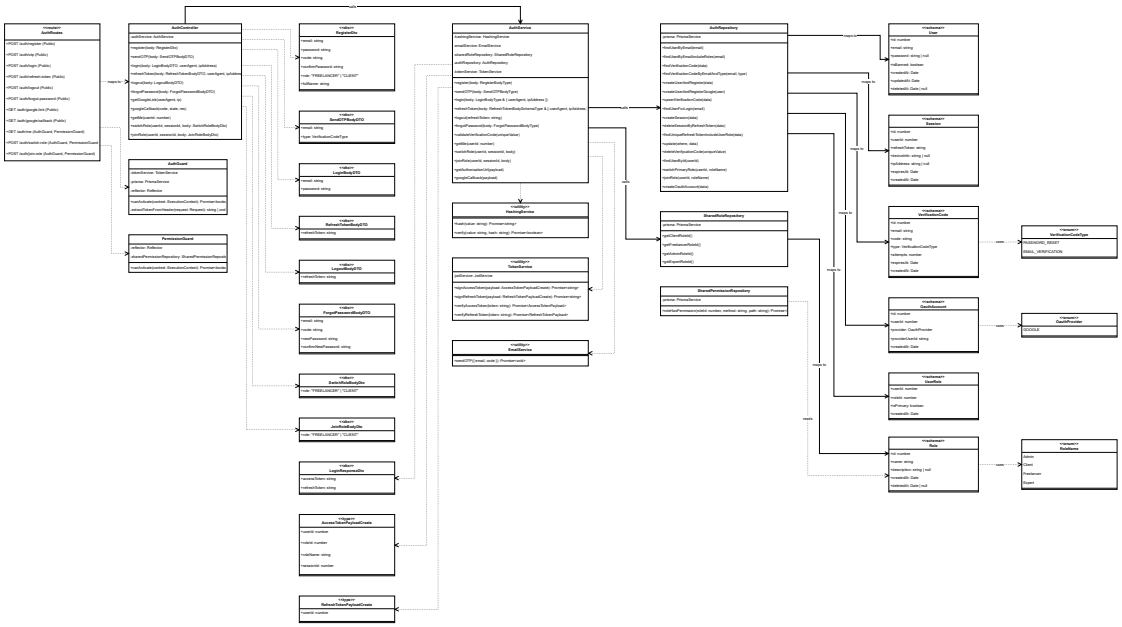

Figure 36: Class diagram of UC-05 Forgot Password

b. Sequence diagram

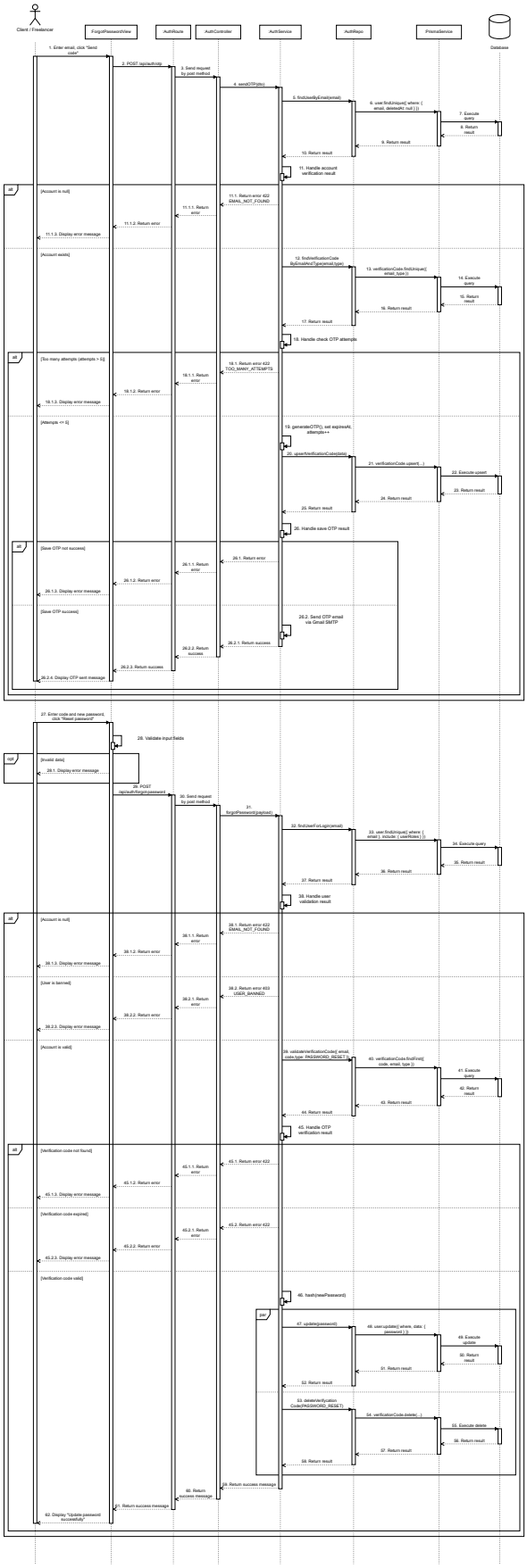

Figure 37: Sequence diagram of UC-05 Forgot Password

3.3 Role Management

Role Management lets an administrator inspect the role catalogue, maintain role metadata, and replace the permission set assigned to a role.

3.3.1 UC-06.01 – View Role List

a. Class diagram

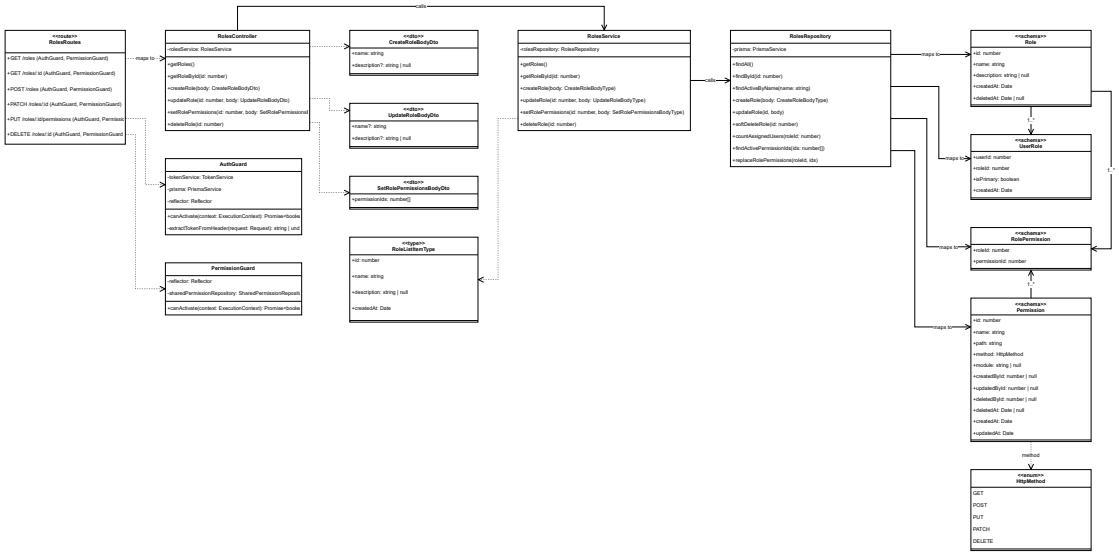

Figure 38: Class diagram of UC-06.01 View Role List

b. Sequence diagram

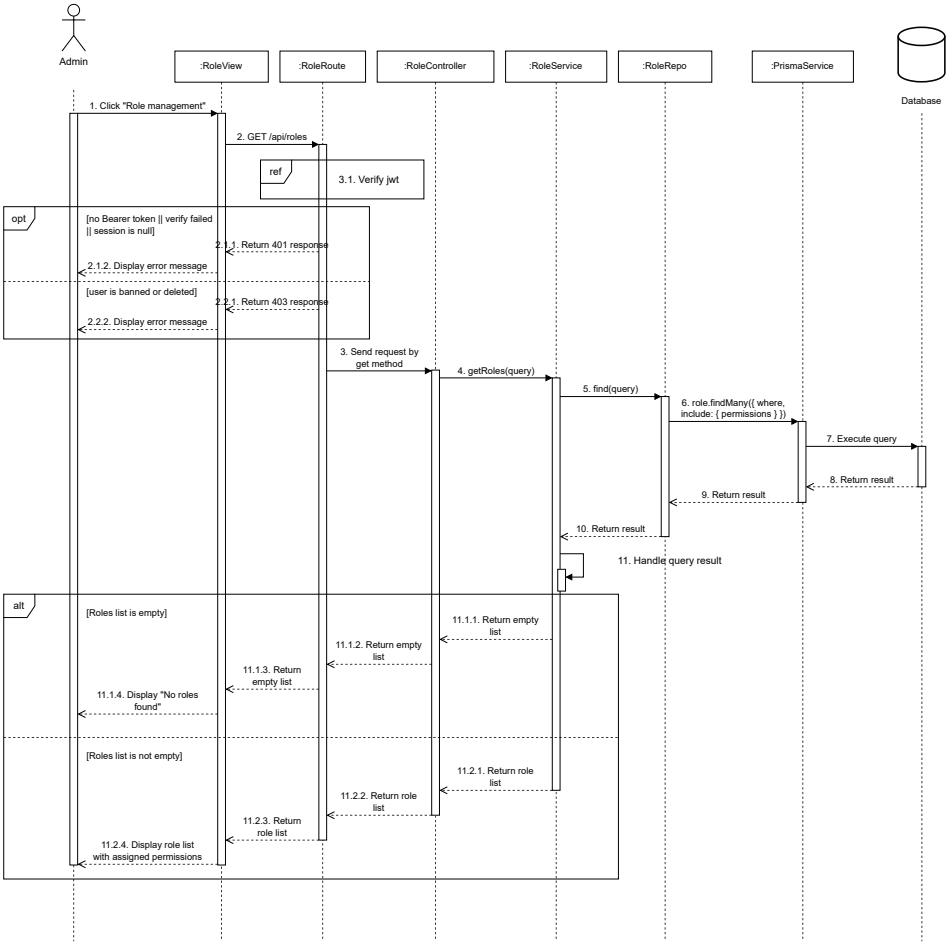

Figure 39: Sequence diagram of UC-06.01 View Role List

3.3.2 UC-06.02 – View Role Detail

a. Class diagram

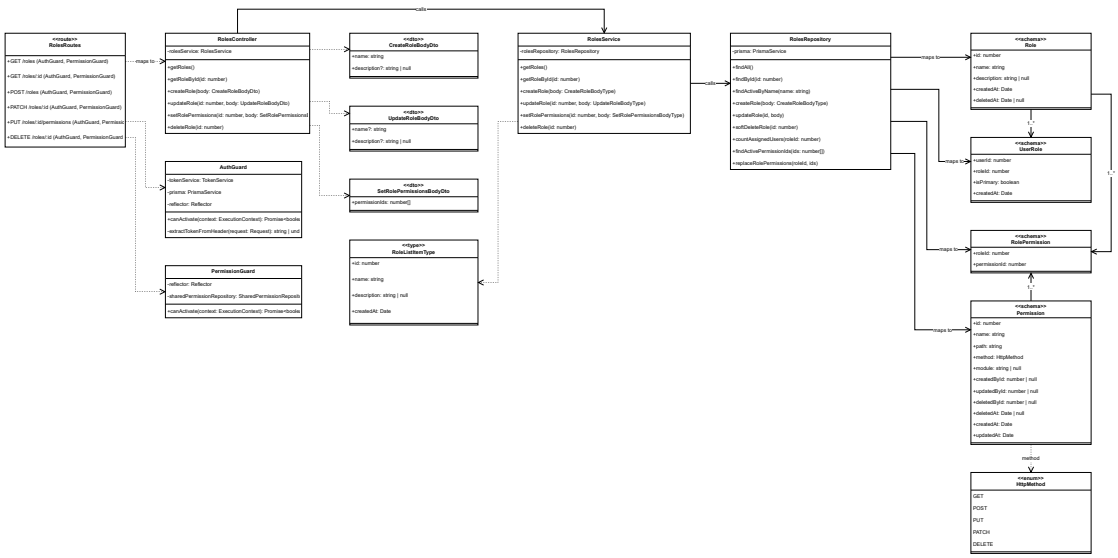

Figure 40: Class diagram of UC-06.02 View Role Detail

b. Sequence diagram

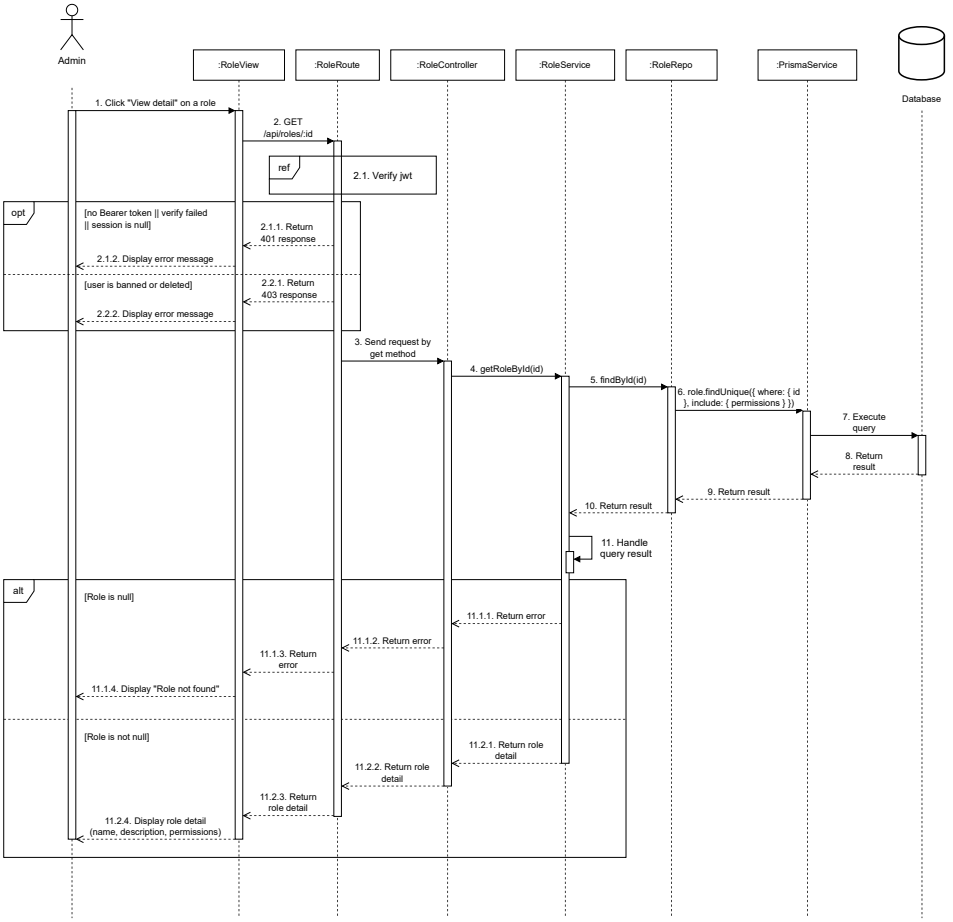

Figure 41: Sequence diagram of UC-06.02 View Role Detail

3.3.3 UC-06.03 – Create Role

a. Class diagram

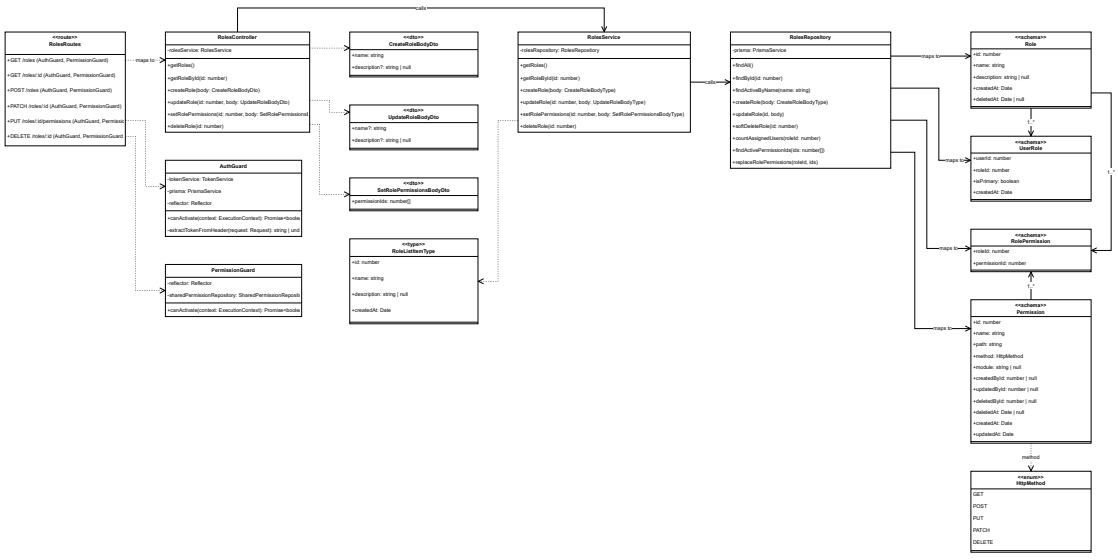

Figure 42: Class diagram of UC-06.03 Create Role

b. Sequence diagram

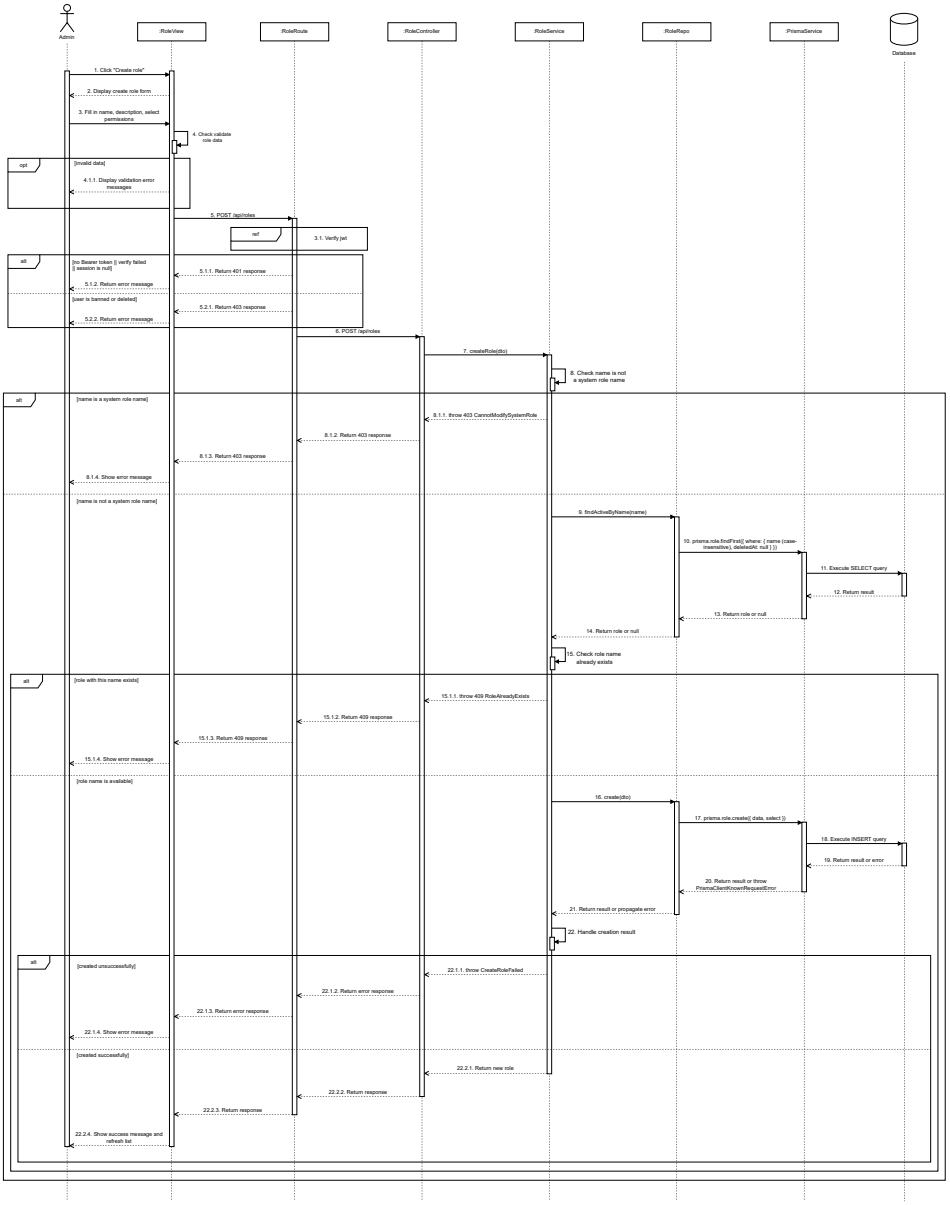

Figure 43: Sequence diagram of UC-06.03 Create Role

3.3.4 UC-06.04 – Update Role

a. Class diagram

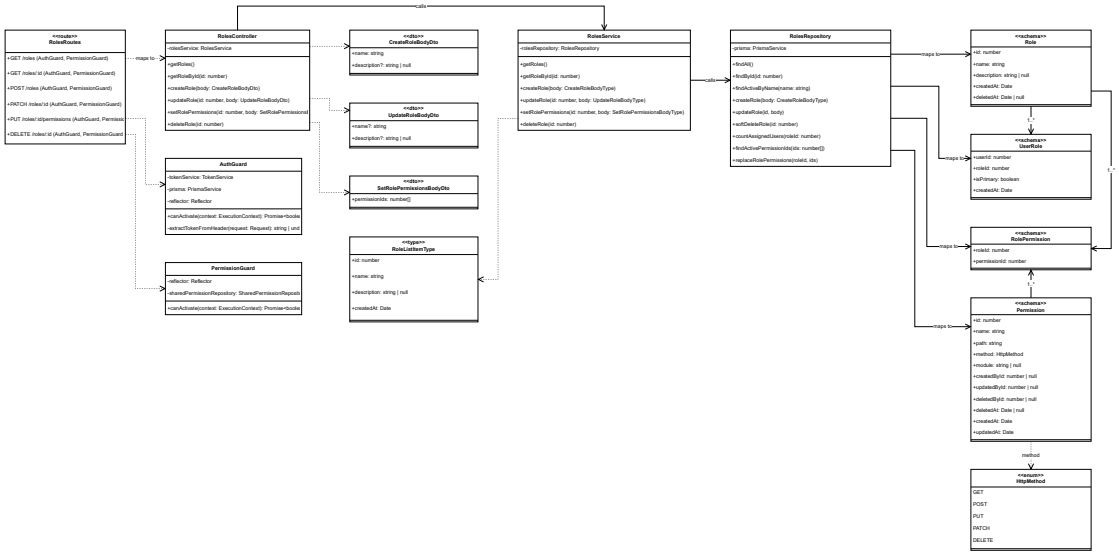

Figure 44: Class diagram of UC-06.04 Update Role

b. Sequence diagram

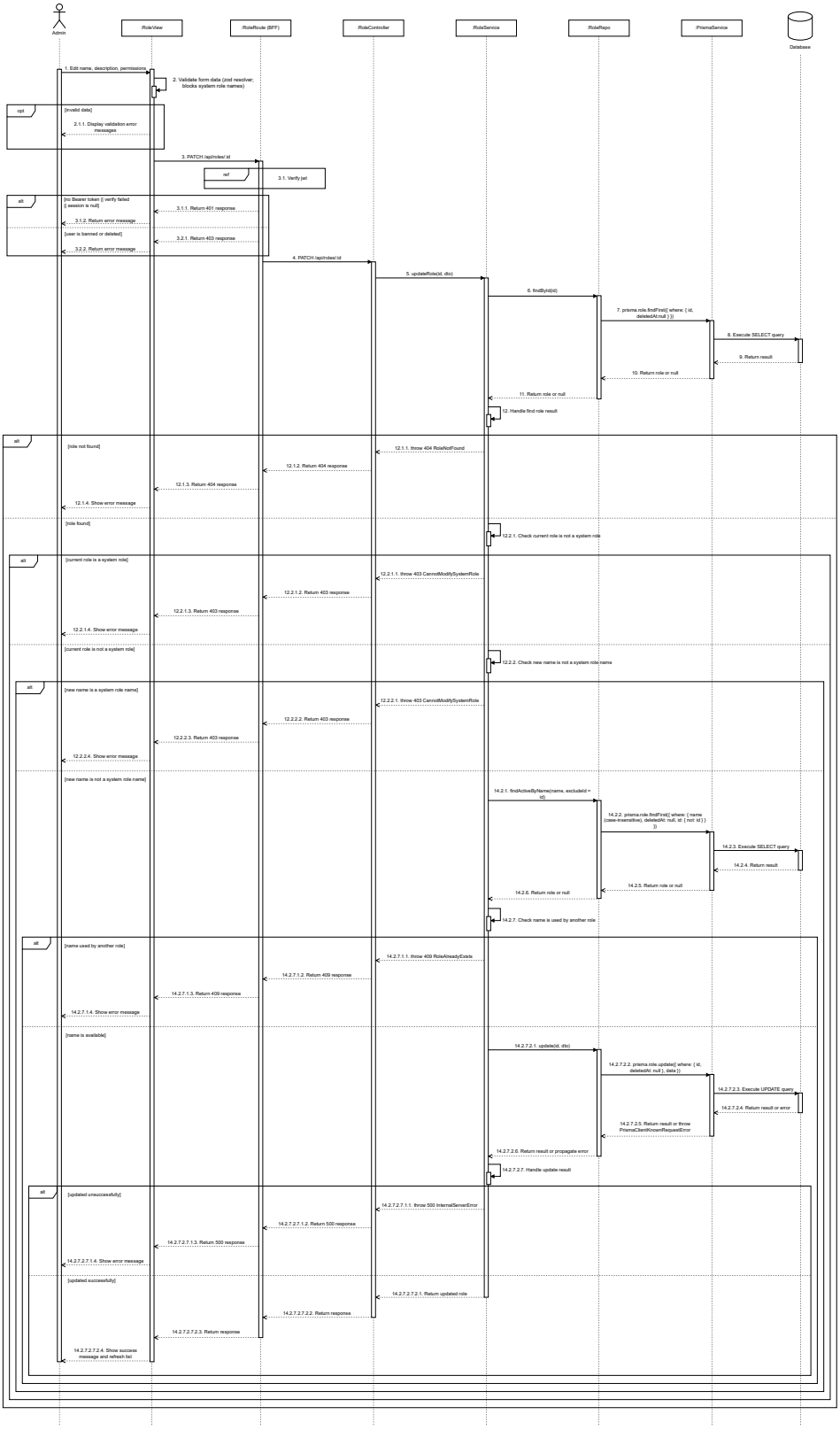

Figure 45: Sequence diagram of UC-06.04 Update Role

3.3.5 UC-06.05 – Delete Role

a. Class diagram

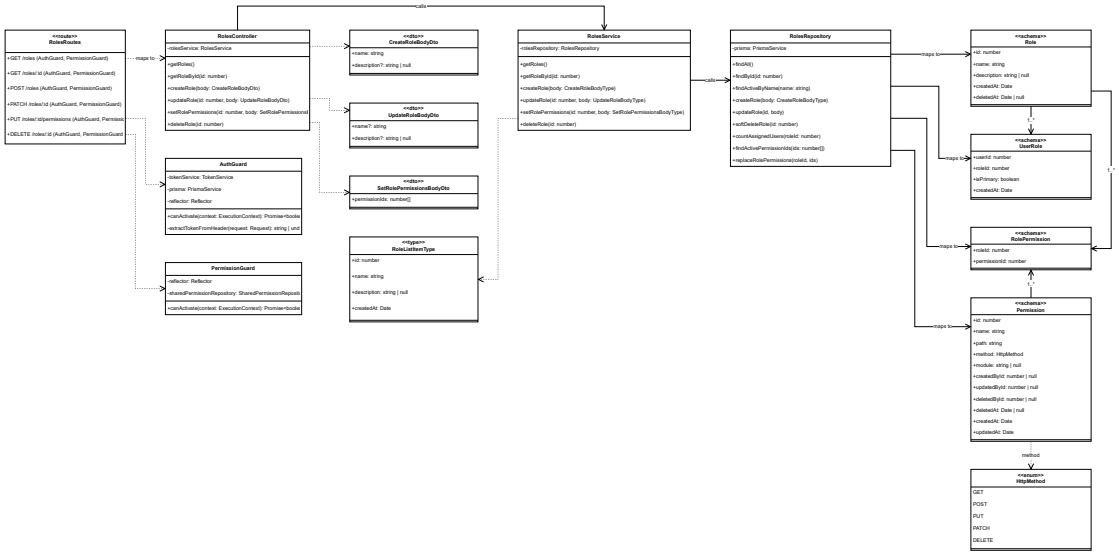

Figure 46: Class diagram of UC-06.05 Delete Role

b. Sequence diagram

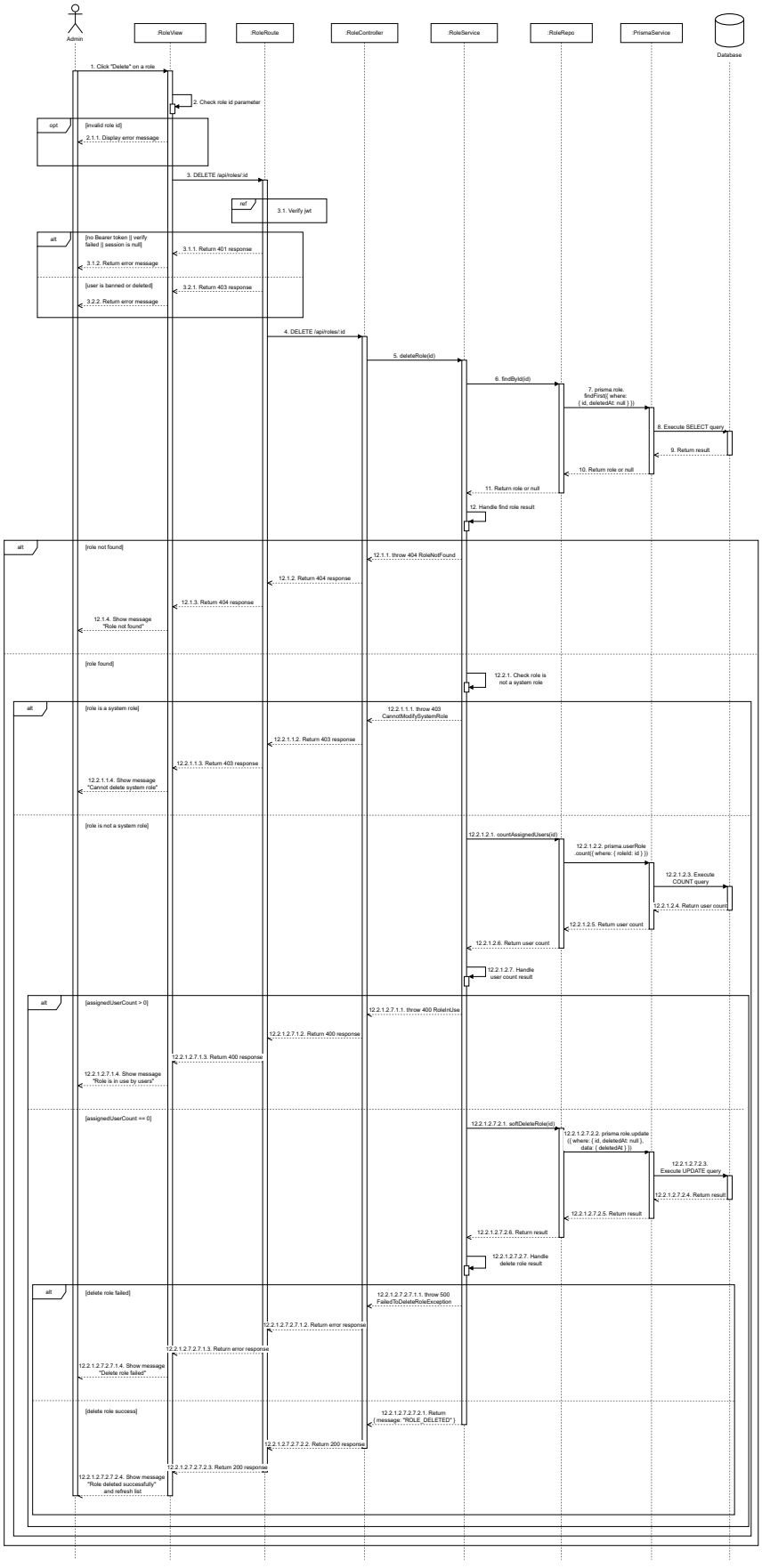

Figure 47: Sequence diagram of UC-06.05 Delete Role

3.3.6 UC-06.06 – Update Role Permissions

a. Class diagram

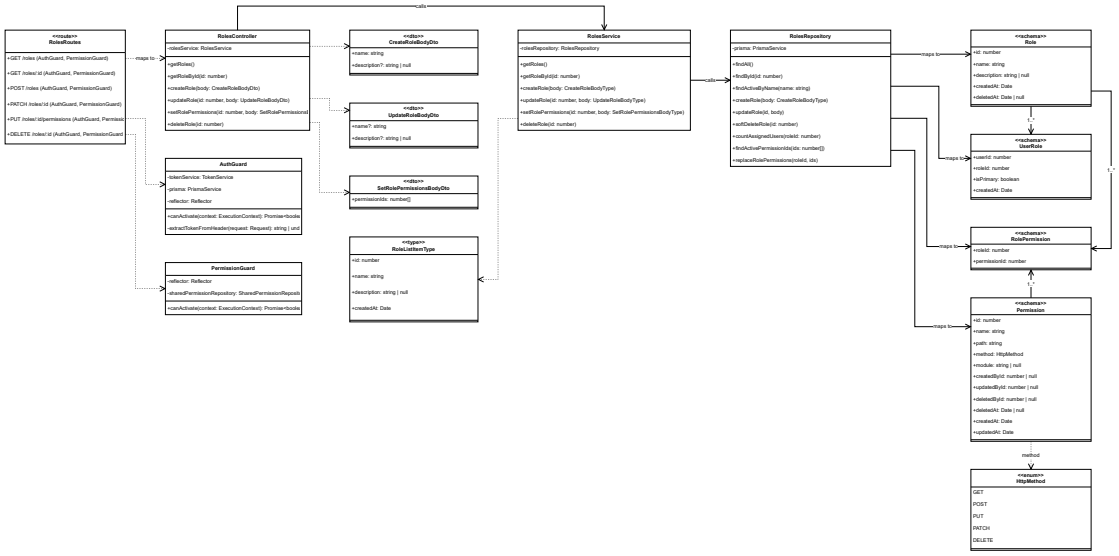

Figure 48: Class diagram of UC-06.06 Update Role Permissions

b. Sequence diagram

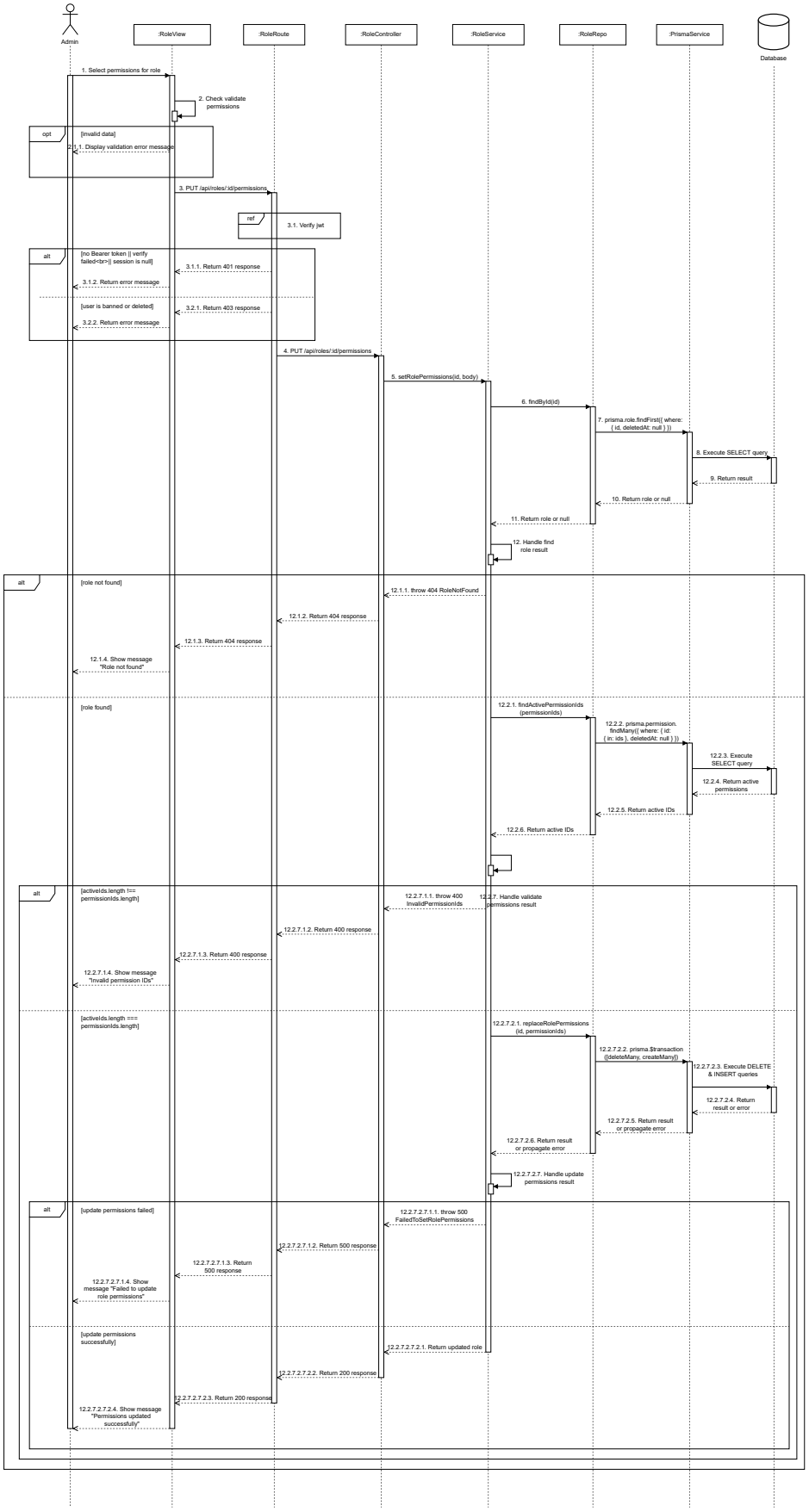

Figure 49: Sequence diagram of UC-06.06 Update Role Permissions

3.4 Job Browsing

Job Browsing allows guests and registered users to search, filter, and view the list of available jobs on the platform, as well as view comprehensive job details.

3.4.1 UC-10.01 – View Job List

a. Class diagram

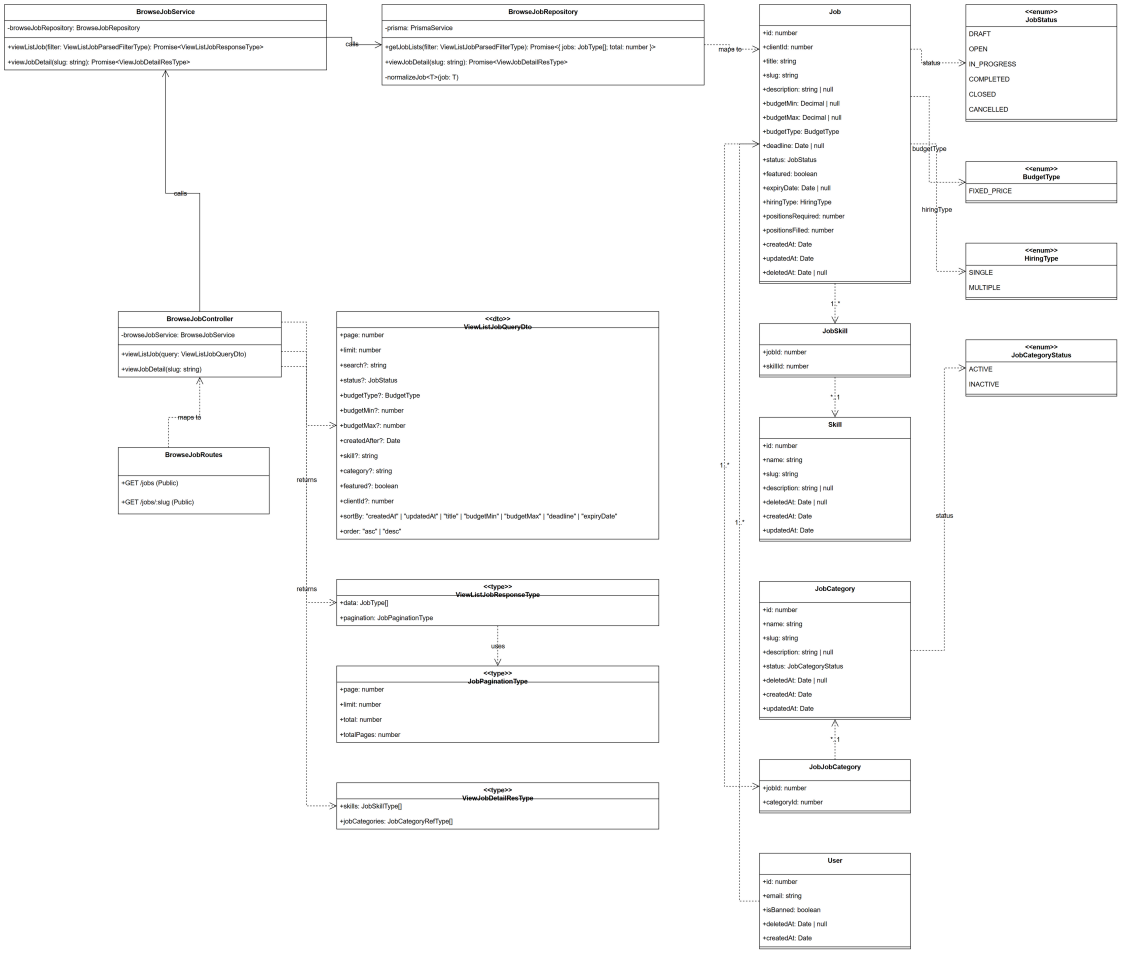

Figure 50: Class diagram of UC-10.01 View Job List

b. Sequence diagram

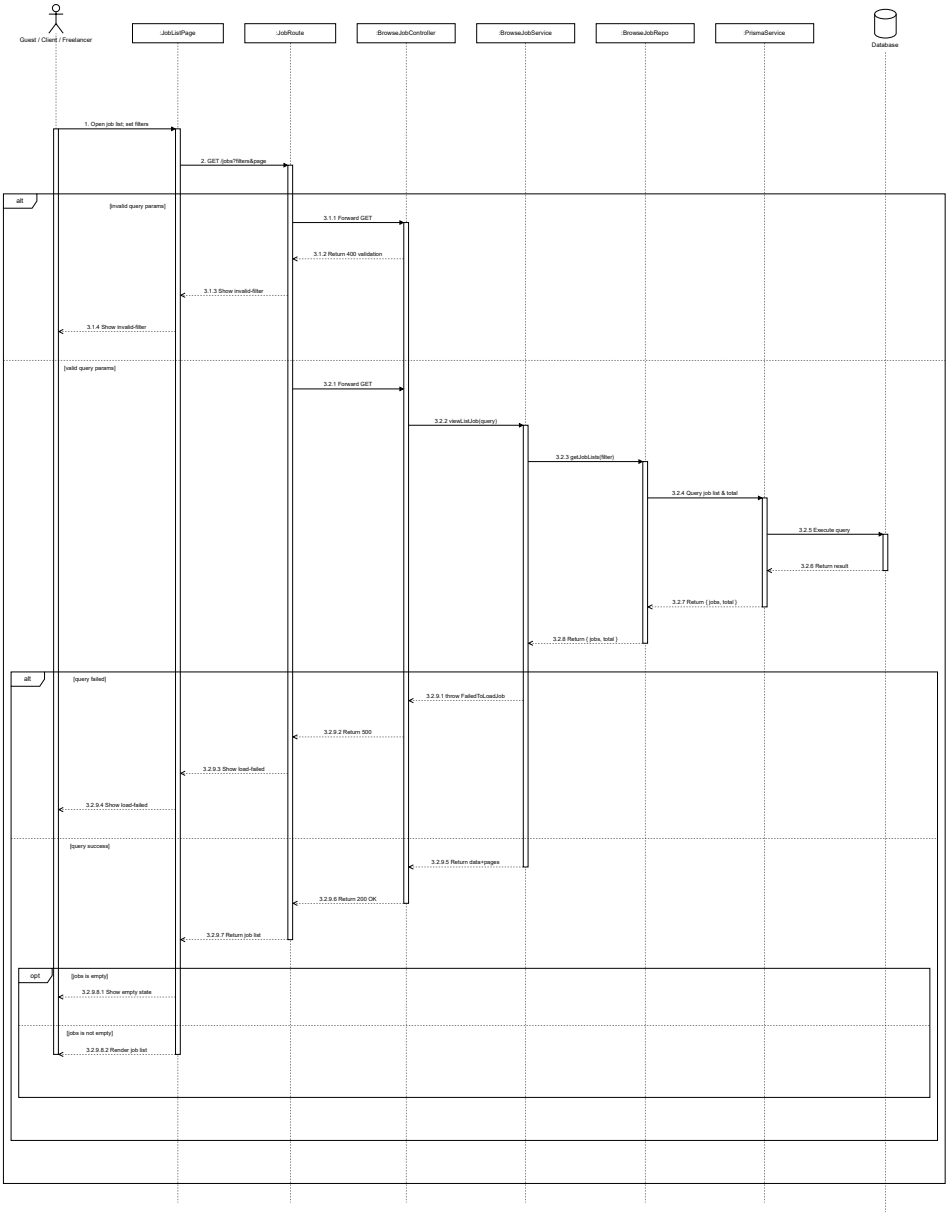

Figure 51: Sequence diagram of UC-10.01 View Job List

3.4.2 UC-10.02 – View Job Detail

a. Class diagram

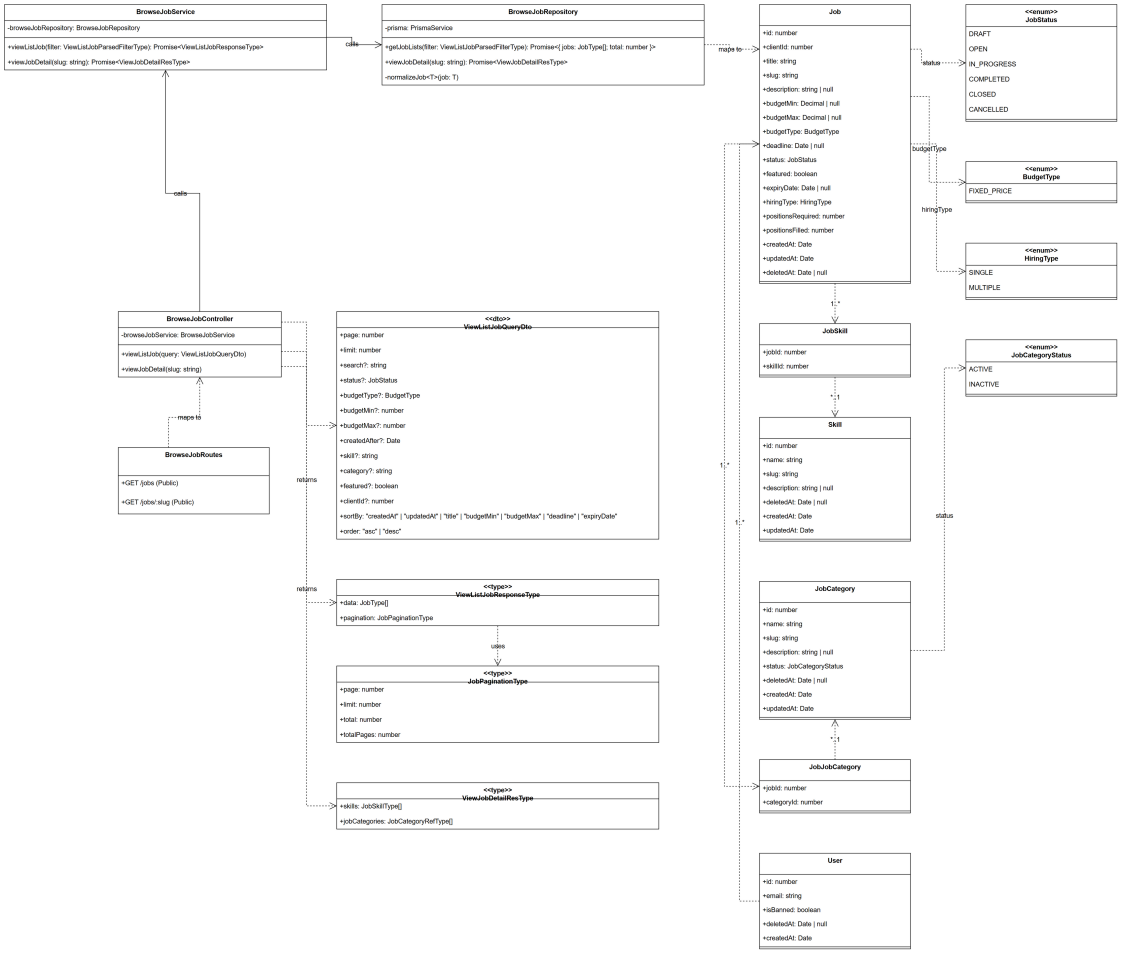

Figure 52: Class diagram of UC-10.02 View Job Detail

b. Sequence diagram

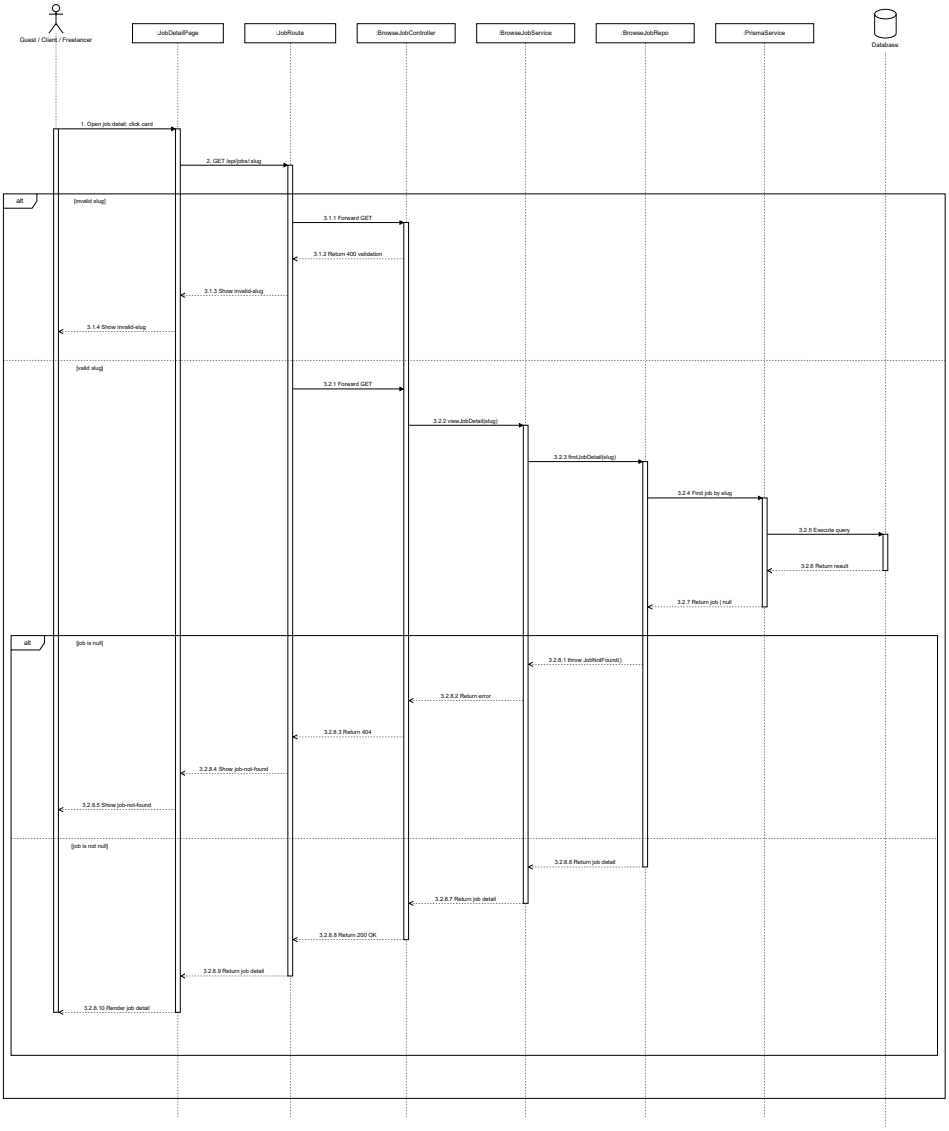

Figure 53: Sequence diagram of UC-10.02 View Job Detail

3.5 Job Management

Job Management enables clients to manage their posted jobs throughout their lifecycle, including creating new jobs, editing job parameters, publishing drafts, closing completed or cancelled jobs, and soft-deleting job postings.

3.5.1 UC-11.01 – View My Jobs

a. Class diagram

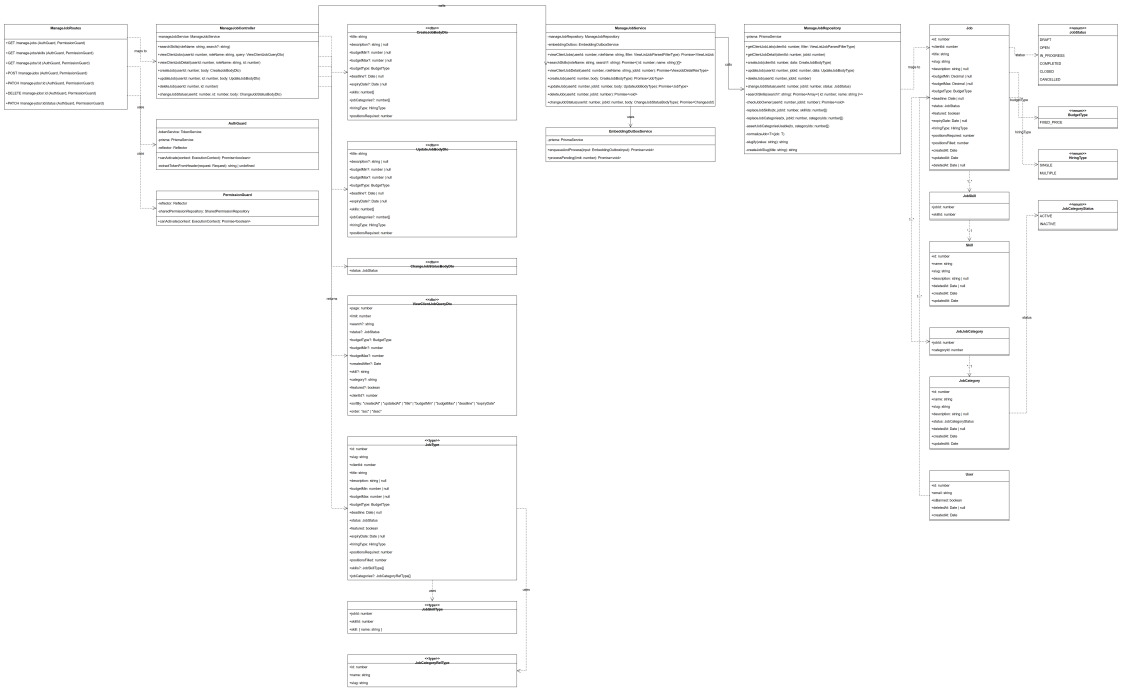

Figure 54: Class diagram of UC-11.01 View My Jobs

b. Sequence diagram

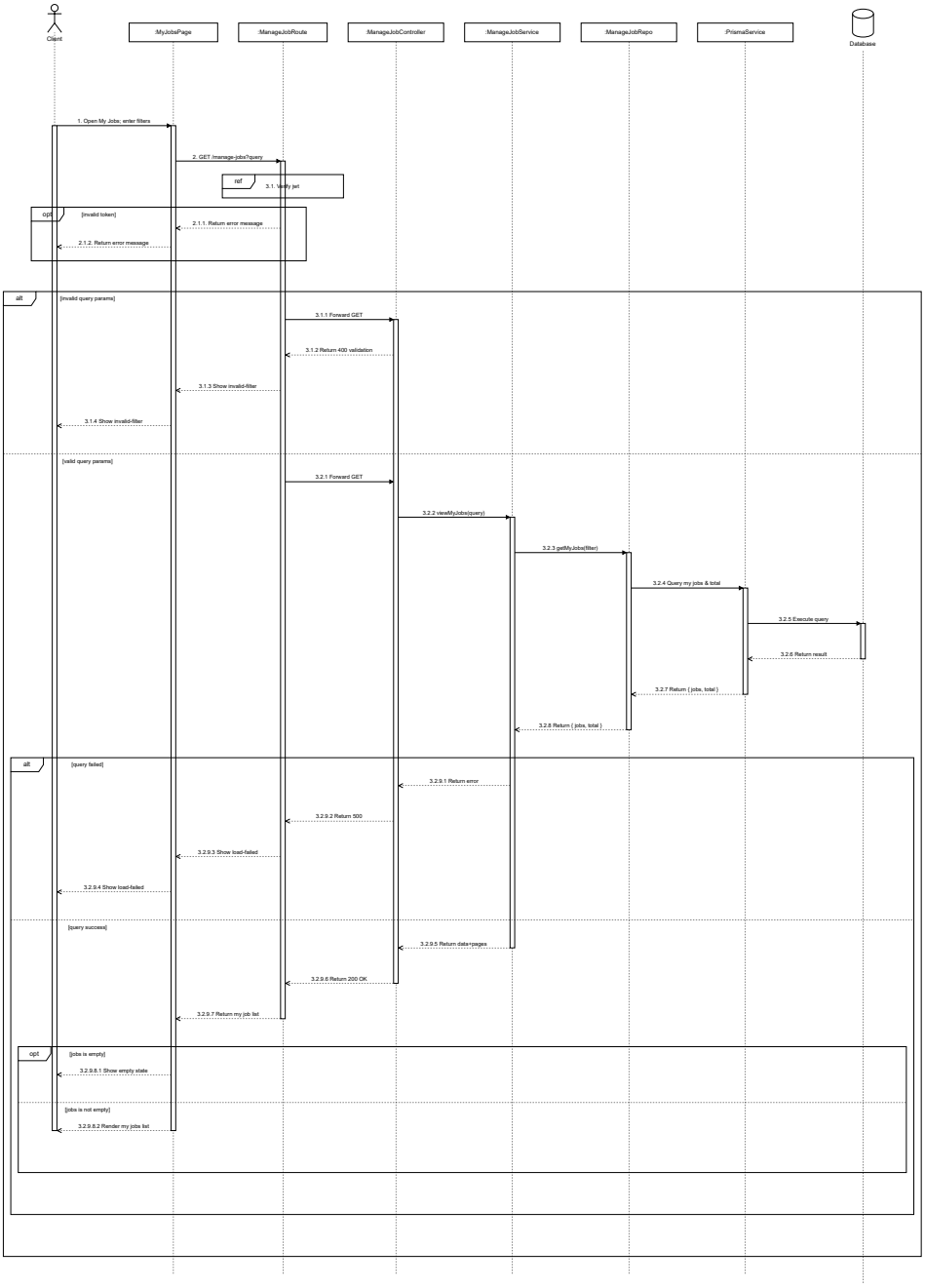

Figure 55: Sequence diagram of UC-11.01 View My Jobs

3.5.2 UC-11.02 – Create Job

a. Class diagram

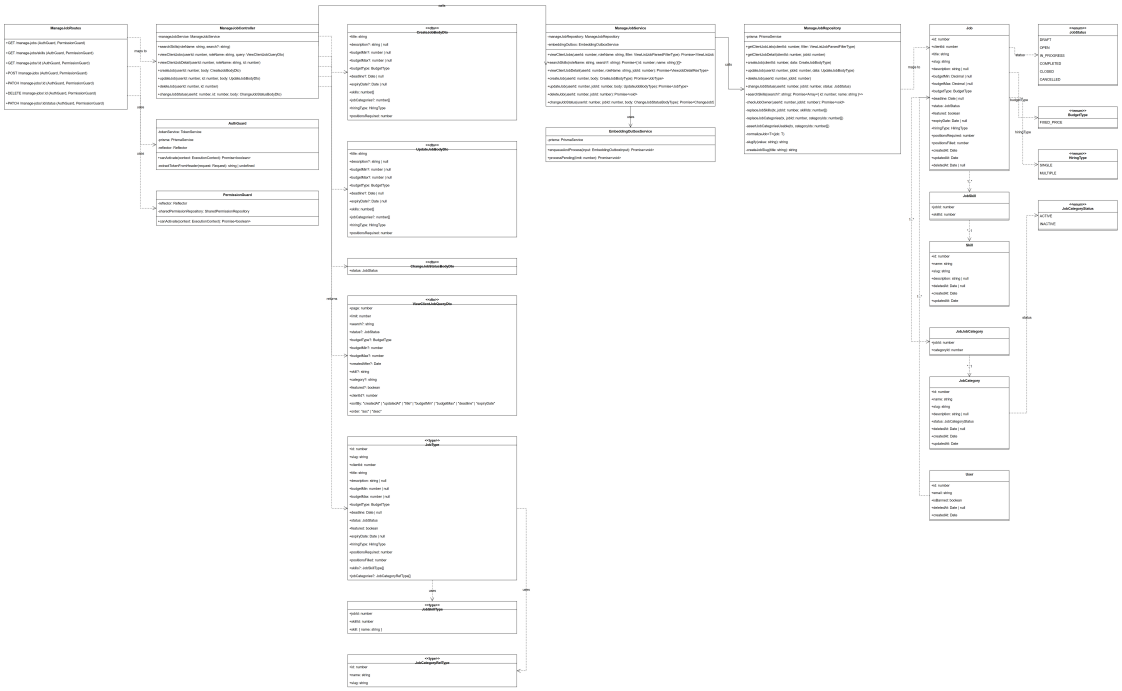

Figure 56: Class diagram of UC-11.02 Create Job

b. Sequence diagram

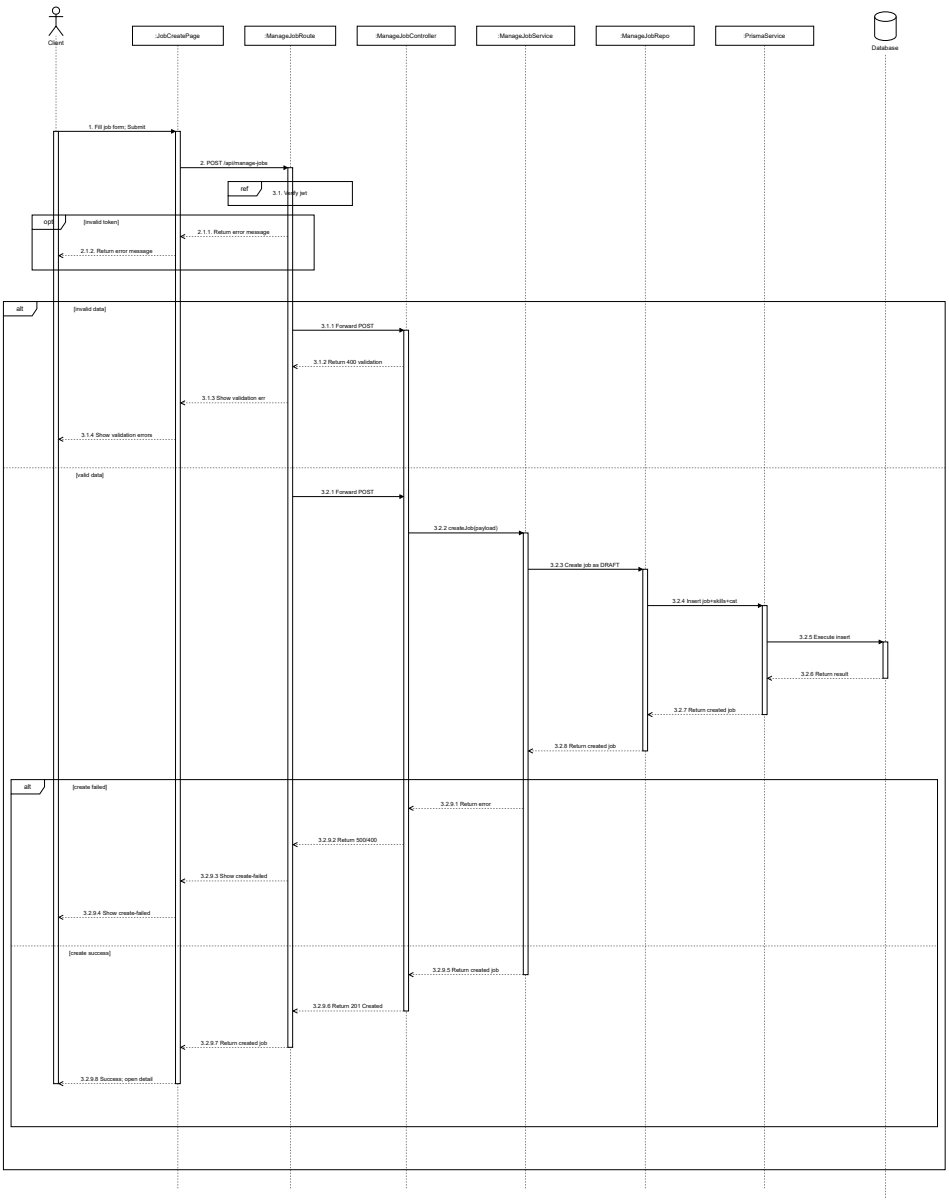

Figure 57: Sequence diagram of UC-11.02 Create Job

3.5.3 UC-11.03 – Publish Job

a. Class diagram

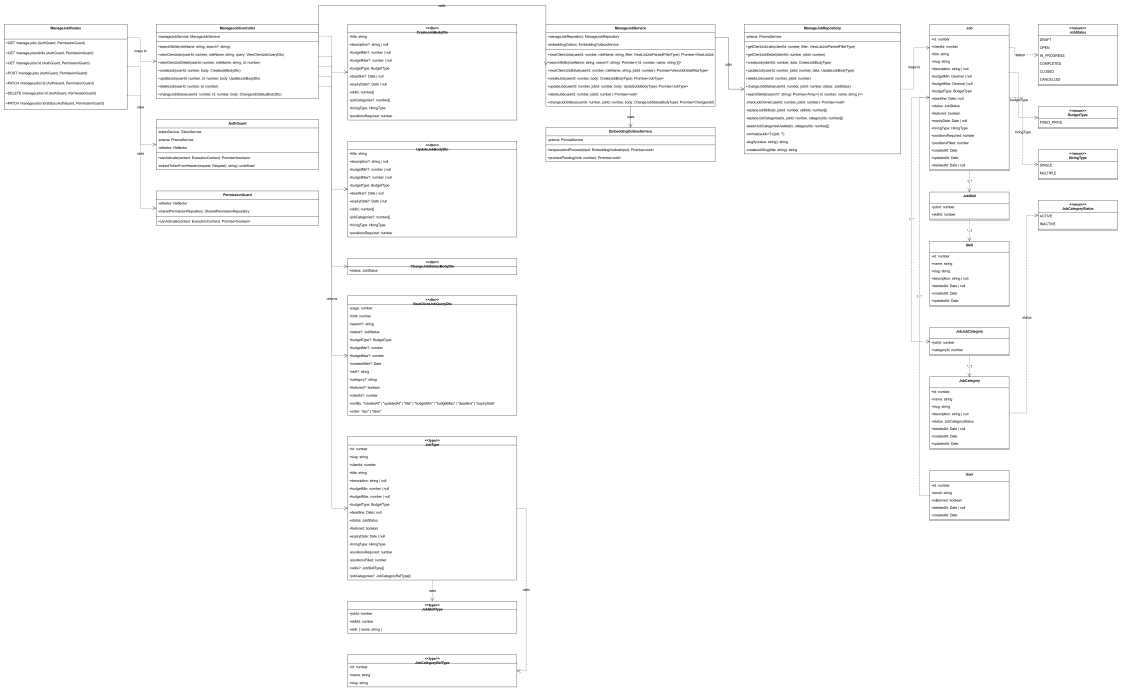

Figure 58: Class diagram of UC-11.03 Publish Job

b. Sequence diagram

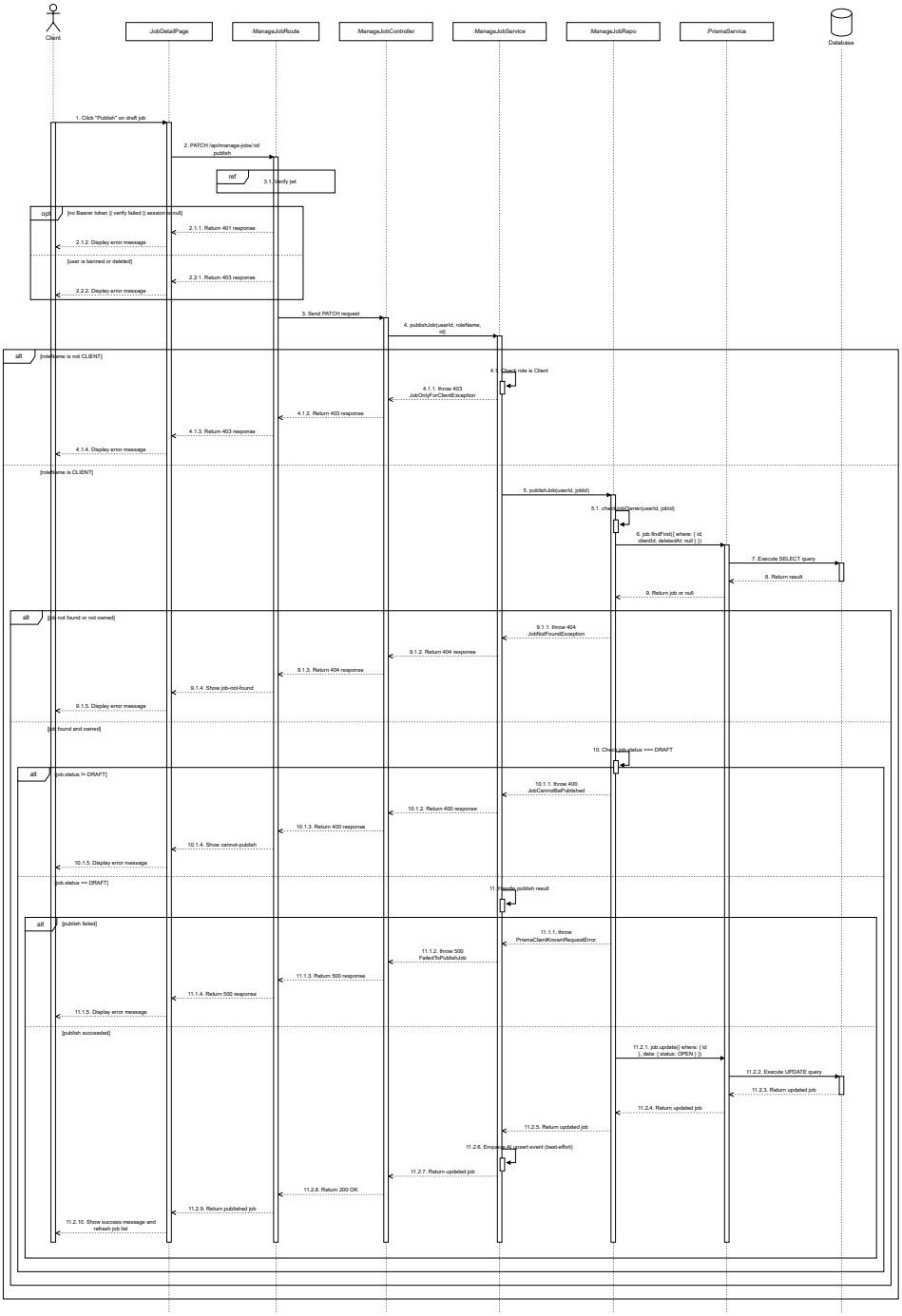

Figure 59: Sequence diagram of UC-11.03 Publish Job

3.5.4 UC-11.04 – Update Job

a. Class diagram

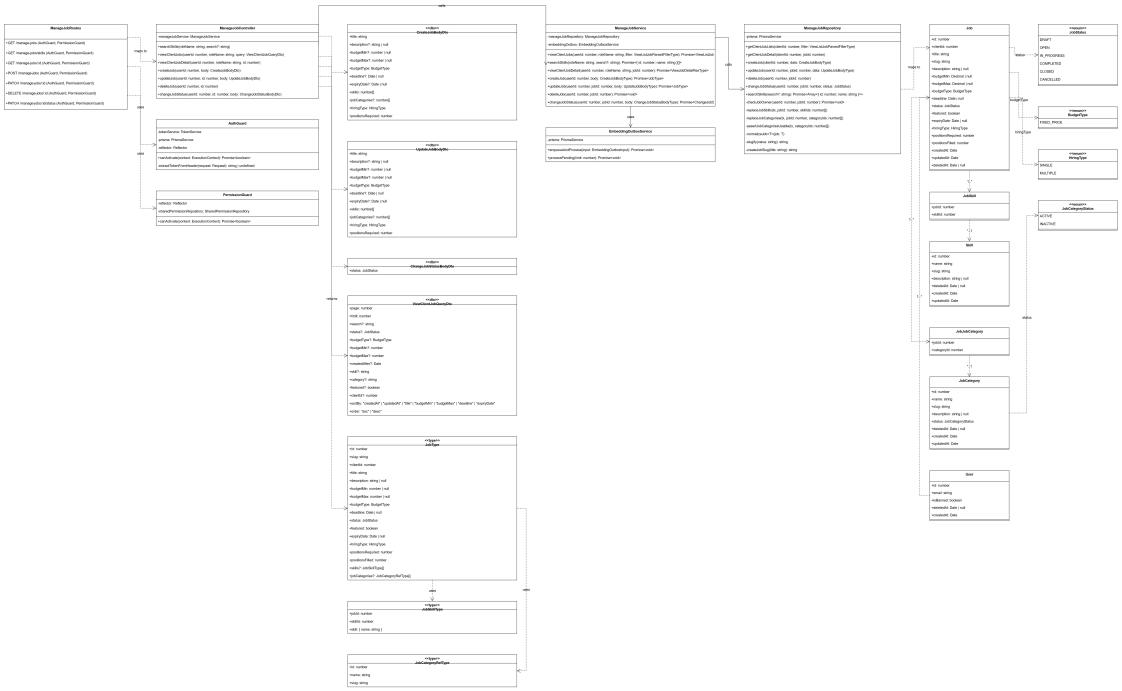

Figure 60: Class diagram of UC-11.04 Update Job

b. Sequence diagram

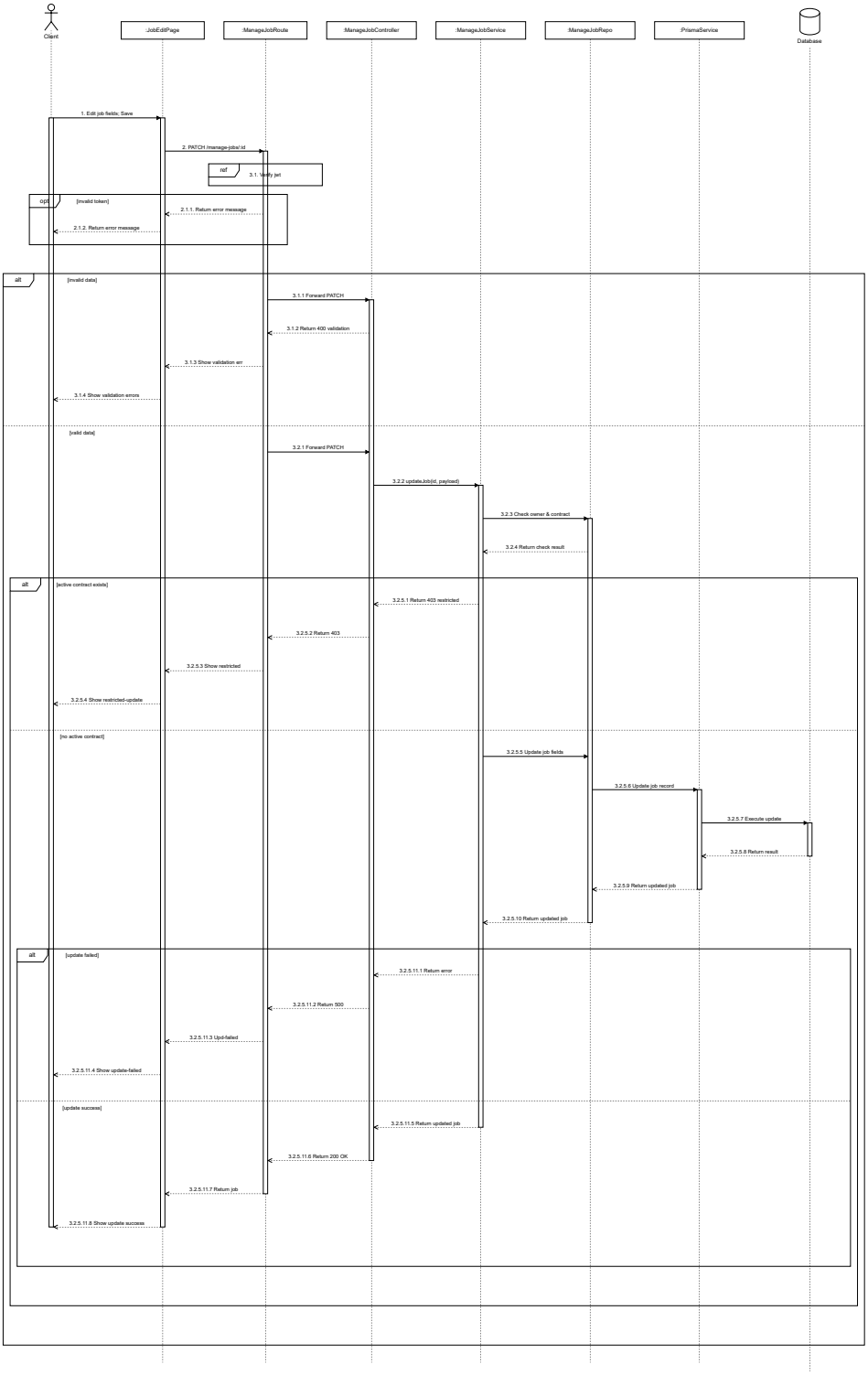

Figure 61: Sequence diagram of UC-11.04 Update Job

3.5.5 UC-11.05 – Delete Job

a. Class diagram

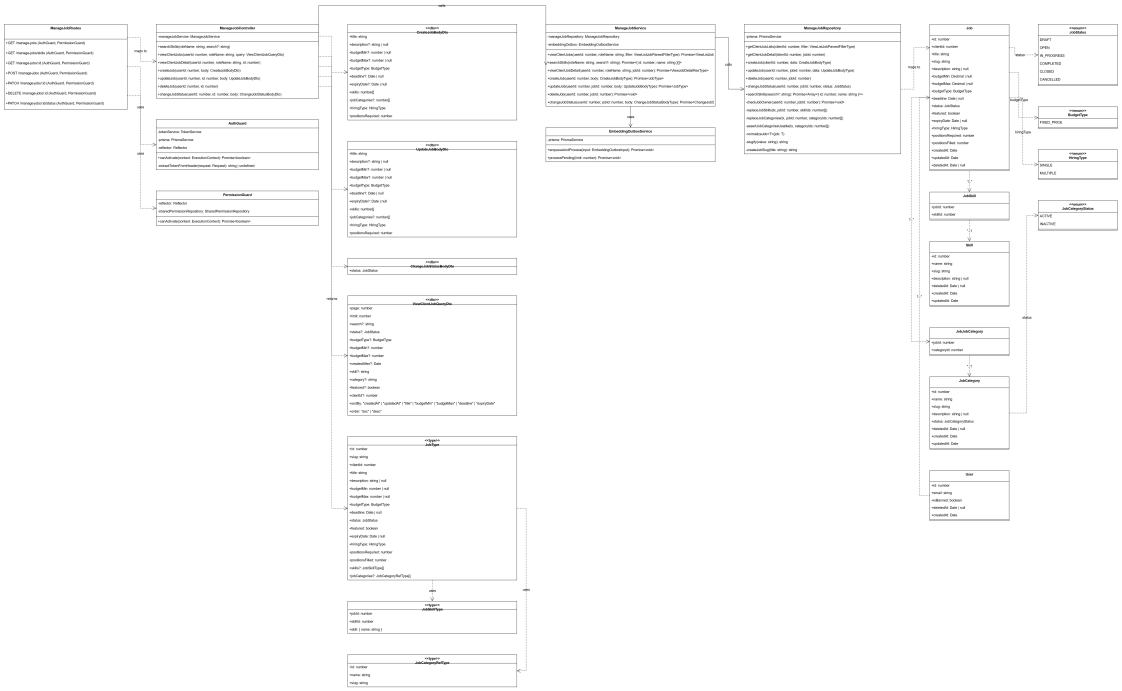

Figure 62: Class diagram of UC-11.05 Delete Job

b. Sequence diagram

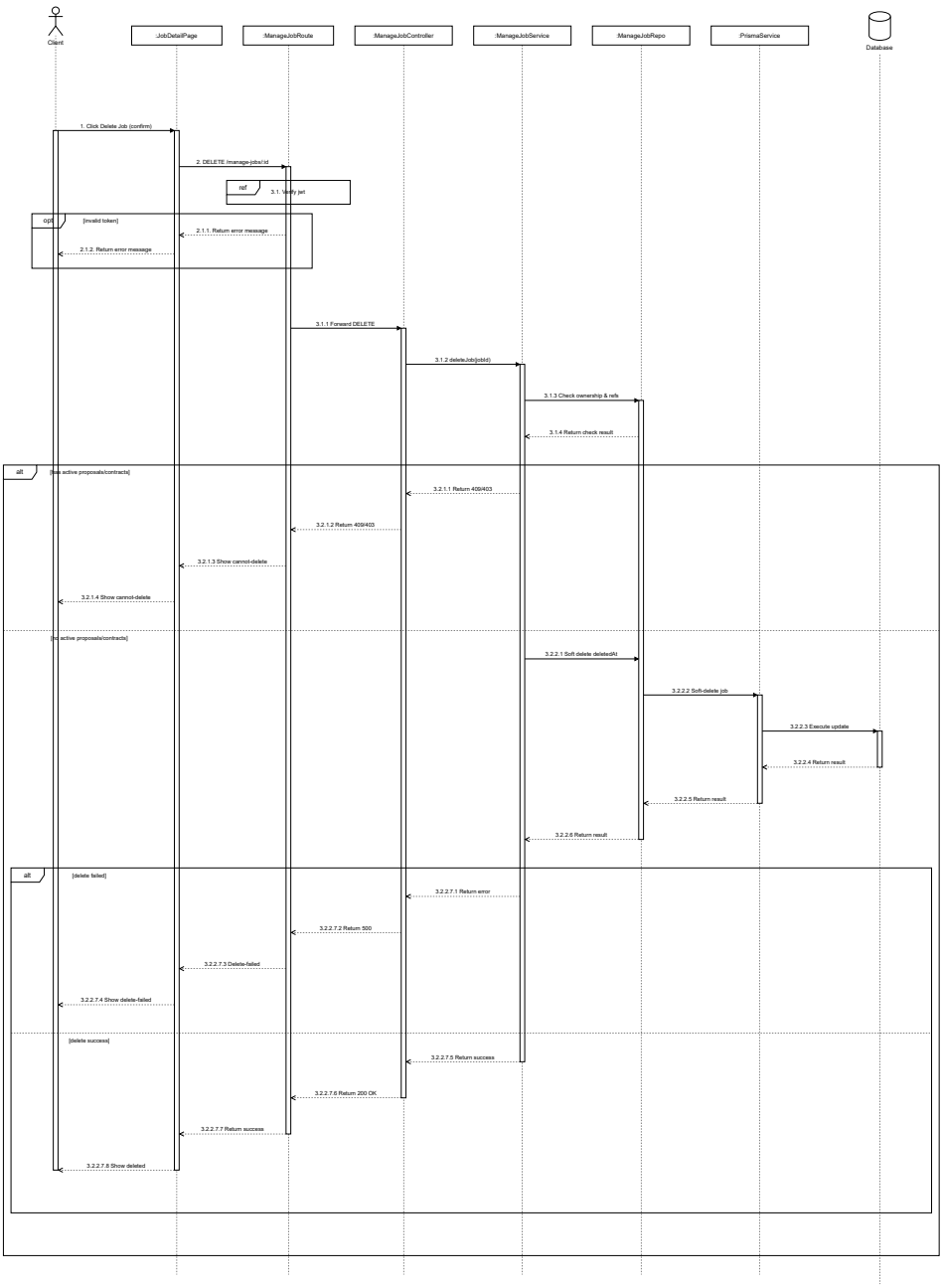

Figure 63: Sequence diagram of UC-11.05 Delete Job

3.5.6 UC-11.06 – Change Job Status

a. Class diagram

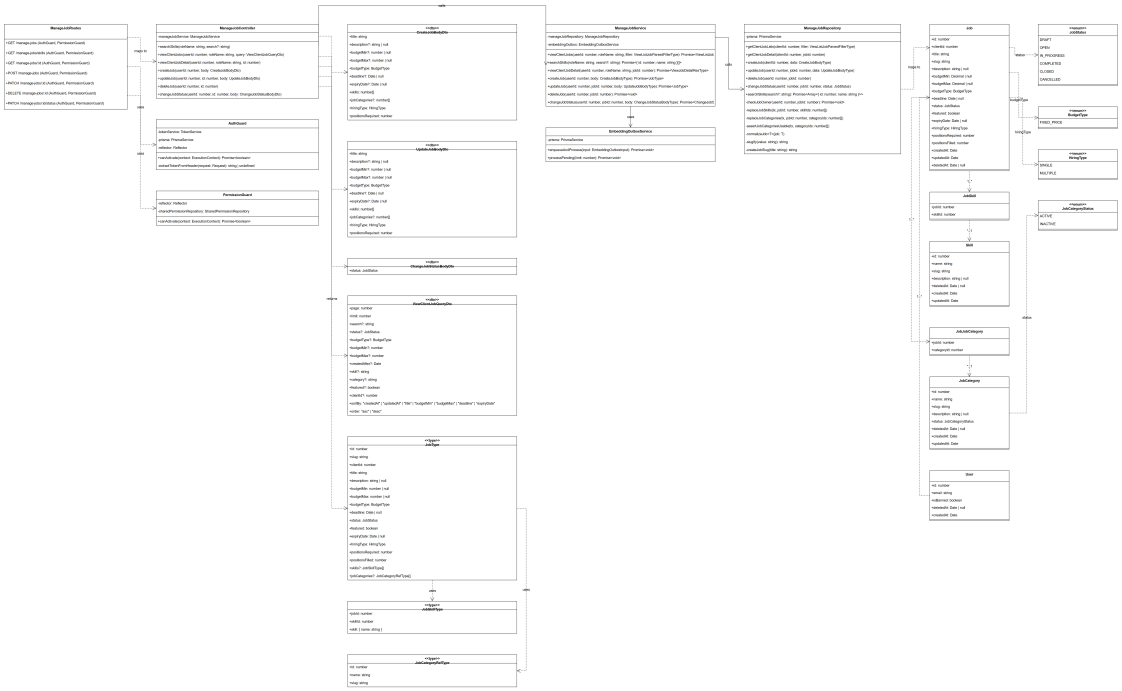

Figure 64: Class diagram of UC-11.06 Change Job Status

b. Sequence diagram

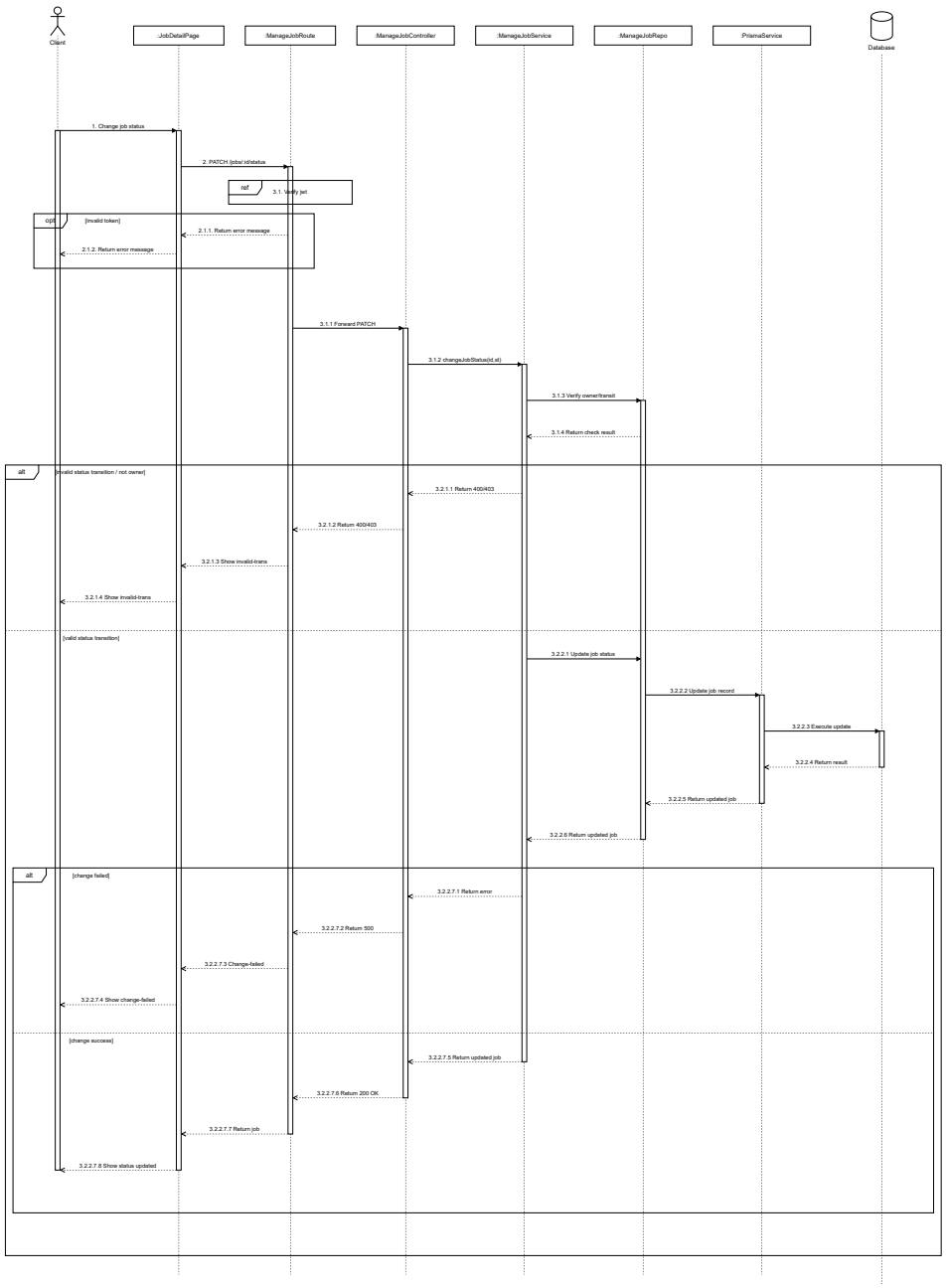

Figure 65: Sequence diagram of UC-11.06 Change Job Status

3.6 Bookmarked Jobs Management

Bookmarked Jobs Management allows freelancers to save interesting job postings for later review, inspect their list of bookmarked jobs, and remove bookmarks when no longer needed.

3.6.1 UC-12.01 – View Bookmarked Job List

a. Class diagram

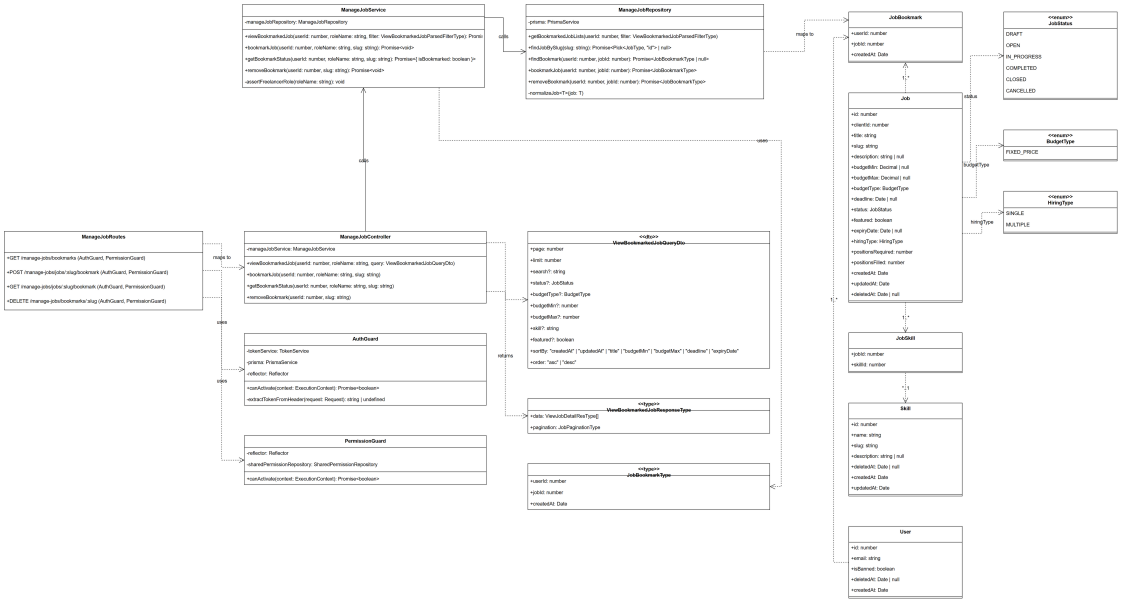

Figure 66: Class diagram of UC-12.01 View Bookmarked Job List

b. Sequence diagram

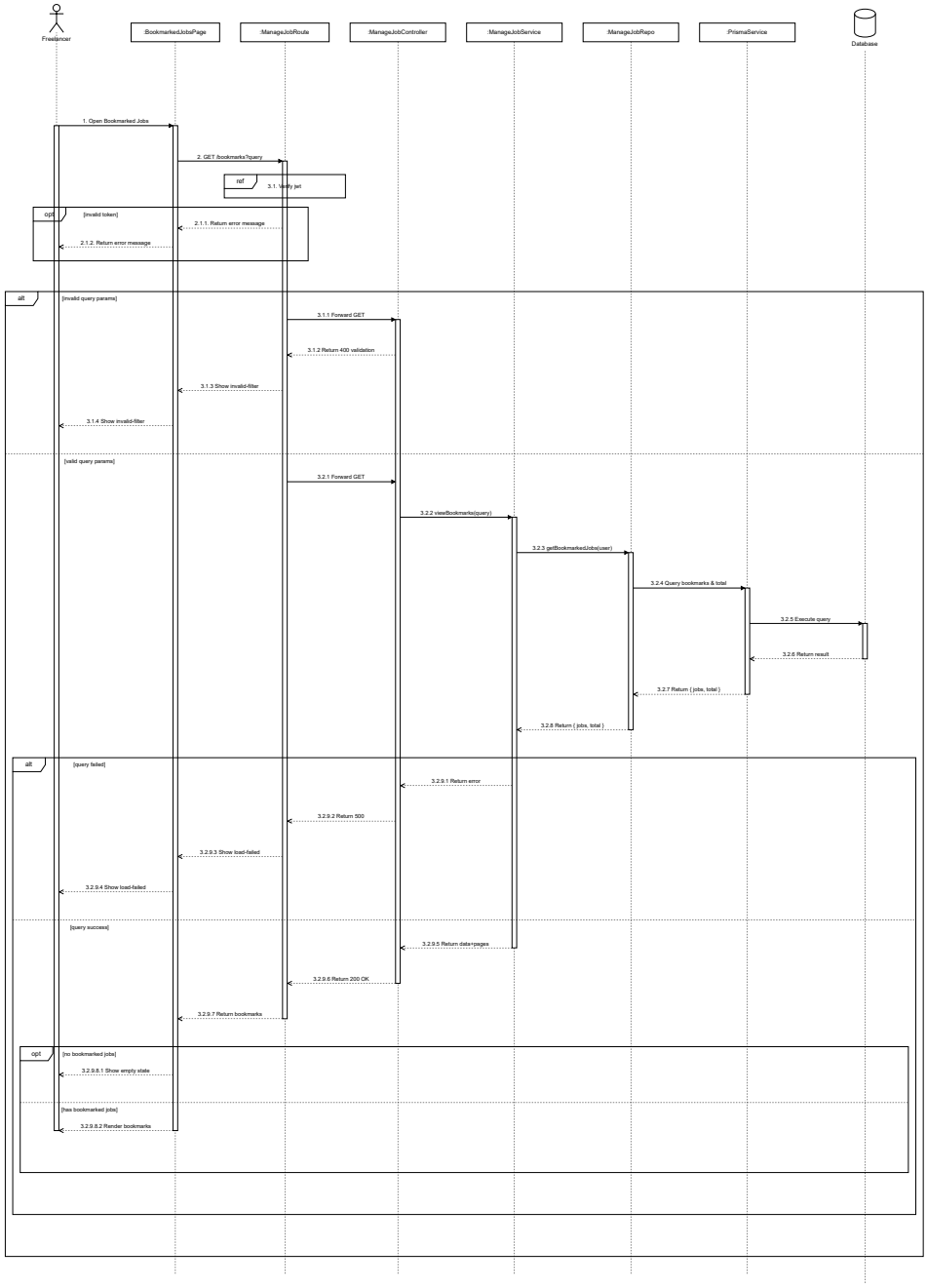

Figure 67: Sequence diagram of UC-12.01 View Bookmarked Job List

3.6.2 UC-12.02 – Bookmark Job

a. Class diagram

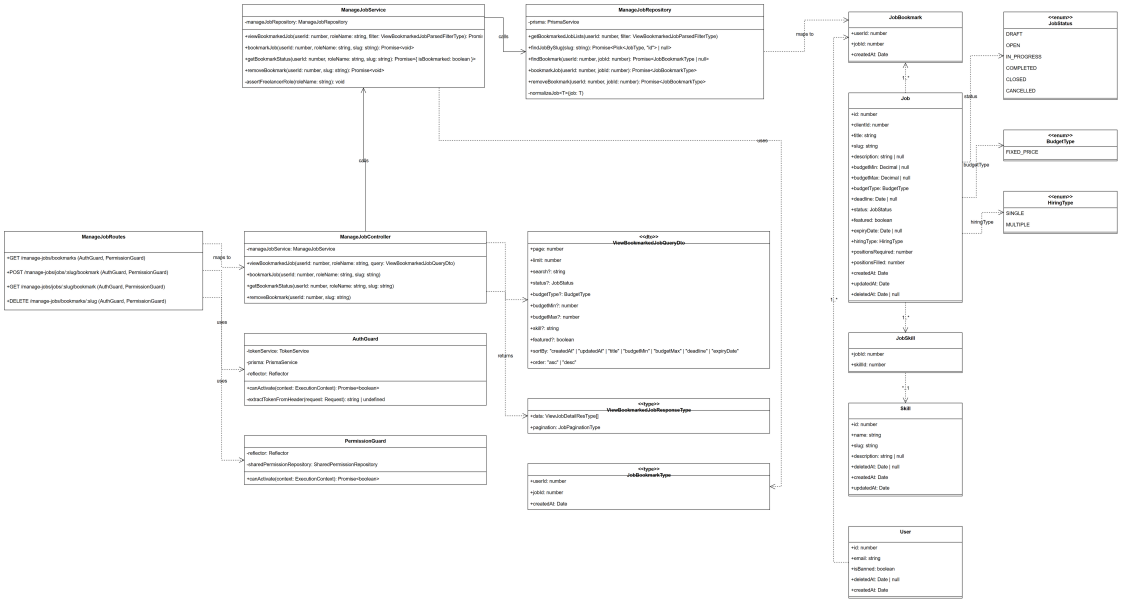

Figure 68: Class diagram of UC-12.02 Bookmark Job

b. Sequence diagram

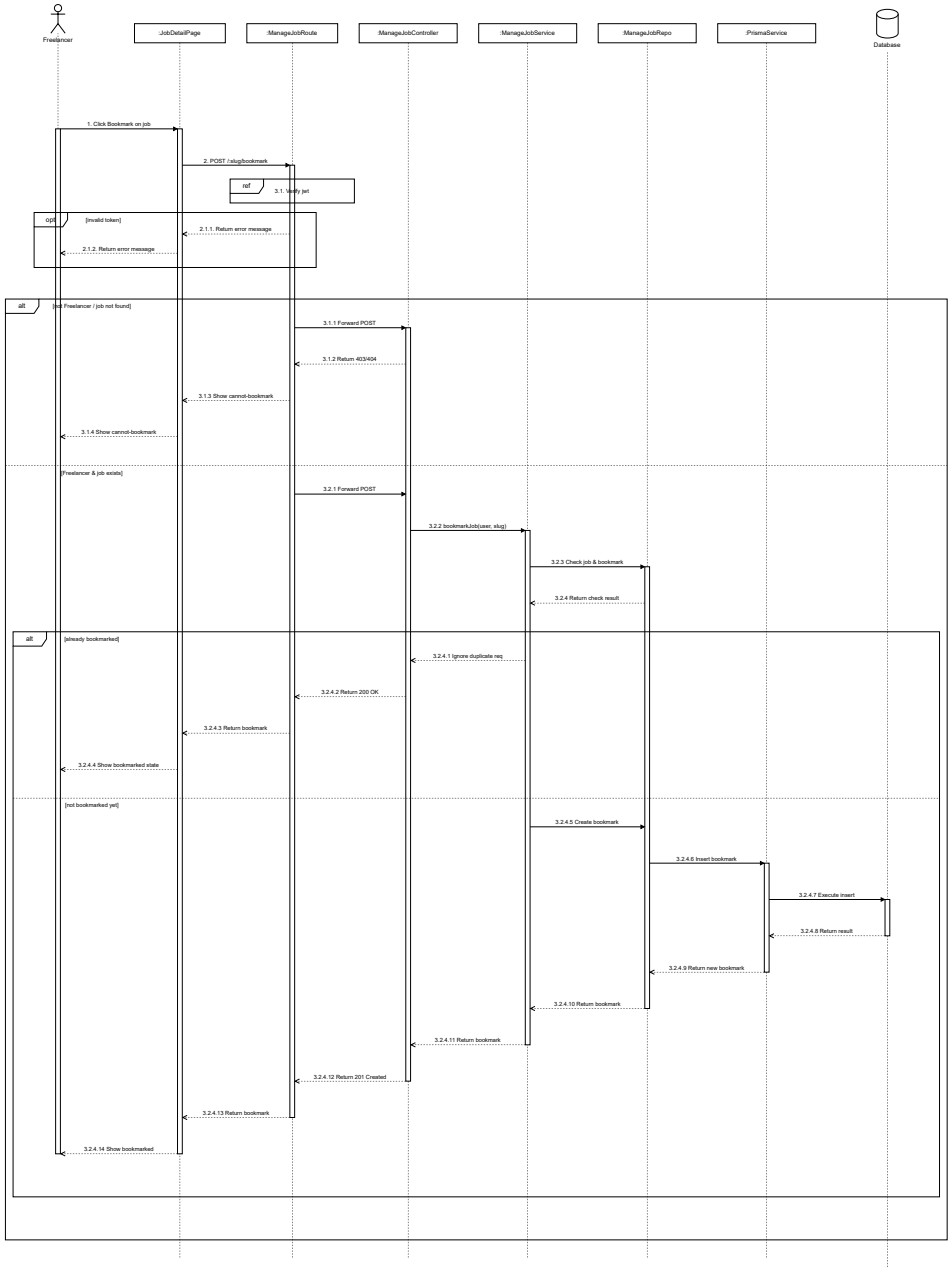

Figure 69: Sequence diagram of UC-12.02 Bookmark Job

3.6.3 UC-12.03 – Remove Bookmark

a. Class diagram

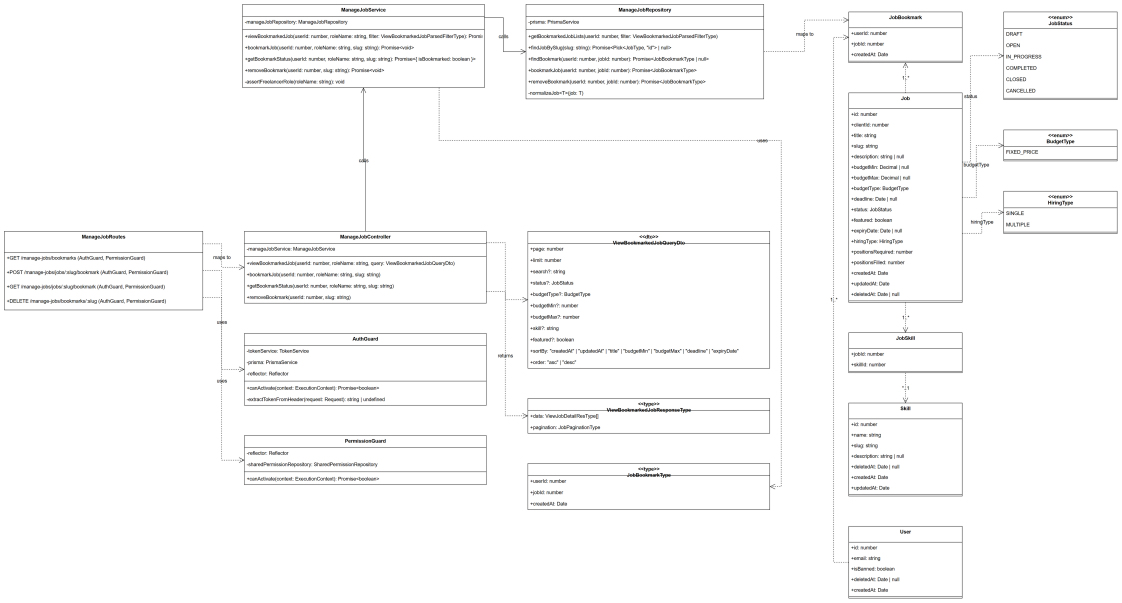

Figure 70: Class diagram of UC-12.03 Remove Bookmark

b. Sequence diagram

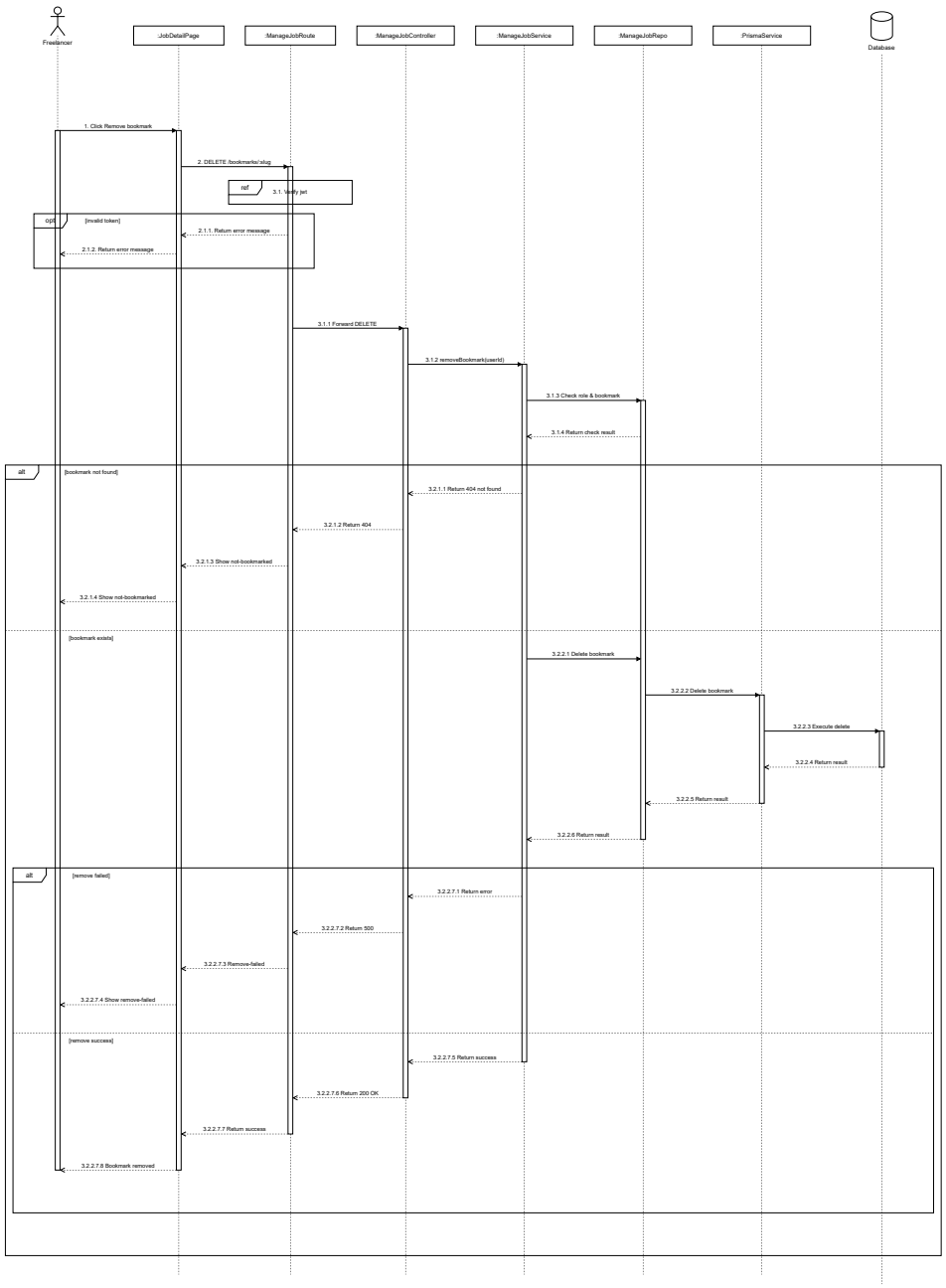

Figure 71: Sequence diagram of UC-12.03 Remove Bookmark

3.7 Forum Management

Forum Management covers the UC-14 community forum: browsing forum categories and posts, publishing and maintaining posts and comments, toggling likes on posts and comments, and reporting content that breaks the forum rules.

3.7.1 UC-14.01 – View Forum Category List

a. Class diagram

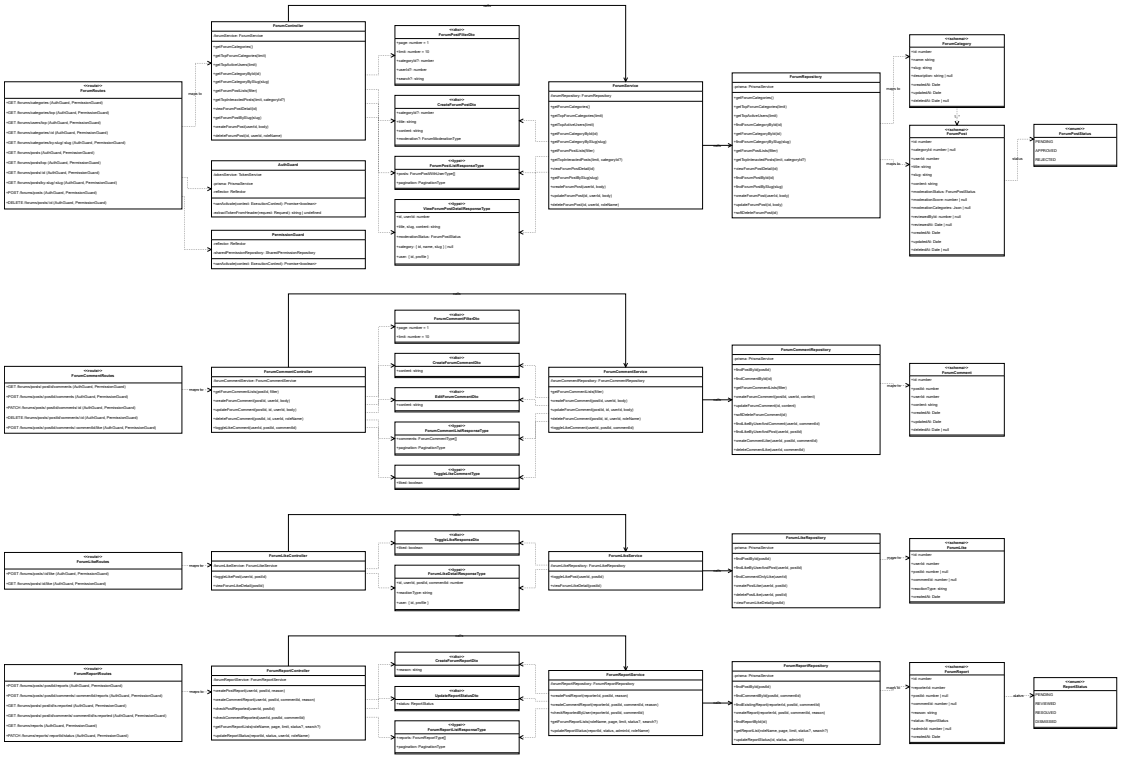

Figure 72: Class diagram of UC-14.01 View Forum Category List

b. Sequence diagram

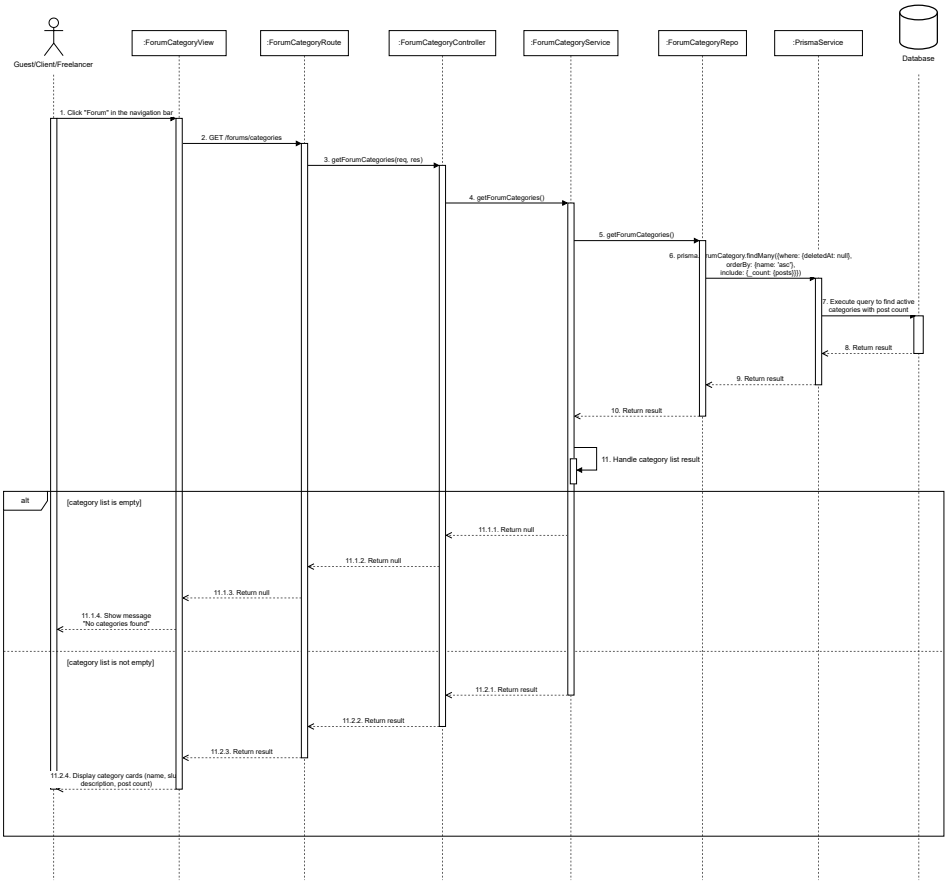

Figure 73: Sequence diagram of UC-14.01 View Forum Category List

3.7.2 UC-14.02 – View Forum Category Detail

a. Class diagram

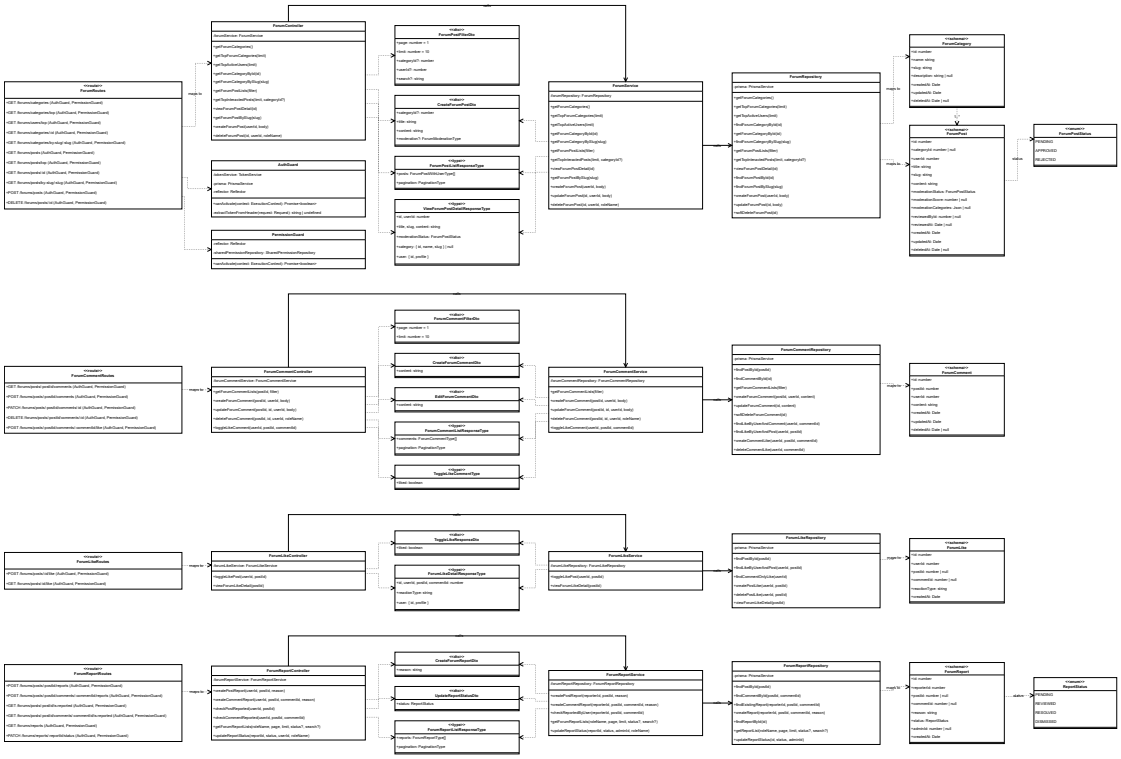

Figure 74: Class diagram of UC-14.02 View Forum Category Detail

b. Sequence diagram

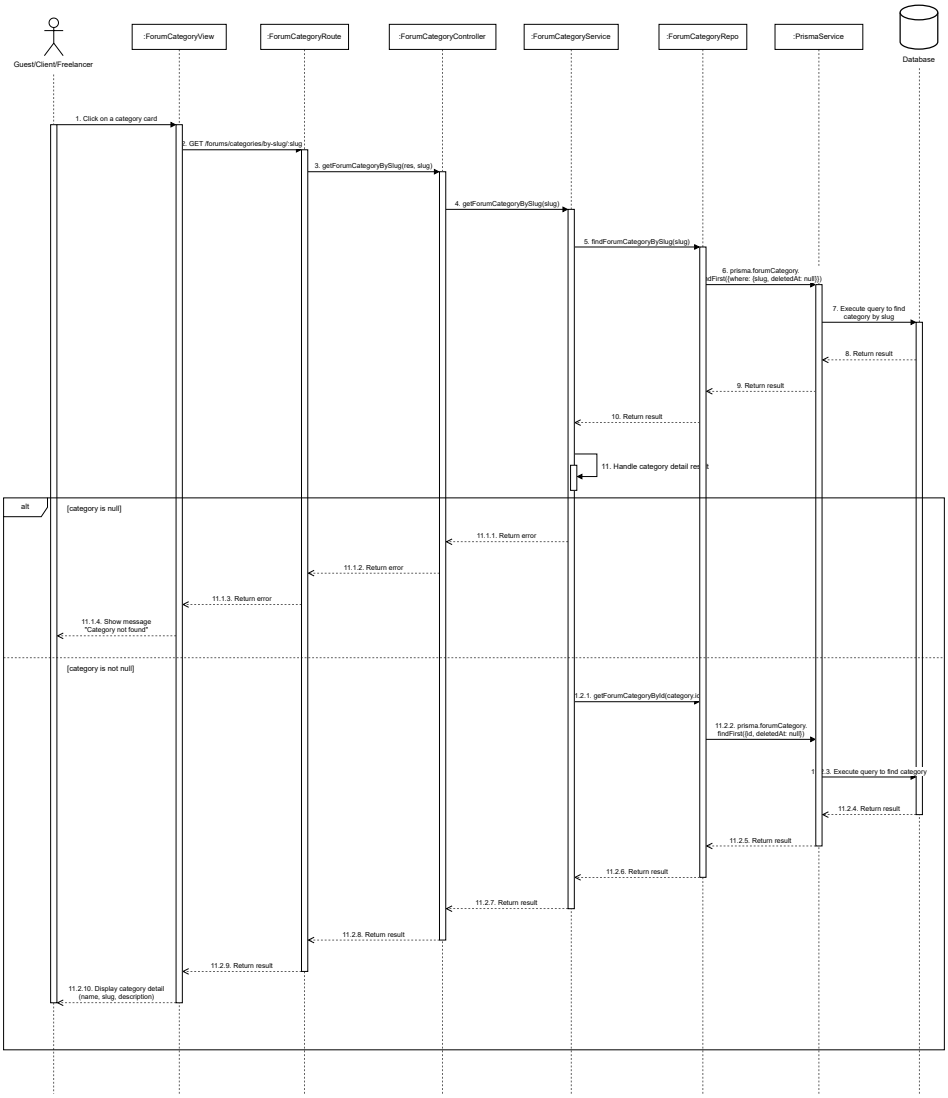

Figure 75: Sequence diagram of UC-14.02 View Forum Category Detail

3.7.3 UC-14.03 – View Post List

a. Class diagram

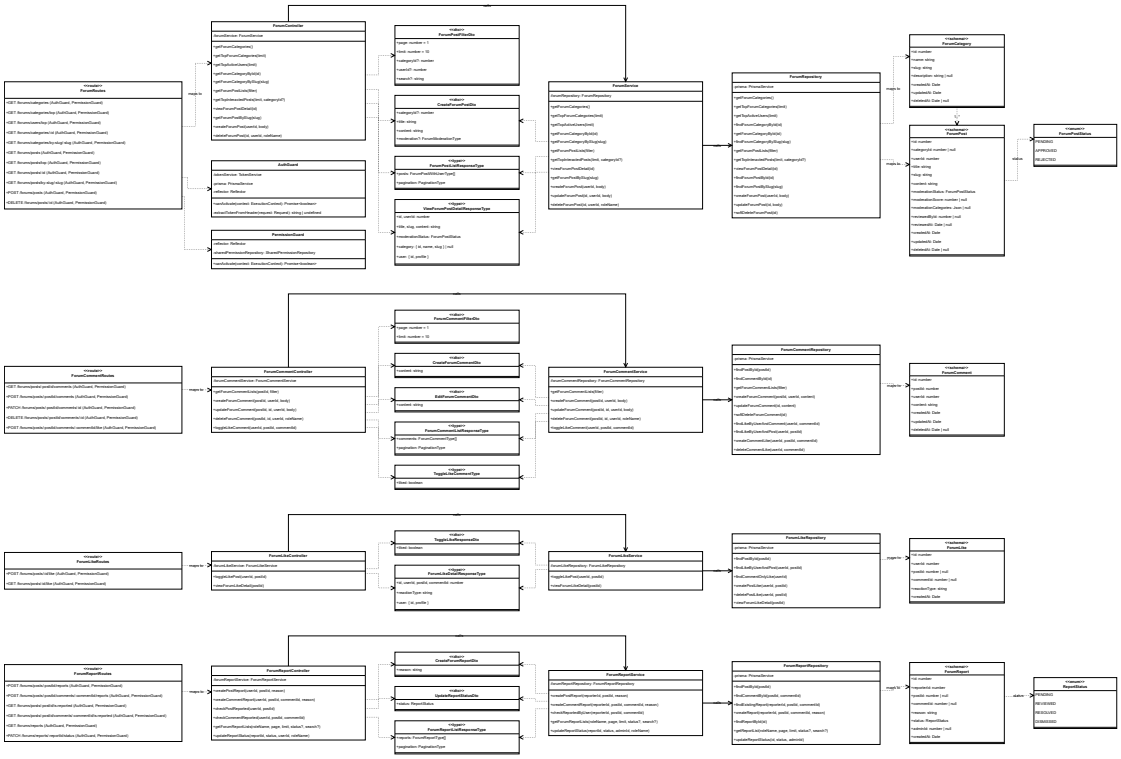

Figure 76: Class diagram of UC-14.03 View Post List

b. Sequence diagram

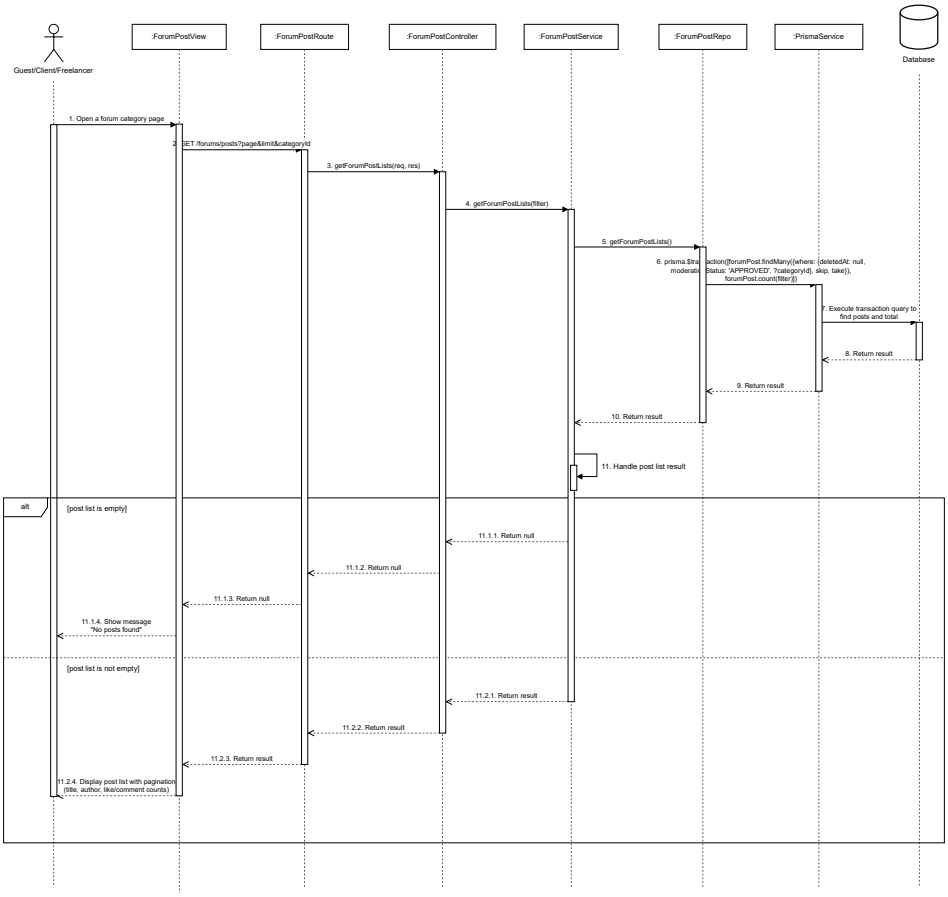

Figure 77: Sequence diagram of UC-14.03 View Post List

3.7.4 UC-14.04 – View Post Detail

a. Class diagram

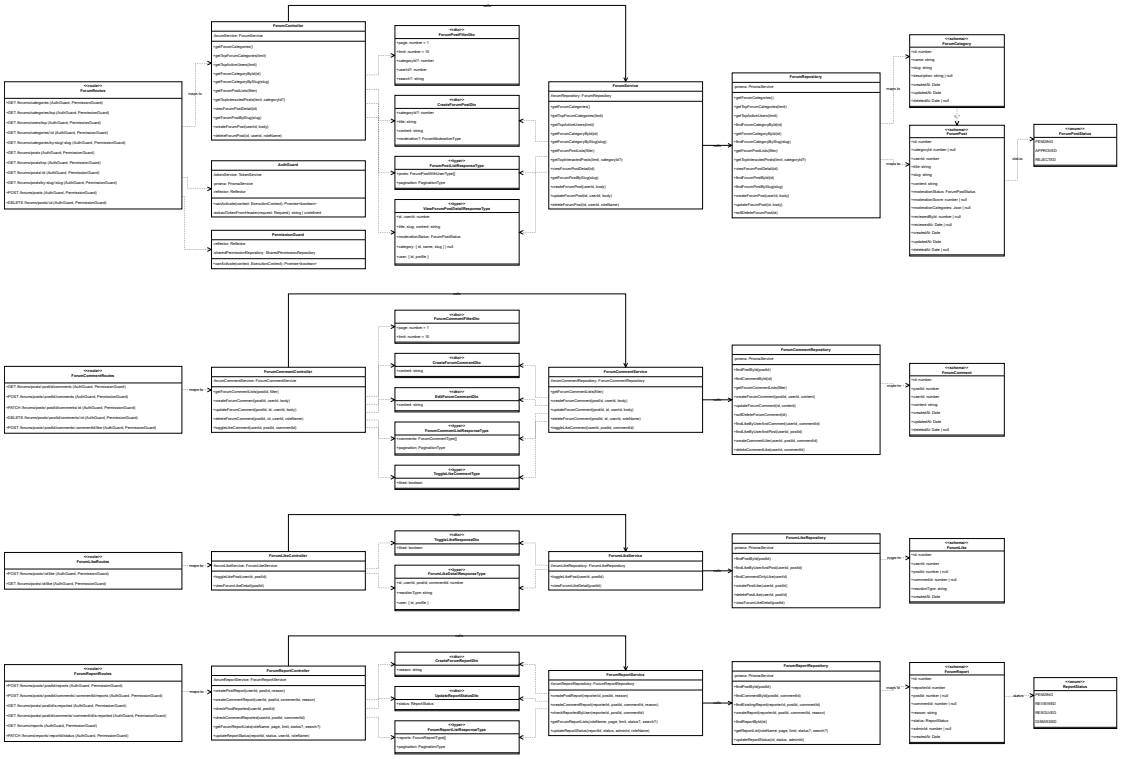

Figure 78: Class diagram of UC-14.04 View Post Detail

b. Sequence diagram

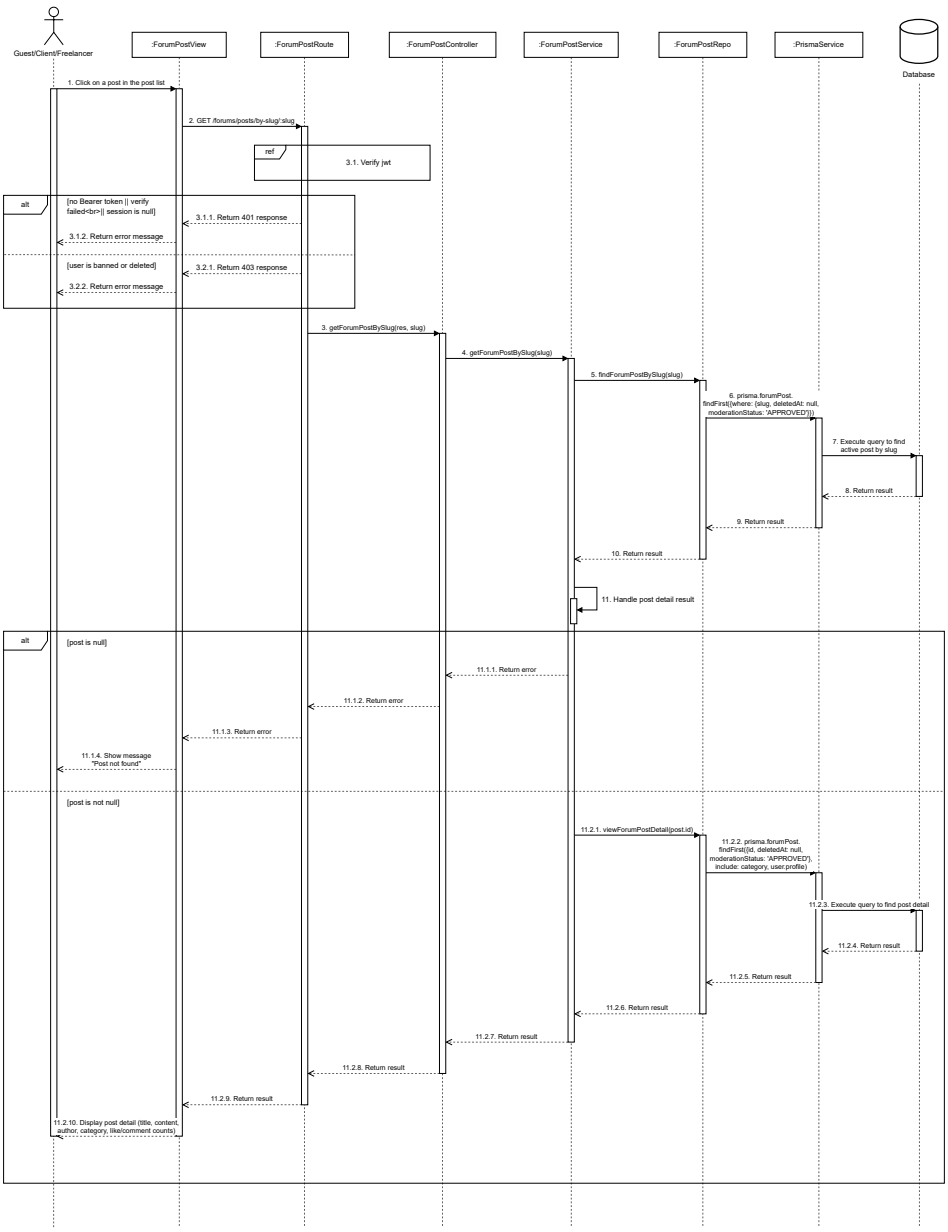

Figure 79: Sequence diagram of UC-14.04 View Post Detail

3.7.5 UC-14.05 – Create Post

a. Class diagram

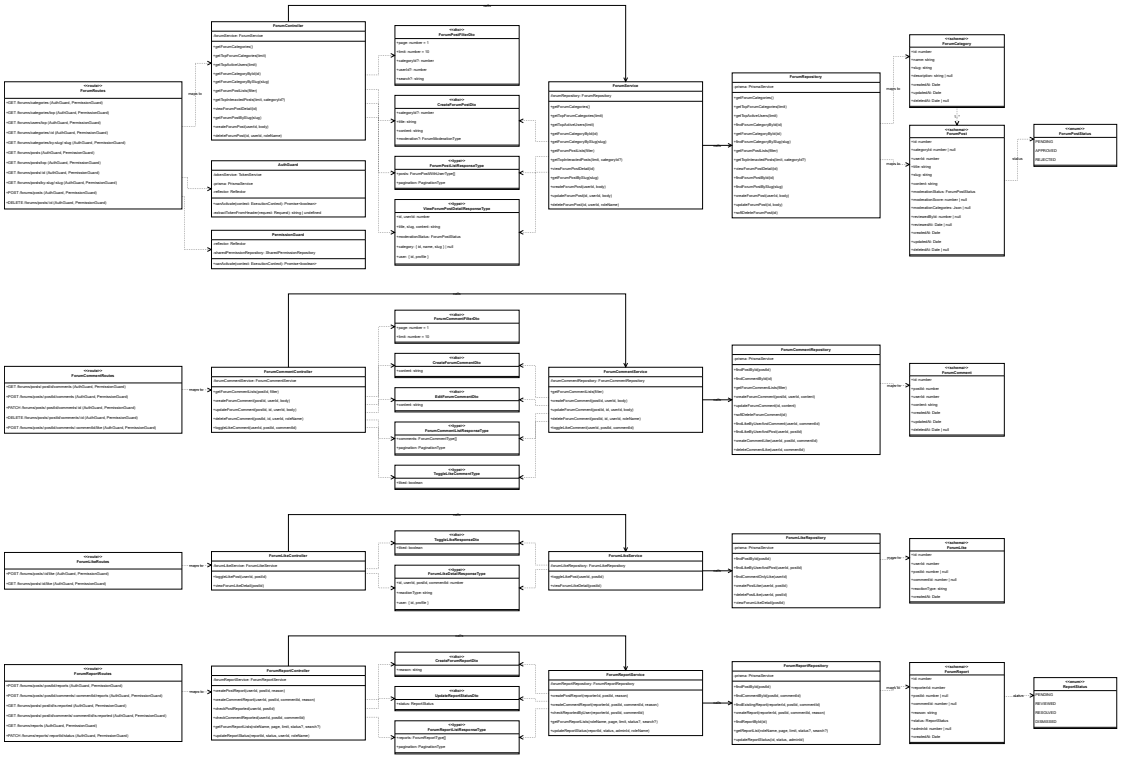

Figure 80: Class diagram of UC-14.05 Create Post

b. Sequence diagram

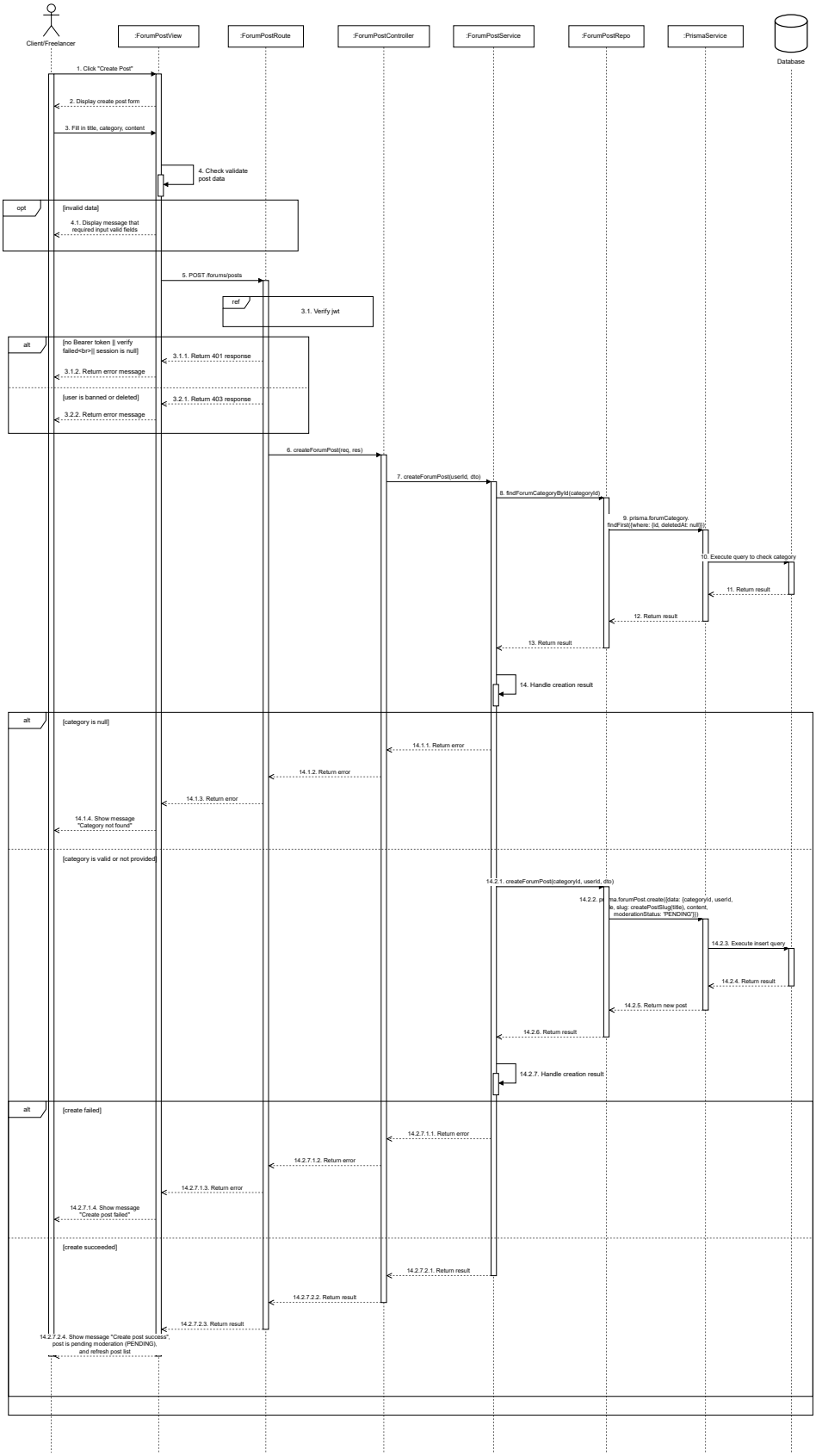

Figure 81: Sequence diagram of UC-14.05 Create Post

3.7.6 UC-14.06 – Update Post

a. Class diagram

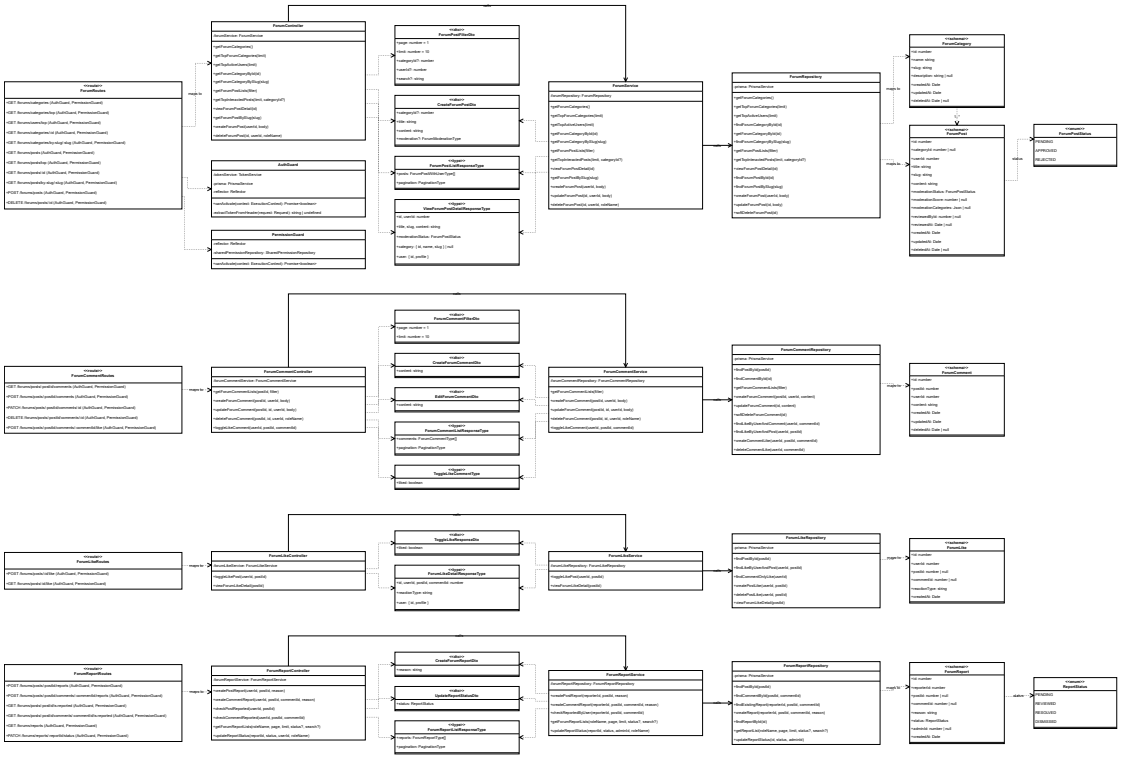

Figure 82: Class diagram of UC-14.06 Update Post

b. Sequence diagram

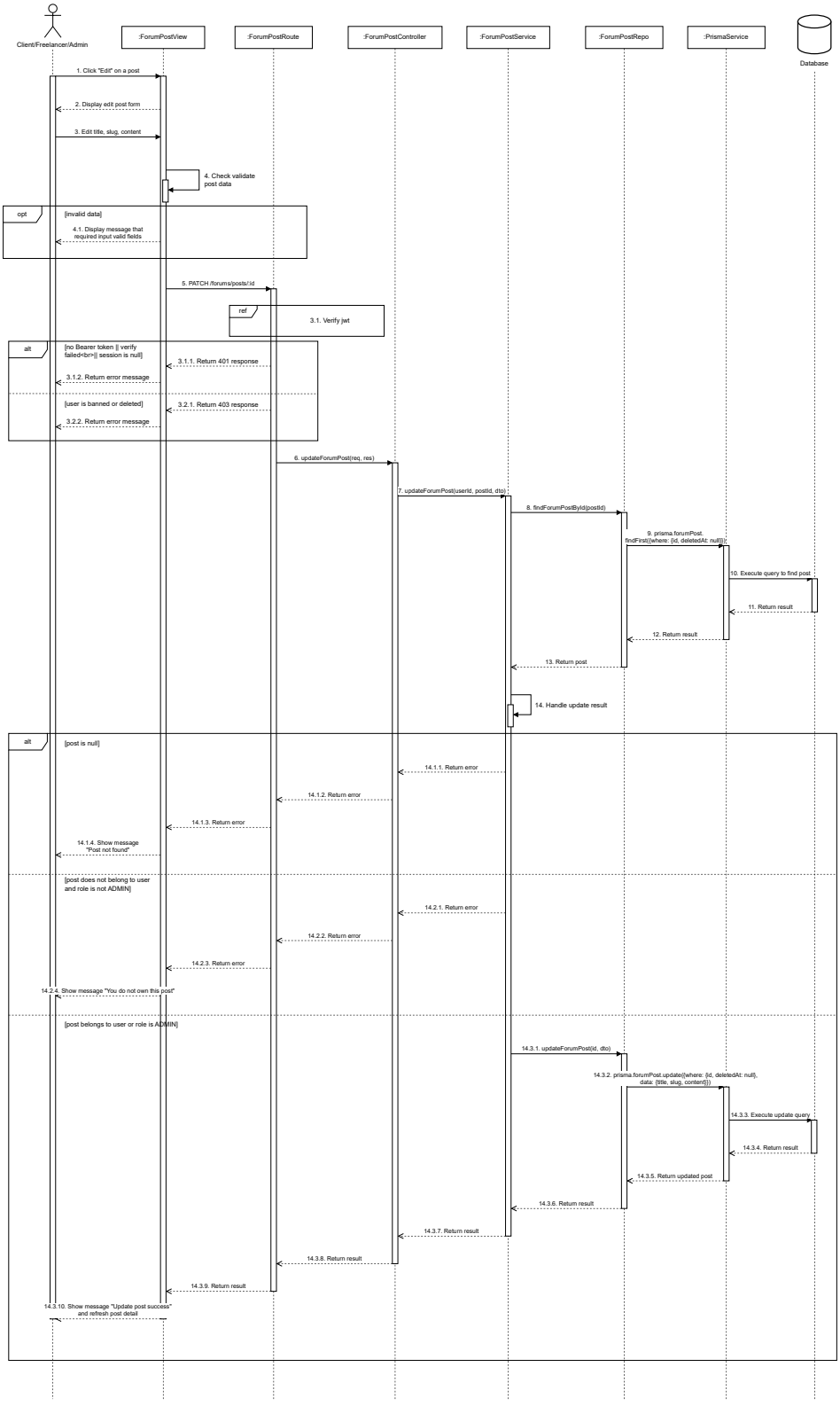

Figure 83: Sequence diagram of UC-14.06 Update Post

3.7.7 UC-14.07 – Delete Post

a. Class diagram

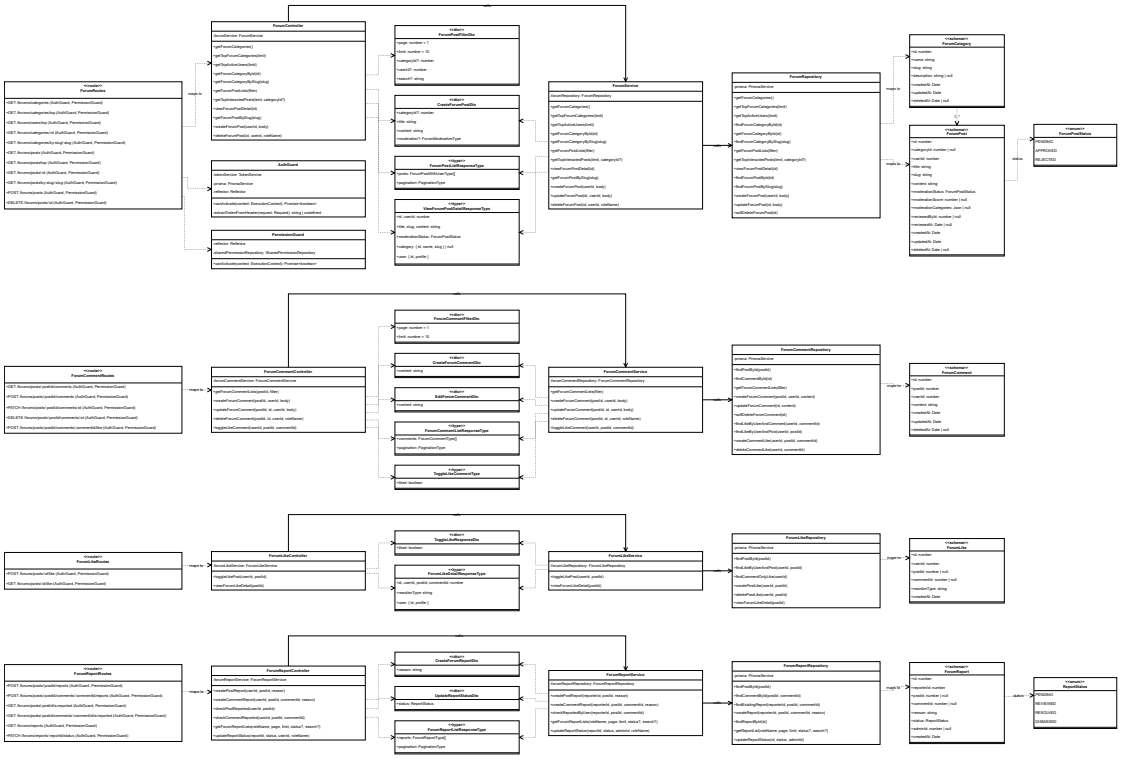

Figure 84: Class diagram of UC-14.07 Delete Post

b. Sequence diagram

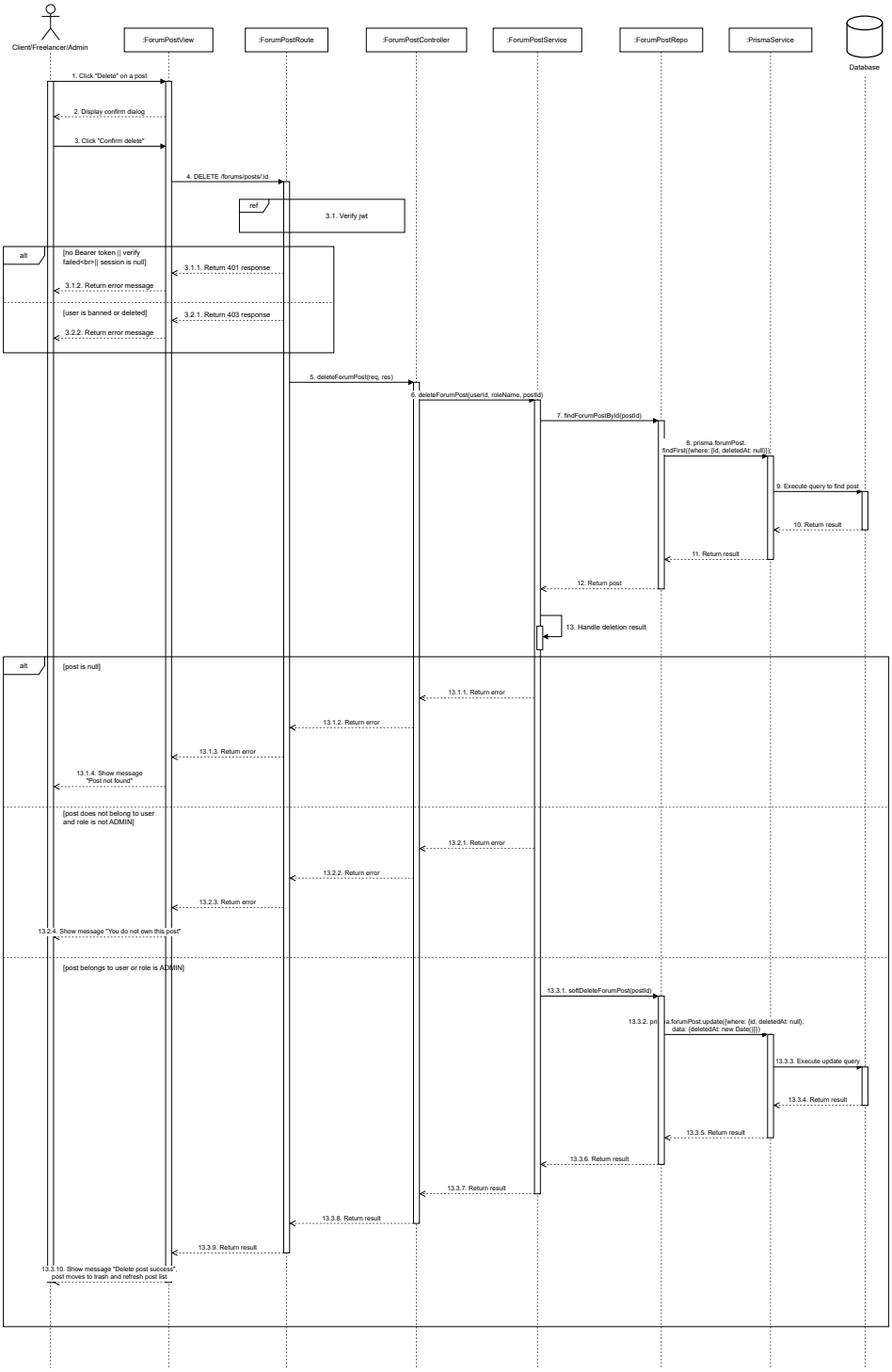

Figure 85: Sequence diagram of UC-14.07 Delete Post

3.7.8 UC-14.08 – Like Post

a. Class diagram

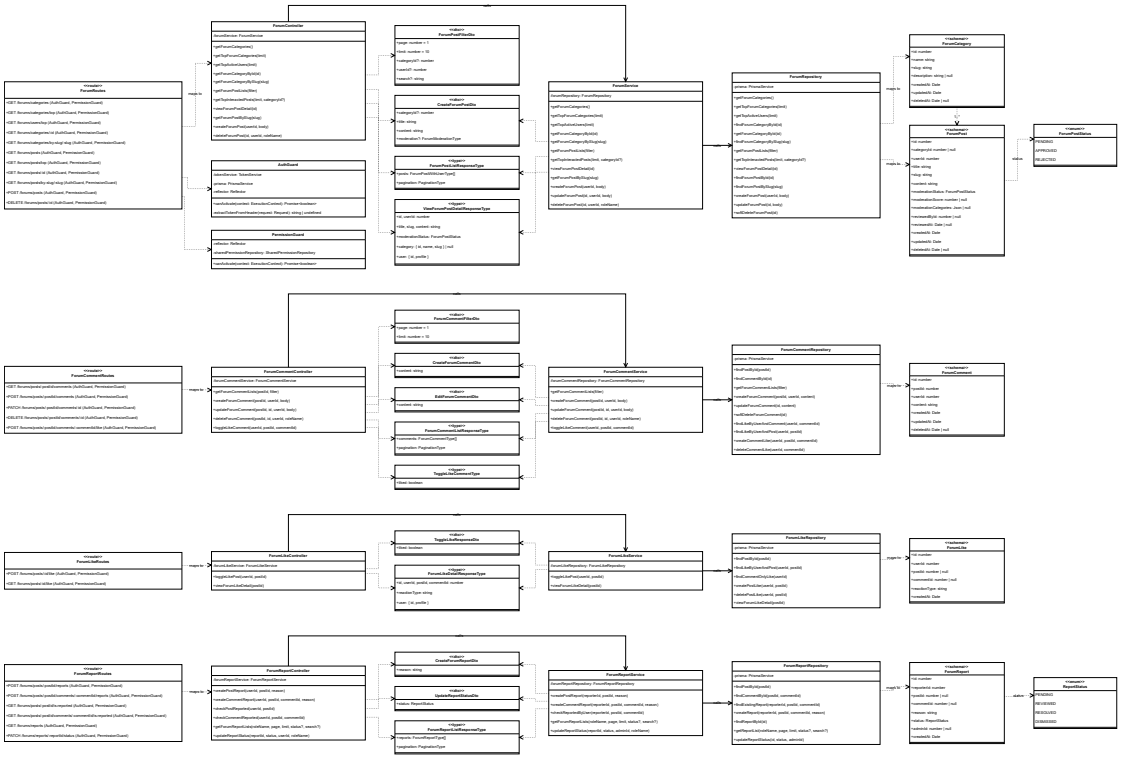

Figure 86: Class diagram of UC-14.08 Like Post

b. Sequence diagram

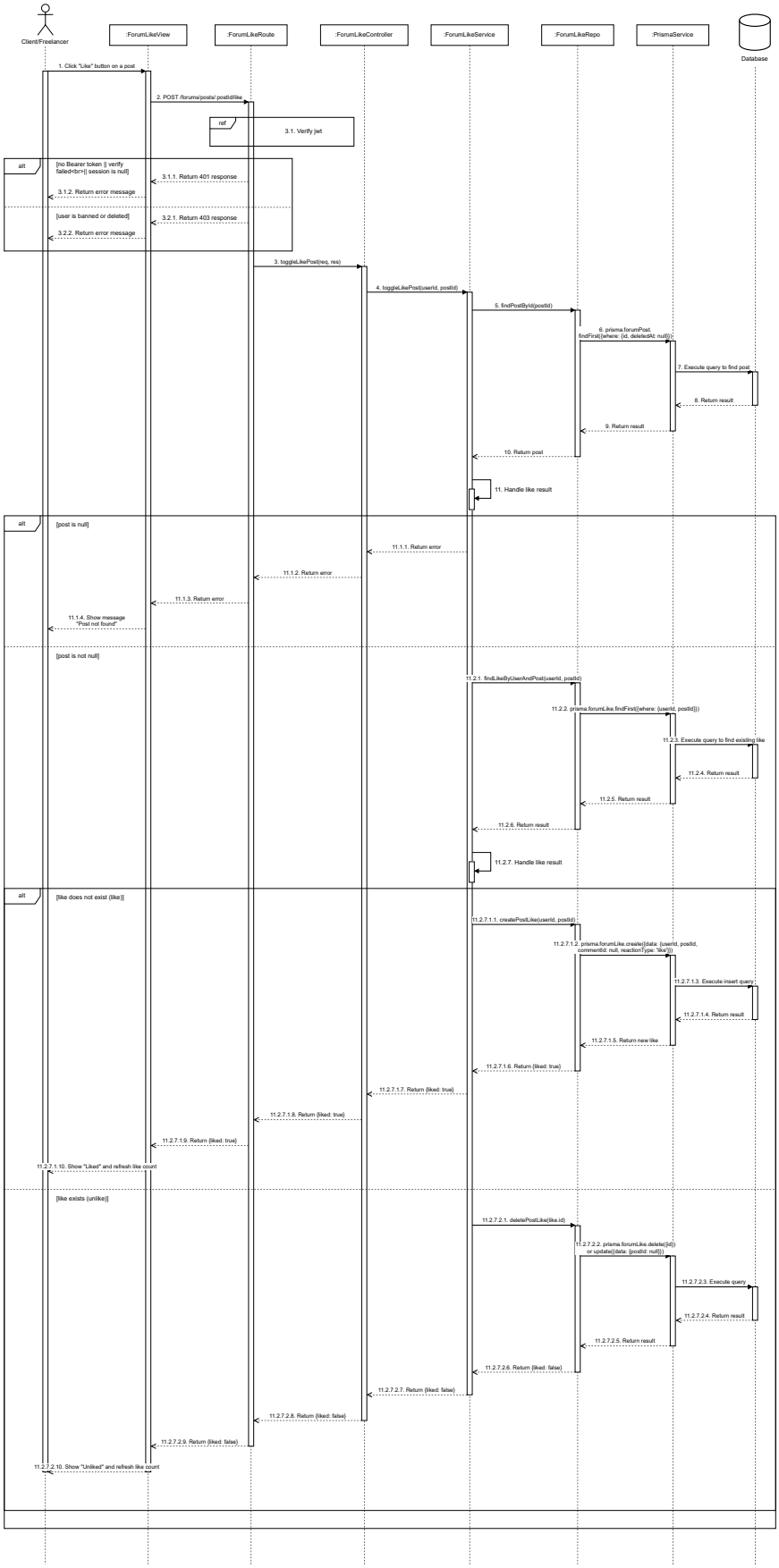

Figure 87: Sequence diagram of UC-14.08 Like Post

3.7.9 UC-14.09 – Unlike Post

a. Class diagram

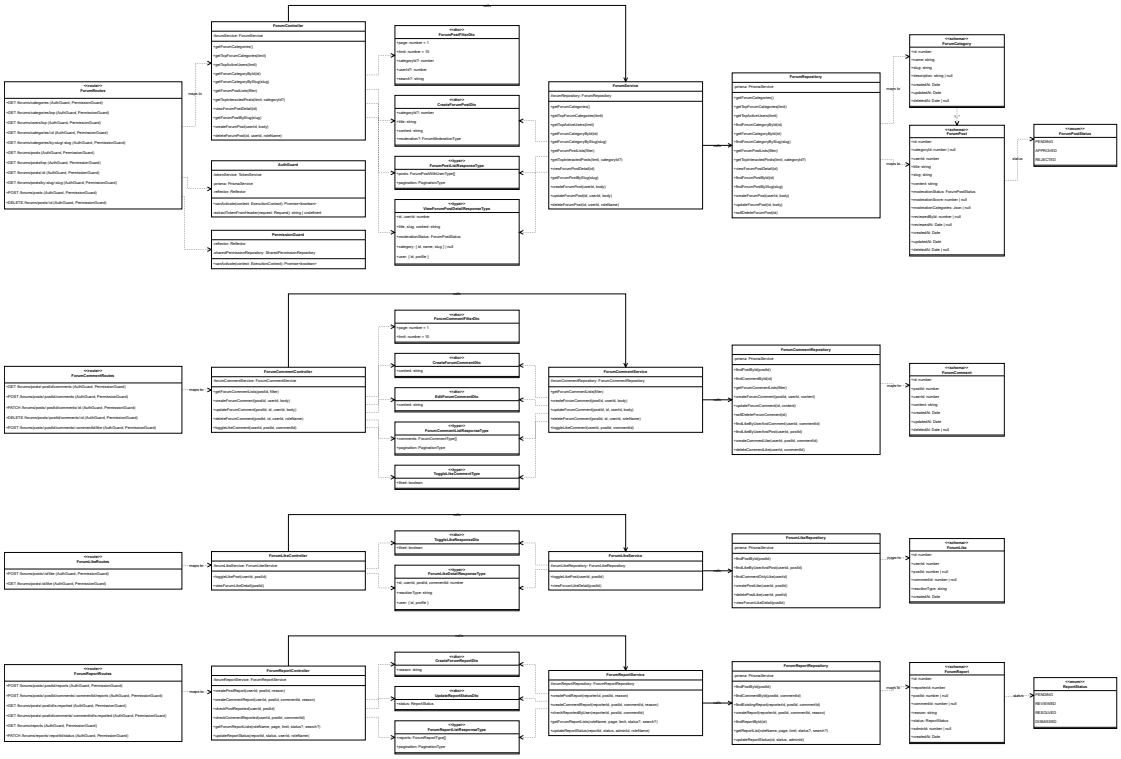

Figure 88: Class diagram of UC-14.09 Unlike Post

b. Sequence diagram

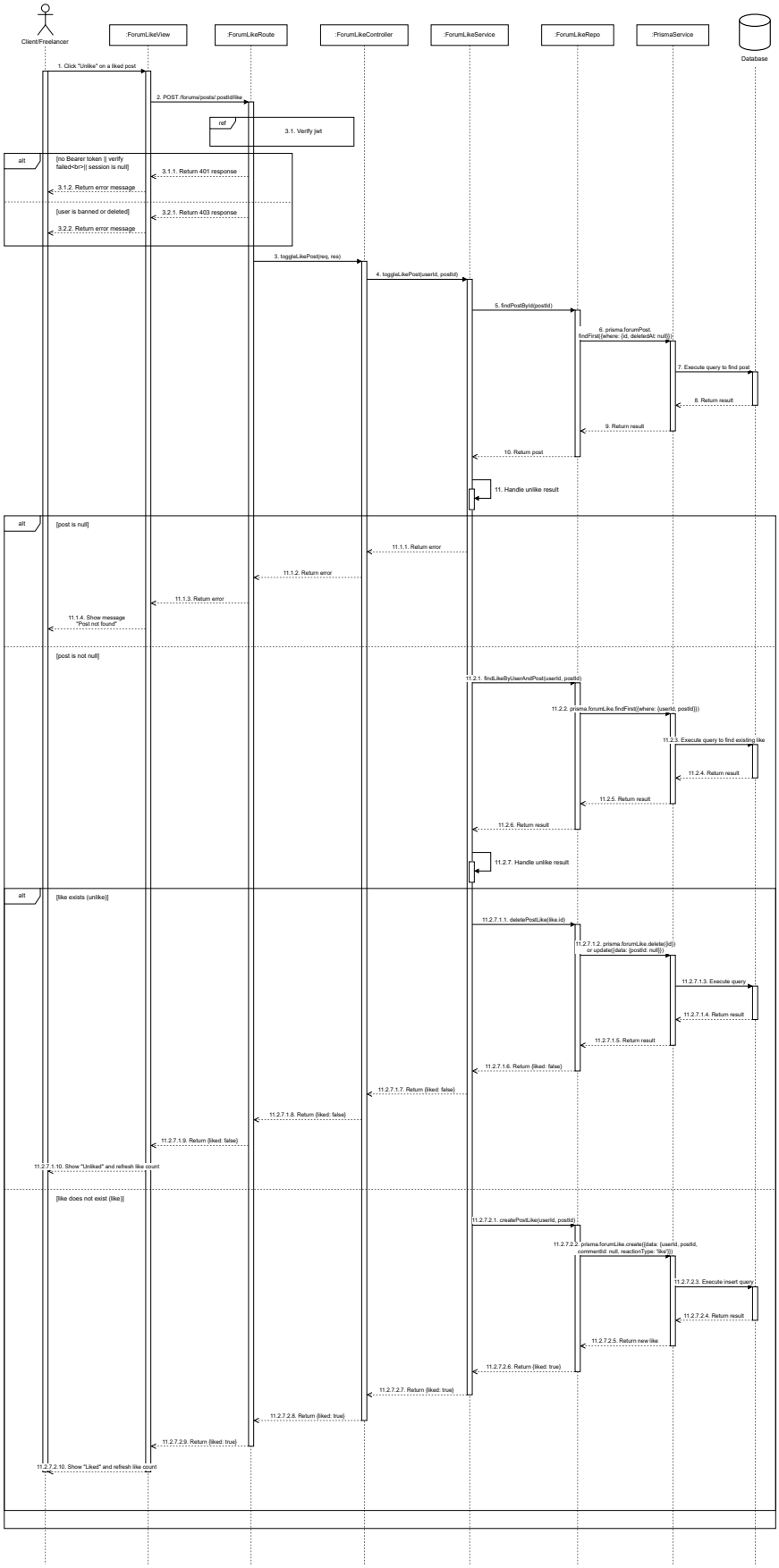

Figure 89: Sequence diagram of UC-14.09 Unlike Post

3.7.10 UC-14.10 – View Post Likes

a. Class diagram

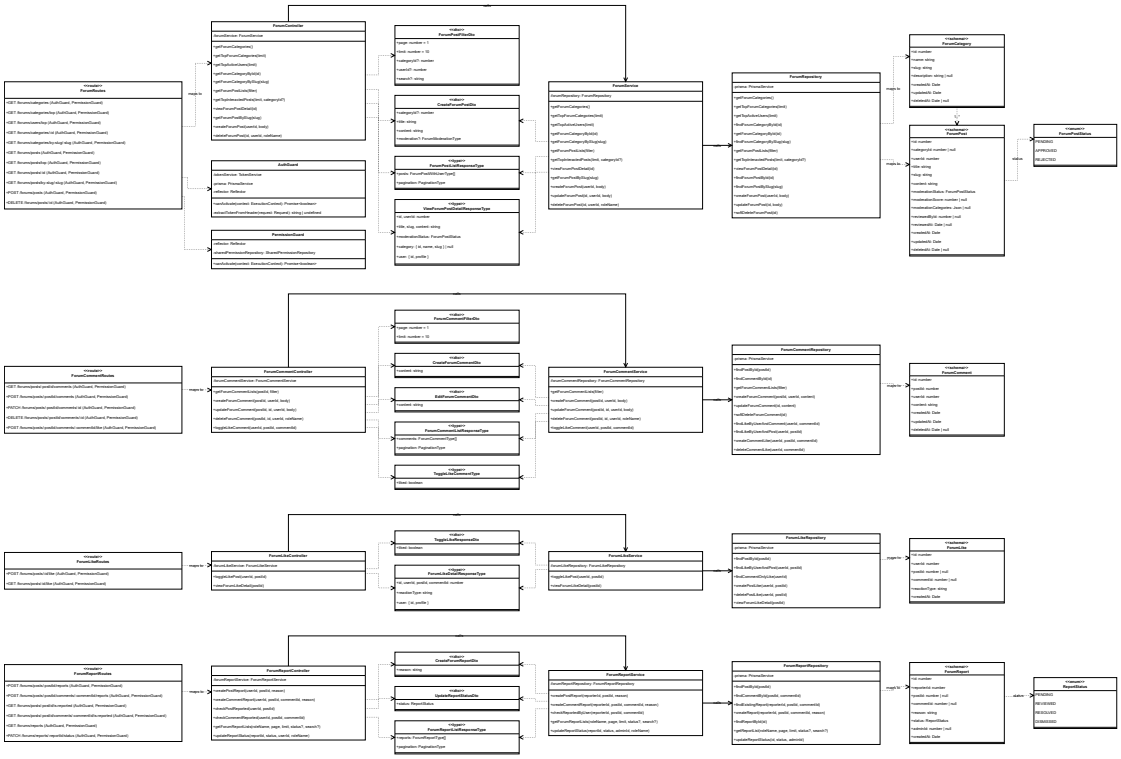

Figure 90: Class diagram of UC-14.10 View Post Likes

b. Sequence diagram

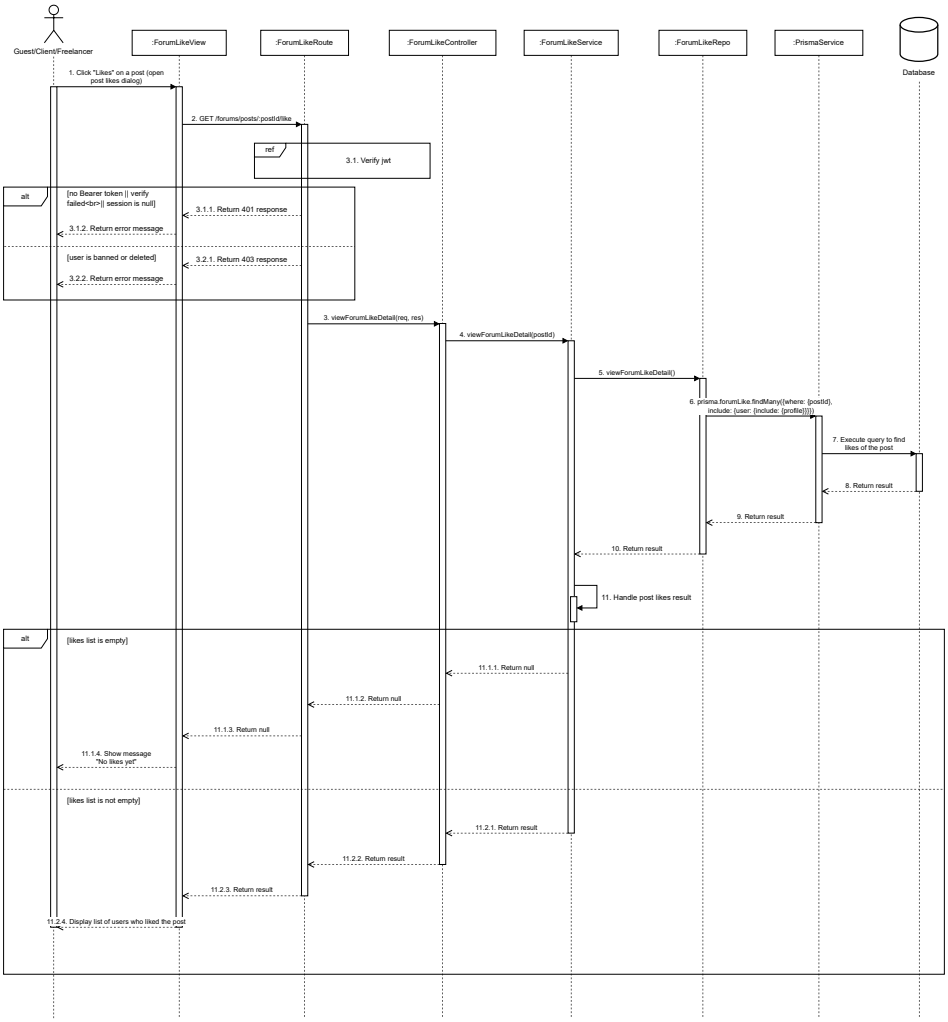

Figure 91: Sequence diagram of UC-14.10 View Post Likes

3.7.11 UC-14.11 – View Post Comments

a. Class diagram

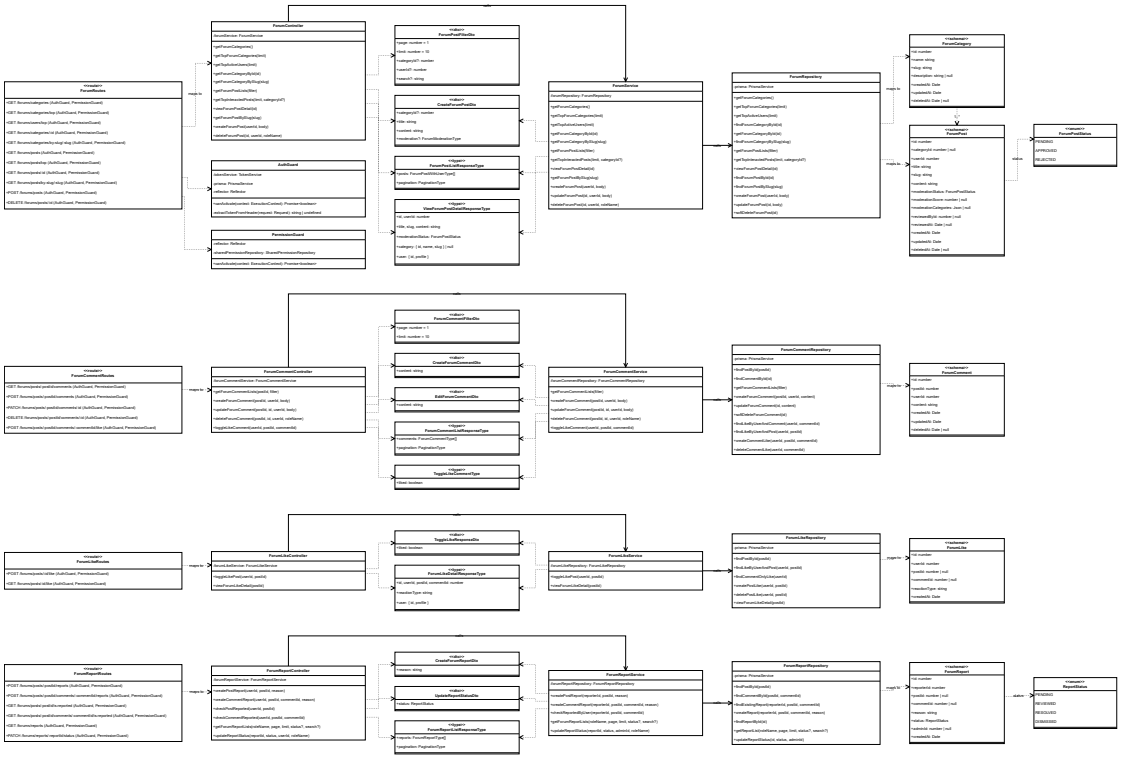

Figure 92: Class diagram of UC-14.11 View Post Comments

b. Sequence diagram

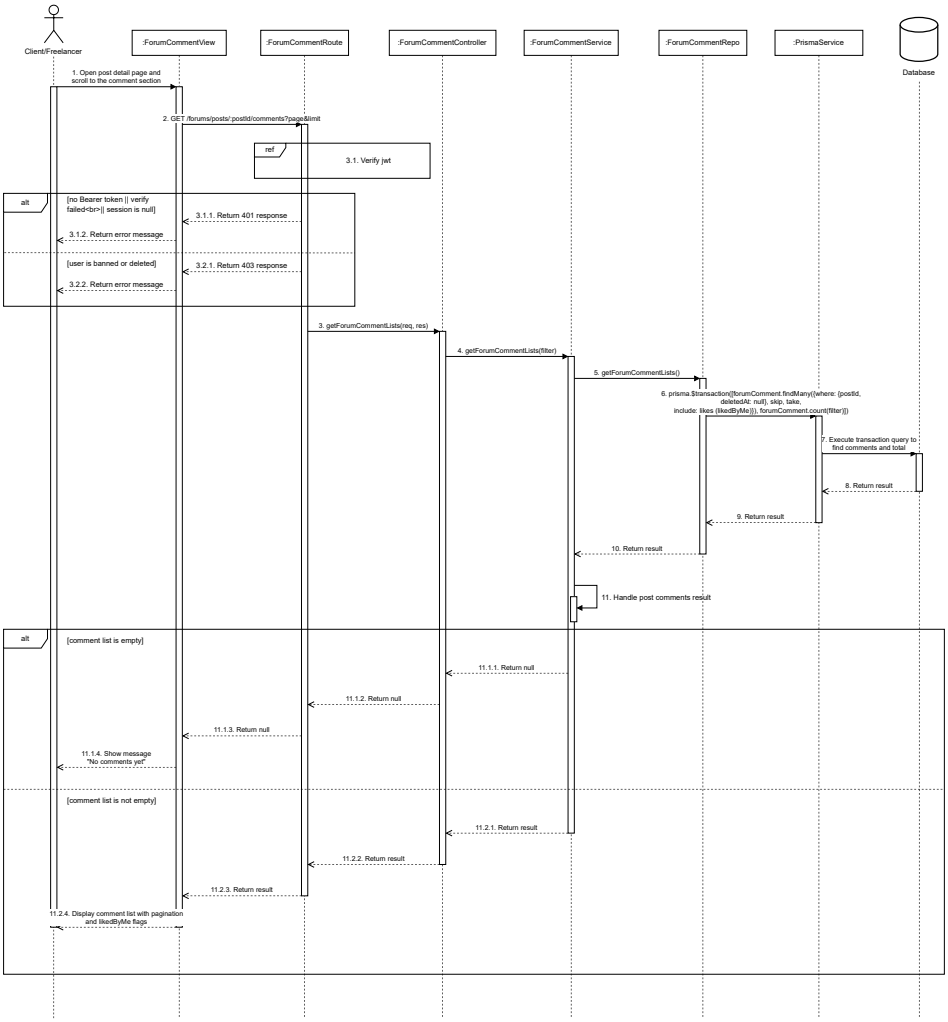

Figure 93: Sequence diagram of UC-14.11 View Post Comments

3.7.12 UC-14.12 – Create Comment

a. Class diagram

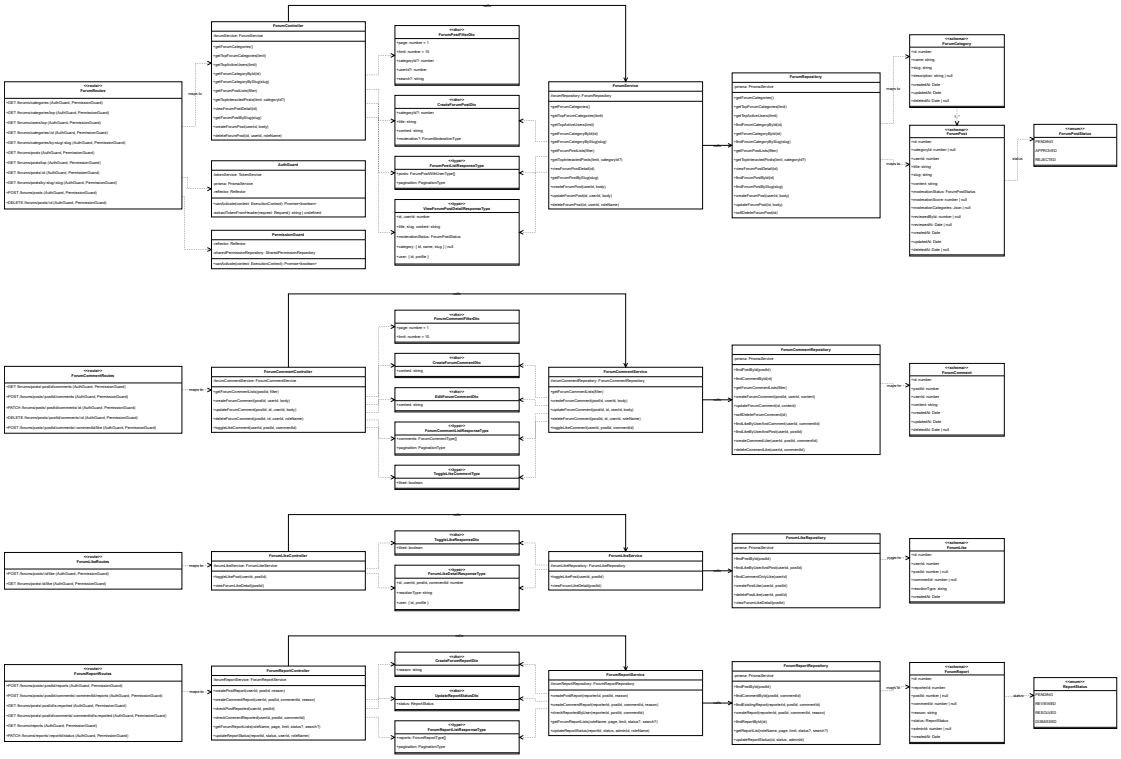

Figure 94: Class diagram of UC-14.12 Create Comment

b. Sequence diagram

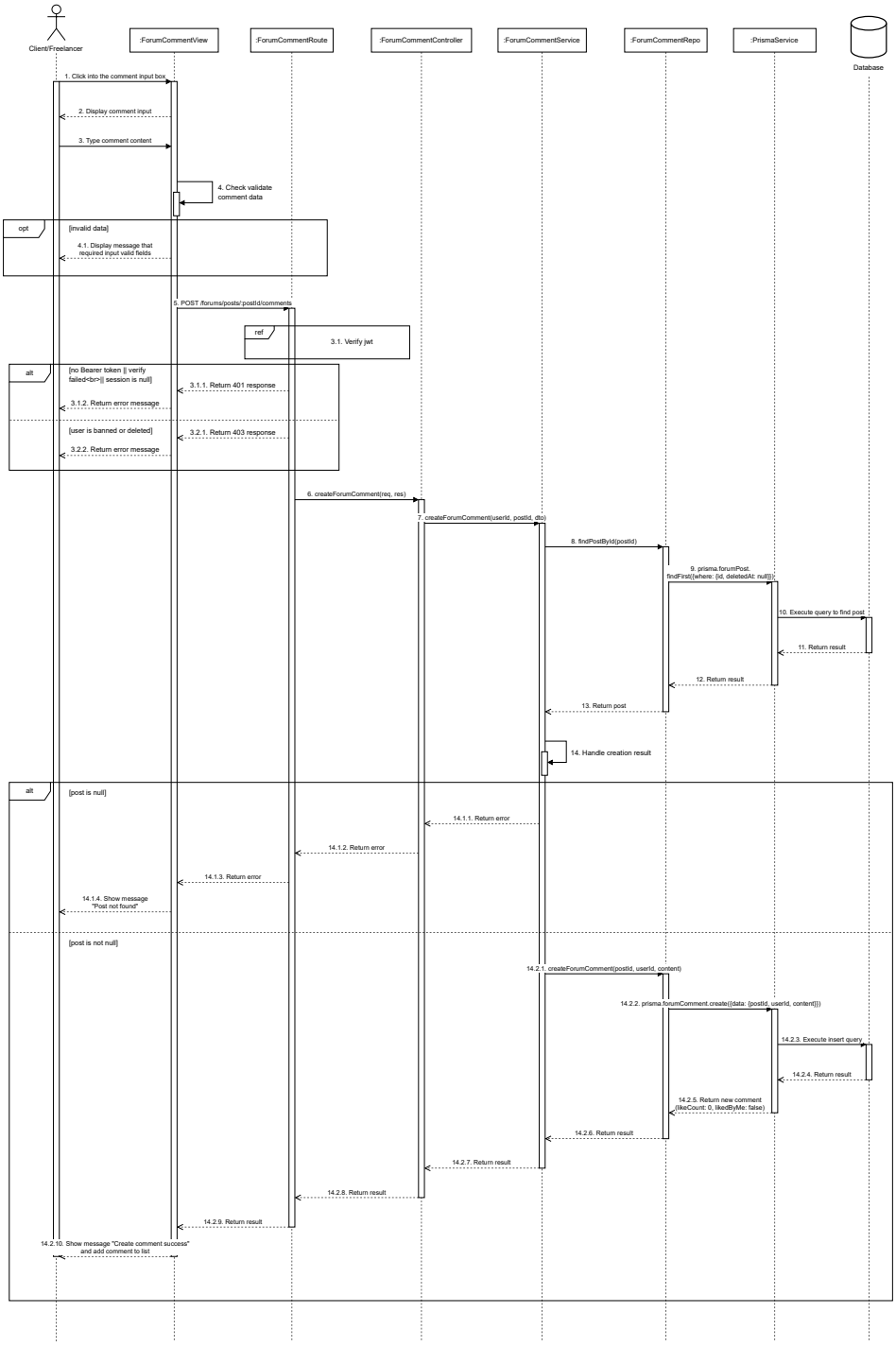

Figure 95: Sequence diagram of UC-14.12 Create Comment

3.7.13 UC-14.13 – Update Comment

a. Class diagram

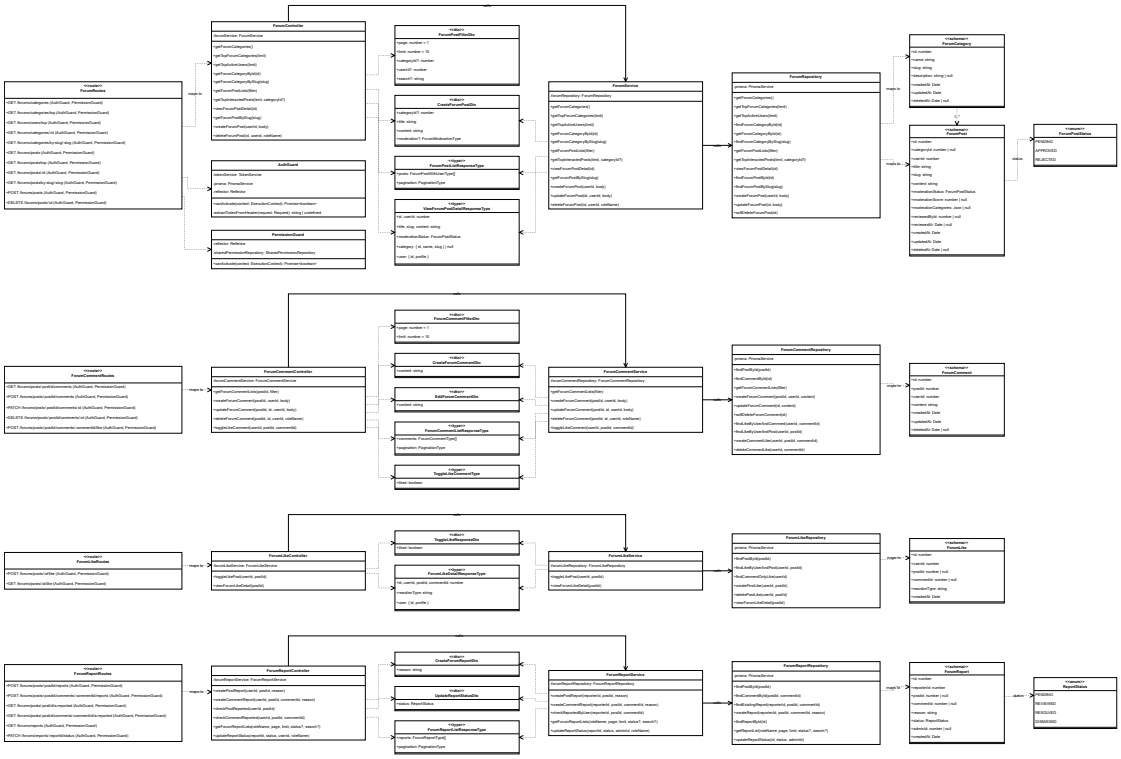

Figure 96: Class diagram of UC-14.13 Update Comment

b. Sequence diagram

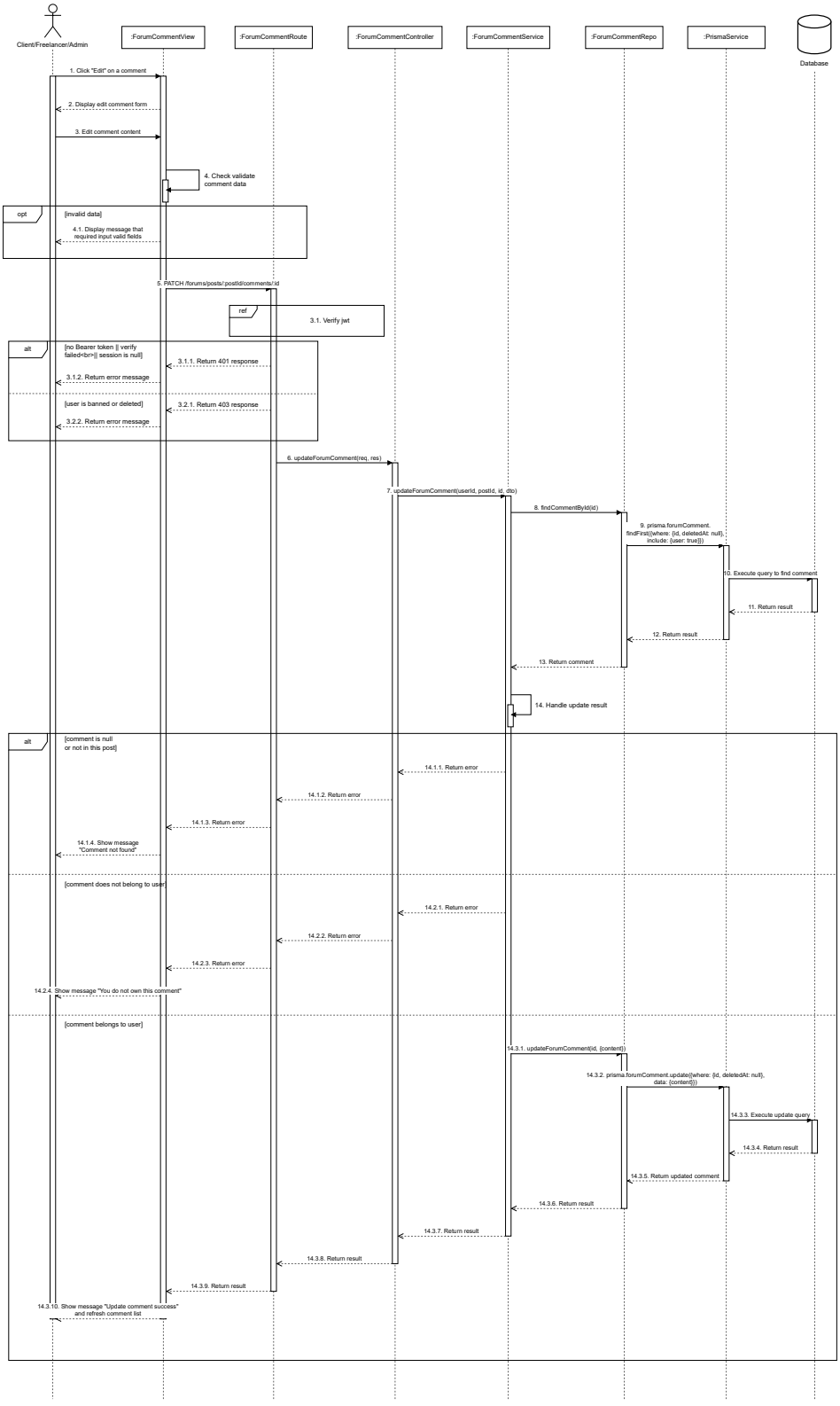

Figure 97: Sequence diagram of UC-14.13 Update Comment

3.7.14 UC-14.14 – Delete Comment

a. Class diagram

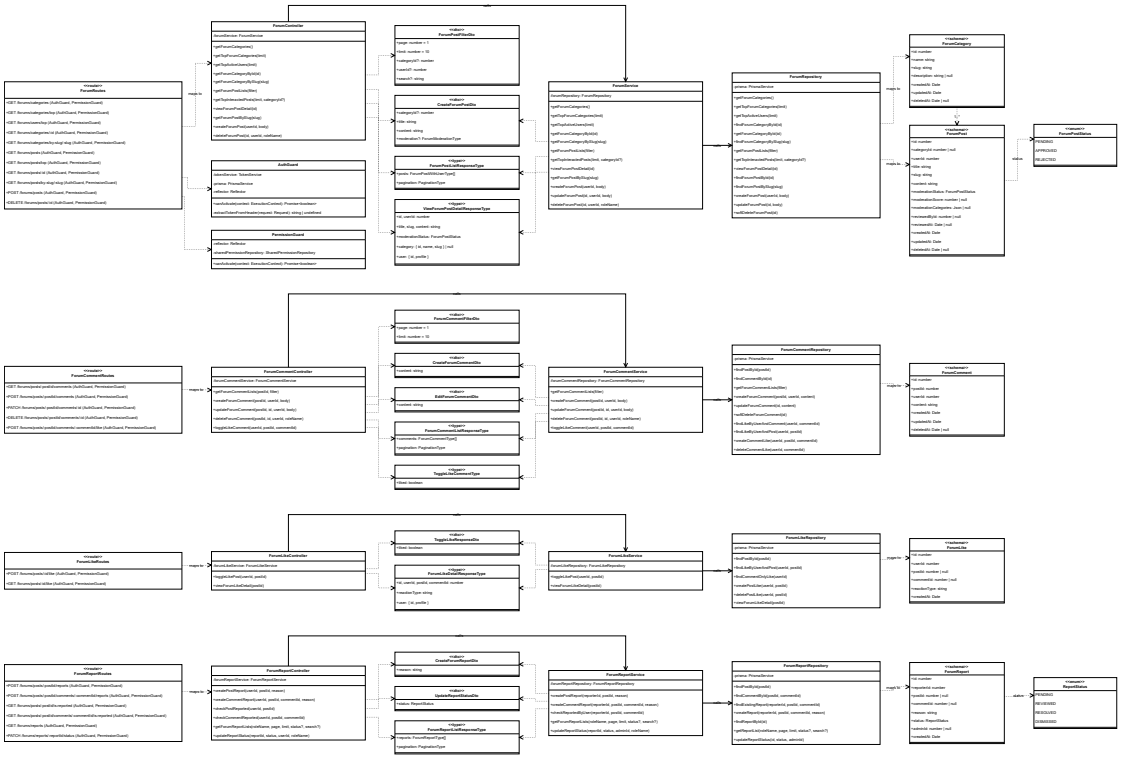

Figure 98: Class diagram of UC-14.14 Delete Comment

b. Sequence diagram

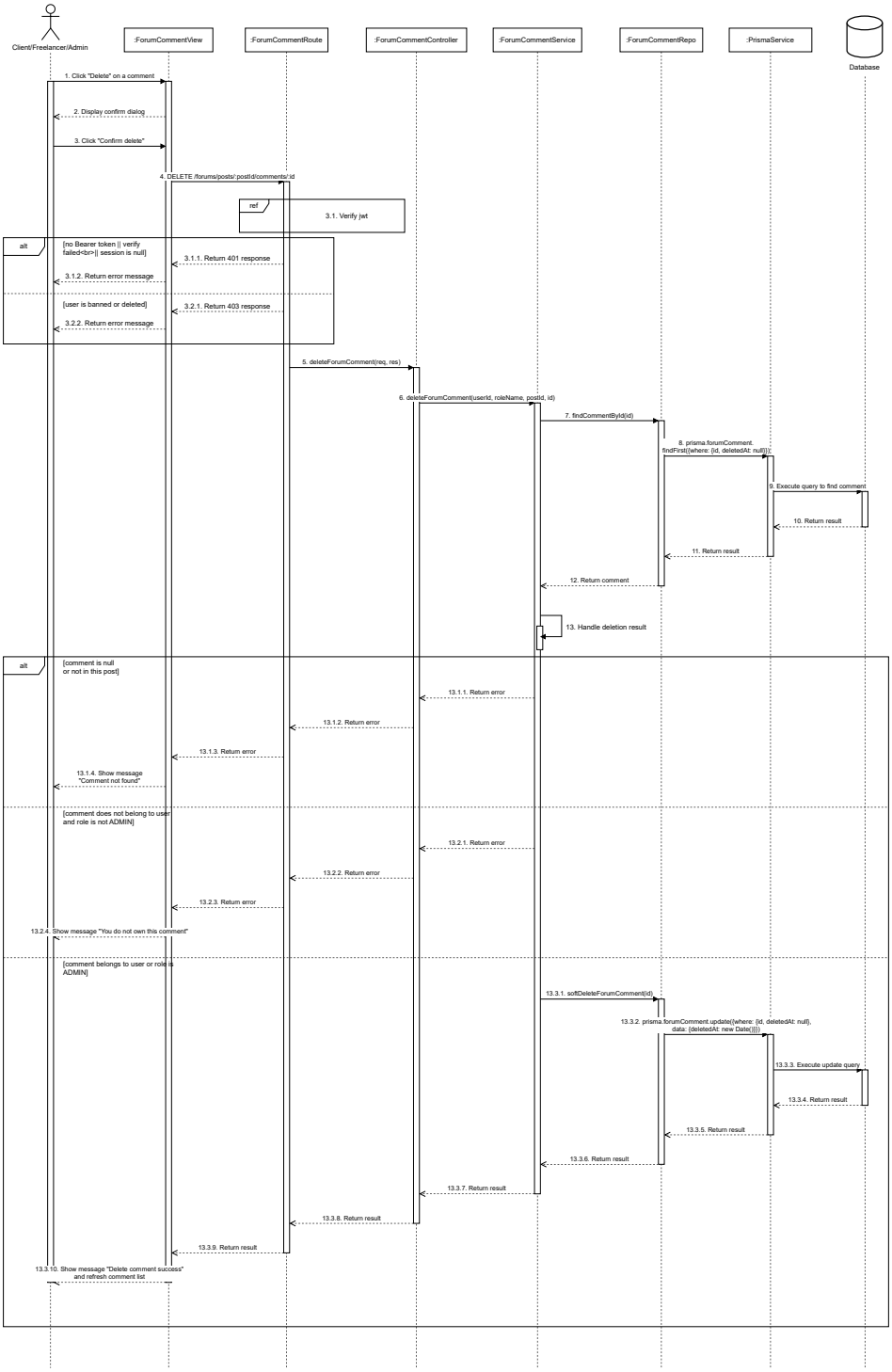

Figure 99: Sequence diagram of UC-14.14 Delete Comment

3.7.15 UC-14.15 – Like Comment

a. Class diagram

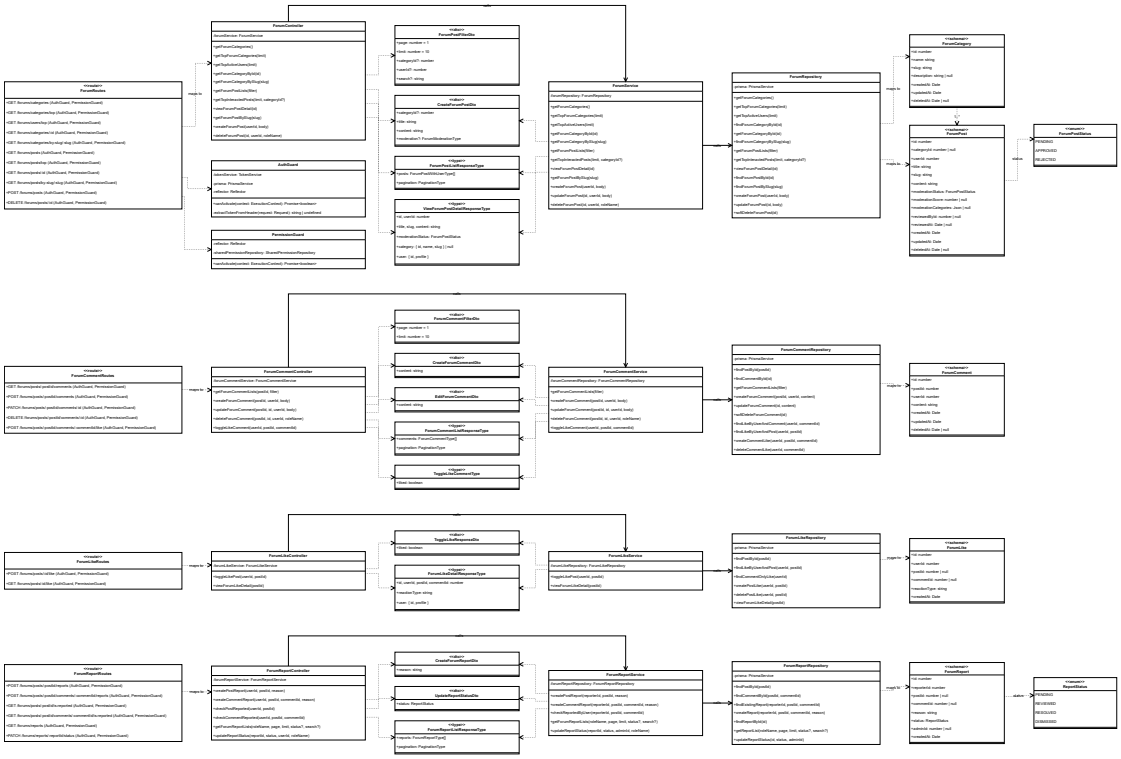

Figure 100: Class diagram of UC-14.15 Like Comment

b. Sequence diagram

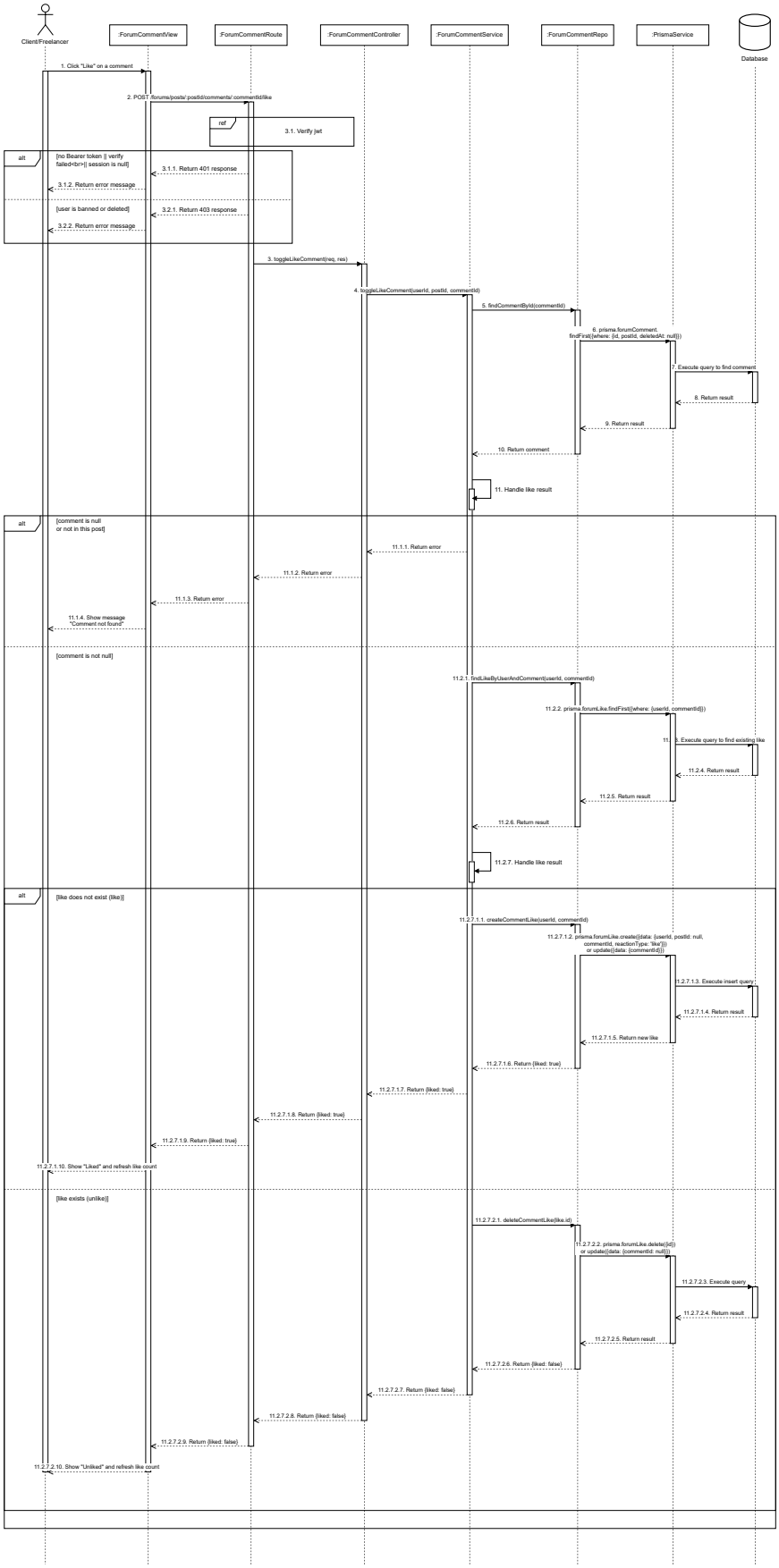

Figure 101: Sequence diagram of UC-14.15 Like Comment

3.7.16 UC-14.16 – Unlike Comment

a. Class diagram

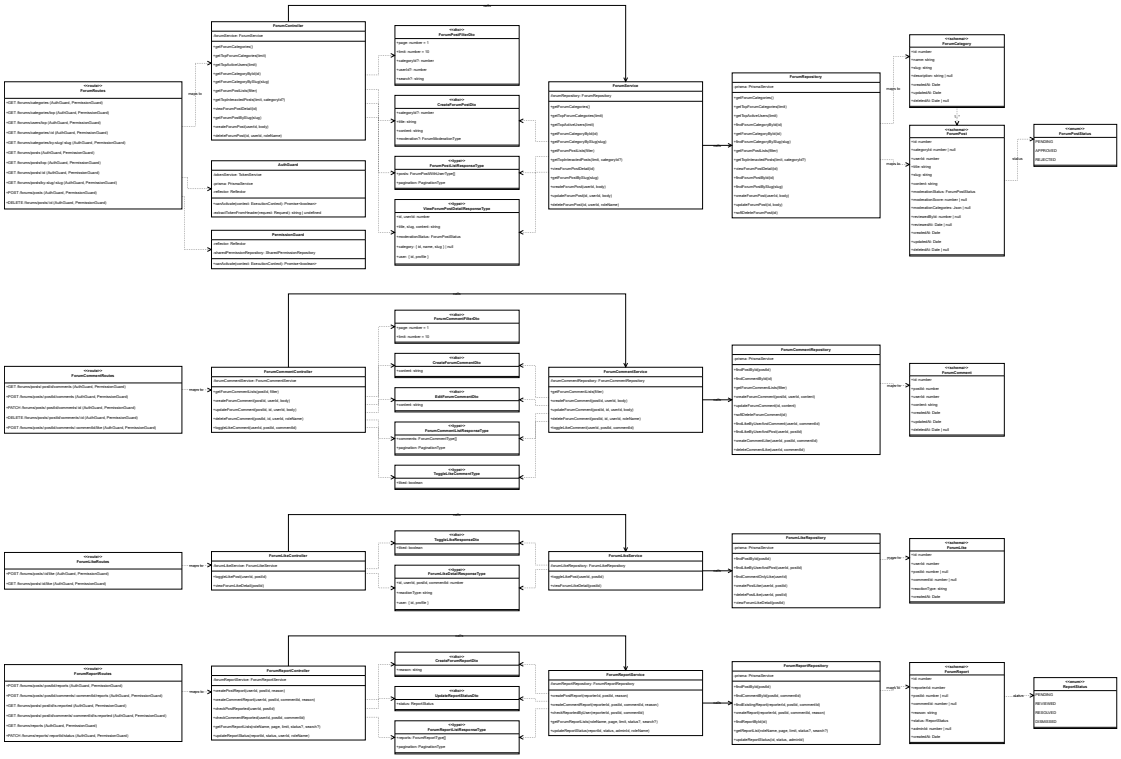

Figure 102: Class diagram of UC-14.16 Unlike Comment

b. Sequence diagram

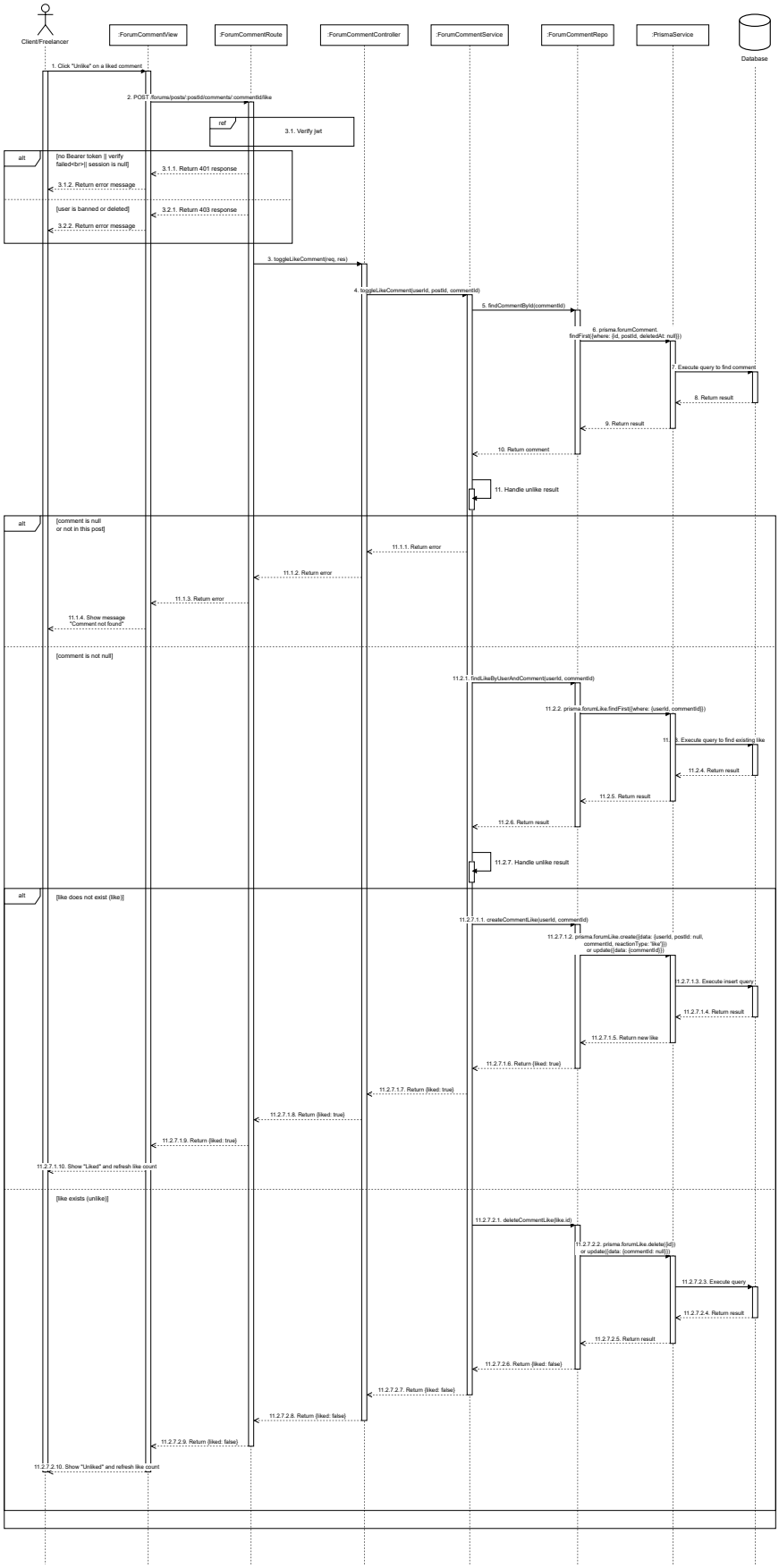

Figure 103: Sequence diagram of UC-14.16 Unlike Comment

3.7.17 UC-14.17 – Report Content

a. Class diagram

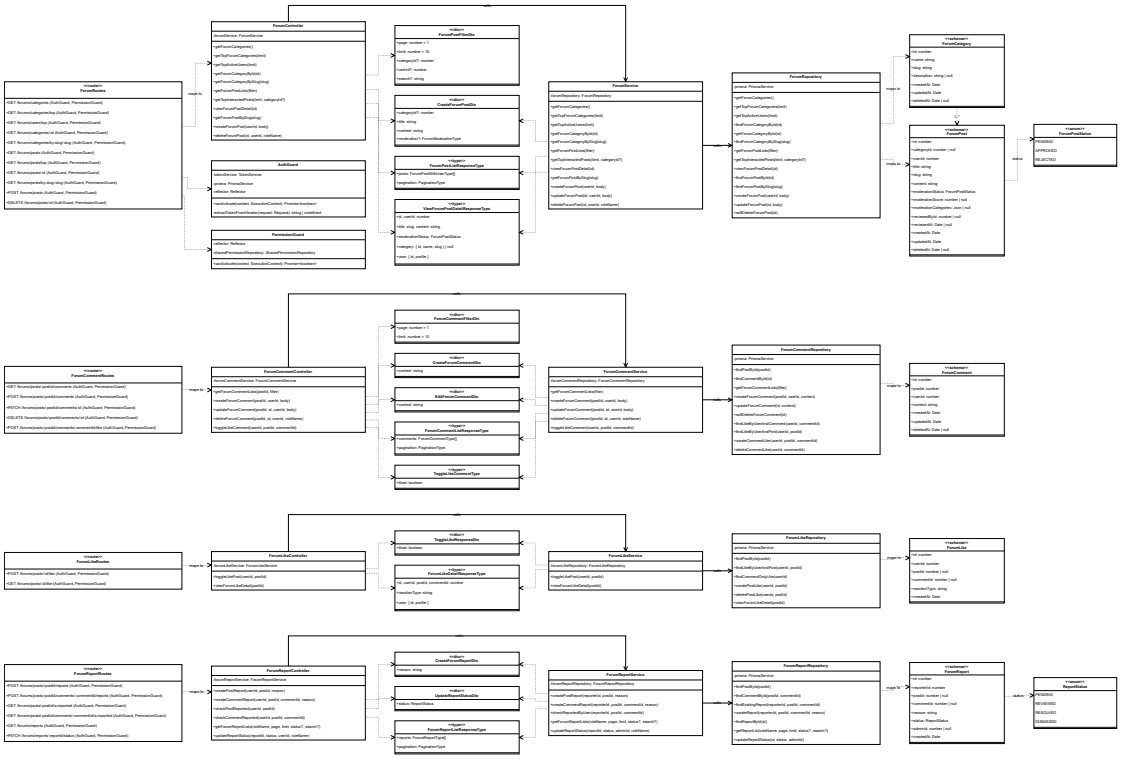

Figure 104: Class diagram of UC-14.17 Report Content

b. Sequence diagram

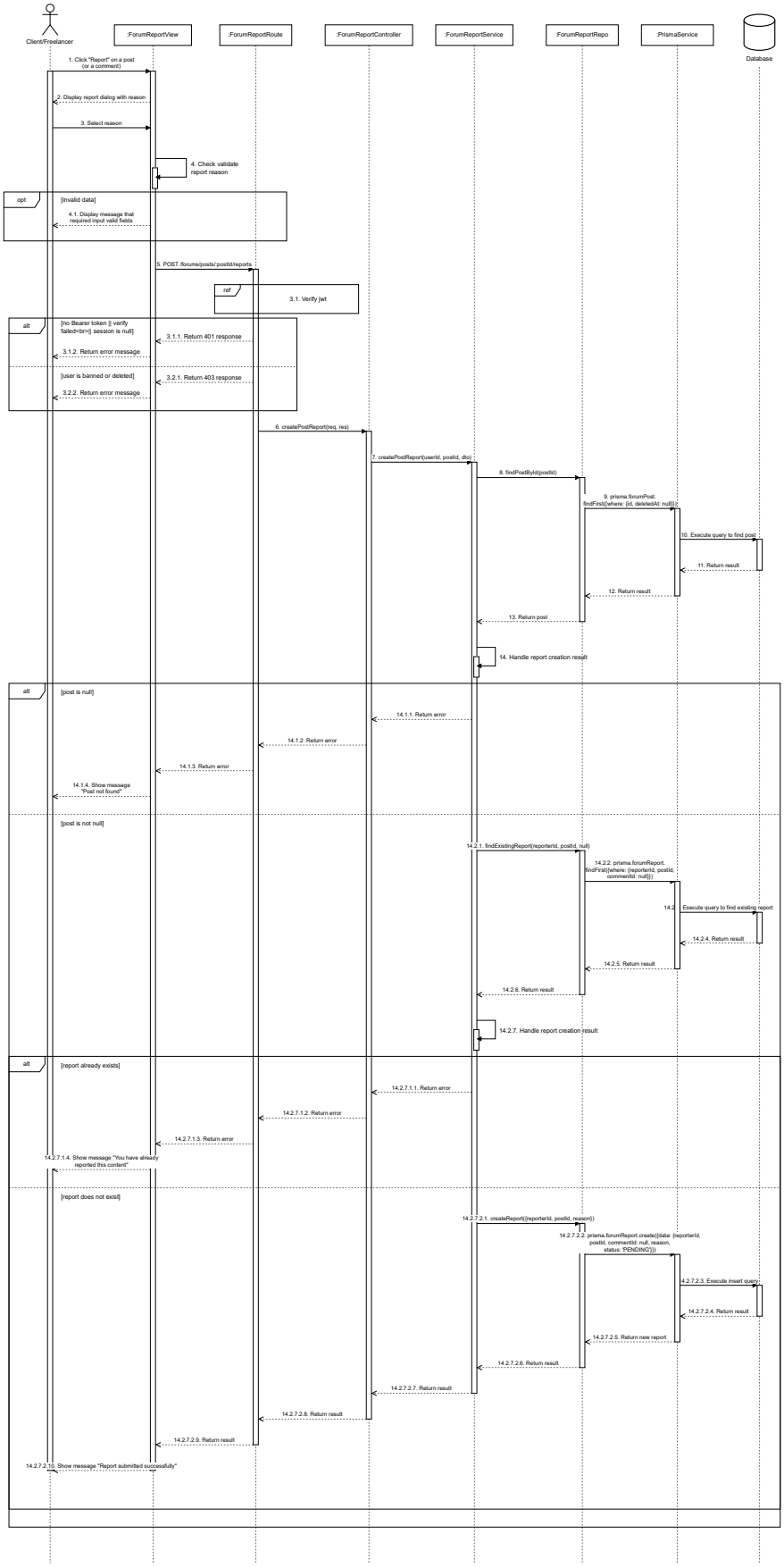

Figure 105: Sequence diagram of UC-14.17 Report Content

3.8 Manage Conversation

Manage Conversation covers the UC-27 private messaging module: listing and starting one-to-one conversations, opening a conversation, sending text messages and files, marking messages as read, recalling a message, and pinning or unpinning a conversation. Delivery is served both by the REST endpoints and by the realtime gateway in the same namespace.

3.8.1 UC-27.01 – View Conversation List

a. Class diagram

Figure 106: Class diagram of UC-27.01 View Conversation List

b. Sequence diagram

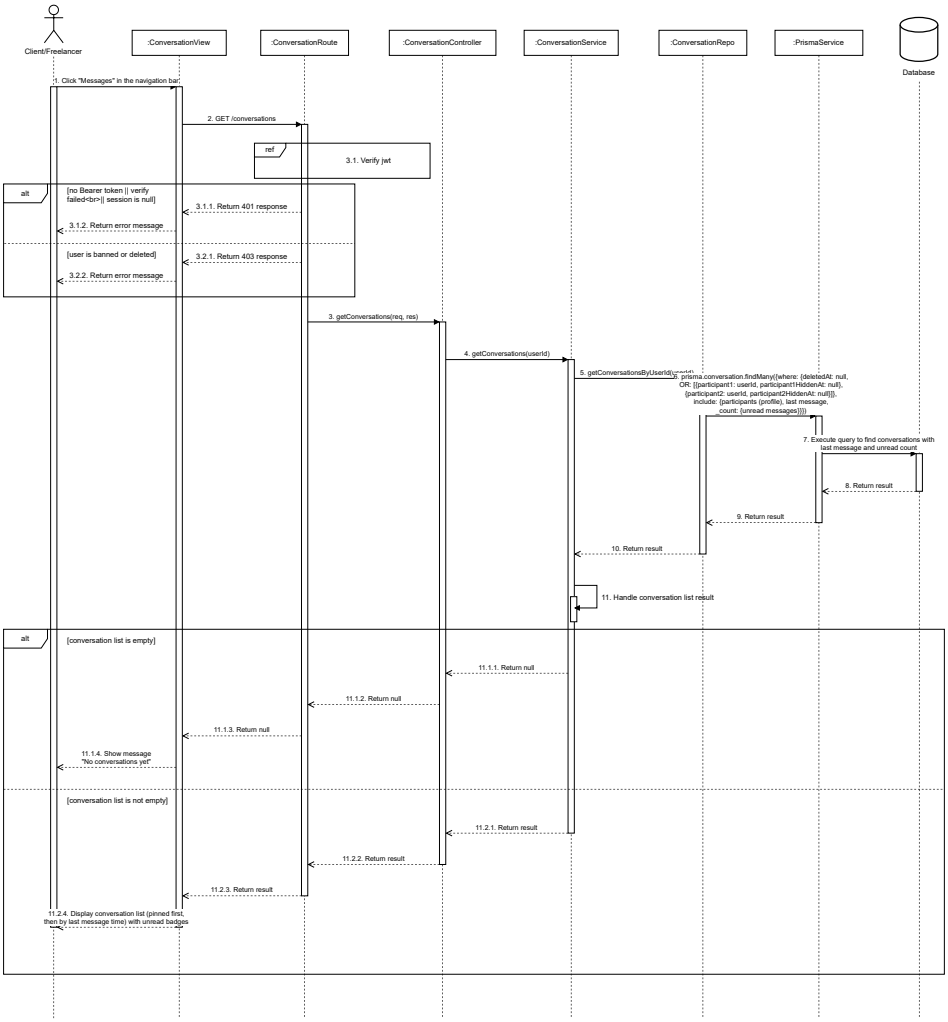

Figure 107: Sequence diagram of UC-27.01 View Conversation List

3.8.2 UC-27.02 – Start Conversation

a. Class diagram

Figure 108: Class diagram of UC-27.02 Start Conversation

b. Sequence diagram

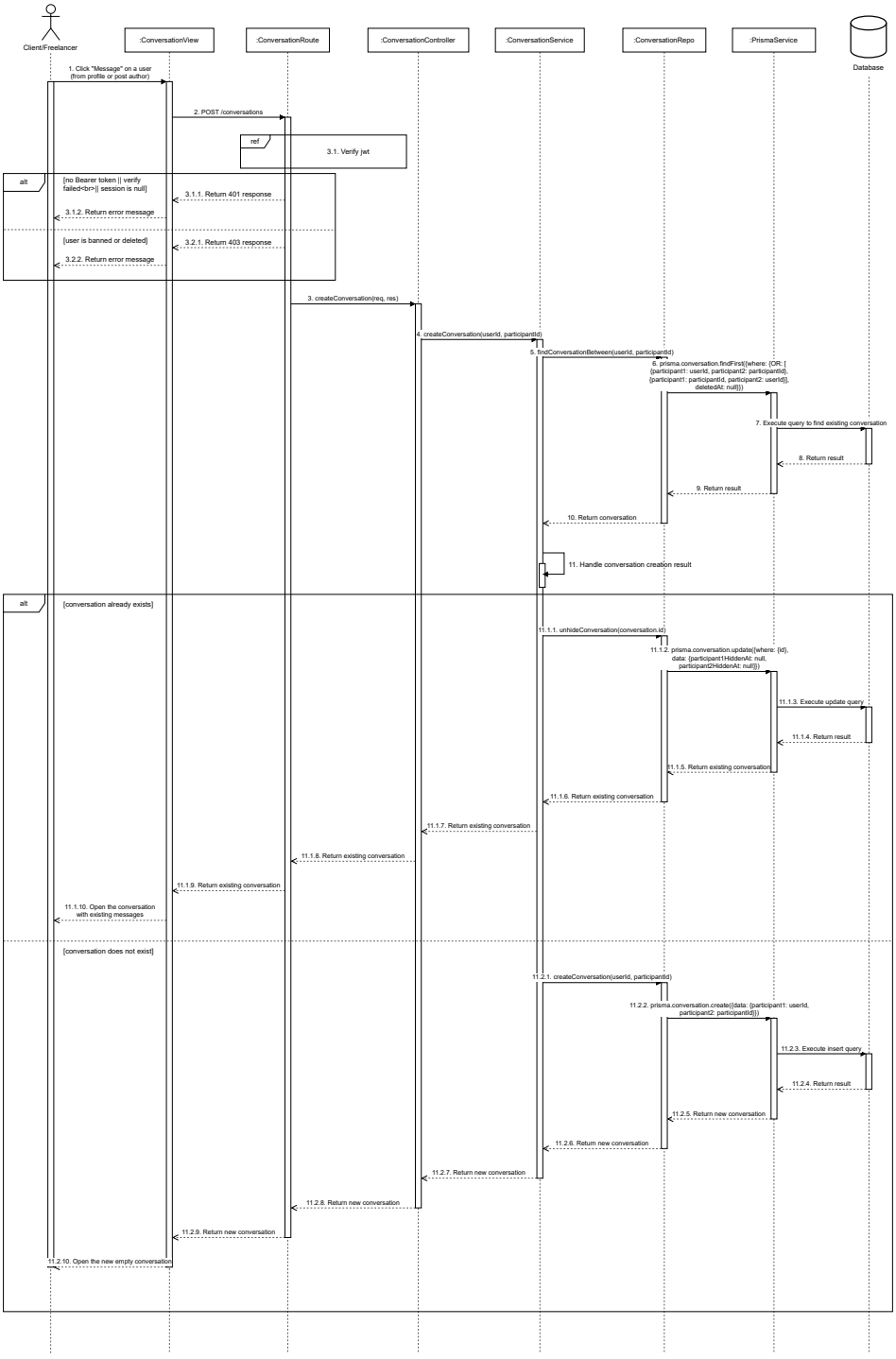

Figure 109: Sequence diagram of UC-27.02 Start Conversation

3.8.3 UC-27.03 – View Conversation Detail

a. Class diagram

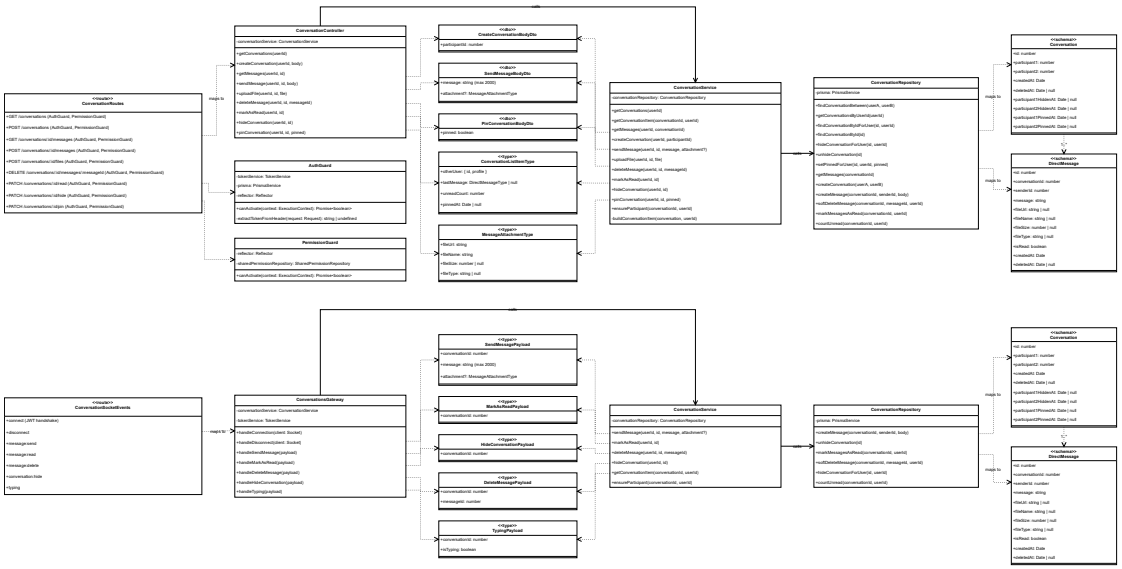

Figure 110: Class diagram of UC-27.03 View Conversation Detail

b. Sequence diagram

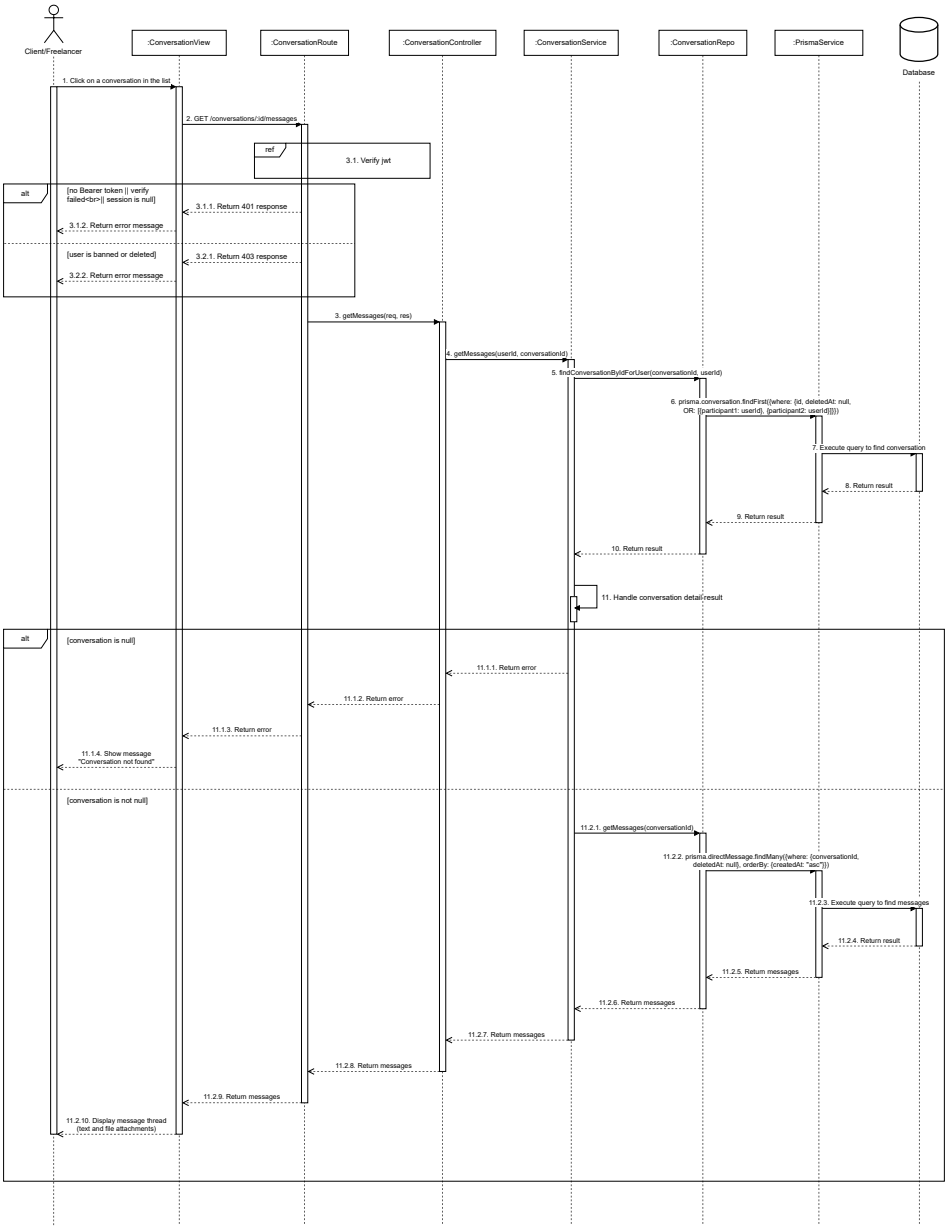

Figure 111: Sequence diagram of UC-27.03 View Conversation Detail

3.8.4 UC-27.04 – Send Message

a. Class diagram

Figure 112: Class diagram of UC-27.04 Send Message

b. Sequence diagram

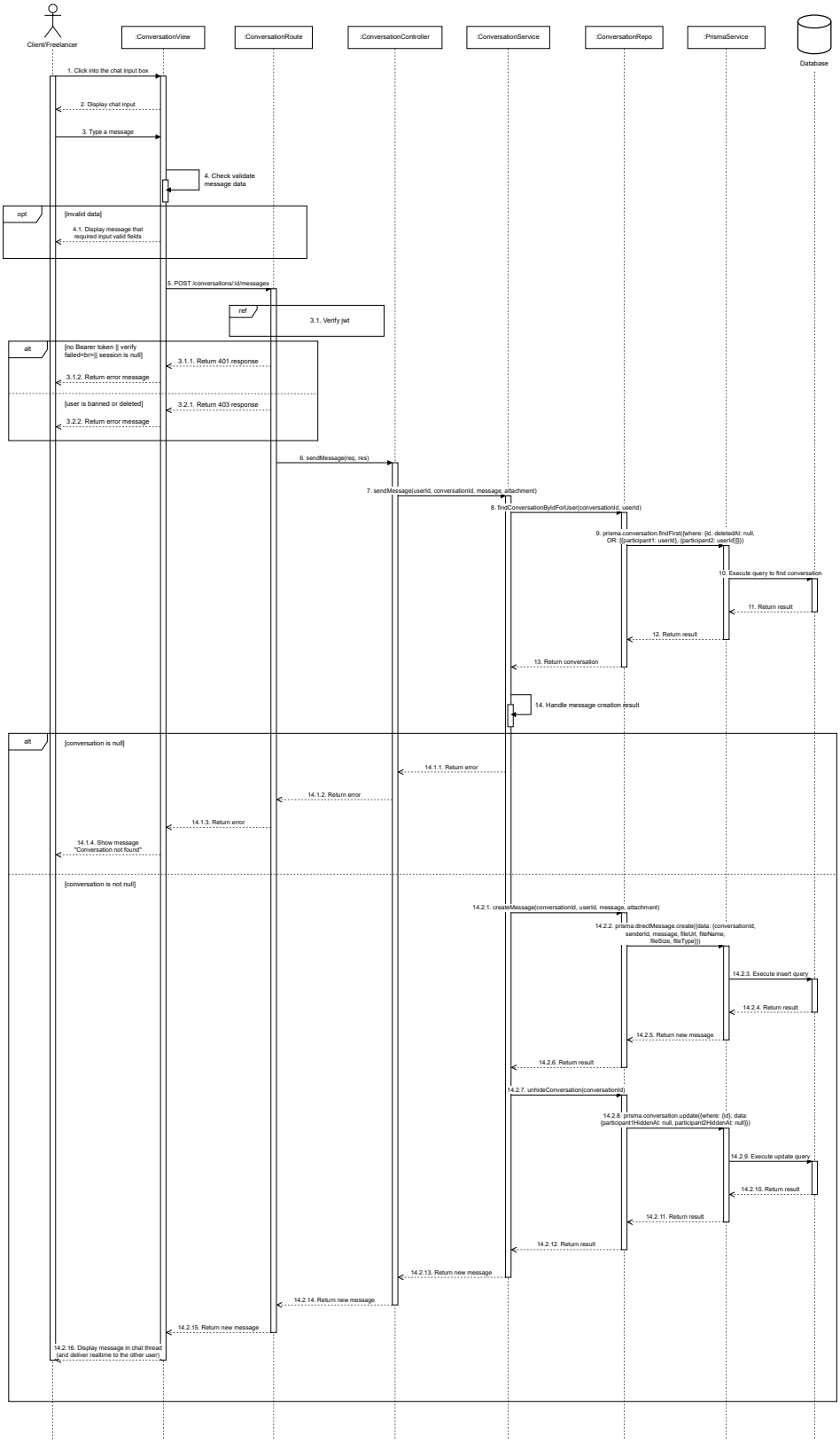

Figure 113: Sequence diagram of UC-27.04 Send Message

3.8.5 UC-27.05 – Mark Message as Read

a. Class diagram

Figure 114: Class diagram of UC-27.05 Mark Message as Read

b. Sequence diagram

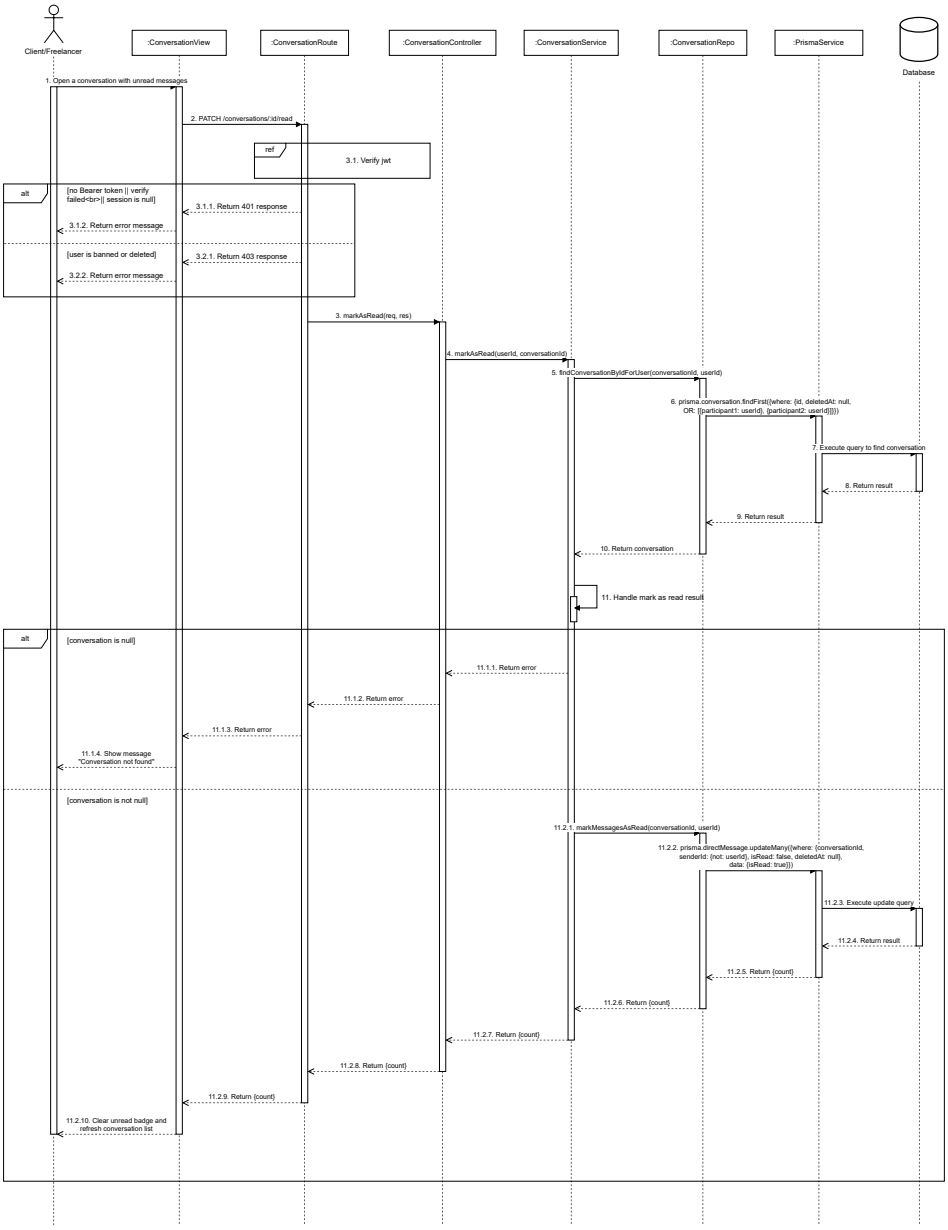

Figure 115: Sequence diagram of UC-27.05 Mark Message as Read

3.8.6 UC-27.06 – Delete/Recall Message

a. Class diagram

Figure 116: Class diagram of UC-27.06 Delete/Recall Message

b. Sequence diagram

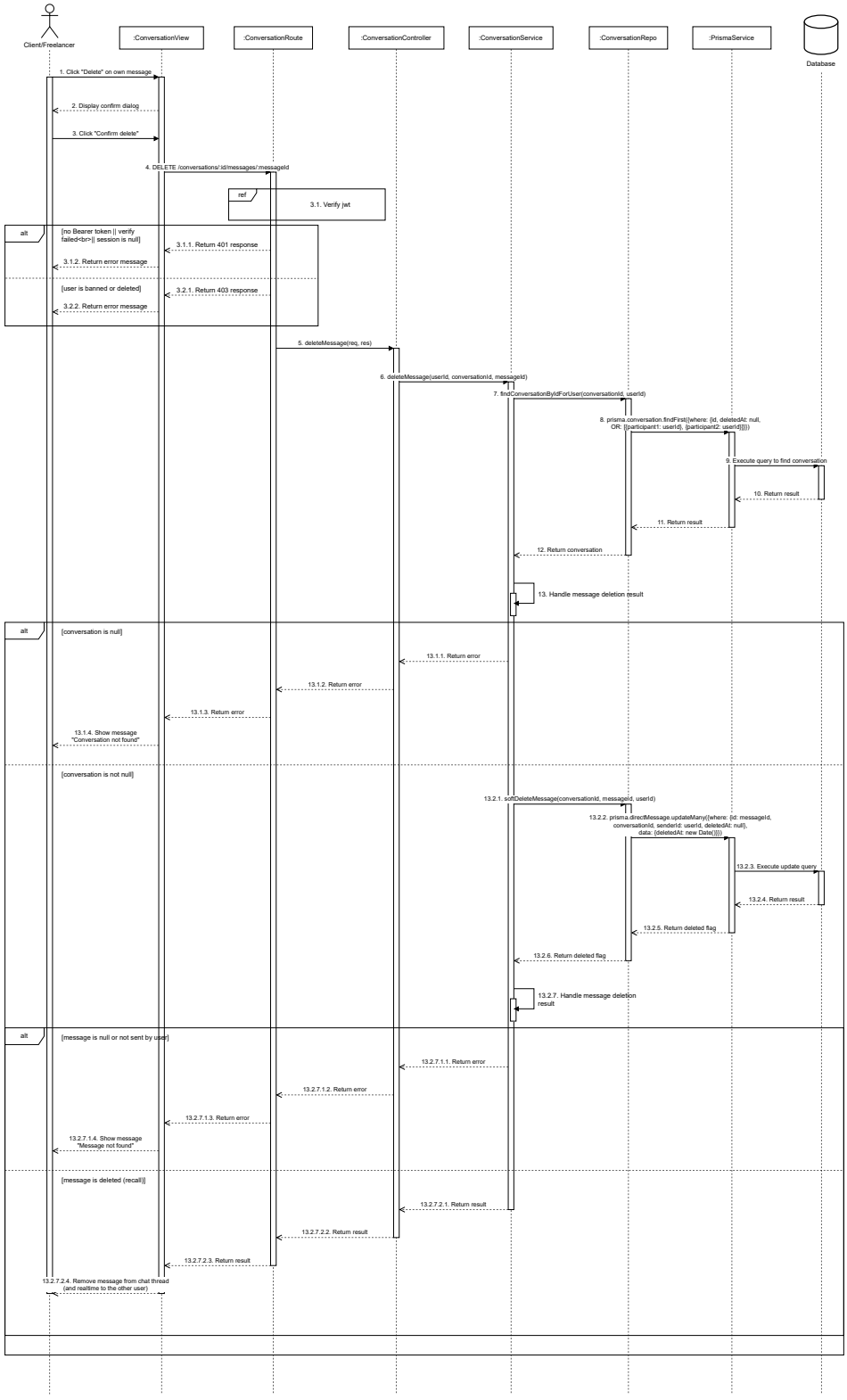

Figure 117: Sequence diagram of UC-27.06 Delete/Recall Message

3.8.7 UC-27.07 – Pin/Unpin Conversation

a. Class diagram

Figure 118: Class diagram of UC-27.07 Pin/Unpin Conversation

b. Sequence diagram

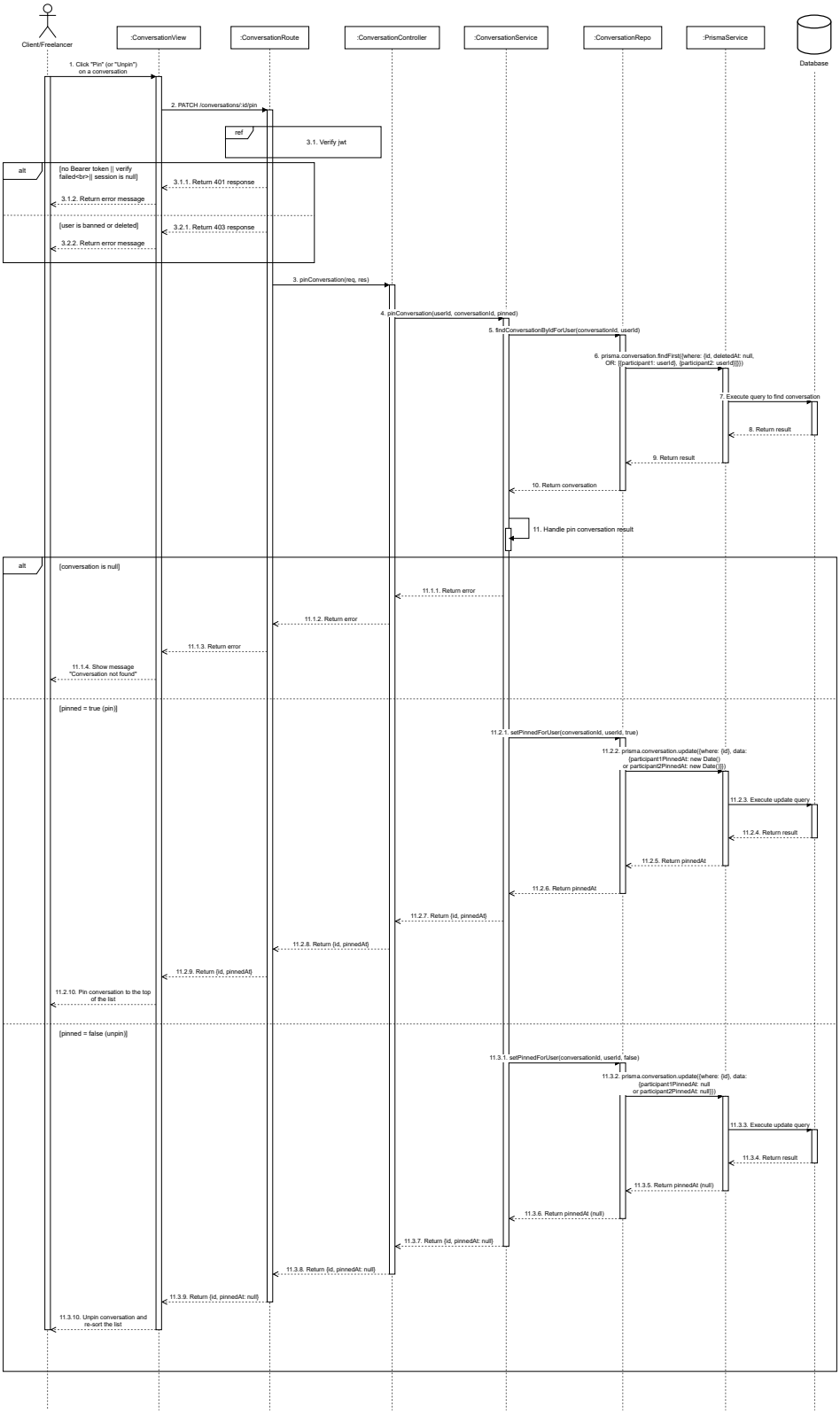

Figure 119: Sequence diagram of UC-27.07 Pin/Unpin Conversation

3.8.8 UC-27.08 – Send File

a. Class diagram

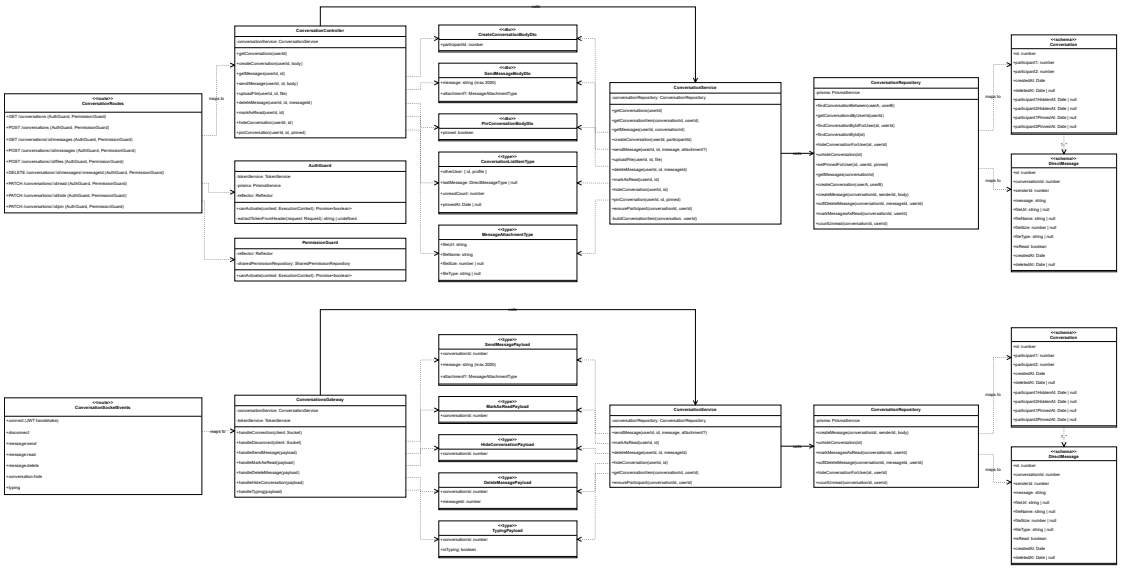

Figure 120: Class diagram of UC-27.08 Send File

b. Sequence diagram

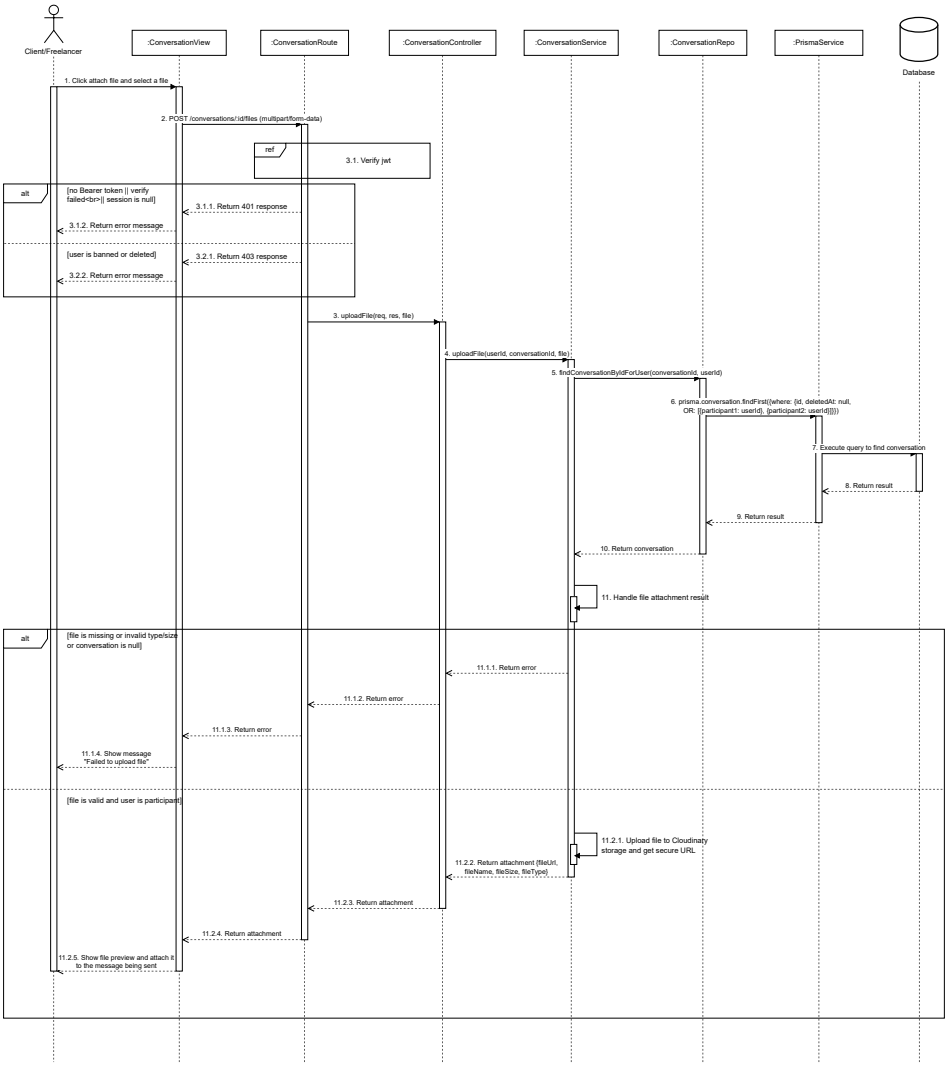

Figure 121: Sequence diagram of UC-27.08 Send File

3.9 Saved Search Management

Saved Search Management allows freelancers to save complex job filter criteria, view their list of saved searches, inspect search details, update filter parameters, and remove obsolete searches.

3.9.1 UC-22.01 – View Saved Search List

a. Class diagram

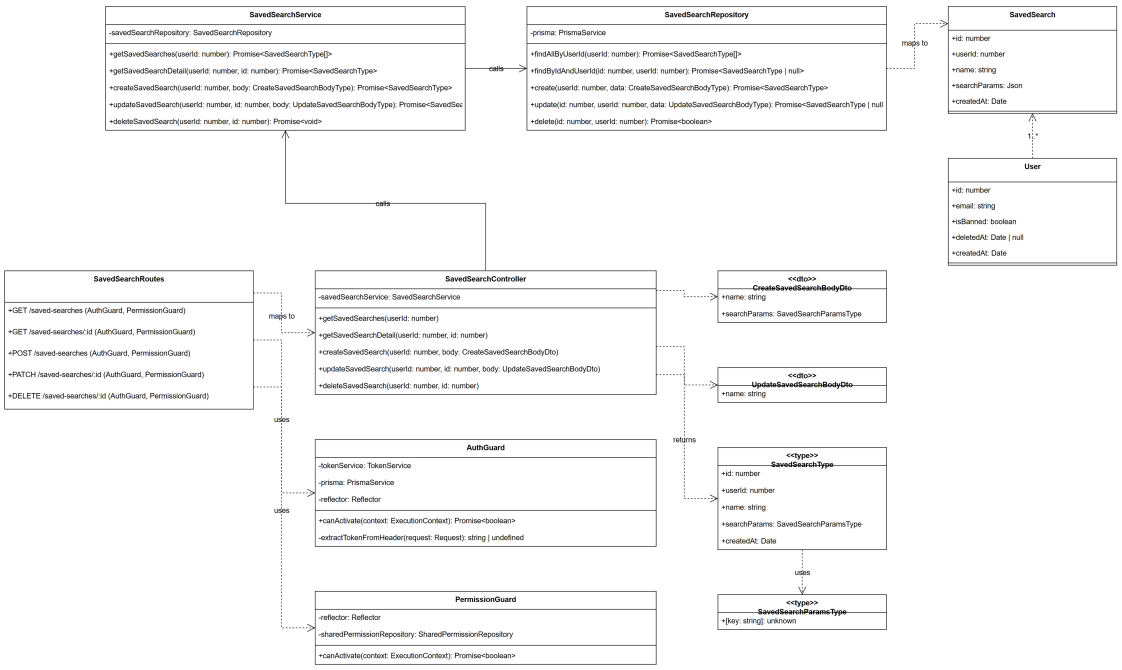

Figure 122: Class diagram of UC-22.01 View Saved Search List

b. Sequence diagram

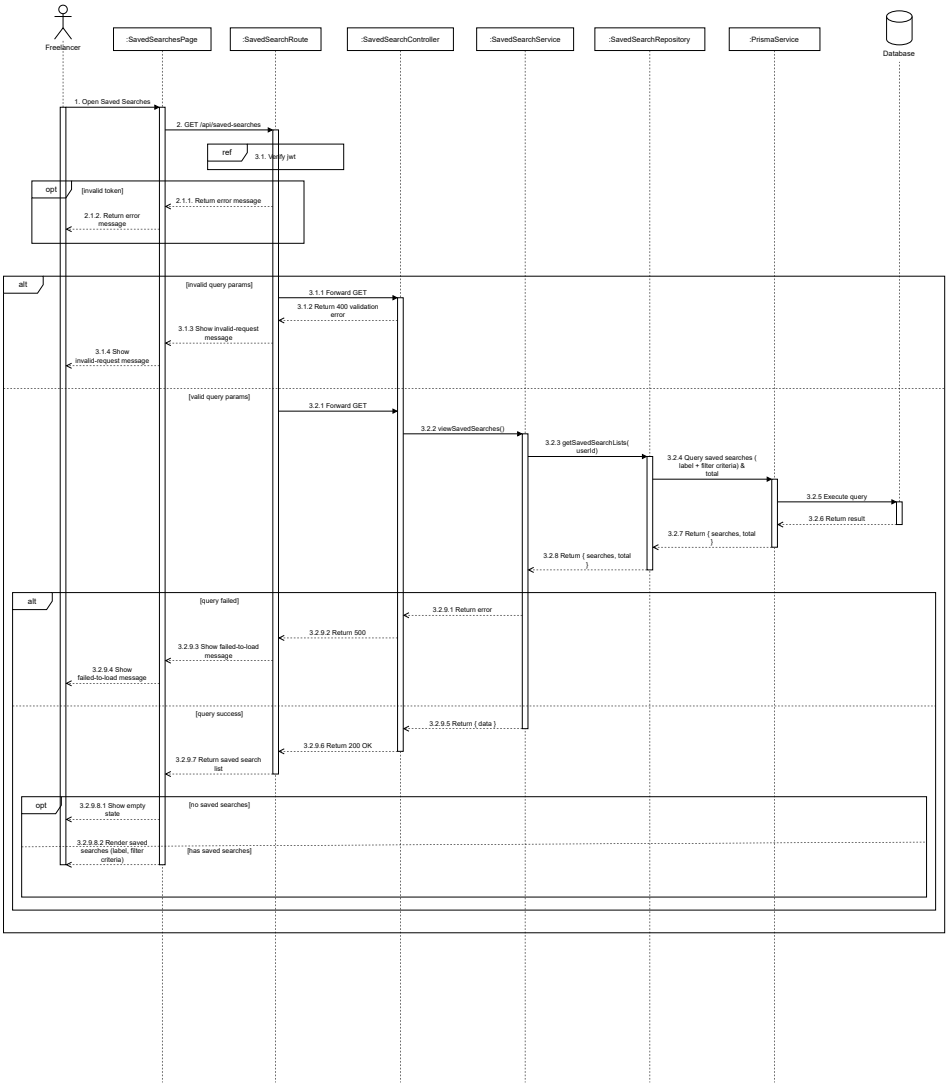

Figure 123: Sequence diagram of UC-22.01 View Saved Search List

3.9.2 UC-22.02 – View Saved Search Detail

a. Class diagram

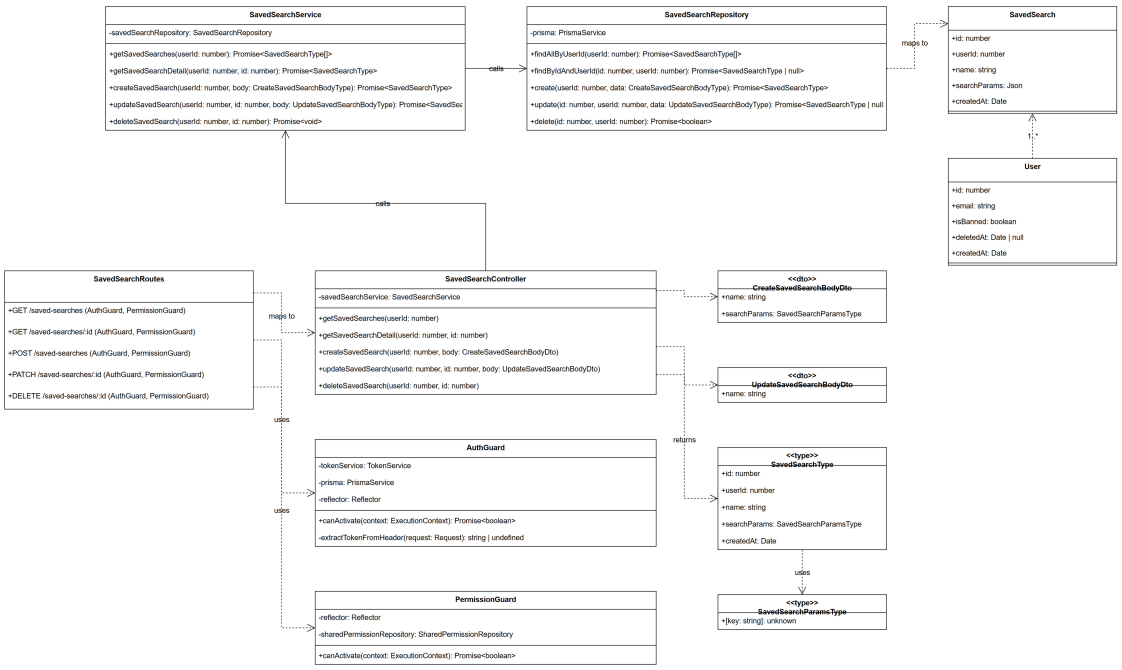

Figure 124: Class diagram of UC-22.02 View Saved Search Detail

b. Sequence diagram

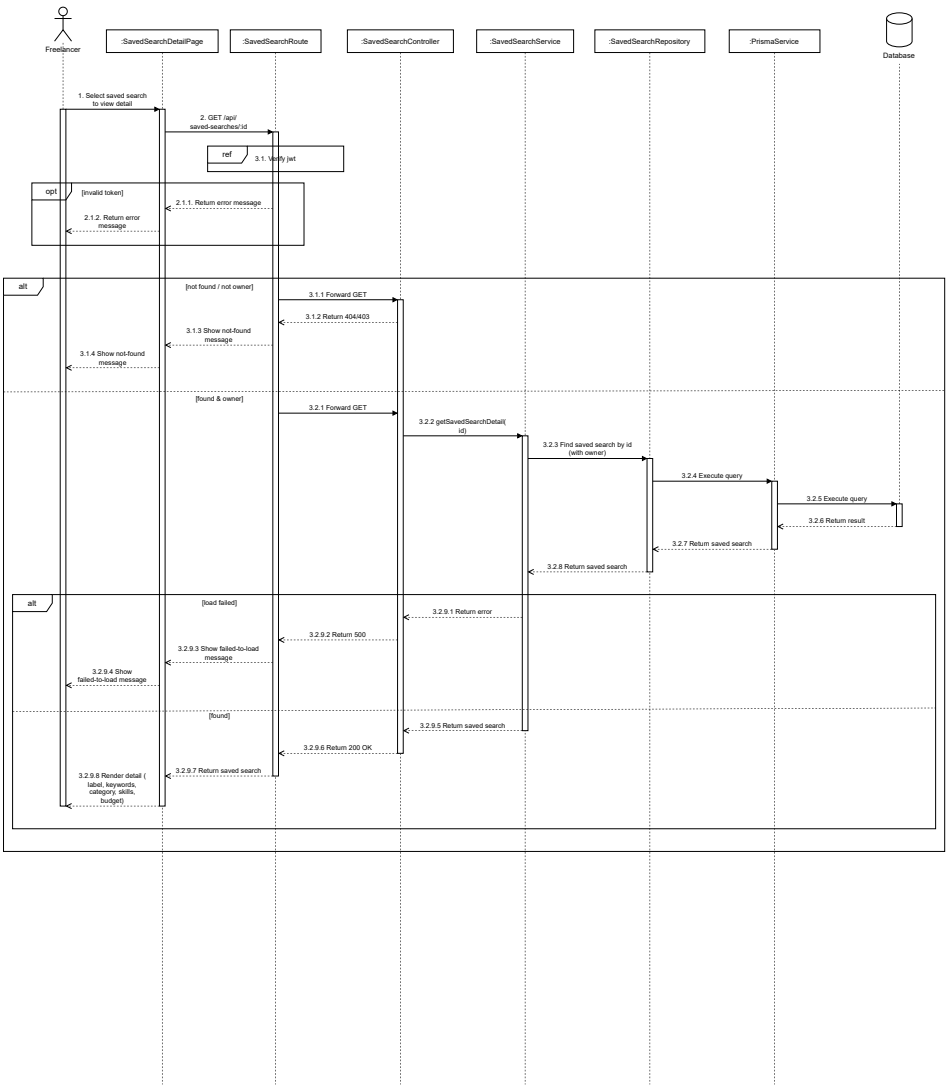

Figure 125: Sequence diagram of UC-22.02 View Saved Search Detail

3.9.3 UC-22.03 – Create Saved Search

a. Class diagram

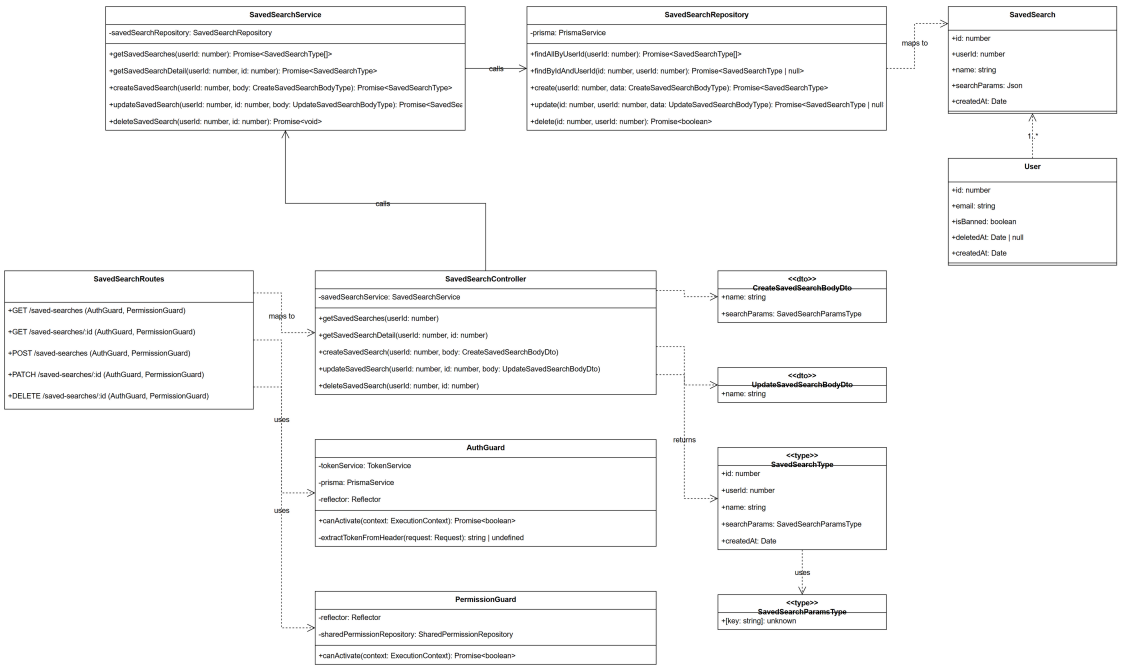

Figure 126: Class diagram of UC-22.03 Create Saved Search

b. Sequence diagram

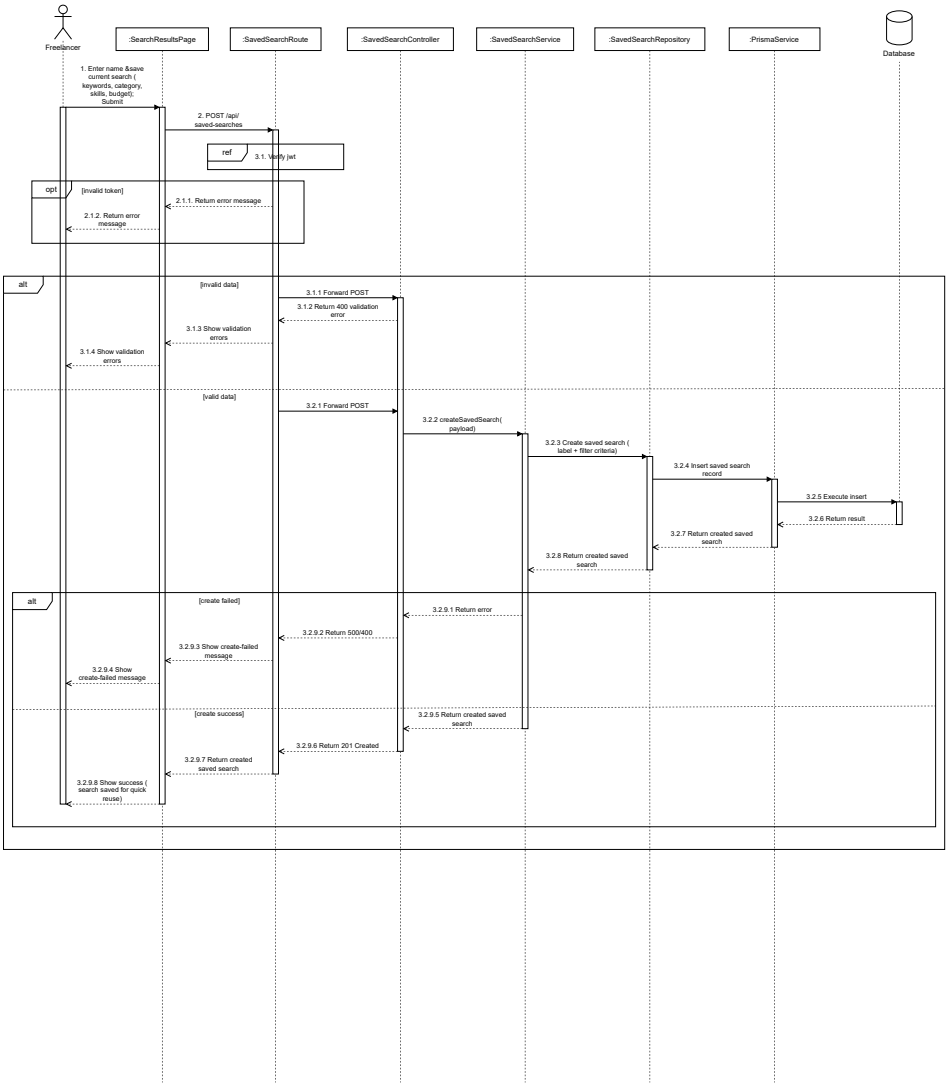

Figure 127: Sequence diagram of UC-22.03 Create Saved Search

3.9.4 UC-22.04 – Update Saved Search

a. Class diagram

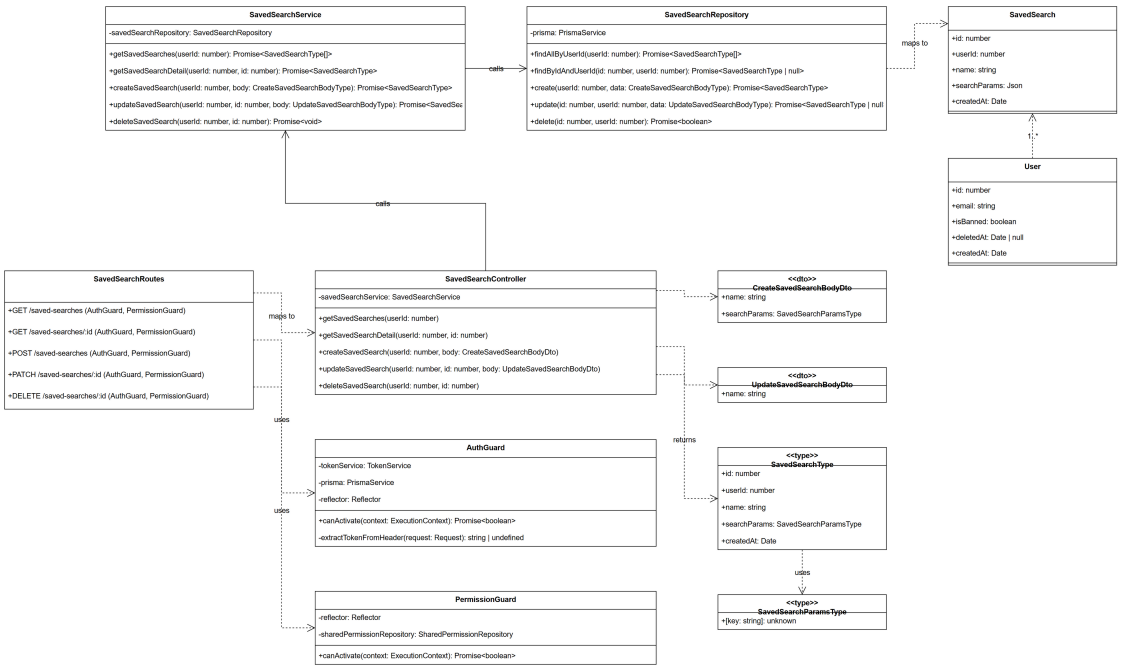

Figure 128: Class diagram of UC-22.04 Update Saved Search

b. Sequence diagram

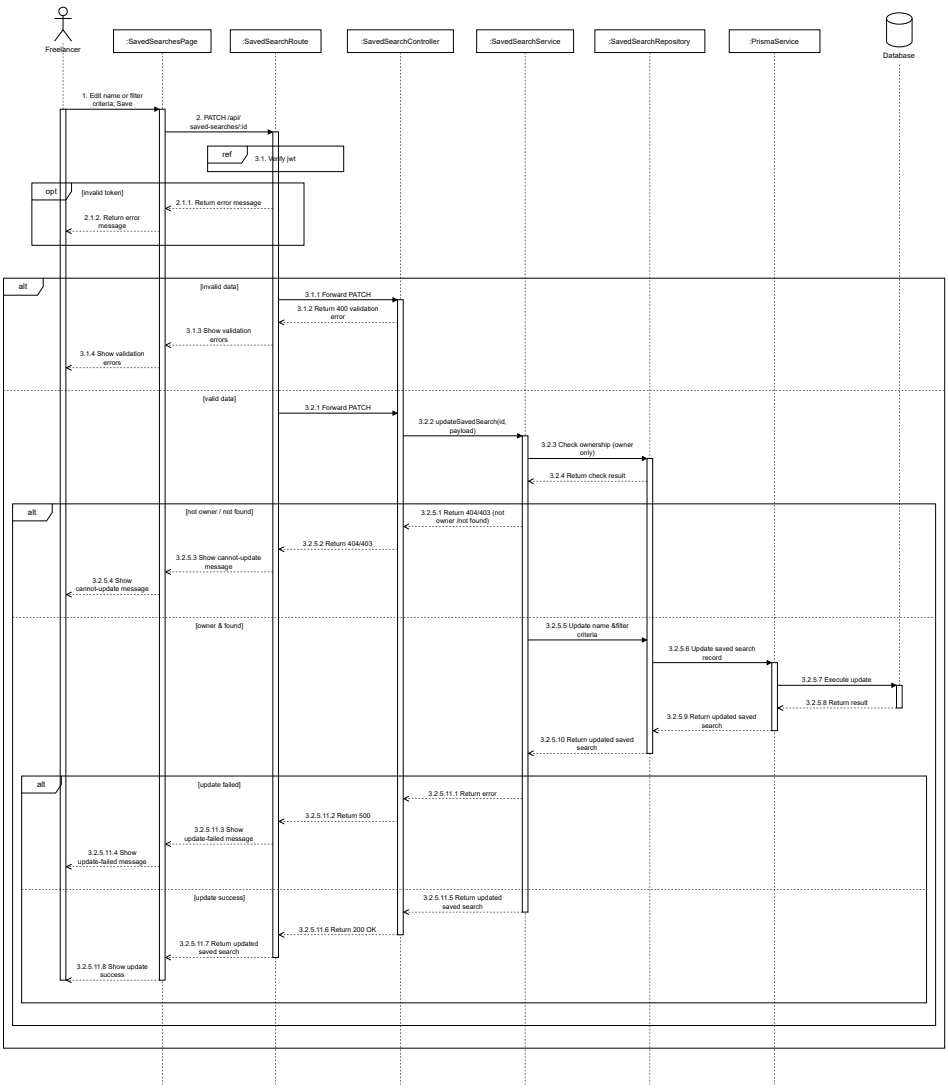

Figure 129: Sequence diagram of UC-22.04 Update Saved Search

3.9.5 UC-22.05 – Delete Saved Search

a. Class diagram

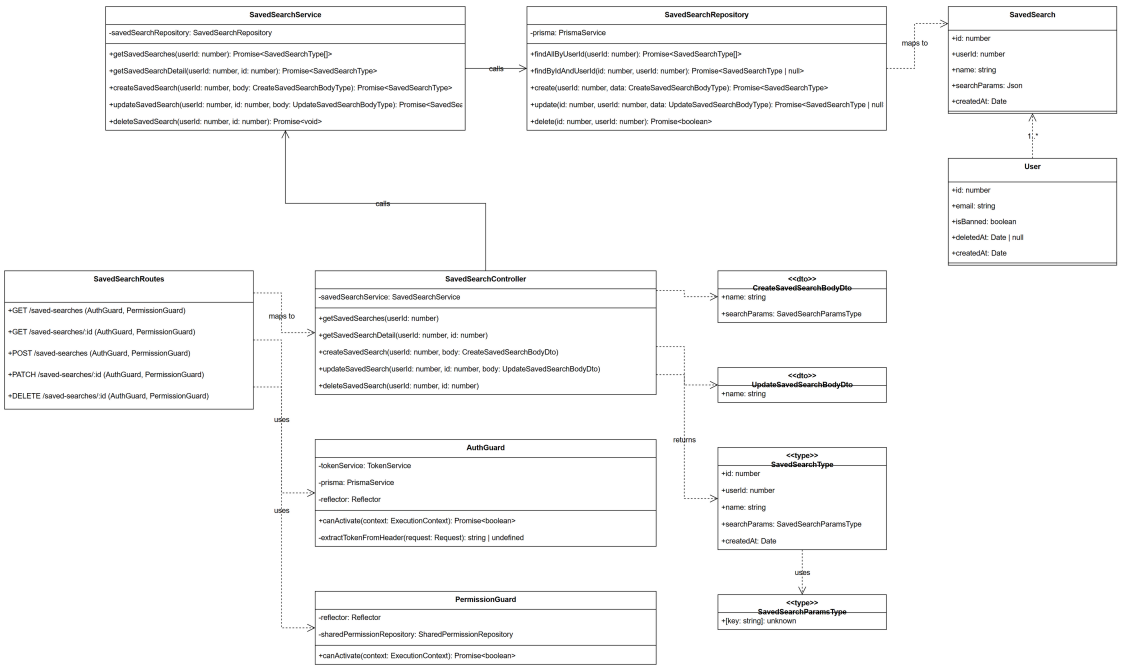

Figure 130: Class diagram of UC-22.05 Delete Saved Search

b. Sequence diagram

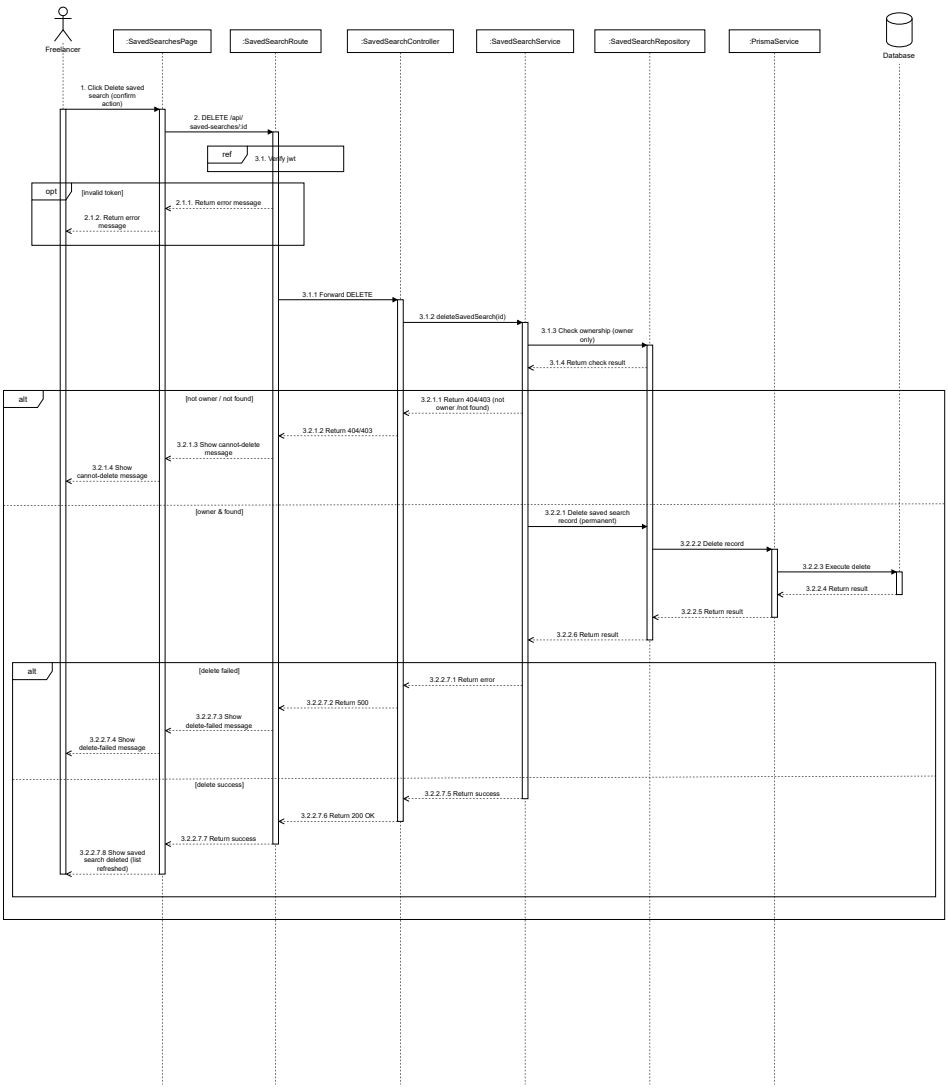

Figure 131: Sequence diagram of UC-22.05 Delete Saved Search

3.10 Job Alert Management

Job Alert Management enables freelancers to configure automated notifications based on job keywords, budget ranges, and category criteria, ensuring they receive prompt alerts when matching jobs are published.

3.10.1 UC-23.01 – View Job Alert List

a. Class diagram

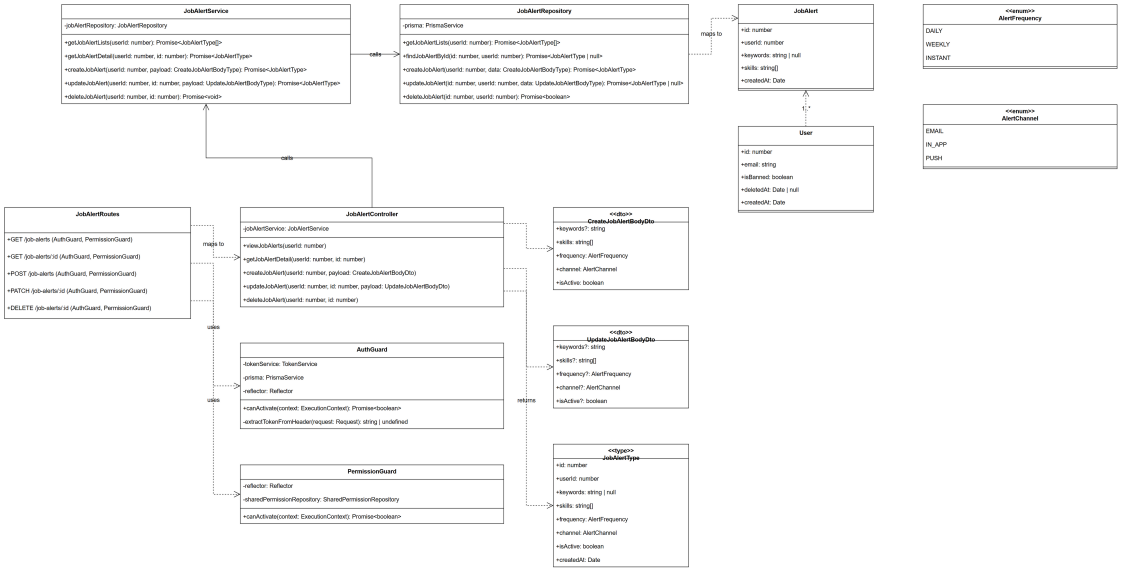

Figure 132: Class diagram of UC-23.01 View Job Alert List

b. Sequence diagram

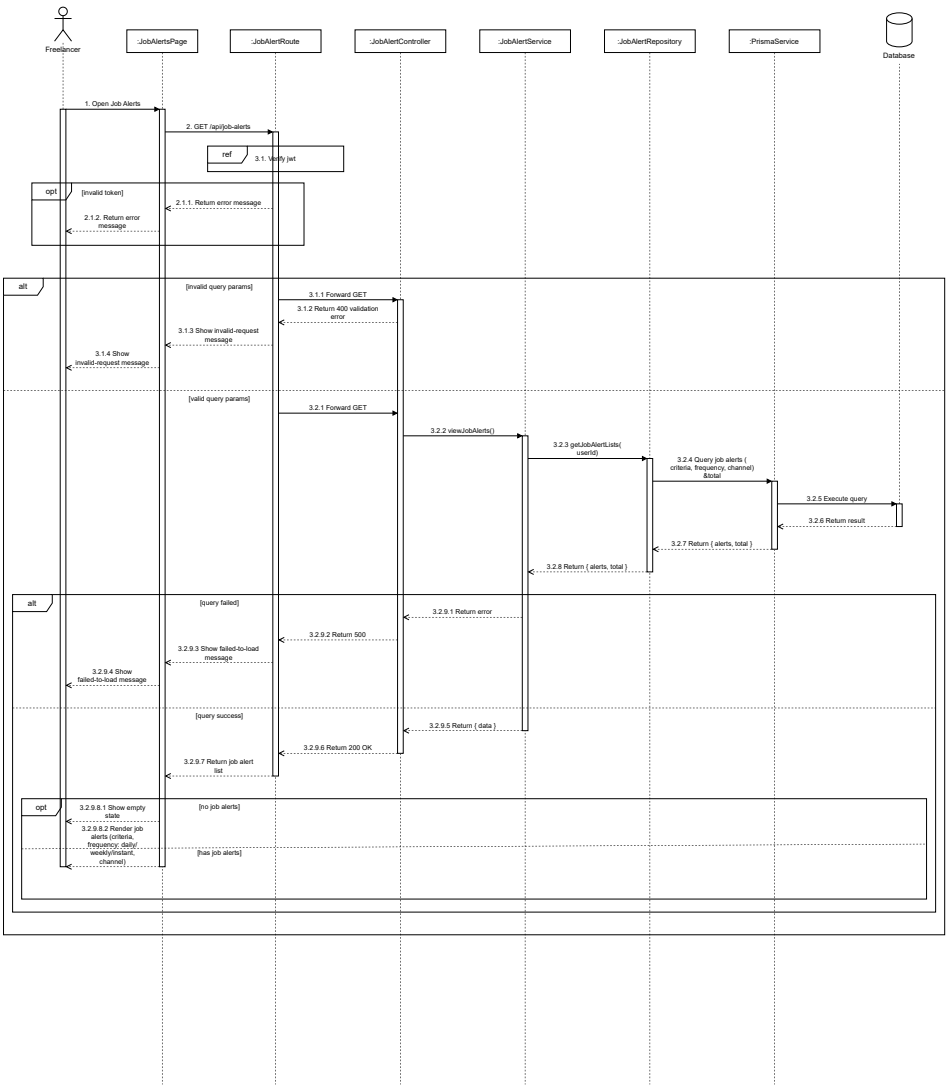

Figure 133: Sequence diagram of UC-23.01 View Job Alert List

3.10.2 UC-23.02 – View Job Alert Detail

a. Class diagram

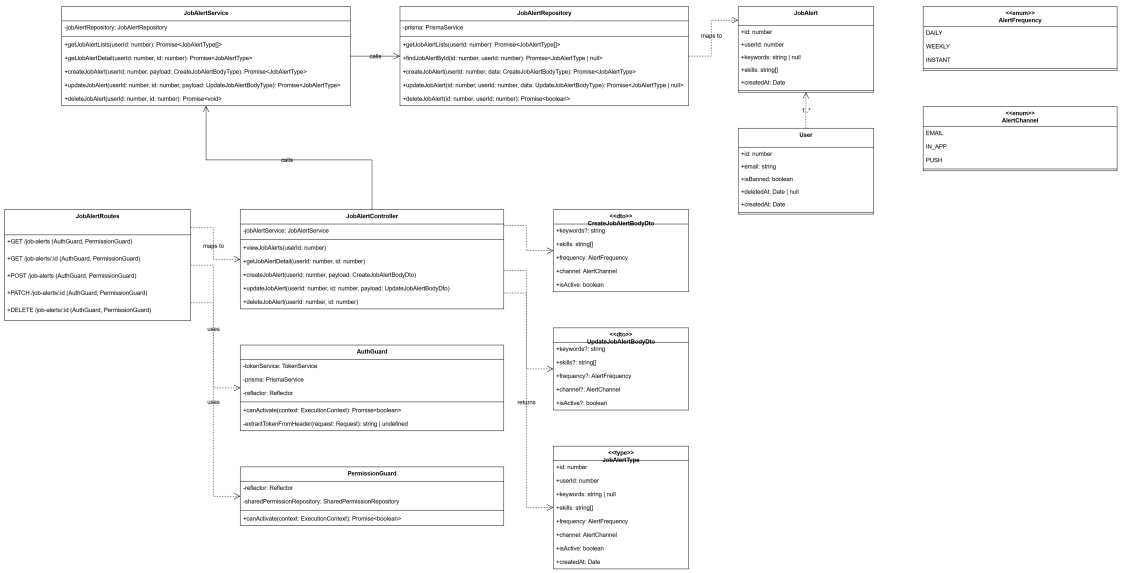

Figure 134: Class diagram of UC-23.02 View Job Alert Detail

b. Sequence diagram

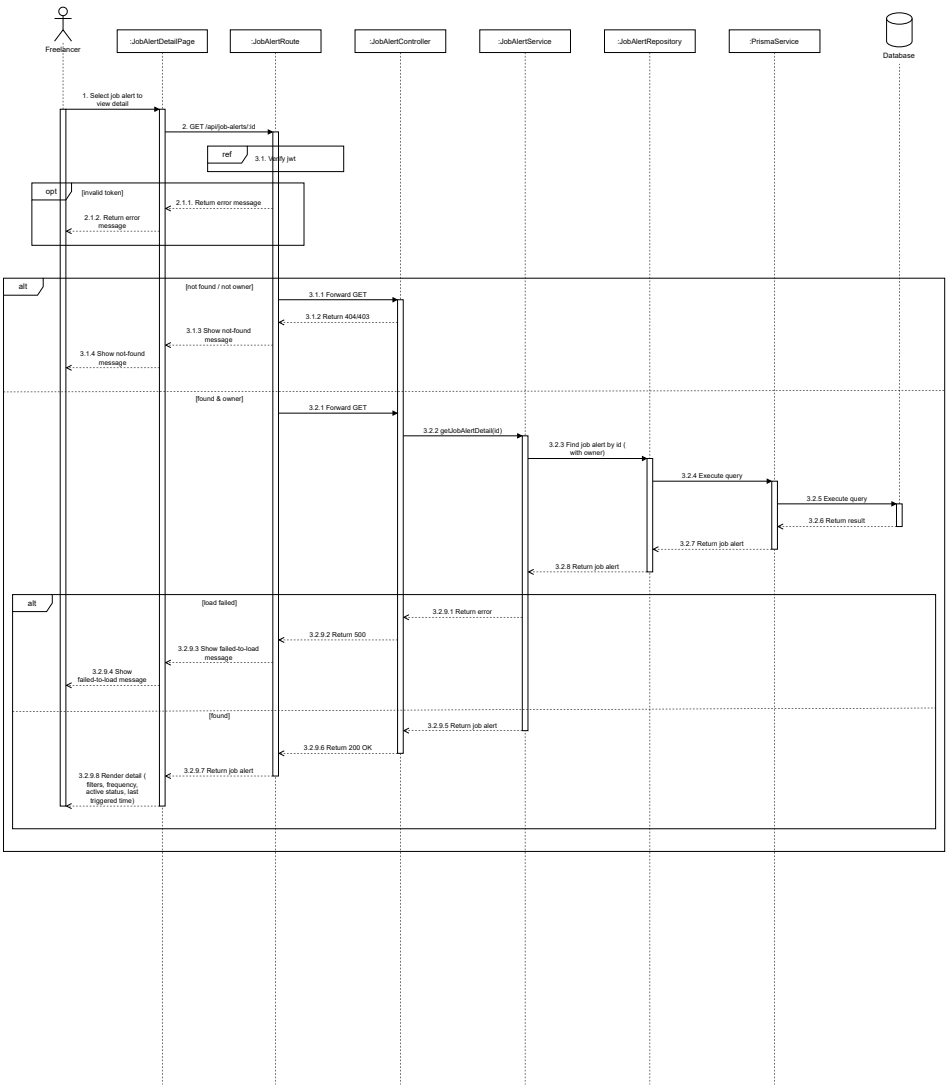

Figure 135: Sequence diagram of UC-23.02 View Job Alert Detail

3.10.3 UC-23.03 – Create Job Alert

a. Class diagram

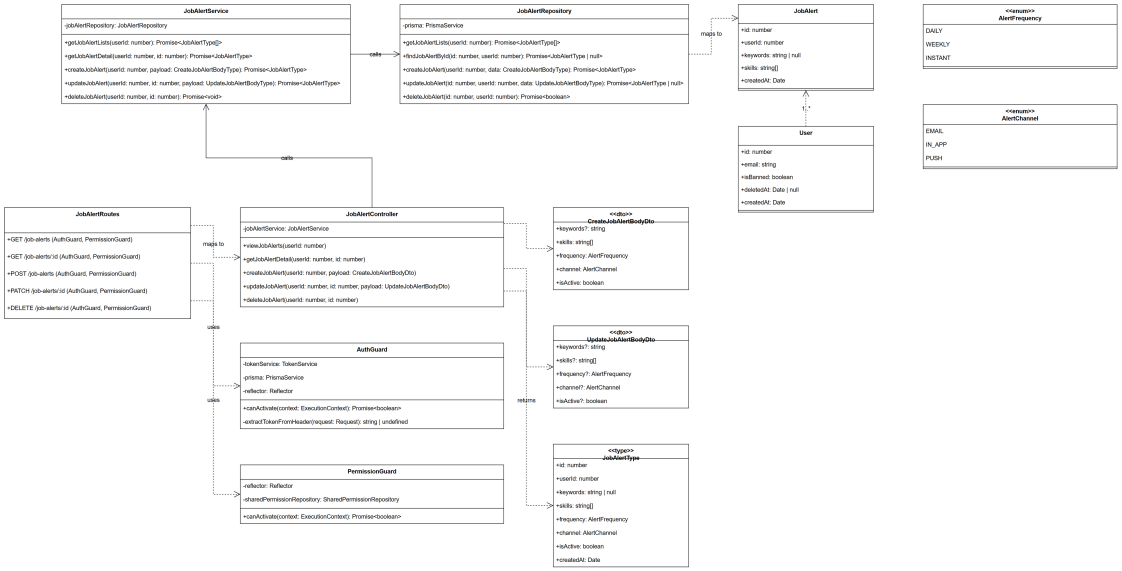

Figure 136: Class diagram of UC-23.03 Create Job Alert

b. Sequence diagram

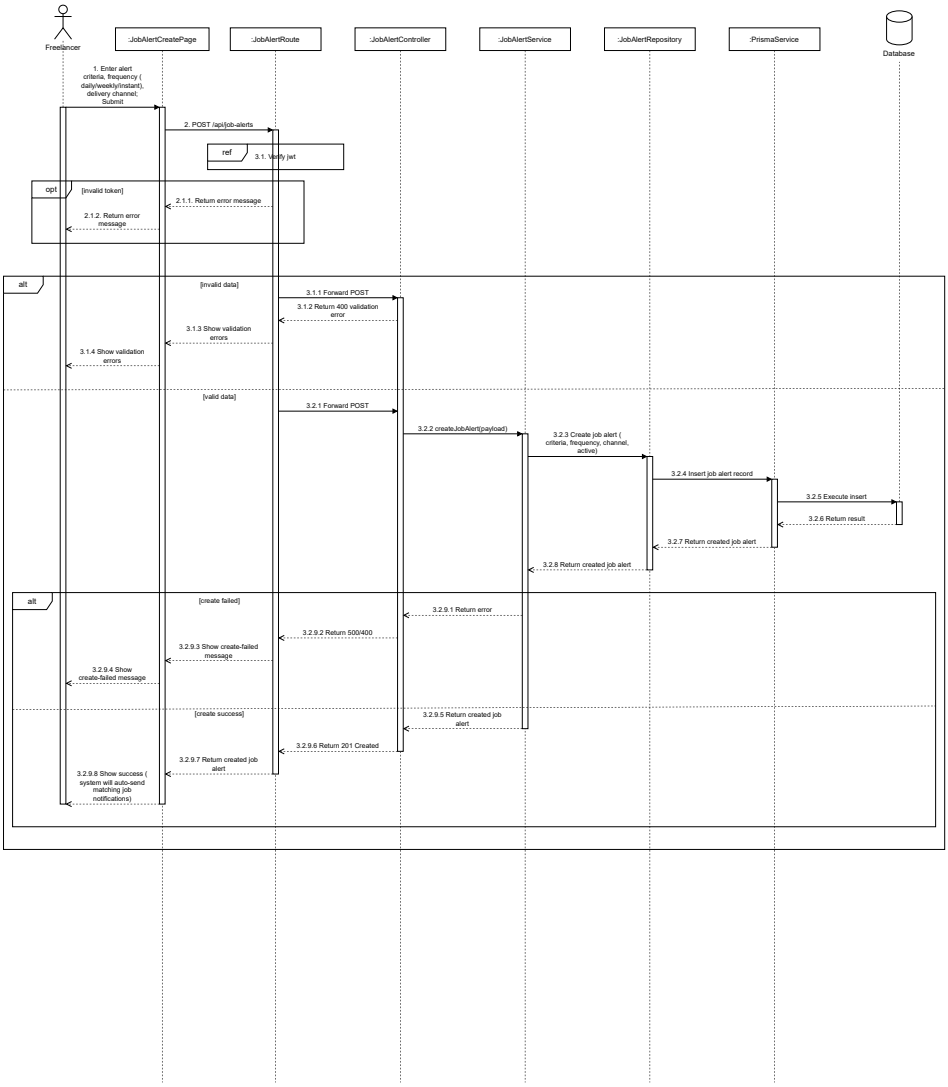

Figure 137: Sequence diagram of UC-23.03 Create Job Alert

3.10.4 UC-23.04 – Update Job Alert

a. Class diagram

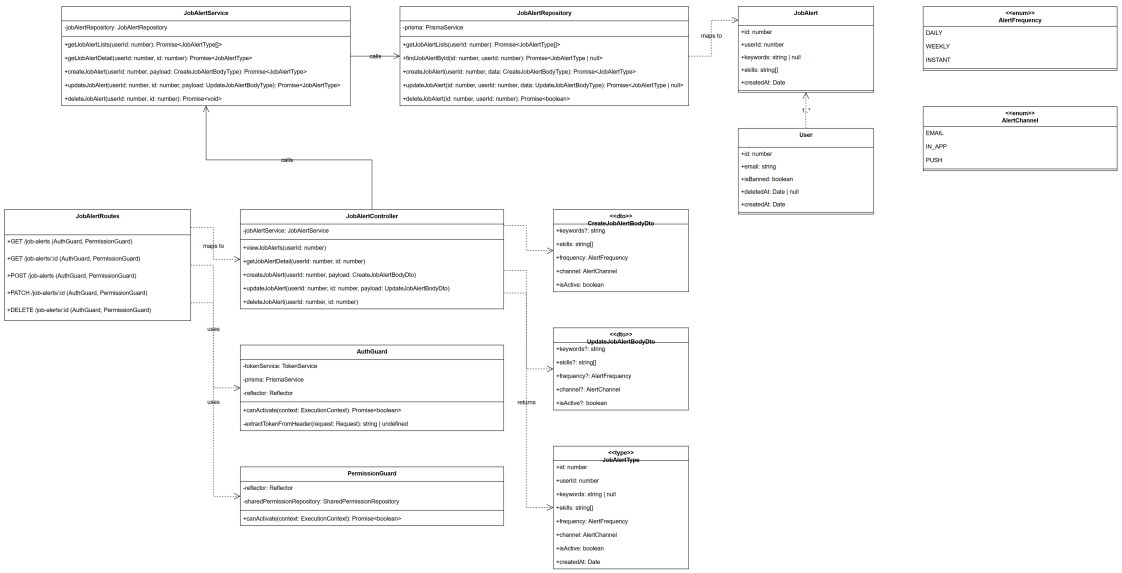

Figure 138: Class diagram of UC-23.04 Update Job Alert

b. Sequence diagram

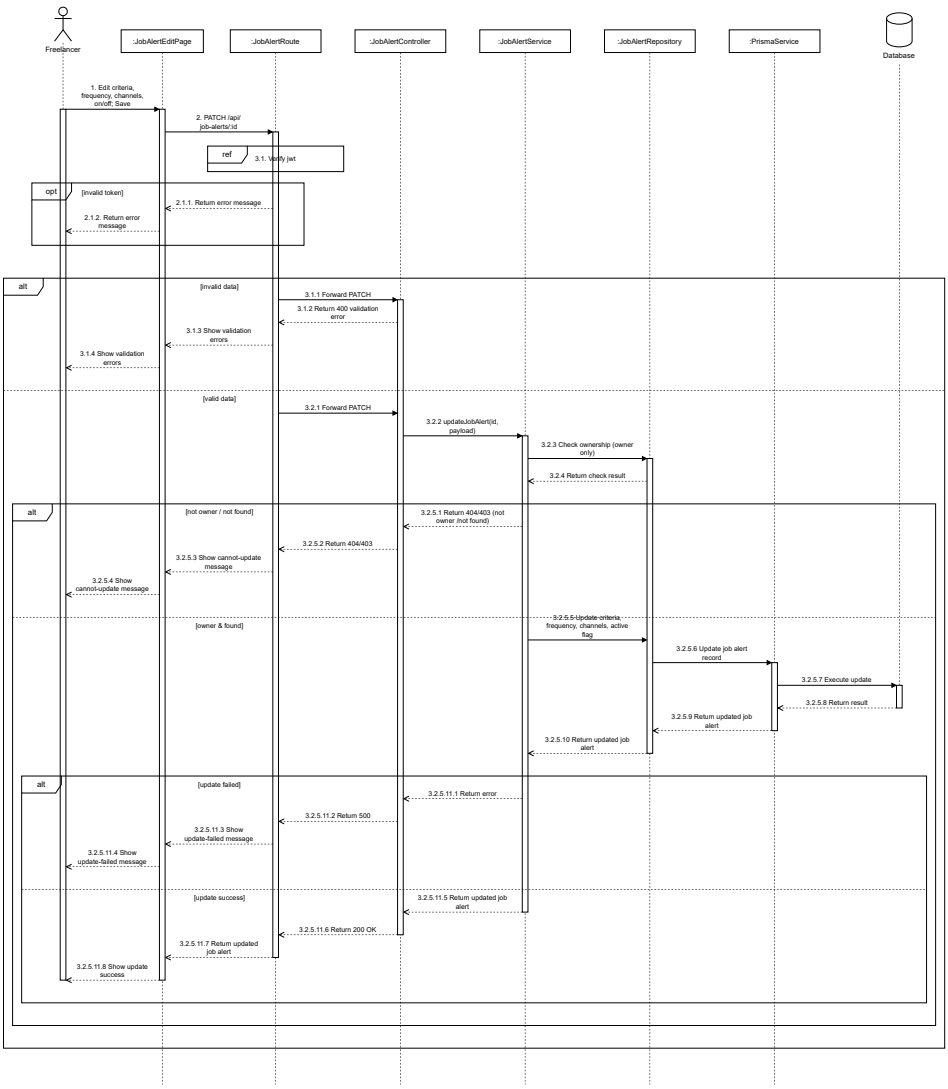

Figure 139: Sequence diagram of UC-23.04 Update Job Alert

3.10.5 UC-23.05 – Delete Job Alert

a. Class diagram

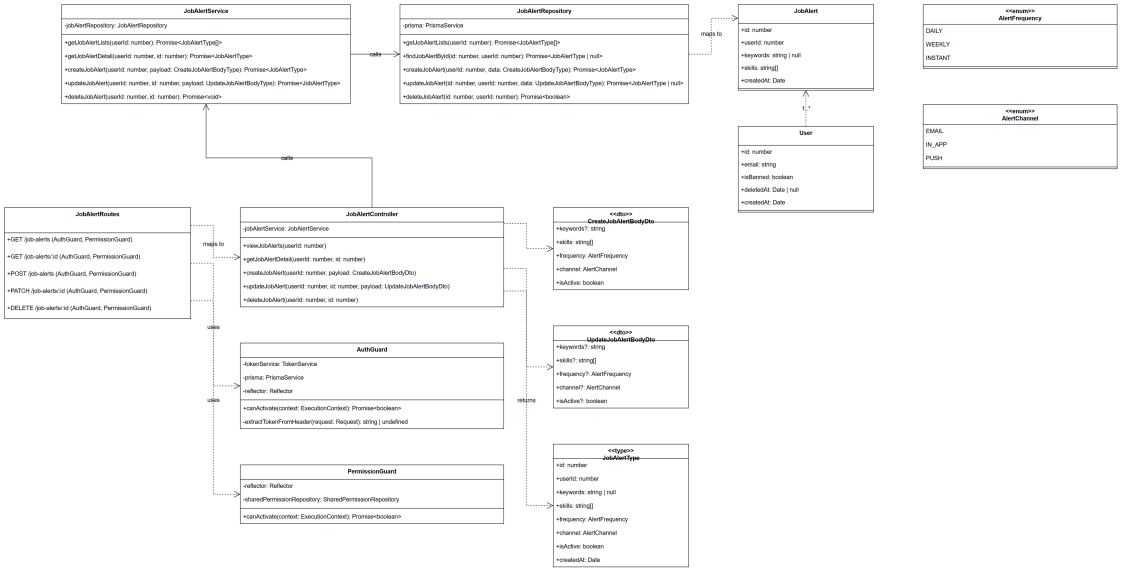

Figure 140: Class diagram of UC-23.05 Delete Job Alert

b. Sequence diagram

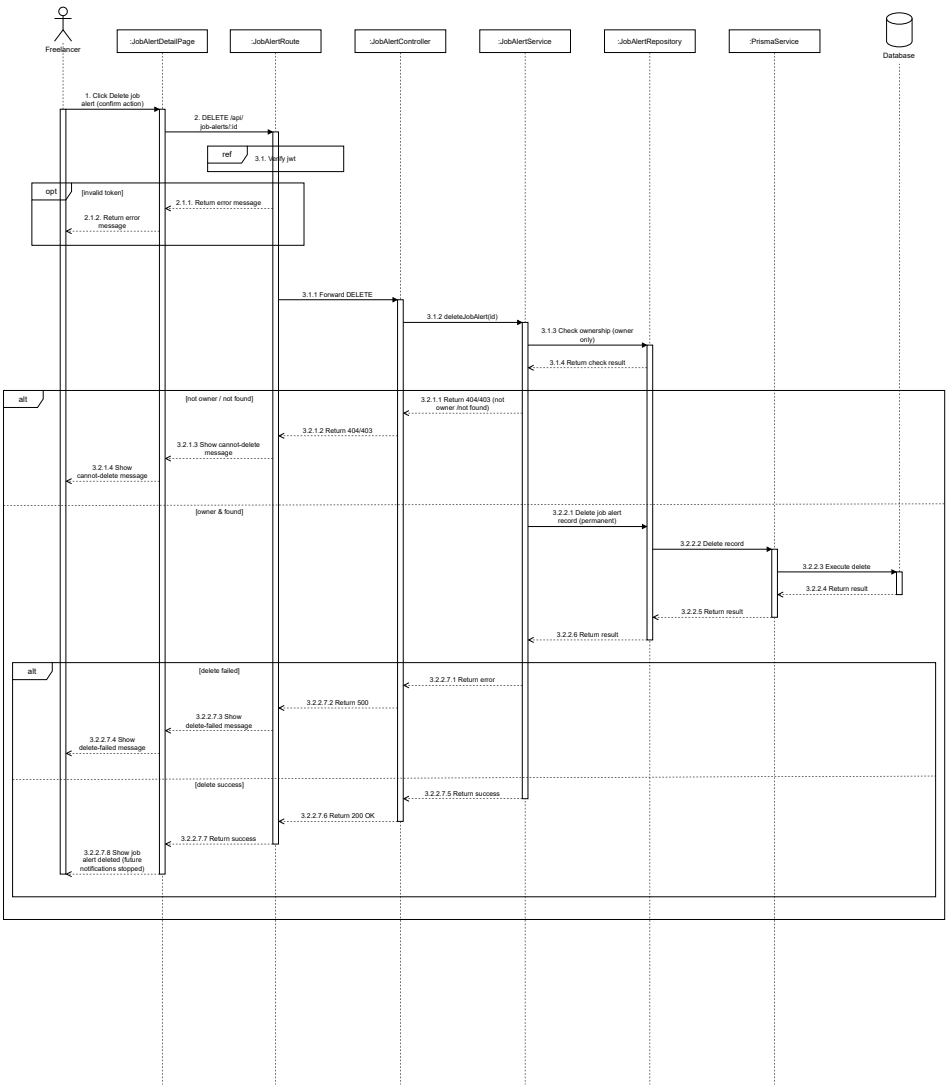

Figure 141: Sequence diagram of UC-23.05 Delete Job Alert

3.11 Permission Management

Permission Management exposes the route-level permission catalogue so that an administrator can inspect the permissions available for role assignment.

3.11.1 UC-15.01 – View Permission List

a. Class diagram

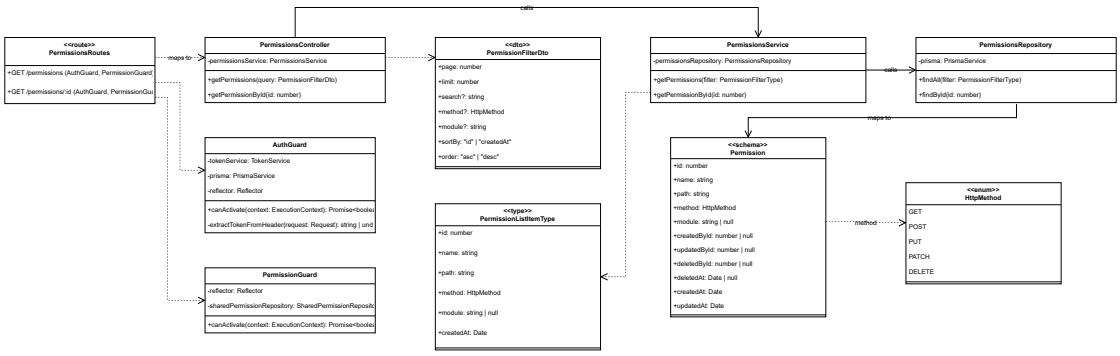

Figure 142: Class diagram of UC-15.01 View Permission List

b. Sequence diagram

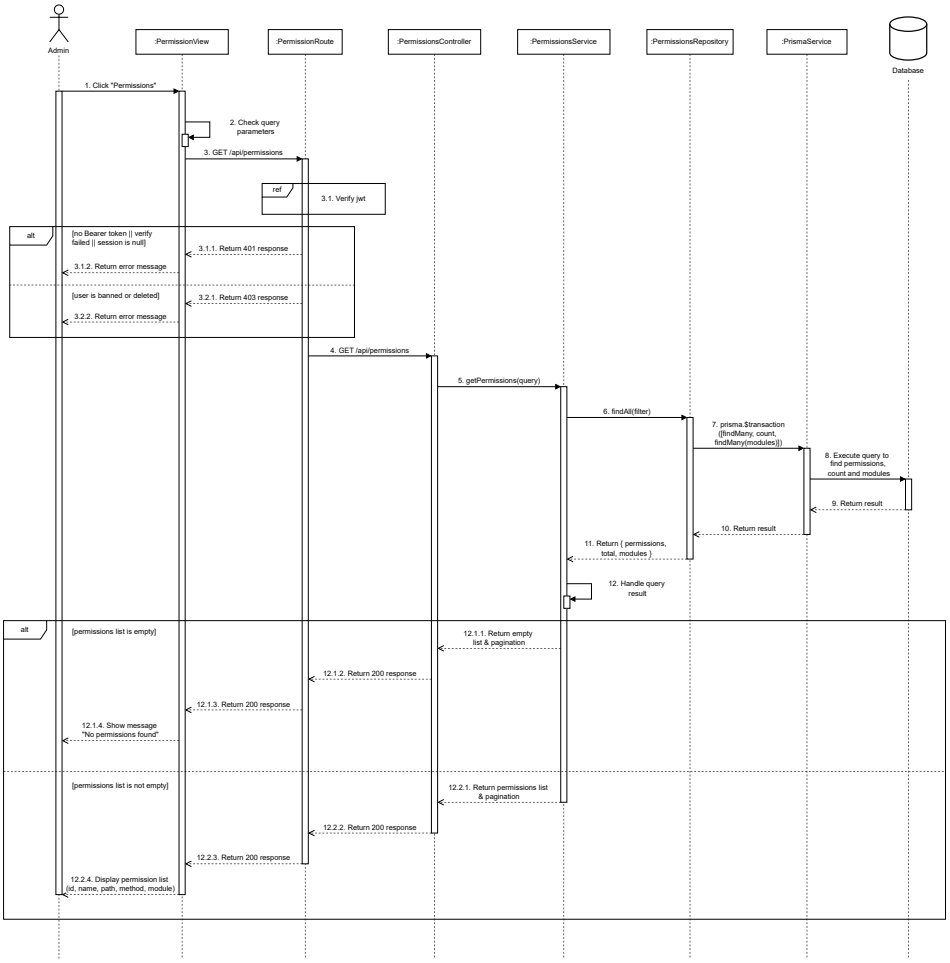

Figure 143: Sequence diagram of UC-15.01 View Permission List

3.11.2 UC-15.02 – View Permission Detail

a. Class diagram

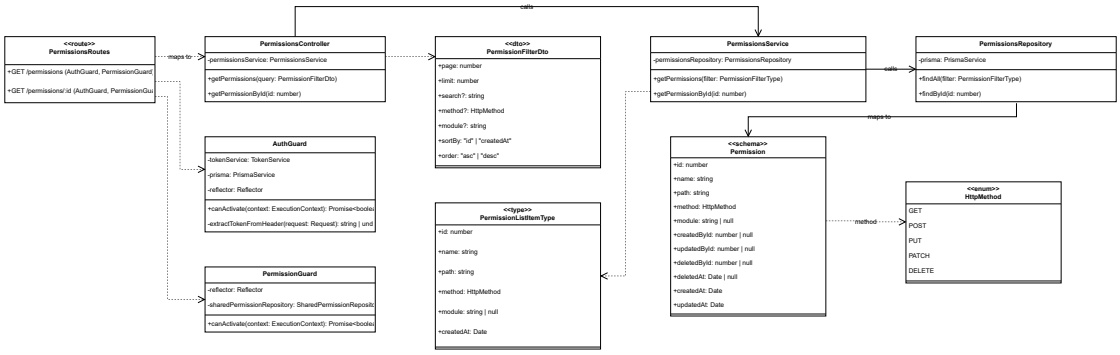

Figure 144: Class diagram of UC-15.02 View Permission Detail

b. Sequence diagram

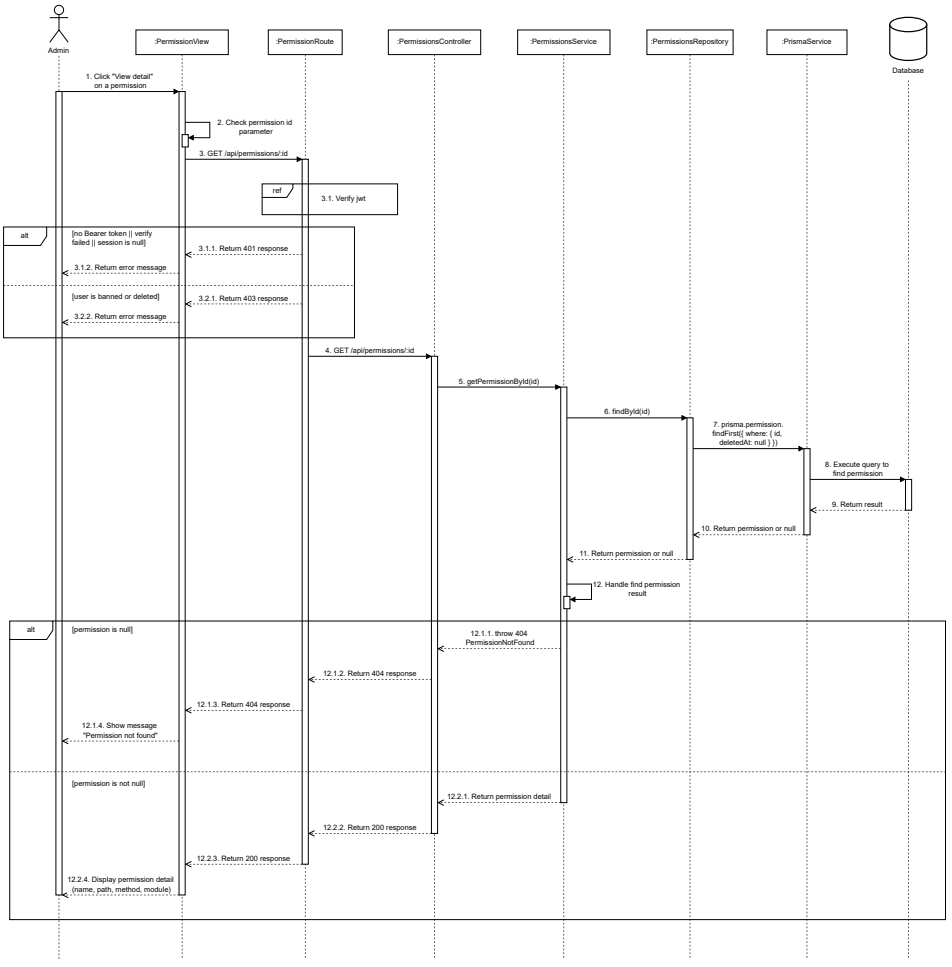

Figure 145: Sequence diagram of UC-15.02 View Permission Detail

3.12 Session Management

Session Management gives an authenticated user visibility over the devices that currently hold an active refresh-token session and allows a session to be revoked remotely, imme-

diately terminating the target device connection via realtime WebSocket events.

3.12.1 UC-16.01 – View Session List

a. Class diagram

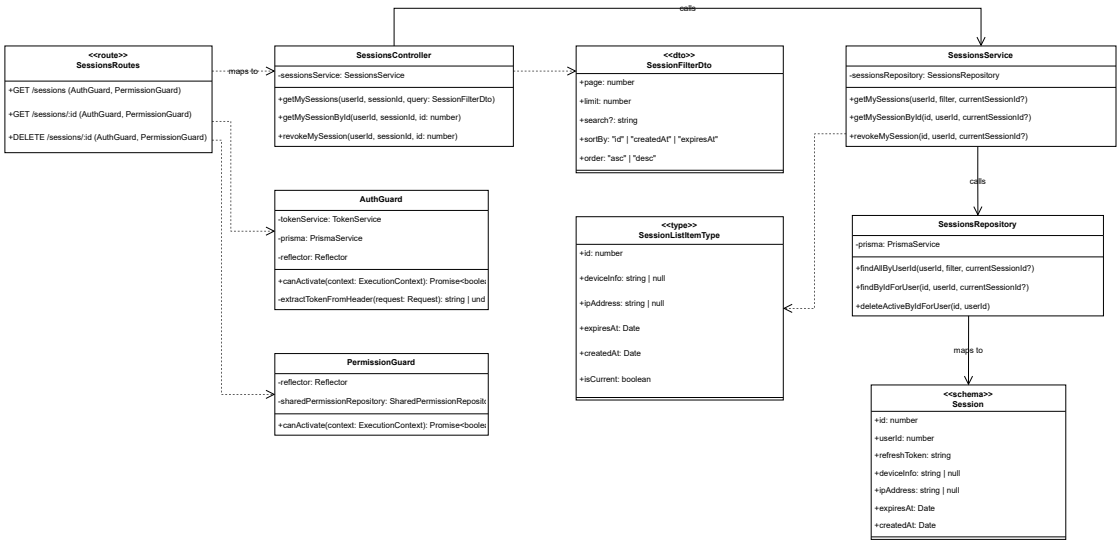

Figure 146: Class diagram of UC-16.01 View Session List

b. Sequence diagram

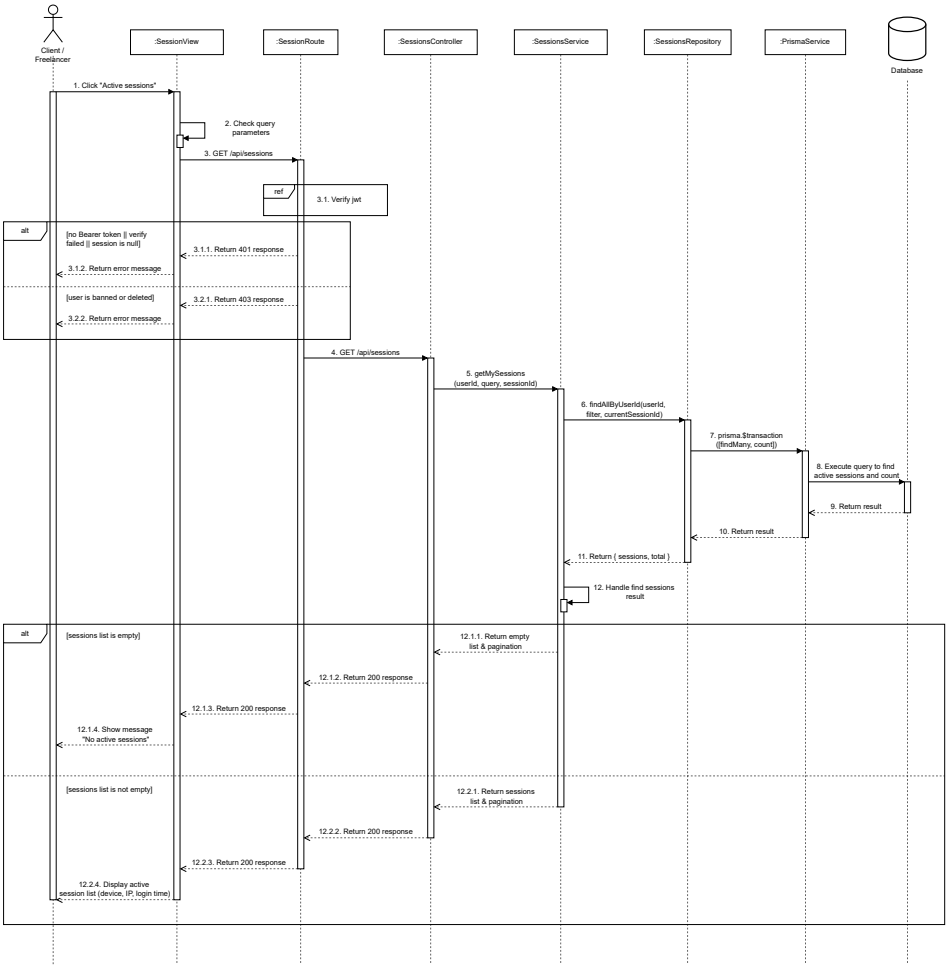

Figure 147: Sequence diagram of UC-16.01 View Session List

3.12.2 UC-16.02 – View Session Detail

a. Class diagram

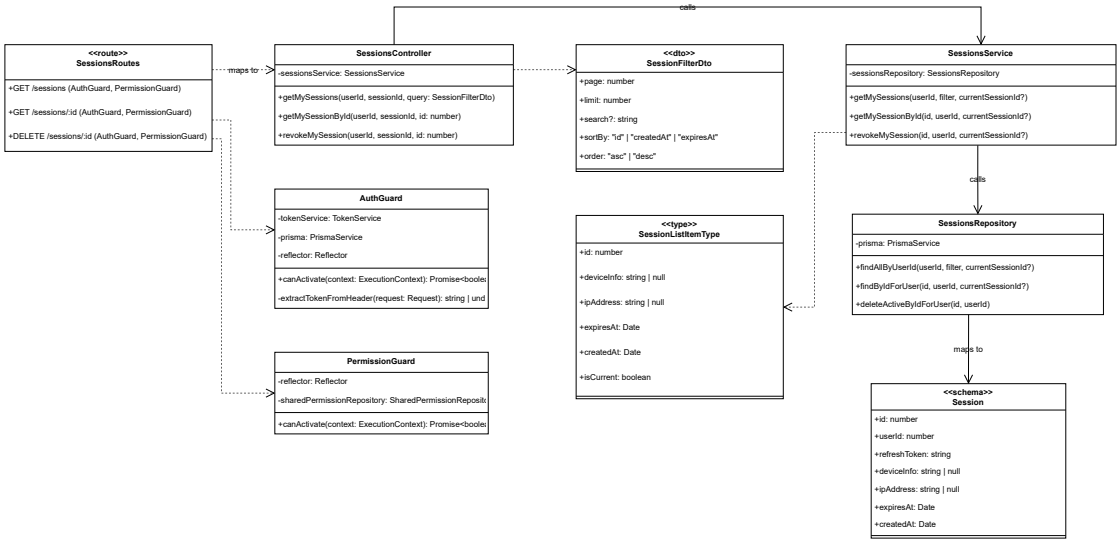

Figure 148: Class diagram of UC-16.02 View Session Detail

b. Sequence diagram

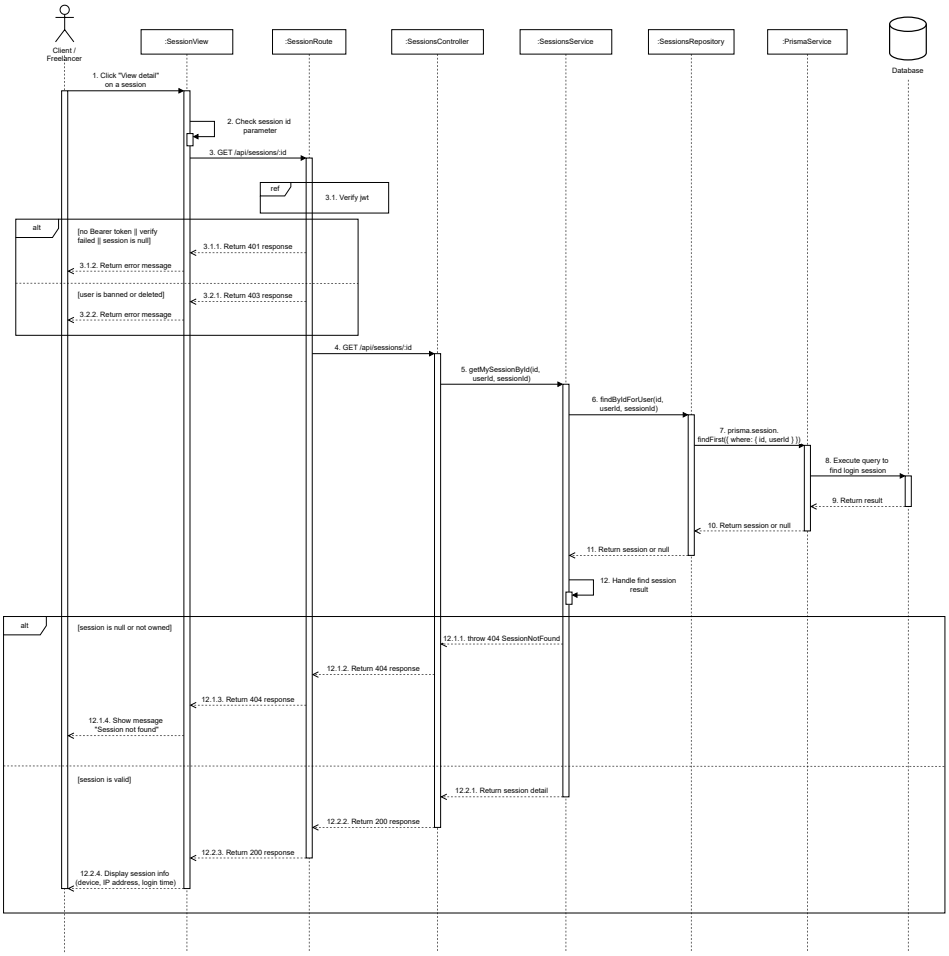

Figure 149: Sequence diagram of UC-16.02 View Session Detail

3.12.3 UC-16.03 – Revoke Session

a. Class diagram

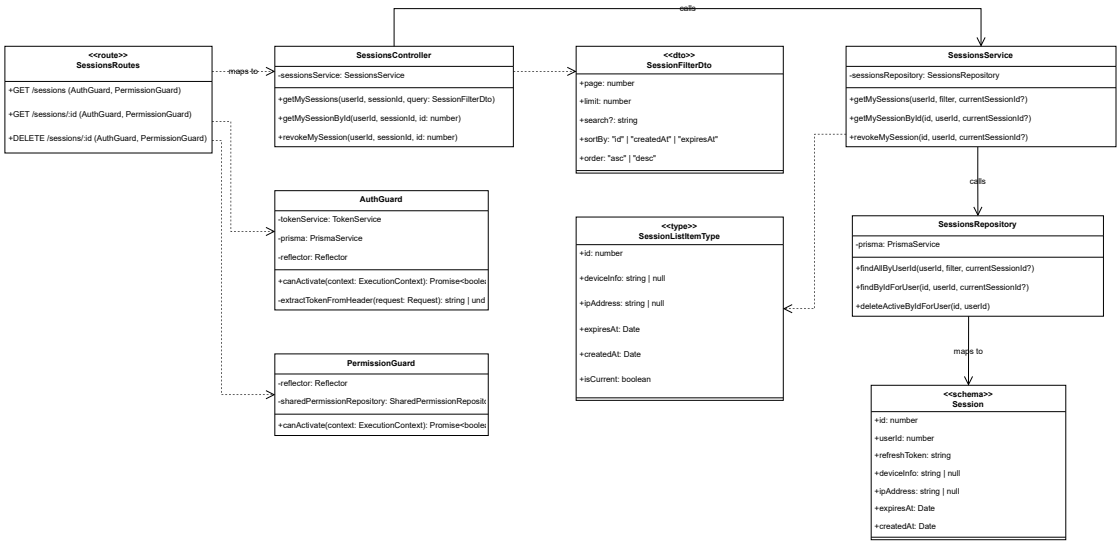

Figure 150: Class diagram of UC-16.03 Revoke Session

b. Sequence diagram

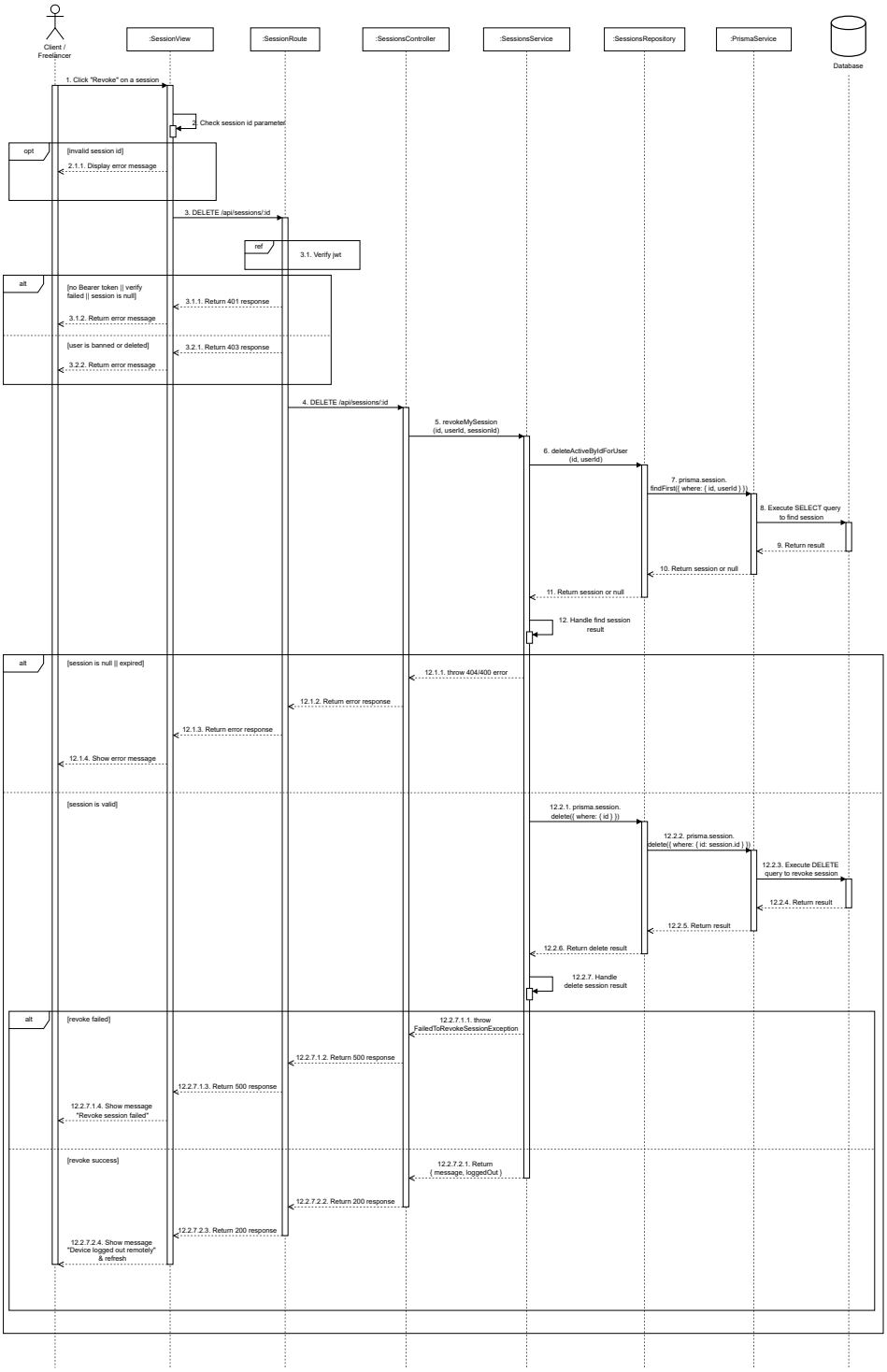

Figure 151: Sequence diagram of UC-16.03 Revoke Session

3.13 Forum Category Management

Forum Category Management covers the administrator-side category catalogue and the public category views used by the forum. Deletion is a soft delete, so a category can be restored later.

3.13.1 UC-17.01 – Create Forum Category

a. Class diagram

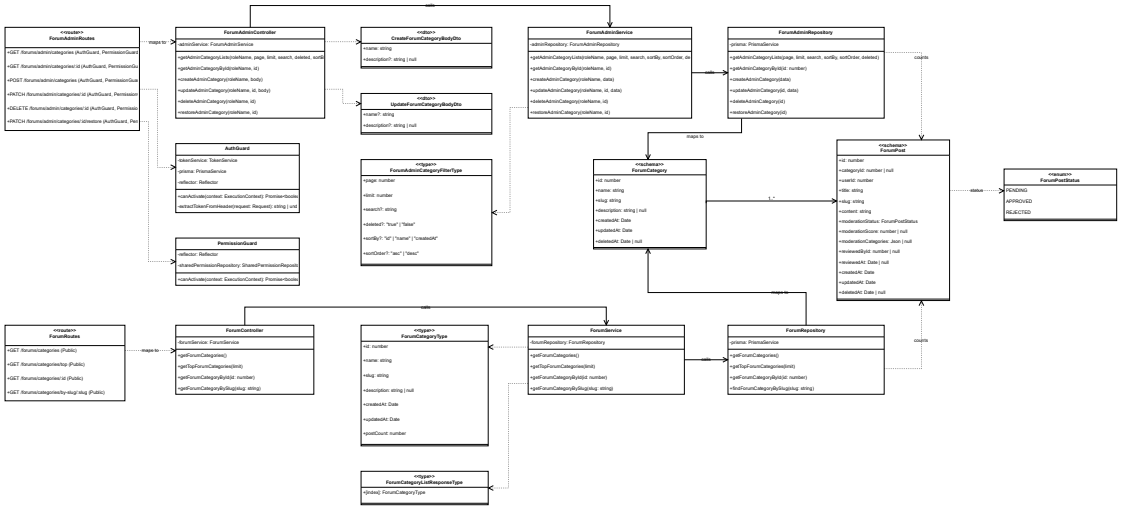

Figure 152: Class diagram of UC-17.01 Create Forum Category

b. Sequence diagram

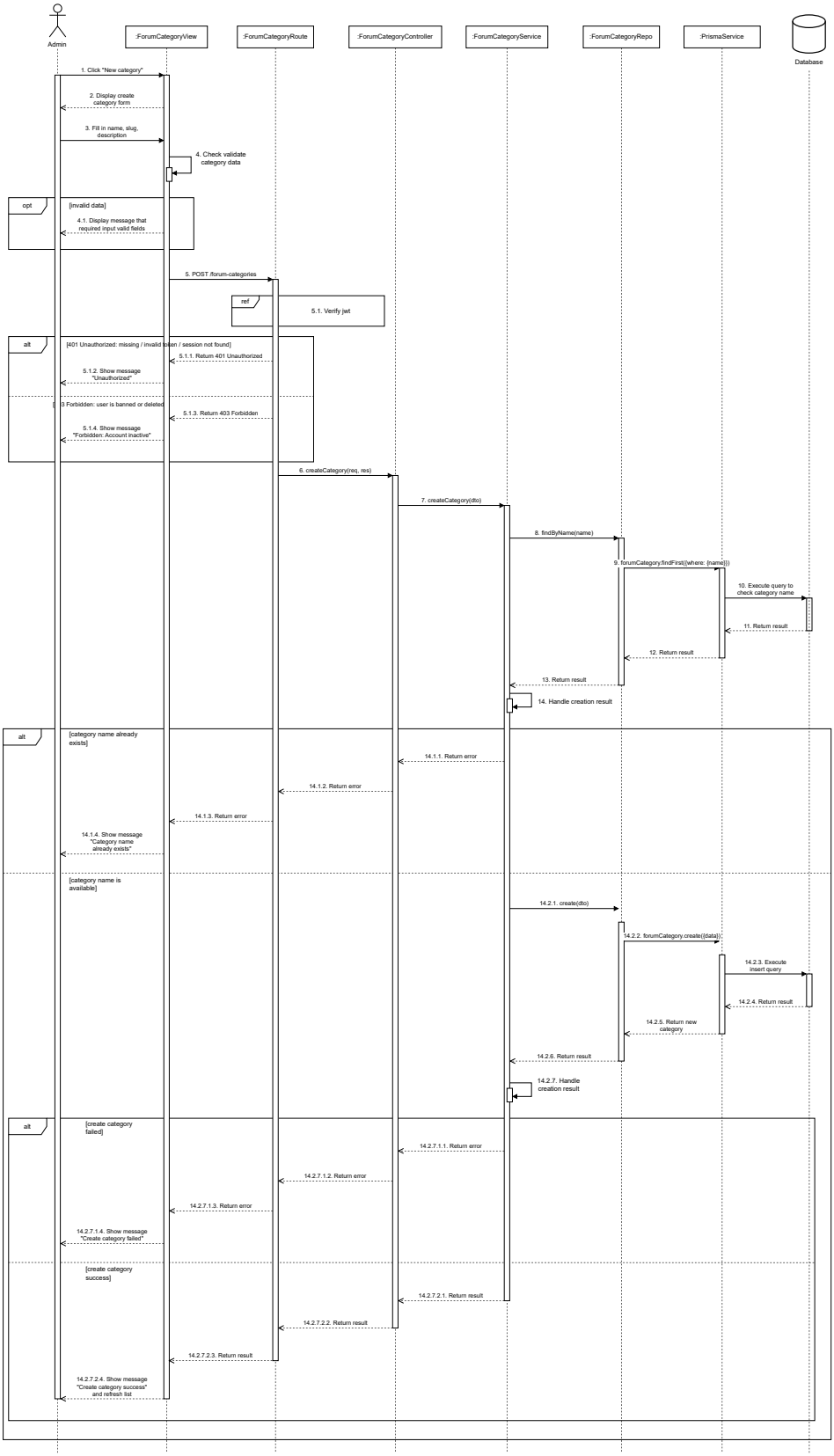

Figure 153: Sequence diagram of UC-17.01 Create Forum Category

3.13.2 UC-17.02 – Update Forum Category

a. Class diagram

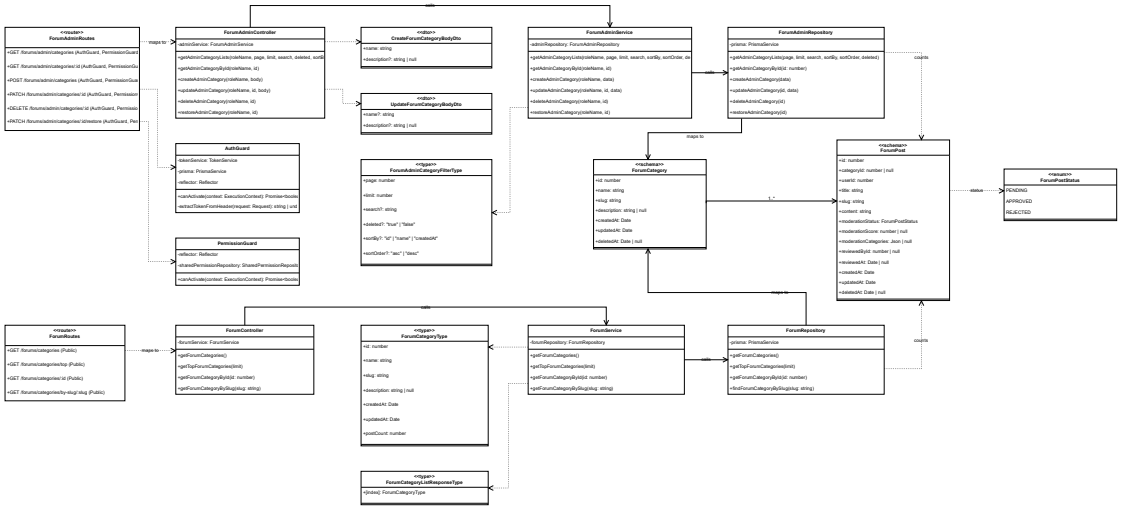

Figure 154: Class diagram of UC-17.02 Update Forum Category

b. Sequence diagram

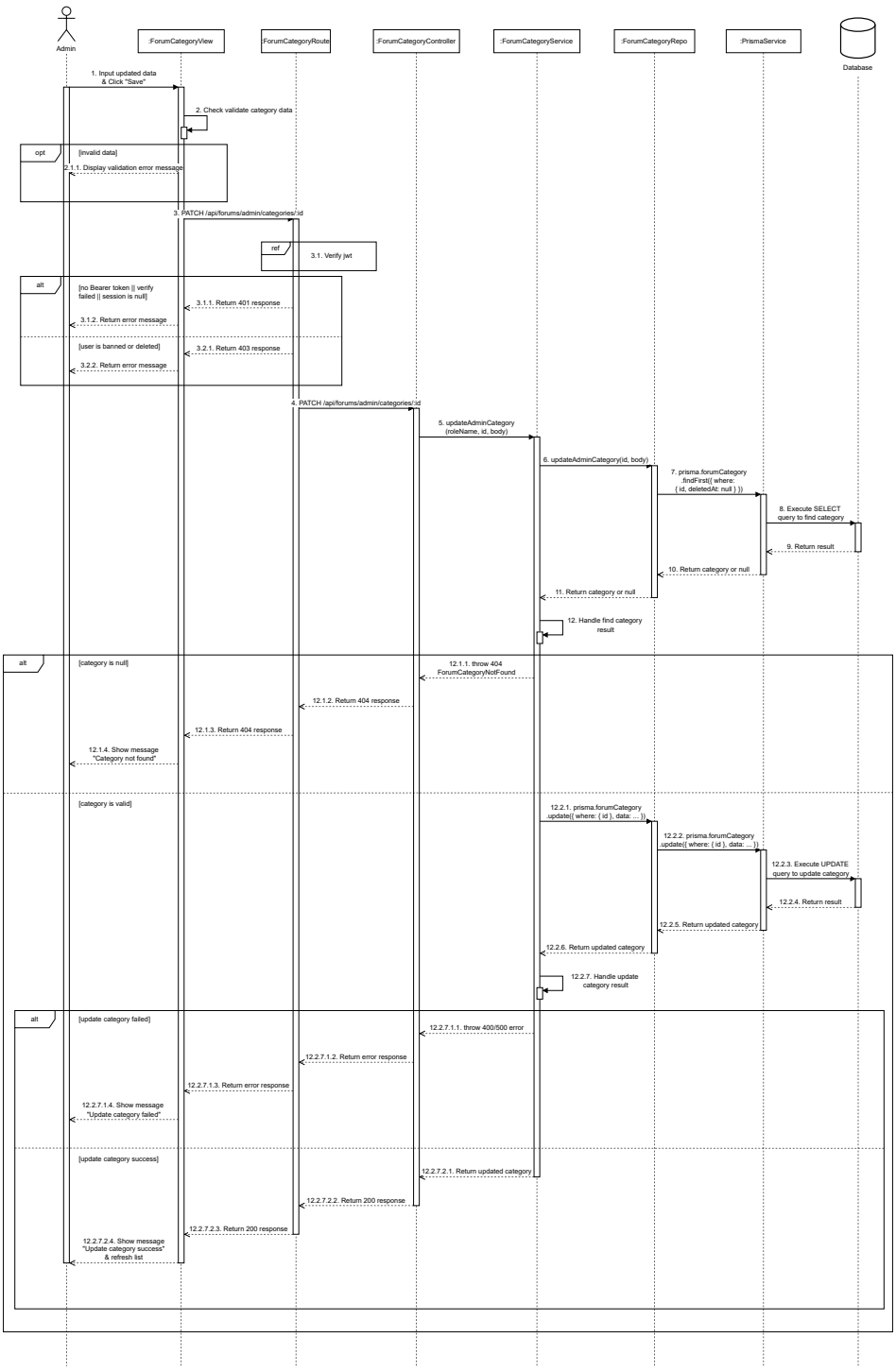

Figure 155: Sequence diagram of UC-17.02 Update Forum Category

3.13.3 UC-17.03 – Delete Forum Category

a. Class diagram

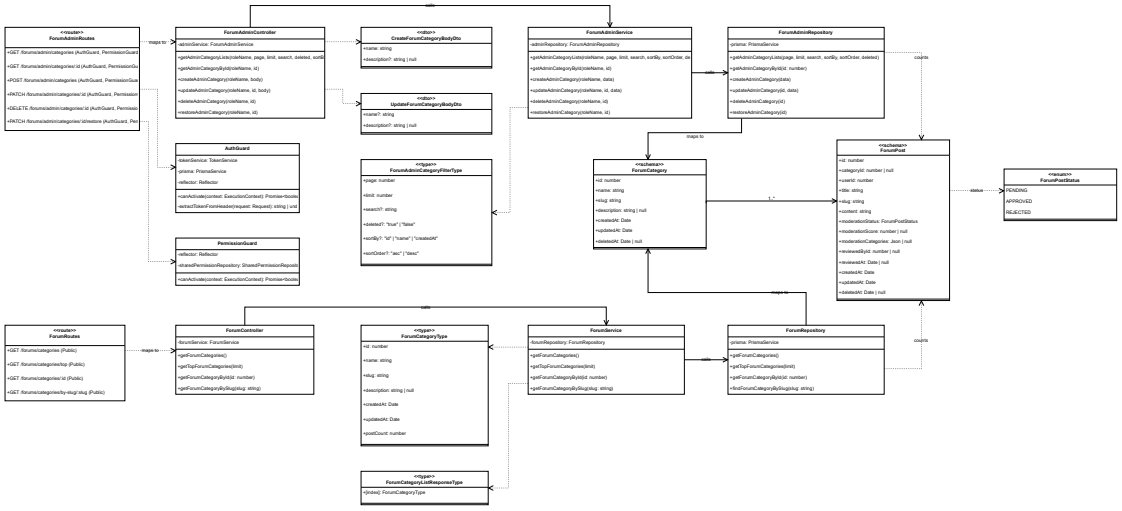

Figure 156: Class diagram of UC-17.03 Delete Forum Category

b. Sequence diagram

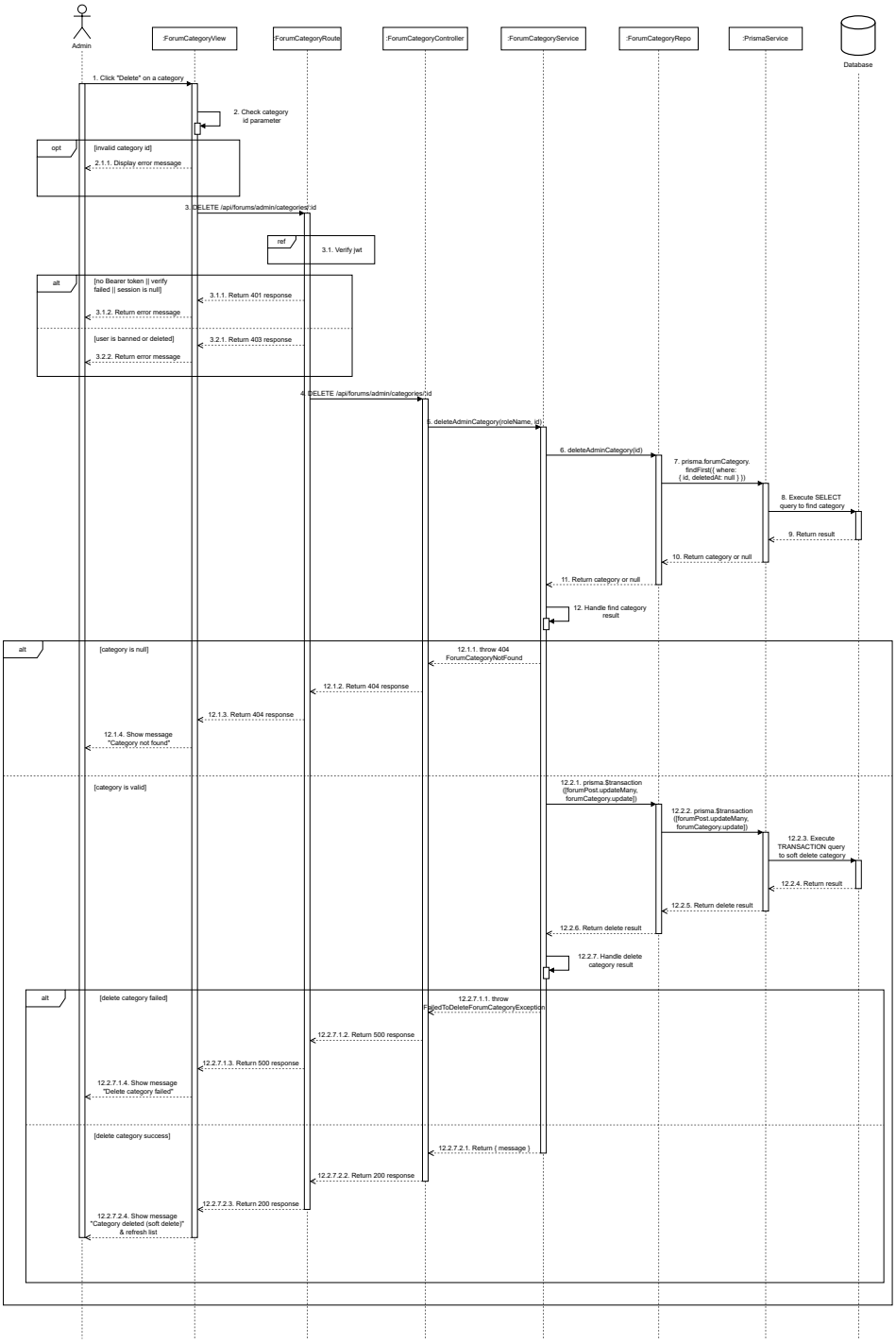

Figure 157: Sequence diagram of UC-17.03 Delete Forum Category

3.13.4 UC-17.06 – Restore Forum Category

a. Class diagram

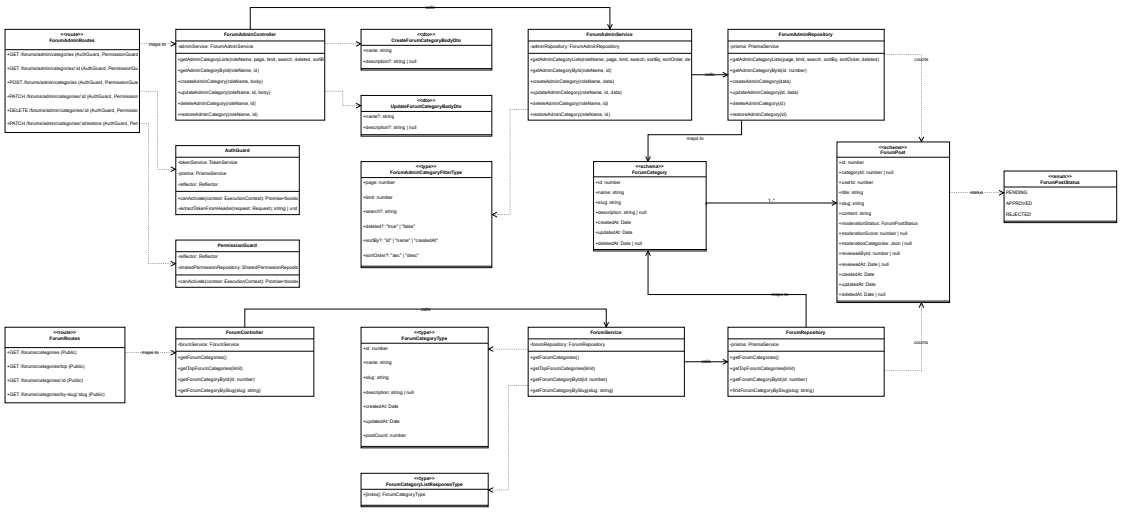

Figure 158: Class diagram of UC-17.06 Restore Forum Category

b. Sequence diagram

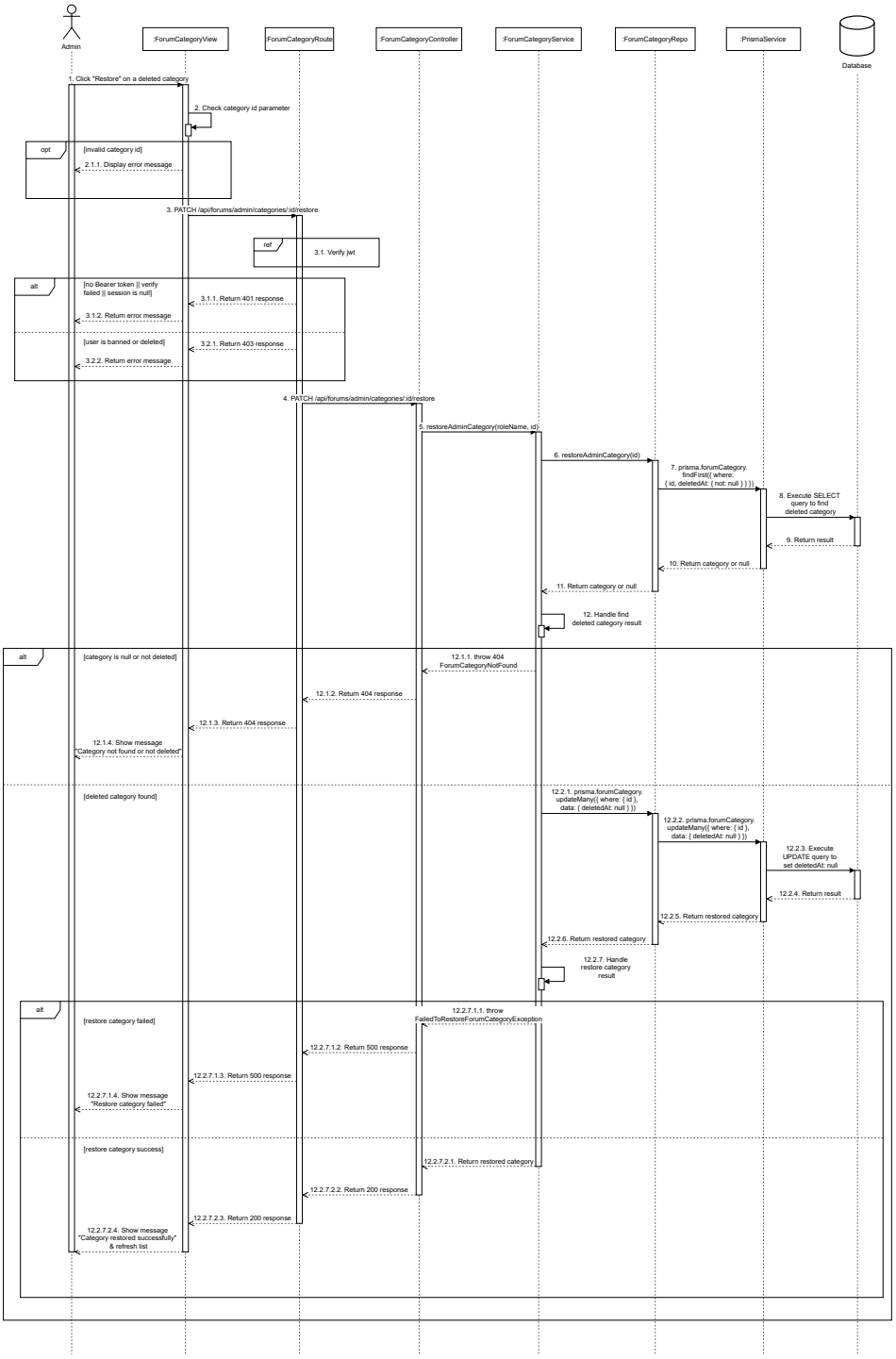

Figure 159: Sequence diagram of UC-17.06 Restore Forum Category

3.14 Manage Moderation

Manage Moderation covers the UC-34 review workflow of the administration console: listing the posts waiting for review, opening a flagged post and approving or rejecting it. A review writes the moderation status, score and reviewer back to the forum post.

3.14.1 UC-34.01 – View Moderation Queue

a. Class diagram

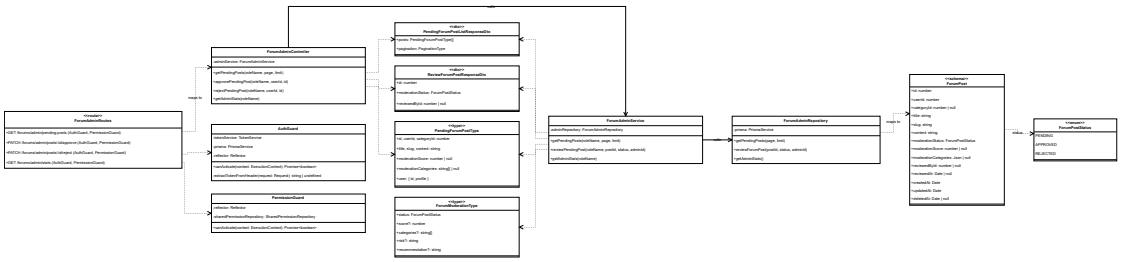

Figure 160: Class diagram of UC-34.01 View Moderation Queue

b. Sequence diagram

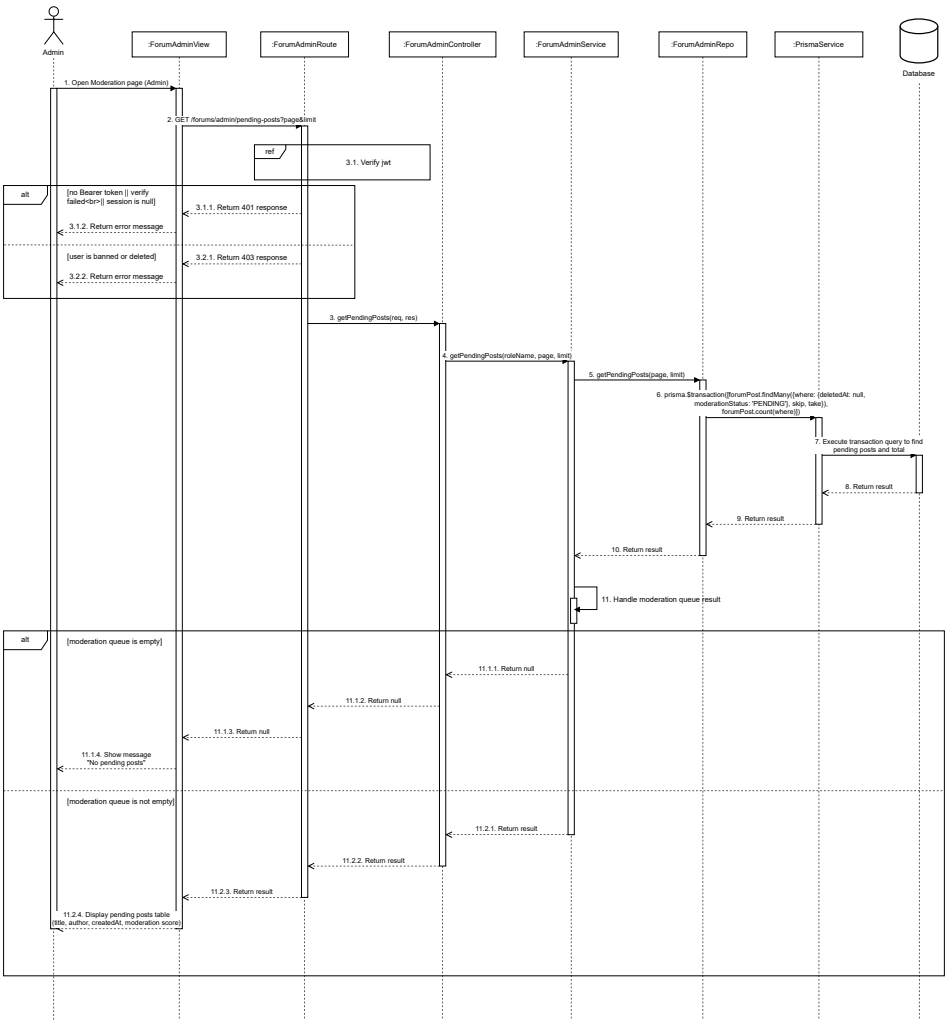

Figure 161: Sequence diagram of UC-34.01 View Moderation Queue

3.14.2 UC-34.02 – View Moderation Item Detail

a. Class diagram

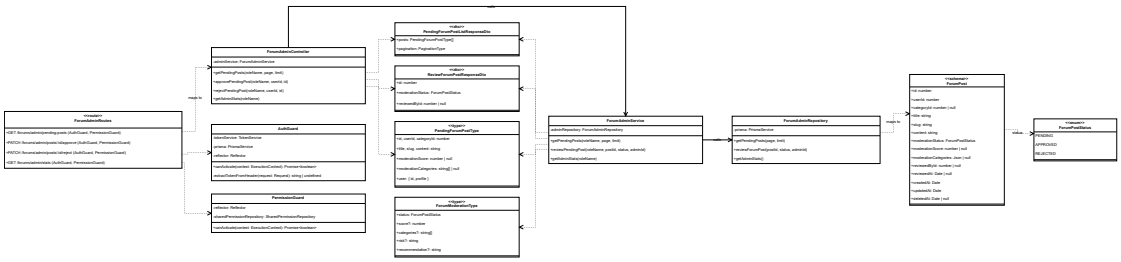

Figure 162: Class diagram of UC-34.02 View Moderation Item Detail

b. Sequence diagram

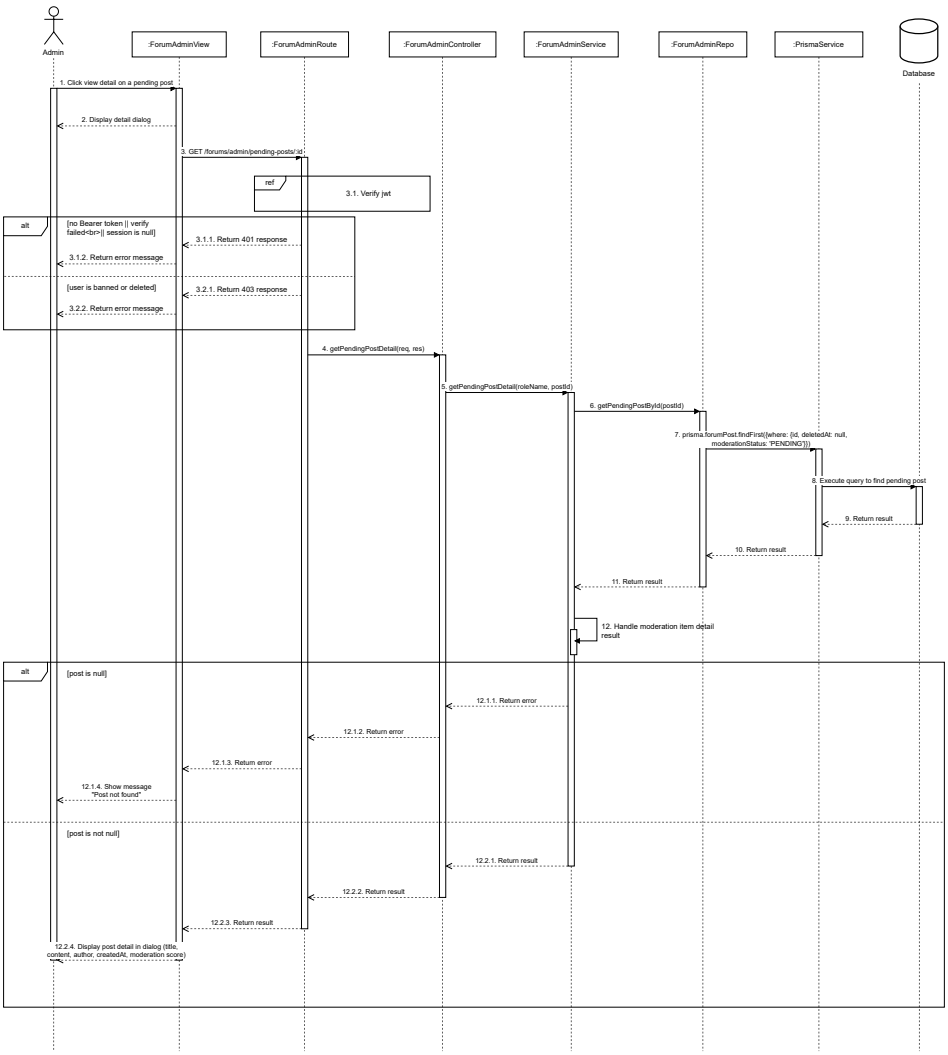

Figure 163: Sequence diagram of UC-34.02 View Moderation Item Detail

3.14.3 UC-34.03 – Resolve Moderation Item

a. Class diagram

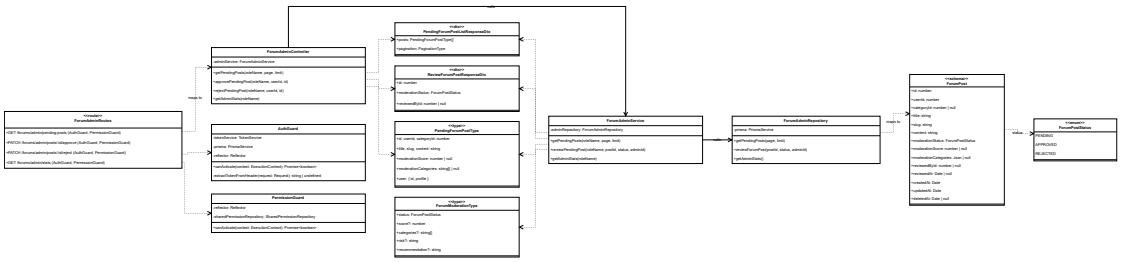

Figure 164: Class diagram of UC-34.03 Resolve Moderation Item

b. Sequence diagram

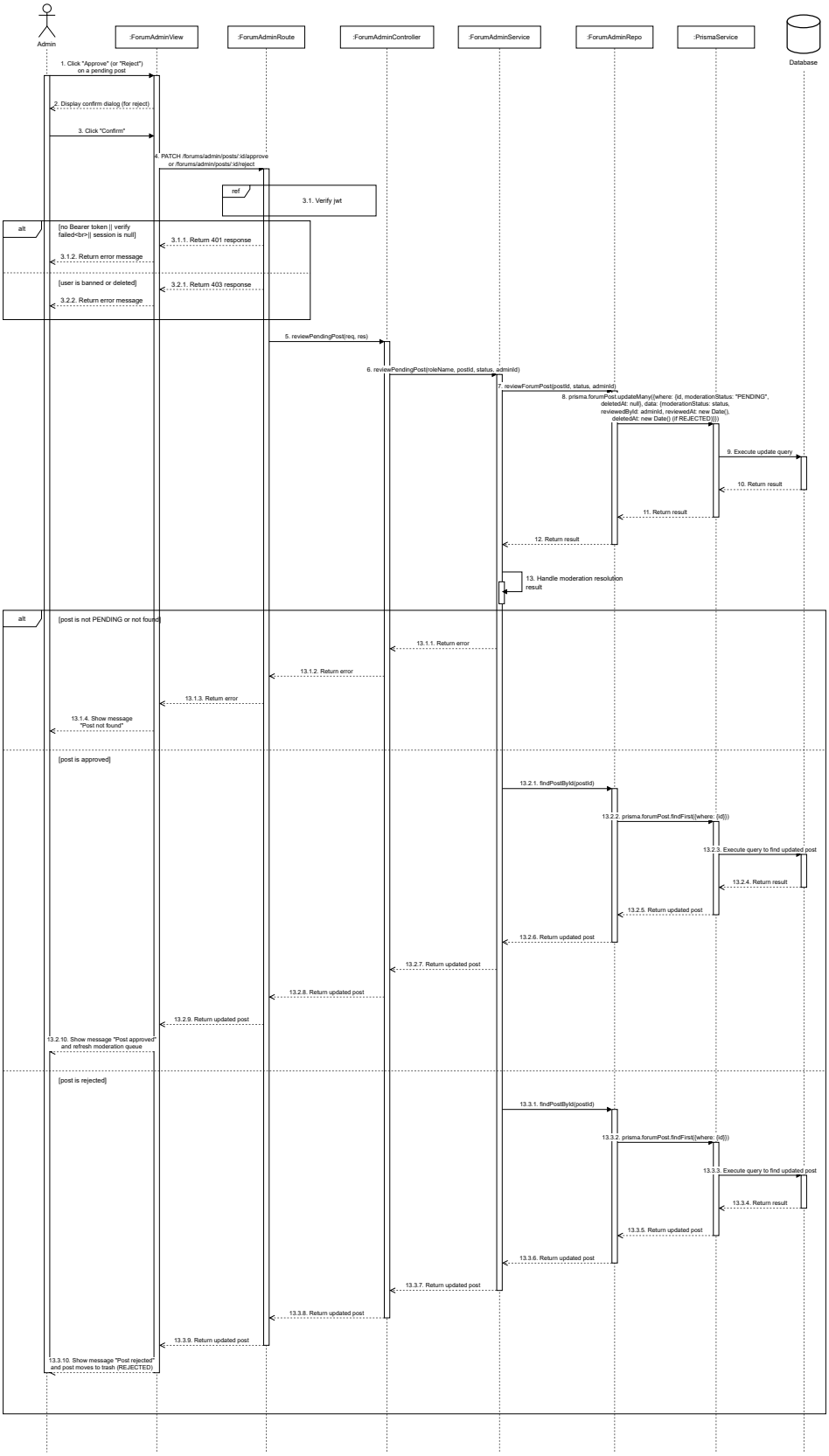

Figure 165: Sequence diagram of UC-34.03 Resolve Moderation Item

3.15 Manage ID Verification

Manage ID Verification covers the UC-35 identity review workflow: listing the submitted identity documents, opening a request with its uploaded file, and approving or rejecting the request. A decision updates the verification status and the reviewer of the identity document.

3.15.1 UC-35.01 – View ID Verification Request List

a. Class diagram

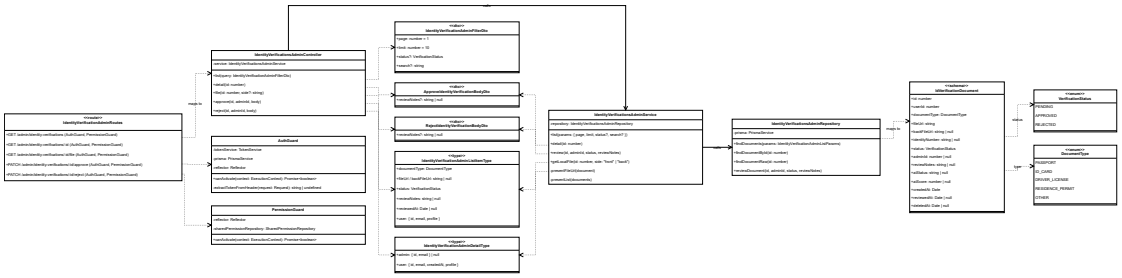

Figure 166: Class diagram of UC-35.01 View ID Verification Request List

b. Sequence diagram

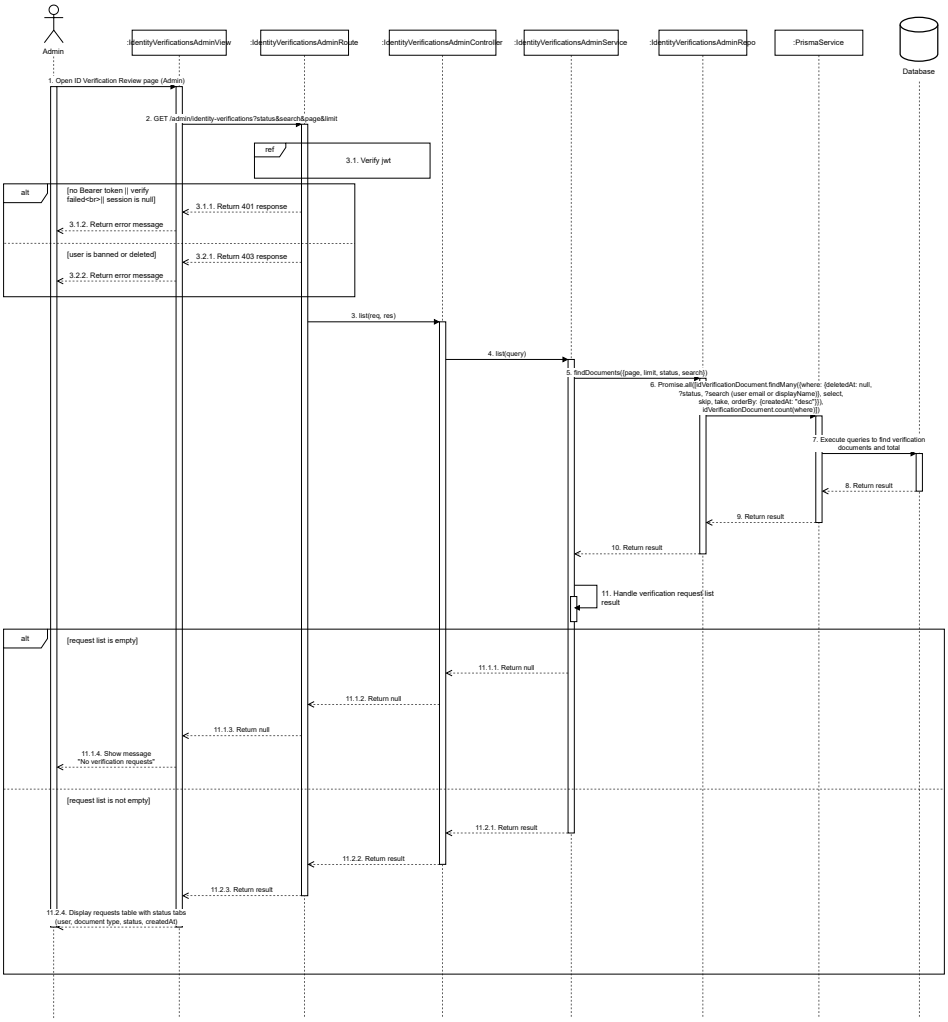

Figure 167: Sequence diagram of UC-35.01 View ID Verification Request List

3.15.2 UC-35.02 – View ID Verification Request Detail

a. Class diagram

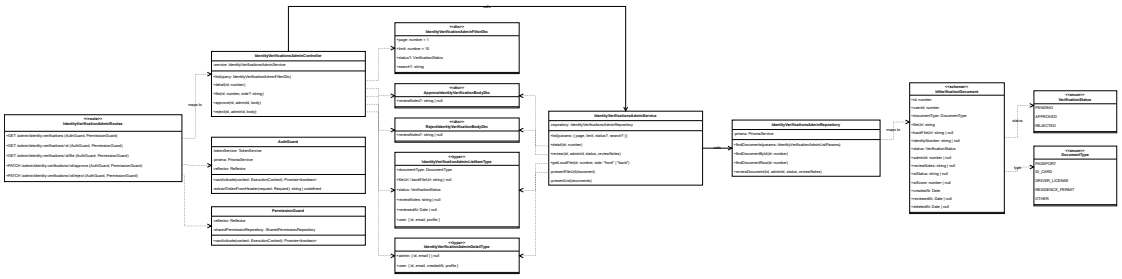

Figure 168: Class diagram of UC-35.02 View ID Verification Request Detail

b. Sequence diagram

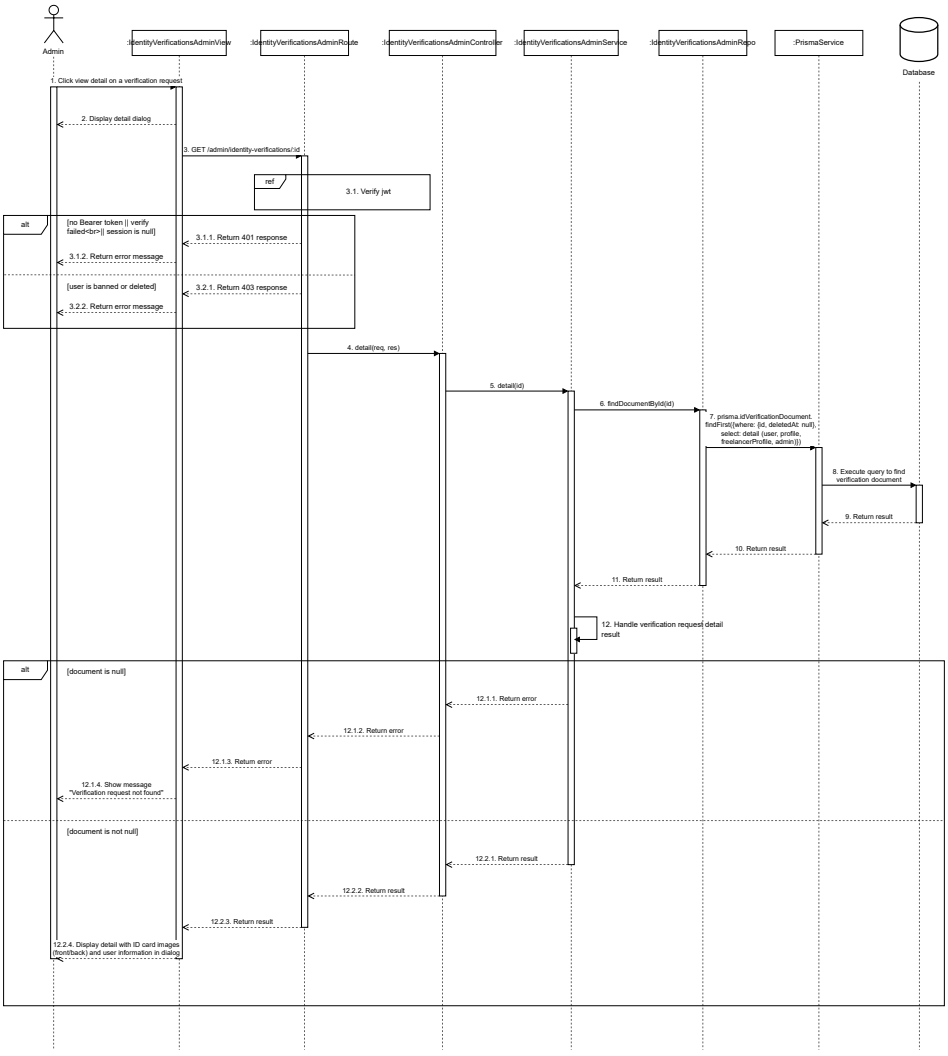

Figure 169: Sequence diagram of UC-35.02 View ID Verification Request Detail

3.15.3 UC-35.03 – Approve ID Verification

a. Class diagram

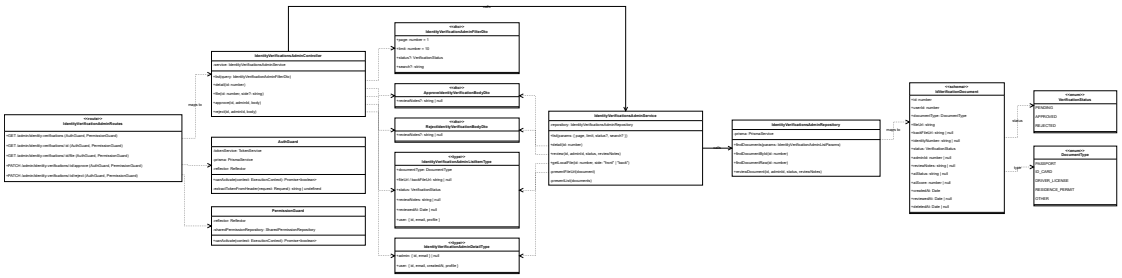

Figure 170: Class diagram of UC-35.03 Approve ID Verification

b. Sequence diagram

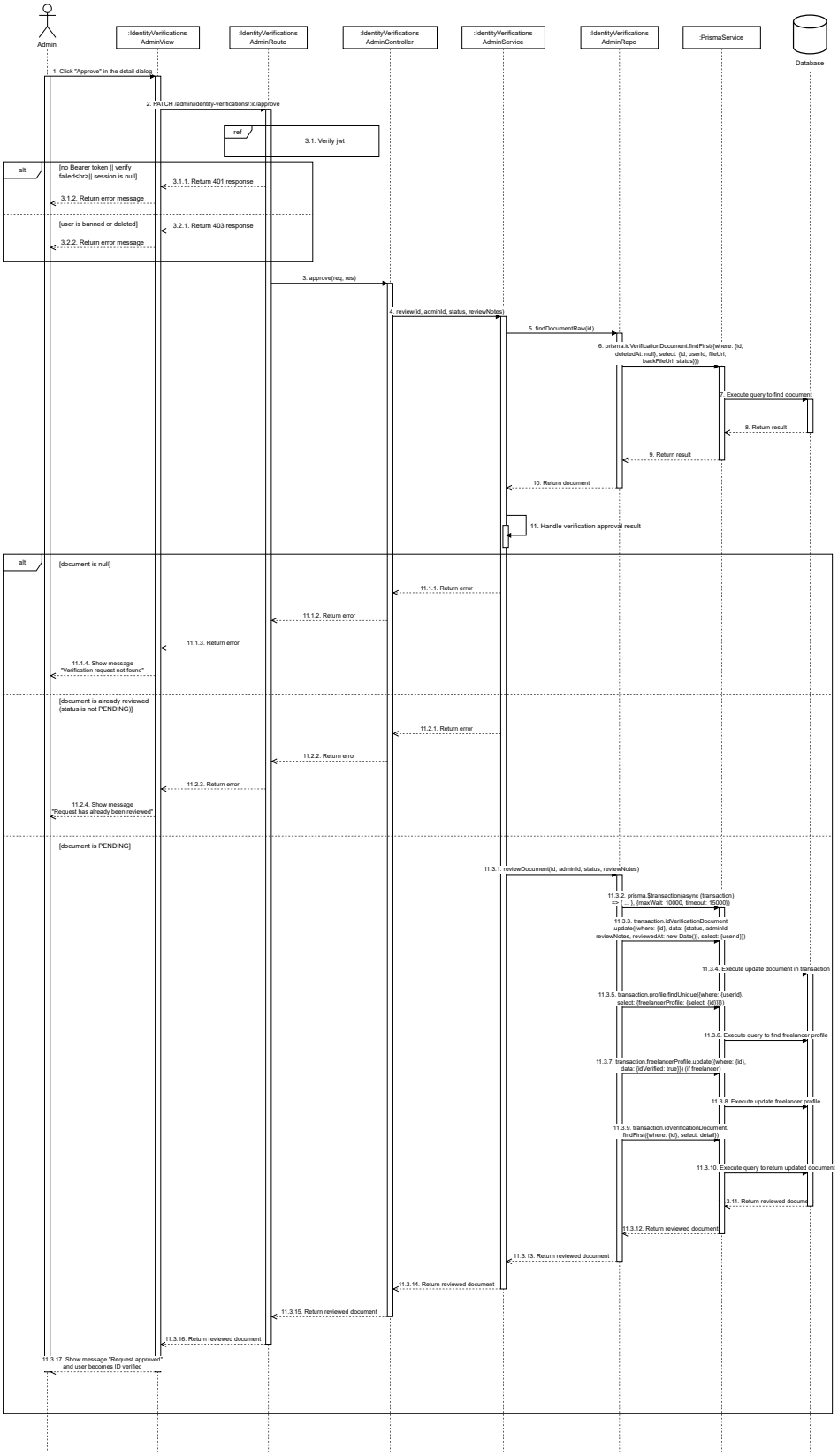

Figure 171: Sequence diagram of UC-35.03 Approve ID Verification

3.15.4 UC-35.04 – Reject ID Verification

a. Class diagram

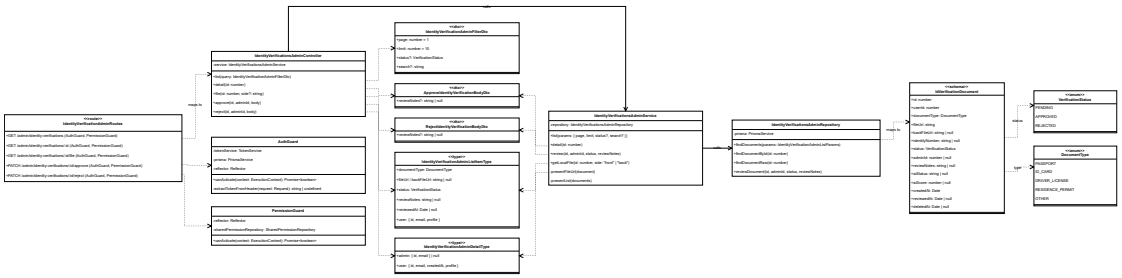

Figure 172: Class diagram of UC-35.04 Reject ID Verification

b. Sequence diagram

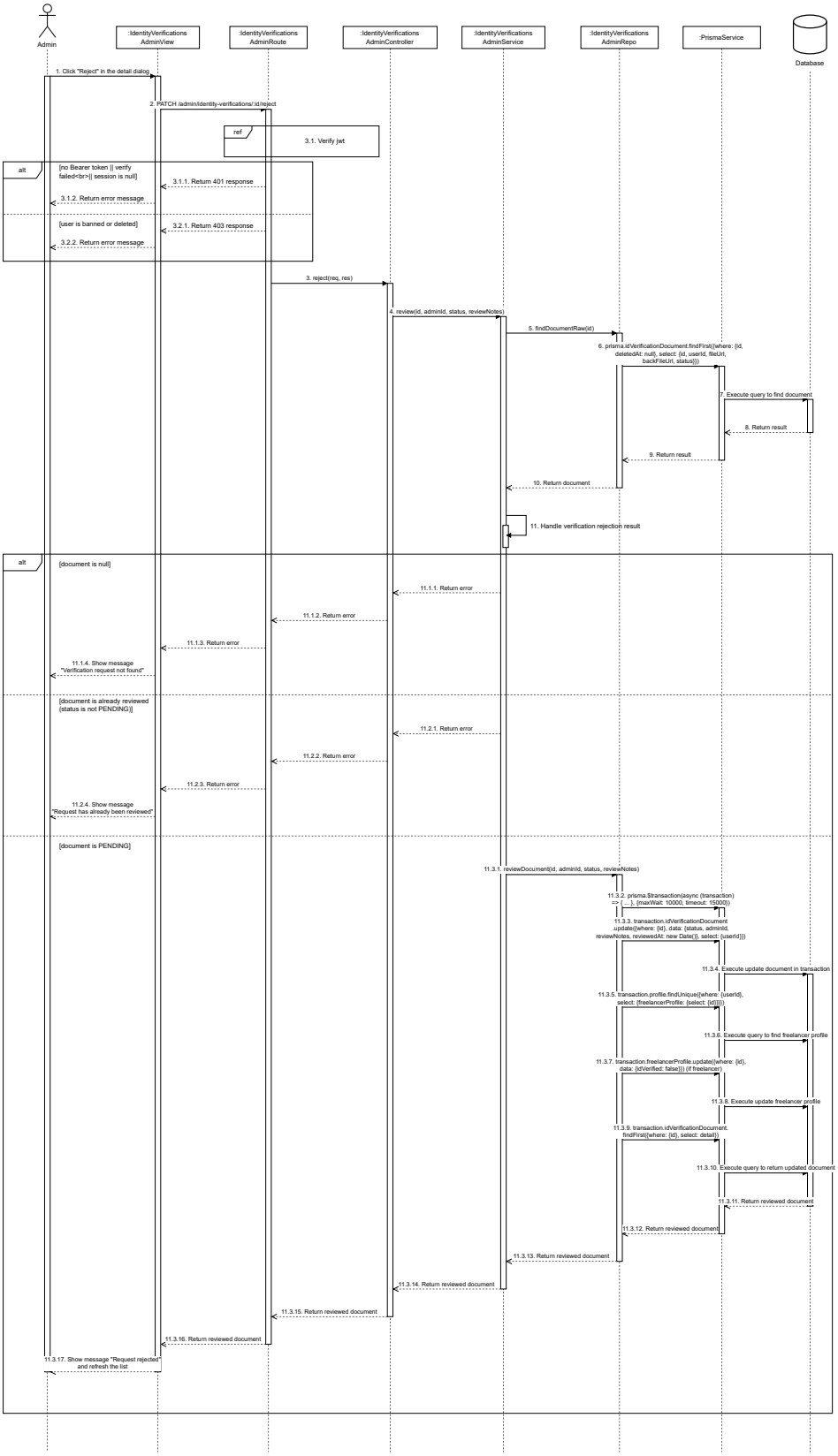

Figure 173: Sequence diagram of UC-35.04 Reject ID Verification

3.16 Manage Forum Trash

Manage Forum Trash covers the UC-36 recovery workflow: listing the soft-deleted posts and comments kept for review, opening one deleted item, and restoring it so that it becomes visible in the forum again.

3.16.1 UC-36.01 – View Deleted Posts/Comments List

a. Class diagram

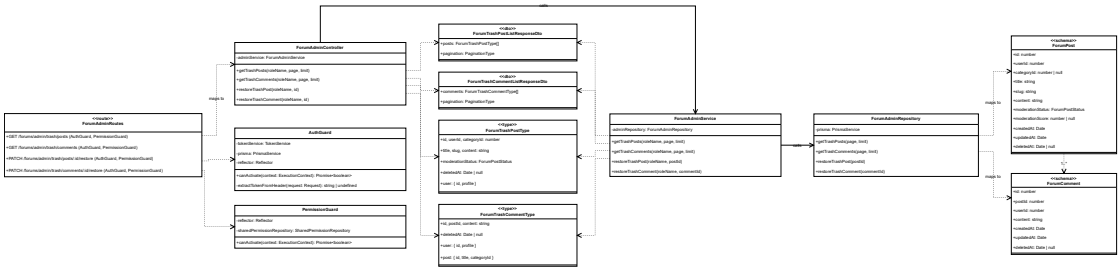

Figure 174: Class diagram of UC-36.01 View Deleted Posts/Comments List

b. Sequence diagram

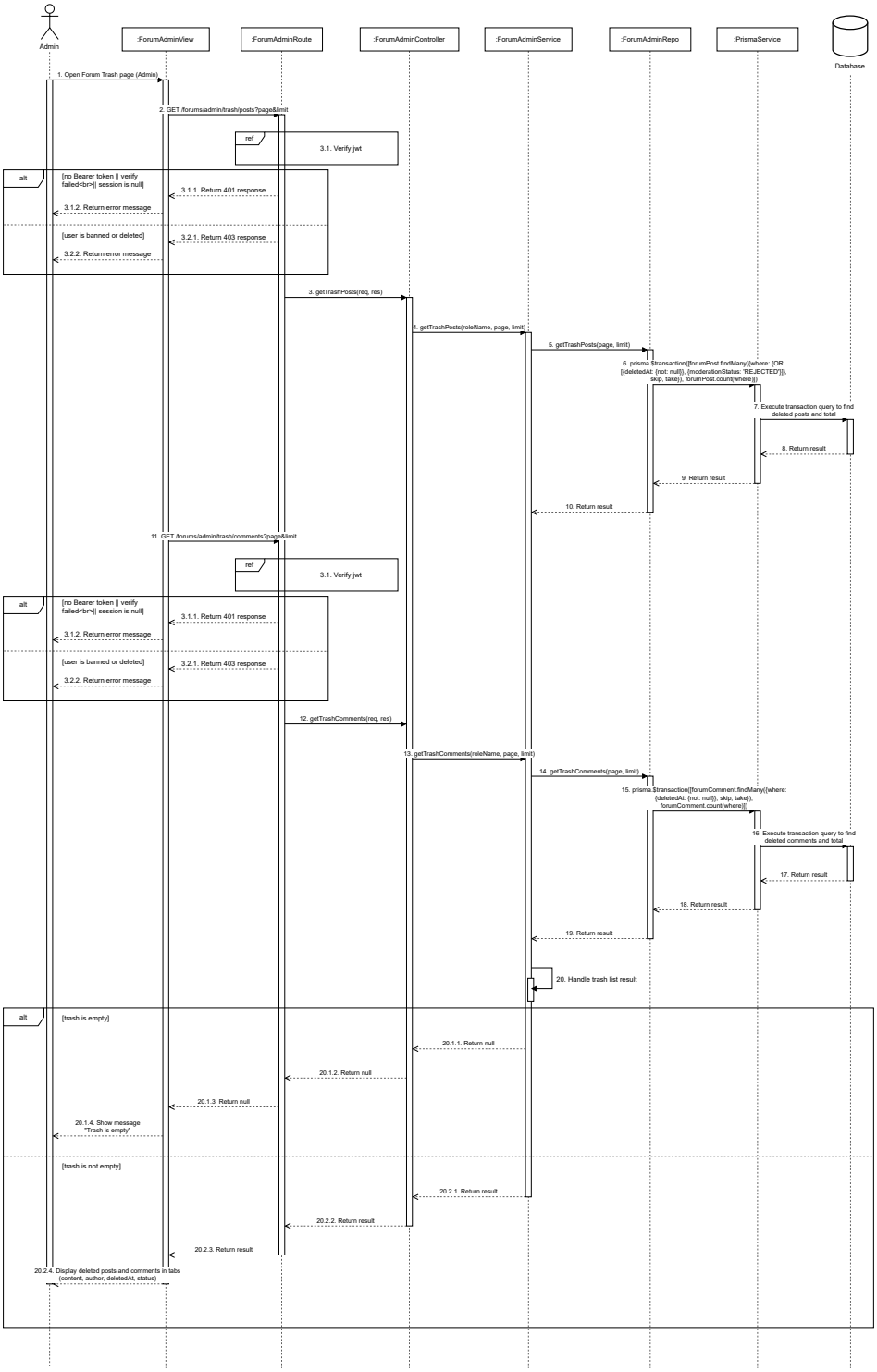

Figure 175: Sequence diagram of UC-36.01 View Deleted Posts/Comments List

3.16.2 UC-36.02 – View Deleted Posts/Comments Detail

a. Class diagram

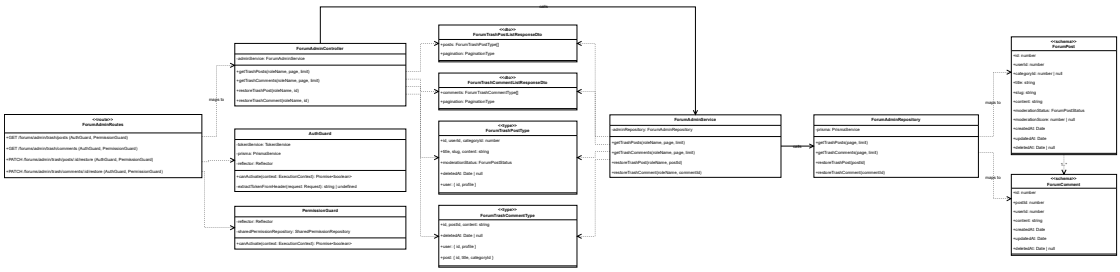

Figure 176: Class diagram of UC-36.02 View Deleted Posts/Comments Detail

b. Sequence diagram

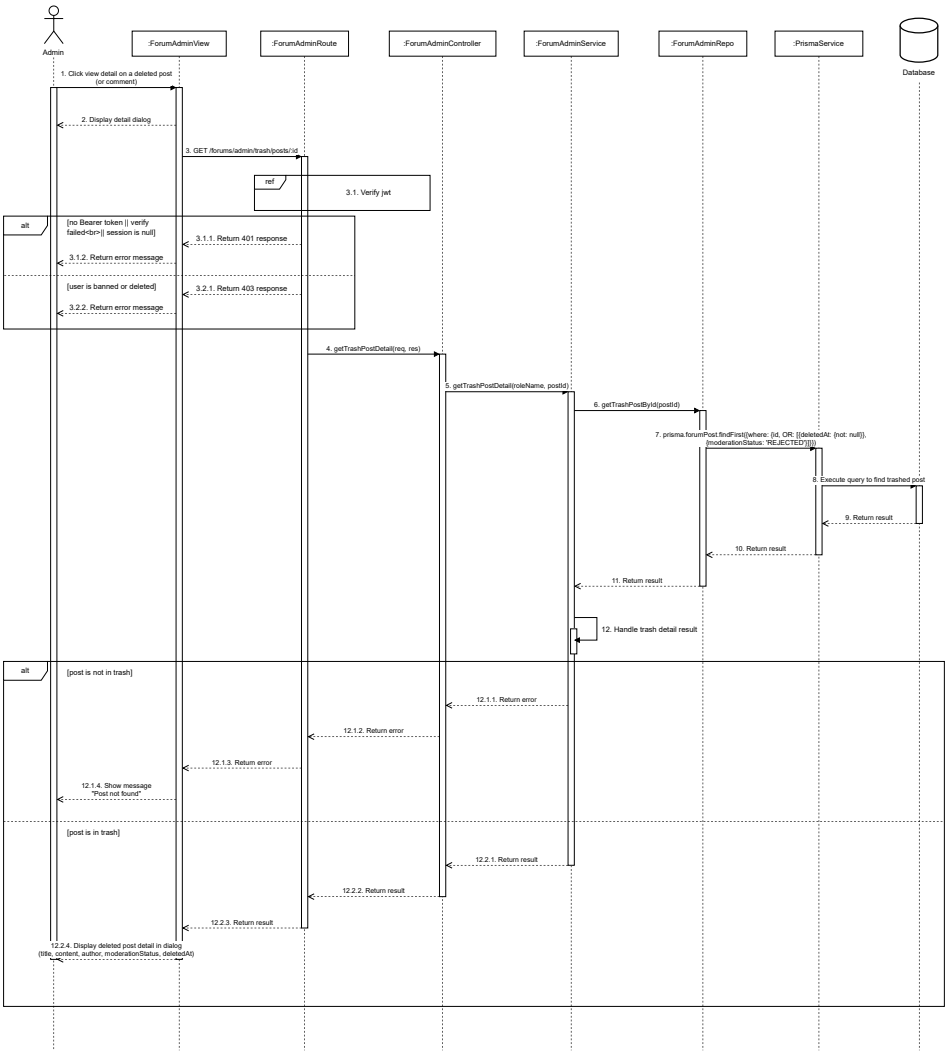

Figure 177: Sequence diagram of UC-36.02 View Deleted Posts/Comments Detail

3.16.3 UC-36.03 – Restore Deleted Posts/Comments

a. Class diagram

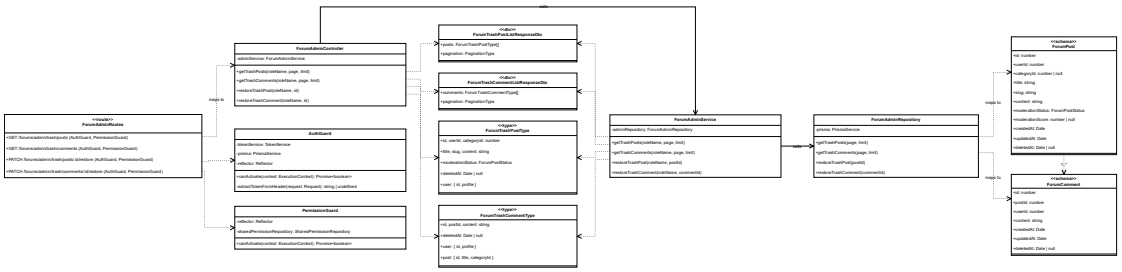

Figure 178: Class diagram of UC-36.03 Restore Deleted Posts/Comments

b. Sequence diagram

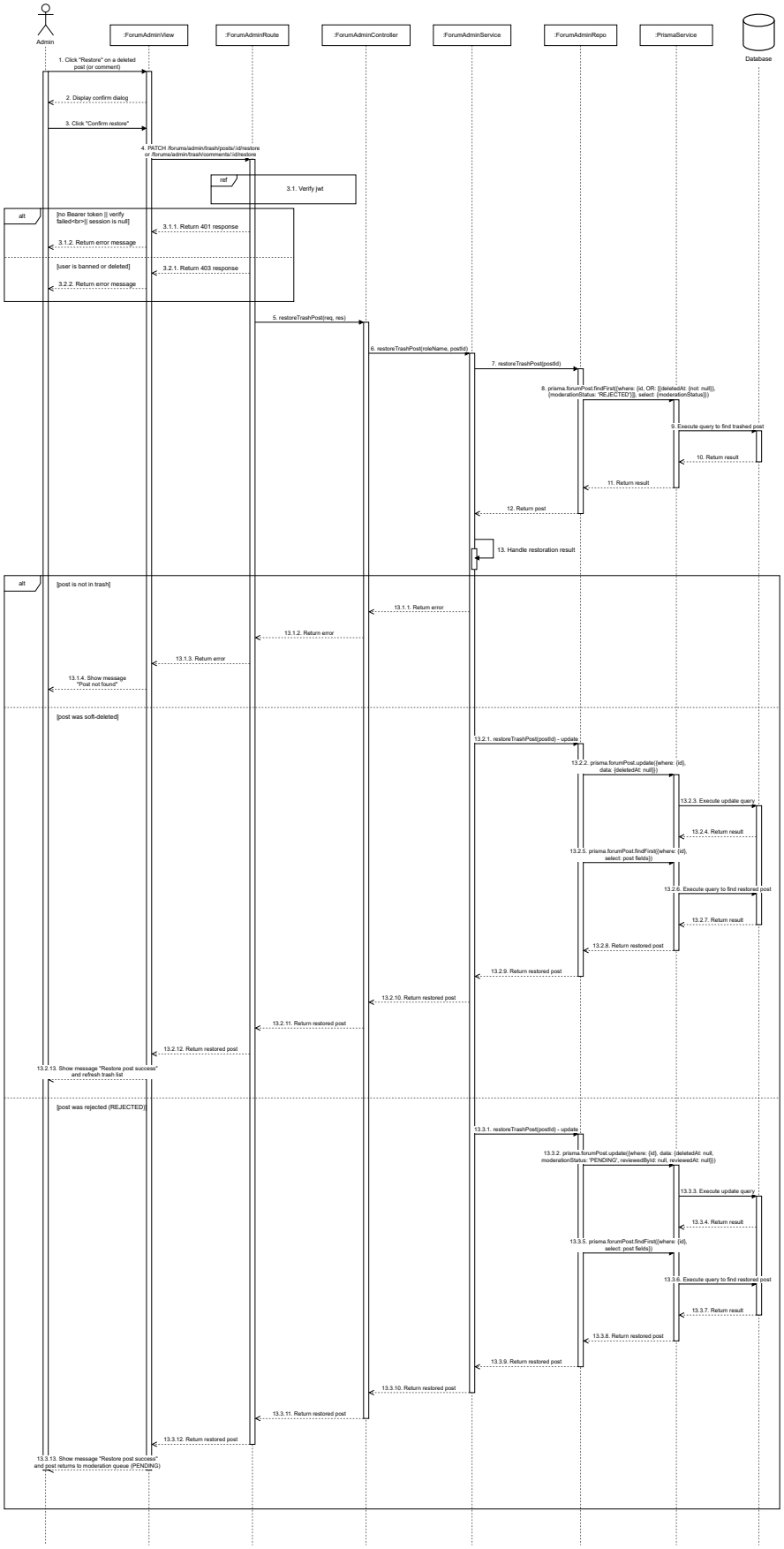

Figure 179: Sequence diagram of UC-36.03 Restore Deleted Posts/Comments

3.17 Proposal Management

Proposal Management covers the full proposal workflow between freelancers and clients. Freelancers can submit proposals with cover letters, bid amounts, and delivery estimates, view their submitted proposals, modify pending proposals, or withdraw them. Clients can view all proposals submitted for their jobs, inspect detailed freelancer bids, and accept or reject proposals to initiate contracting.

3.17.1 UC-32.01 – View My Proposal List

a. Class diagram

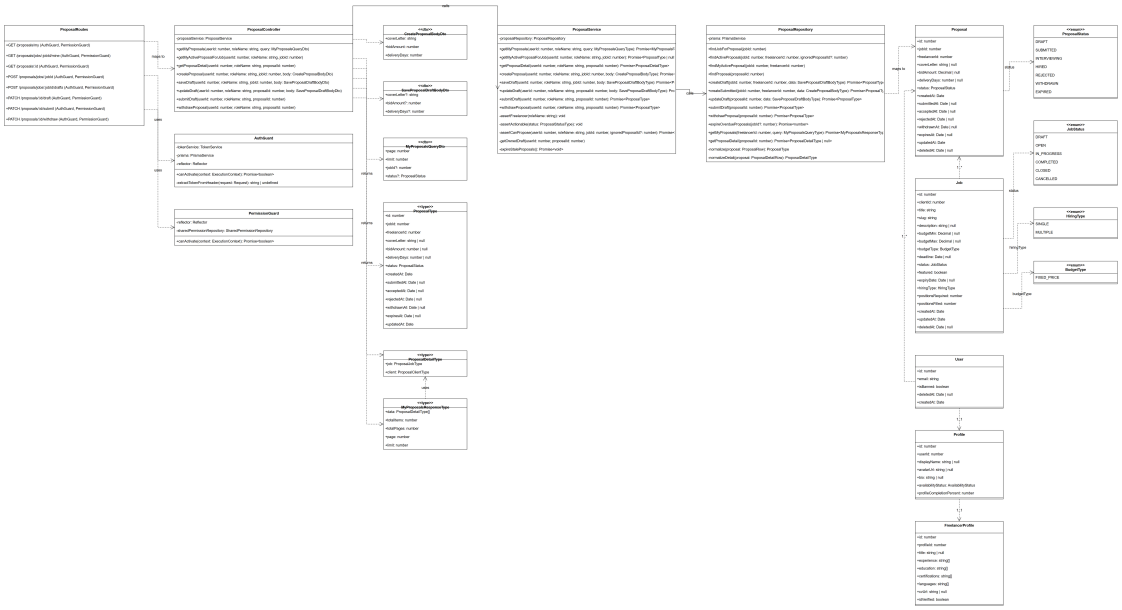

Figure 180: Class diagram of UC-32.01 View My Proposal List

b. Sequence diagram

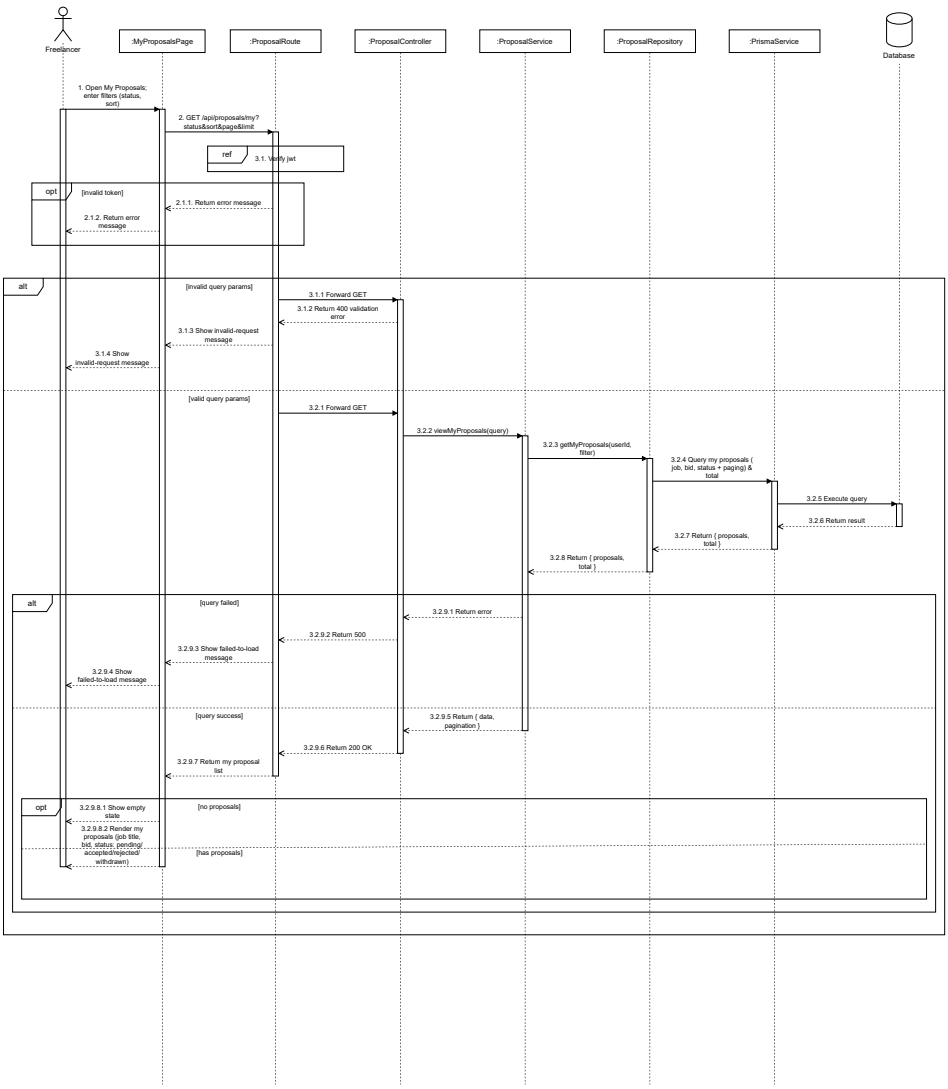

Figure 181: Sequence diagram of UC-32.01 View My Proposal List

3.17.2 UC-32.02 – View My Proposal Detail

a. Class diagram

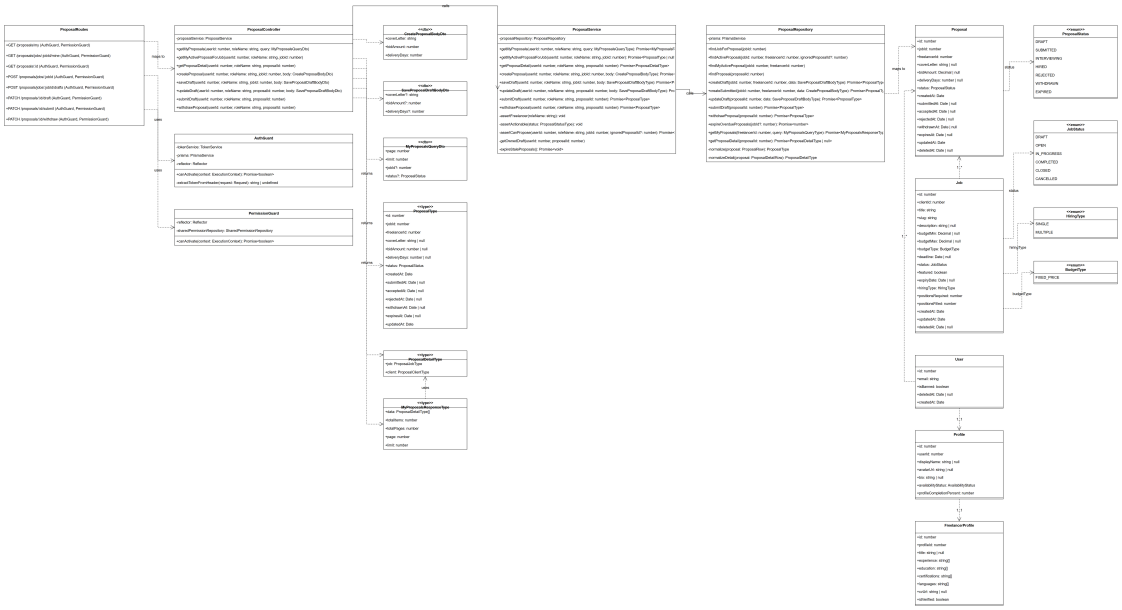

Figure 182: Class diagram of UC-32.02 View My Proposal Detail

b. Sequence diagram

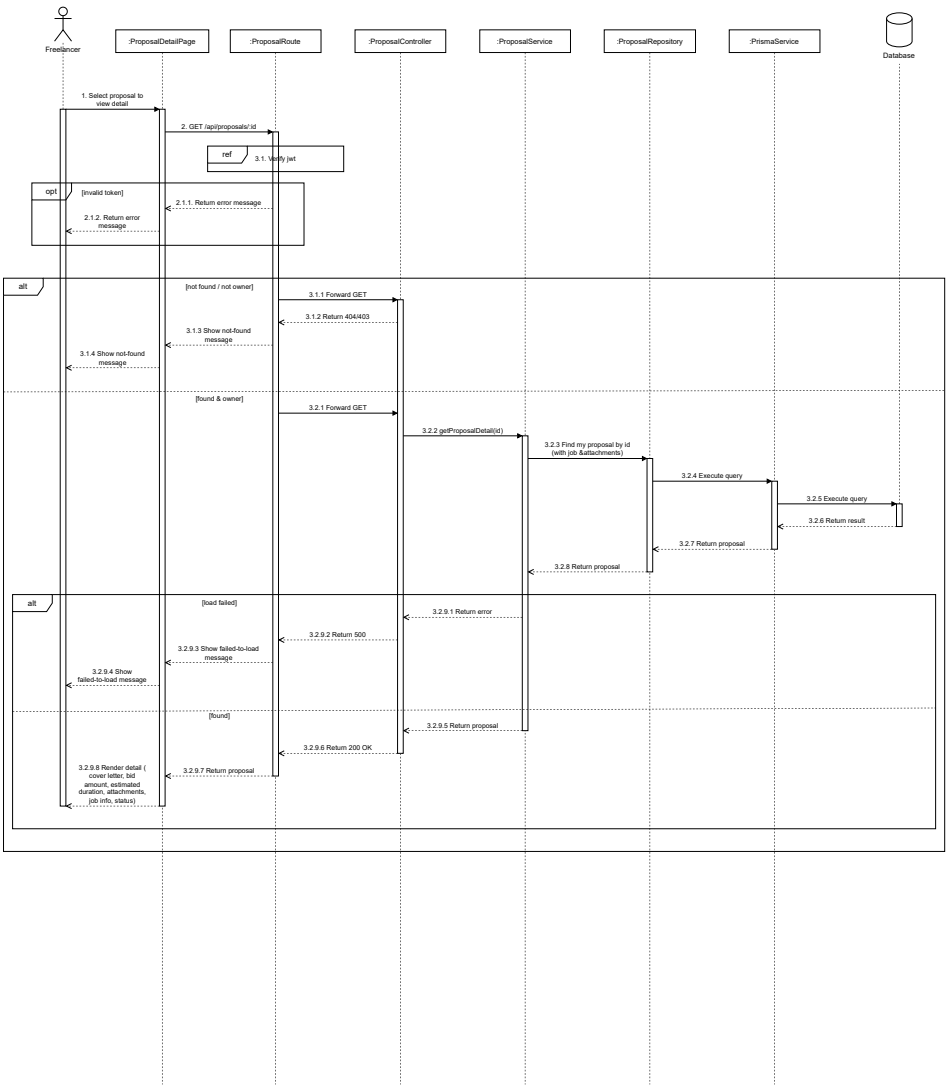

Figure 183: Sequence diagram of UC-32.02 View My Proposal Detail

3.17.3 UC-32.03 – Submit Proposal

a. Class diagram

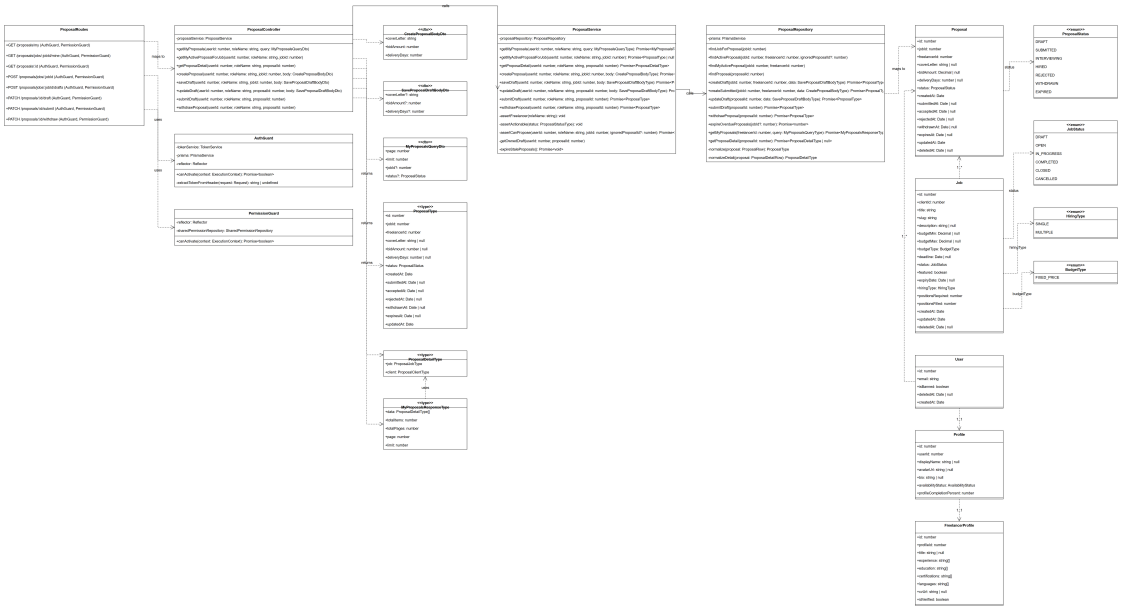

Figure 184: Class diagram of UC-32.03 Submit Proposal

b. Sequence diagram

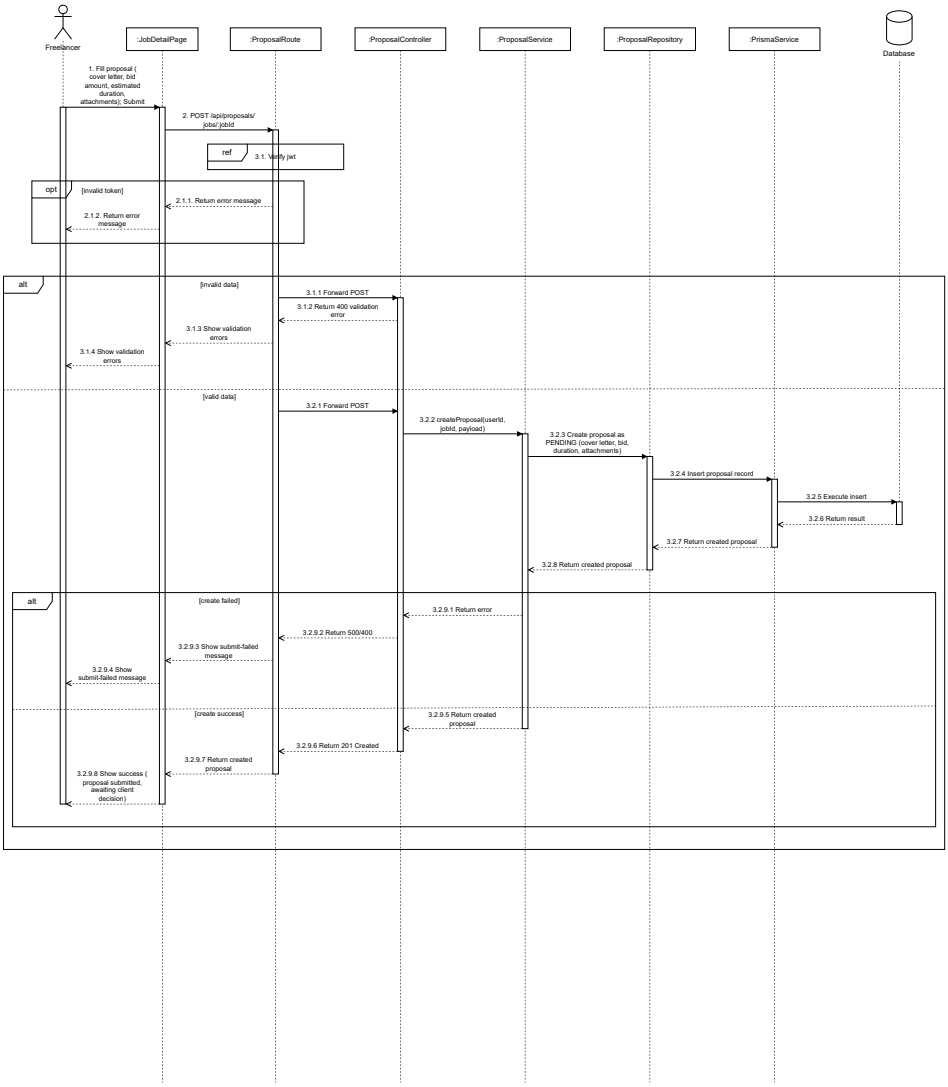

Figure 185: Sequence diagram of UC-32.03 Submit Proposal

3.17.4 UC-32.04 – Update Proposal

a. Class diagram

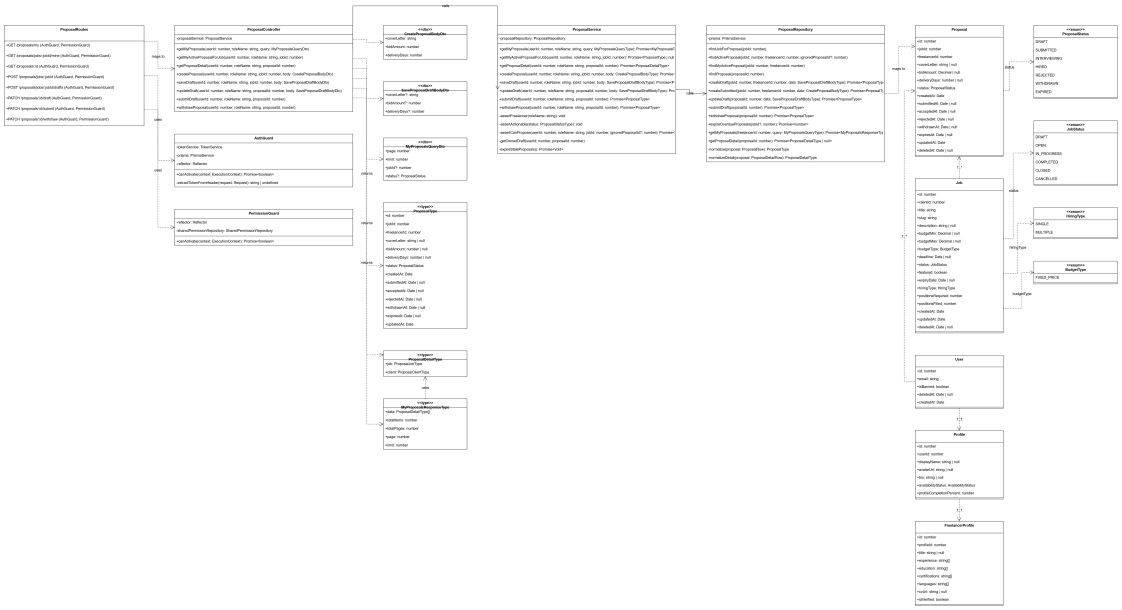

Figure 186: Class diagram of UC-32.04 Update Proposal

b. Sequence diagram

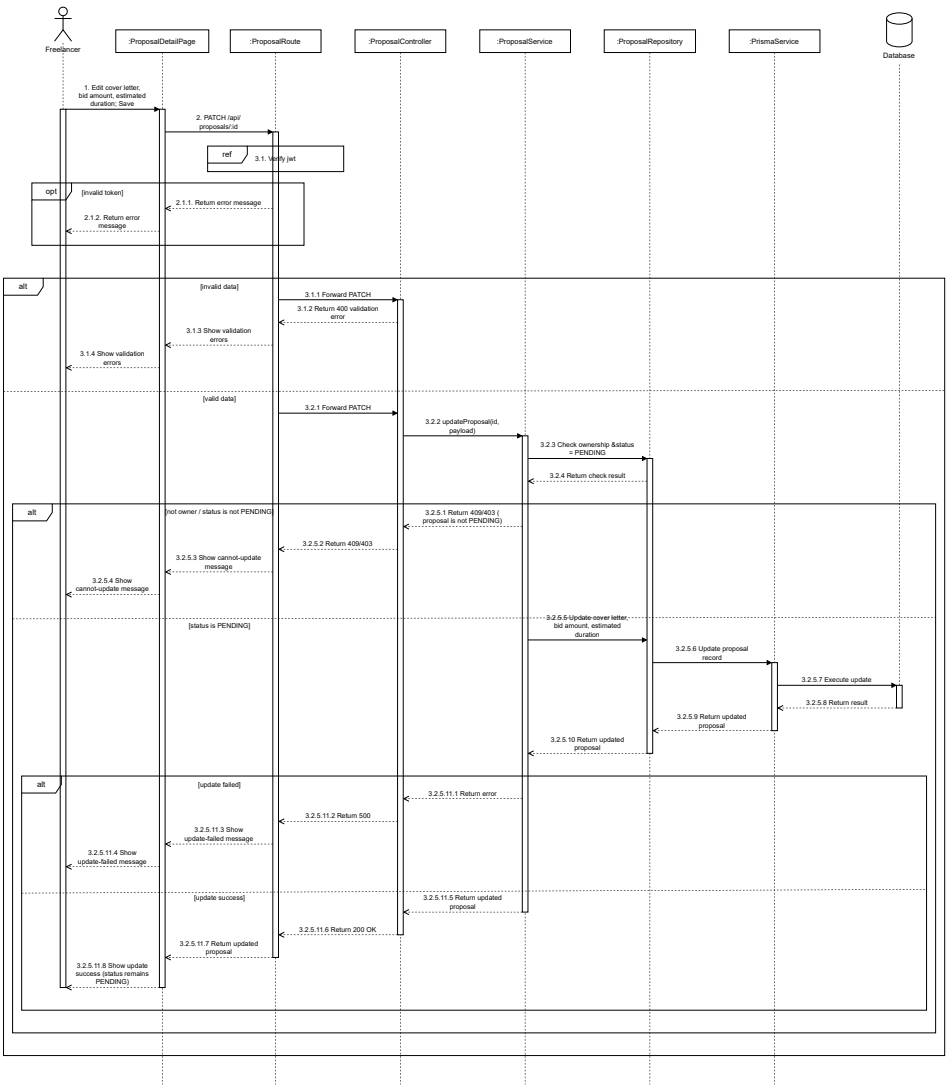

Figure 187: Sequence diagram of UC-32.04 Update Proposal

3.17.5 UC-32.05 – Withdraw Proposal

a. Class diagram

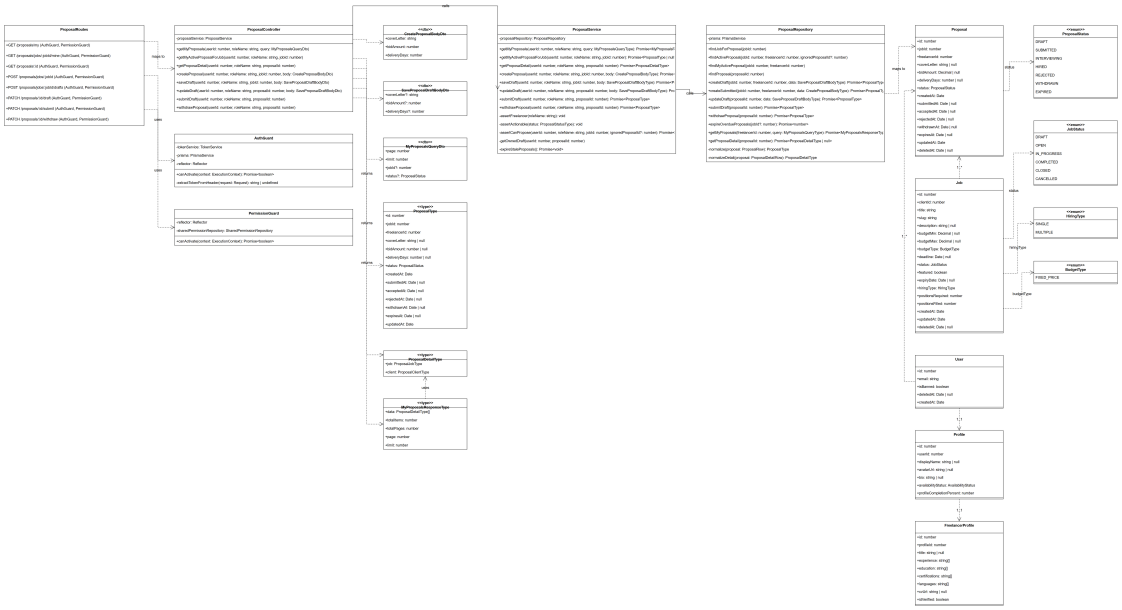

Figure 188: Class diagram of UC-32.05 Withdraw Proposal

b. Sequence diagram

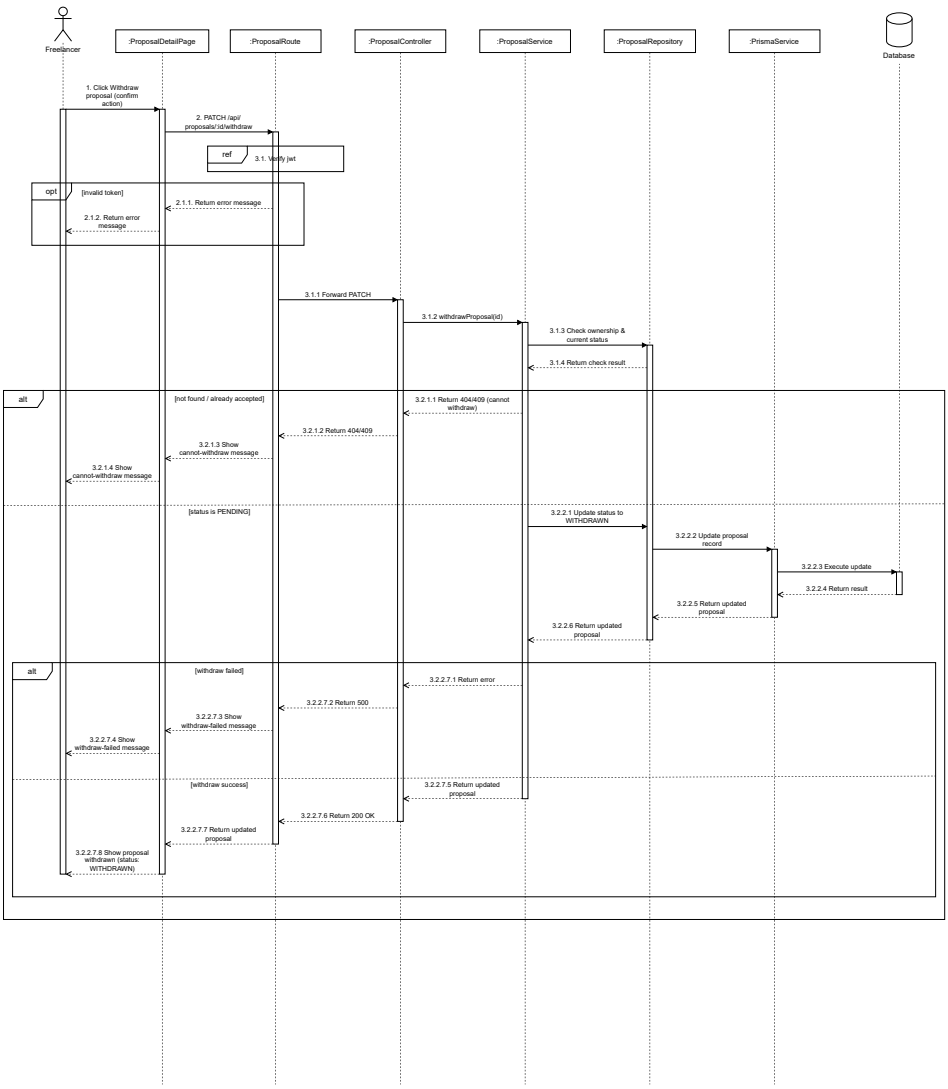

Figure 189: Sequence diagram of UC-32.05 Withdraw Proposal

3.17.6 UC-32.06 – Accept Proposal

a. Class diagram

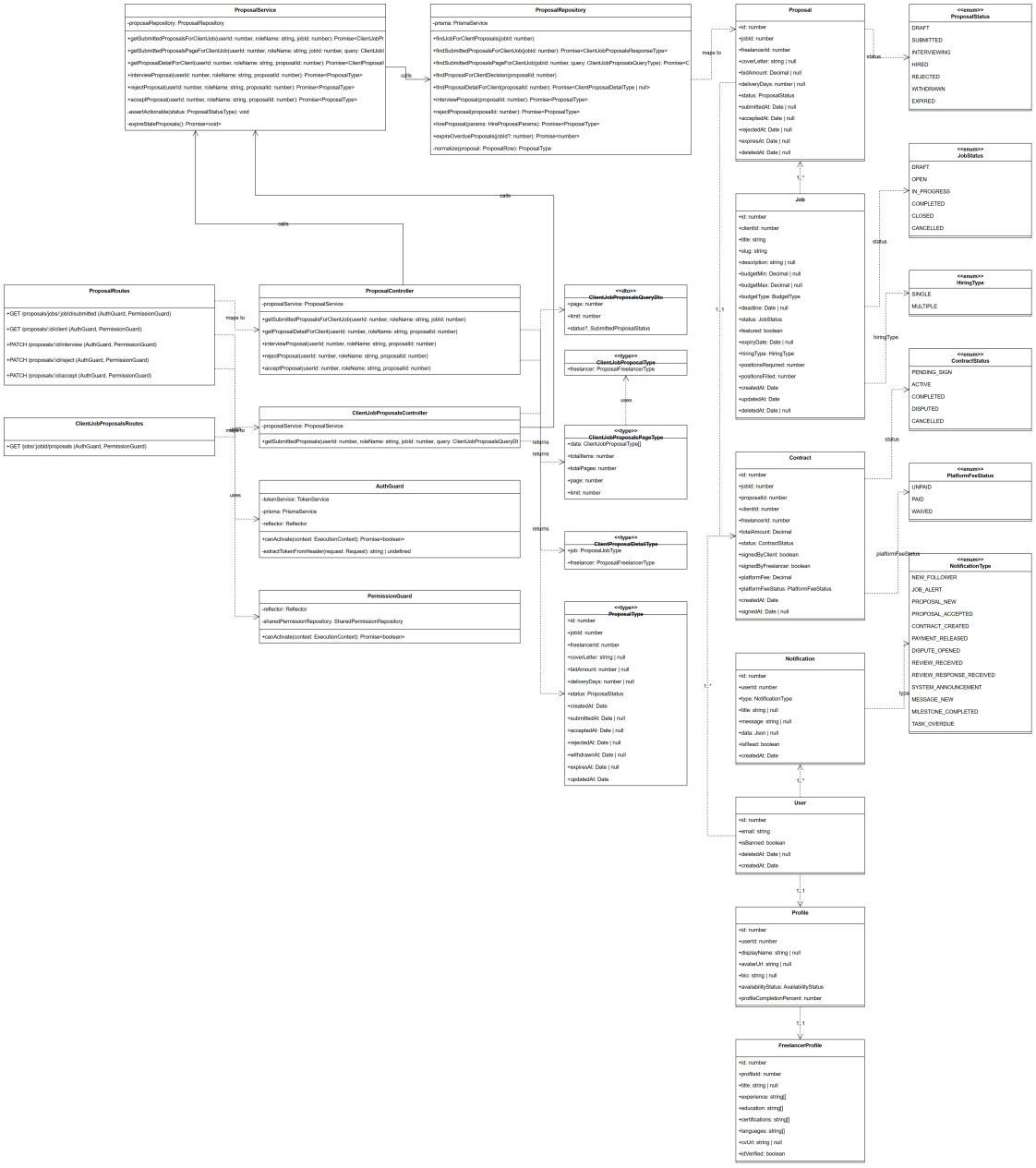

Figure 190: Class diagram of UC-32.06 Accept Proposal

b. Sequence diagram

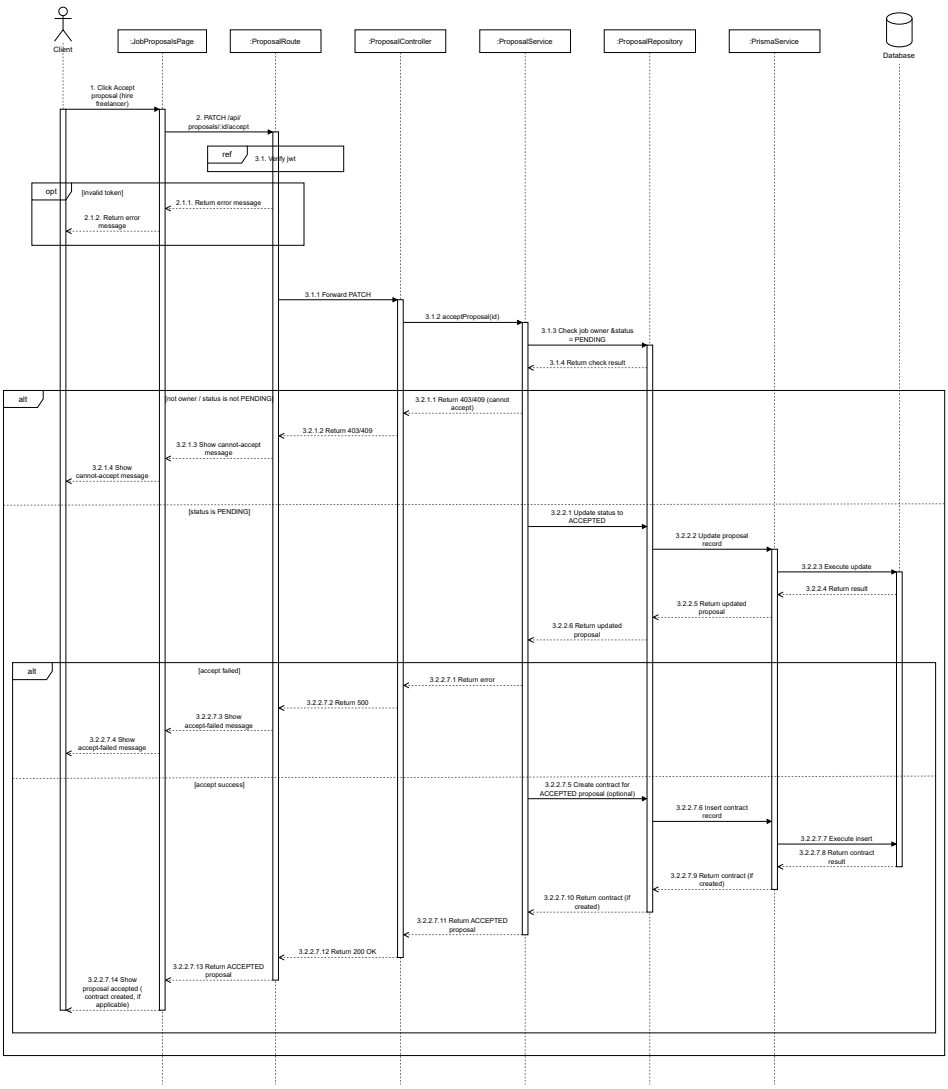

Figure 191: Sequence diagram of UC-32.06 Accept Proposal

3.17.7 UC-32.07 – Reject Proposal

a. Class diagram

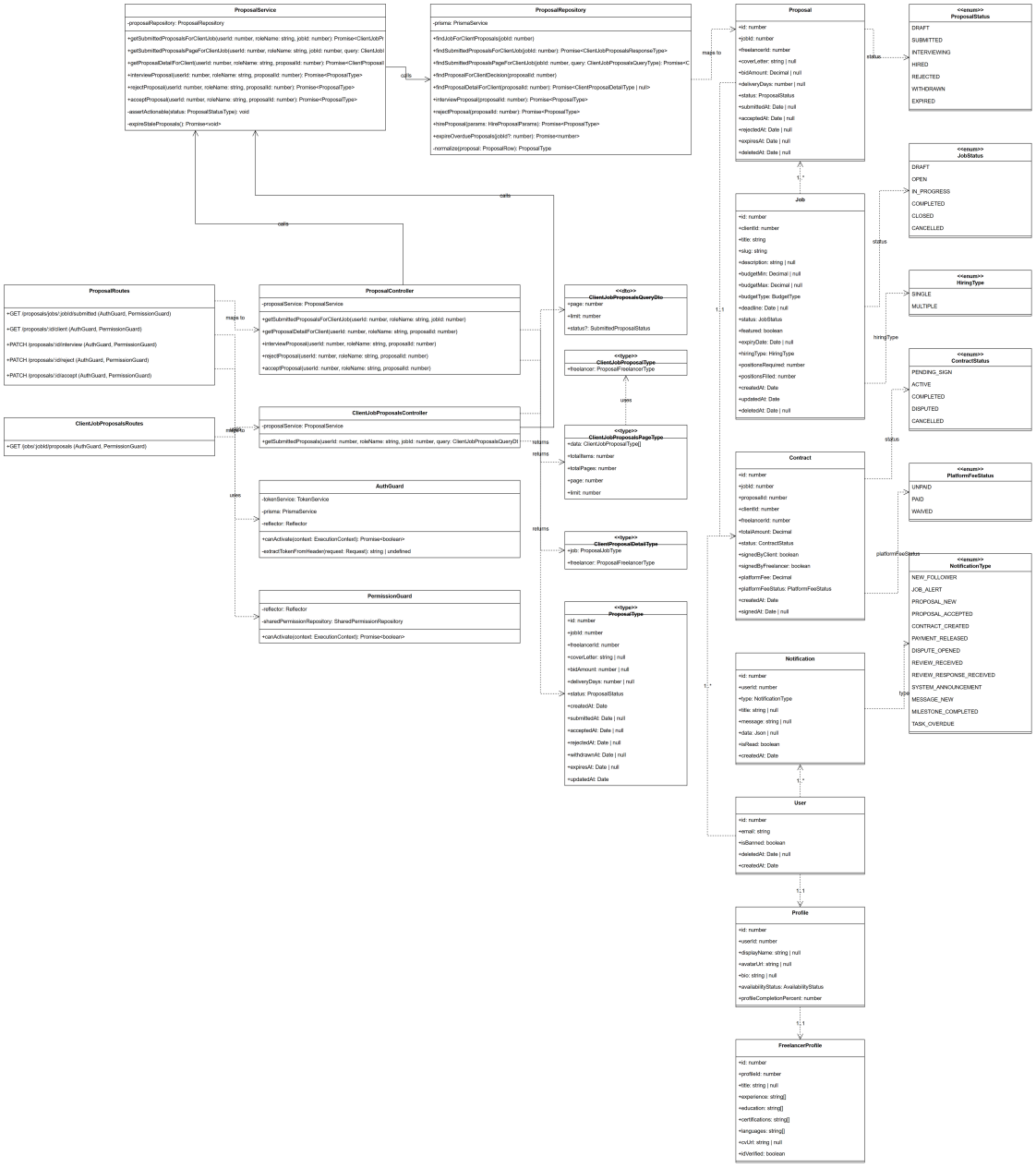

Figure 192: Class diagram of UC-32.07 Reject Proposal

b. Sequence diagram

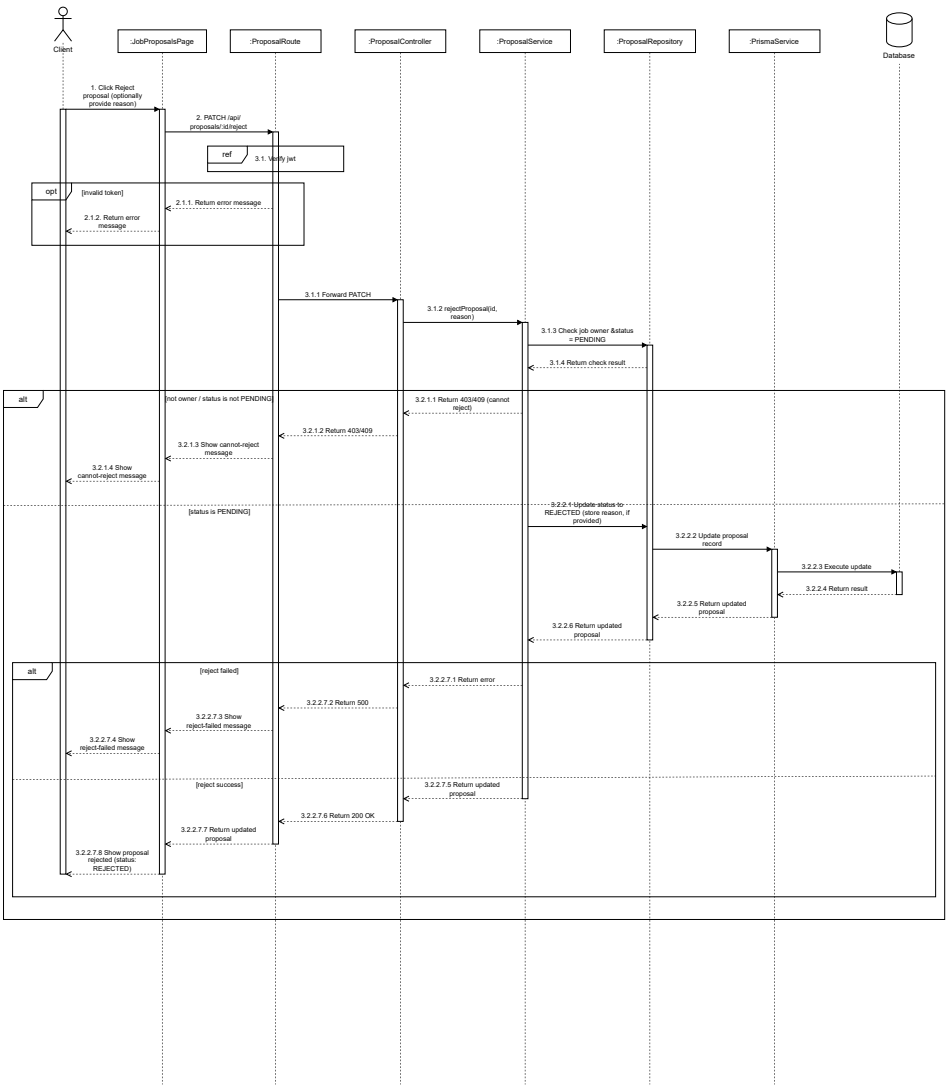

Figure 193: Sequence diagram of UC-32.07 Reject Proposal

3.17.8 UC-32.08 – View Job Proposal List

a. Class diagram

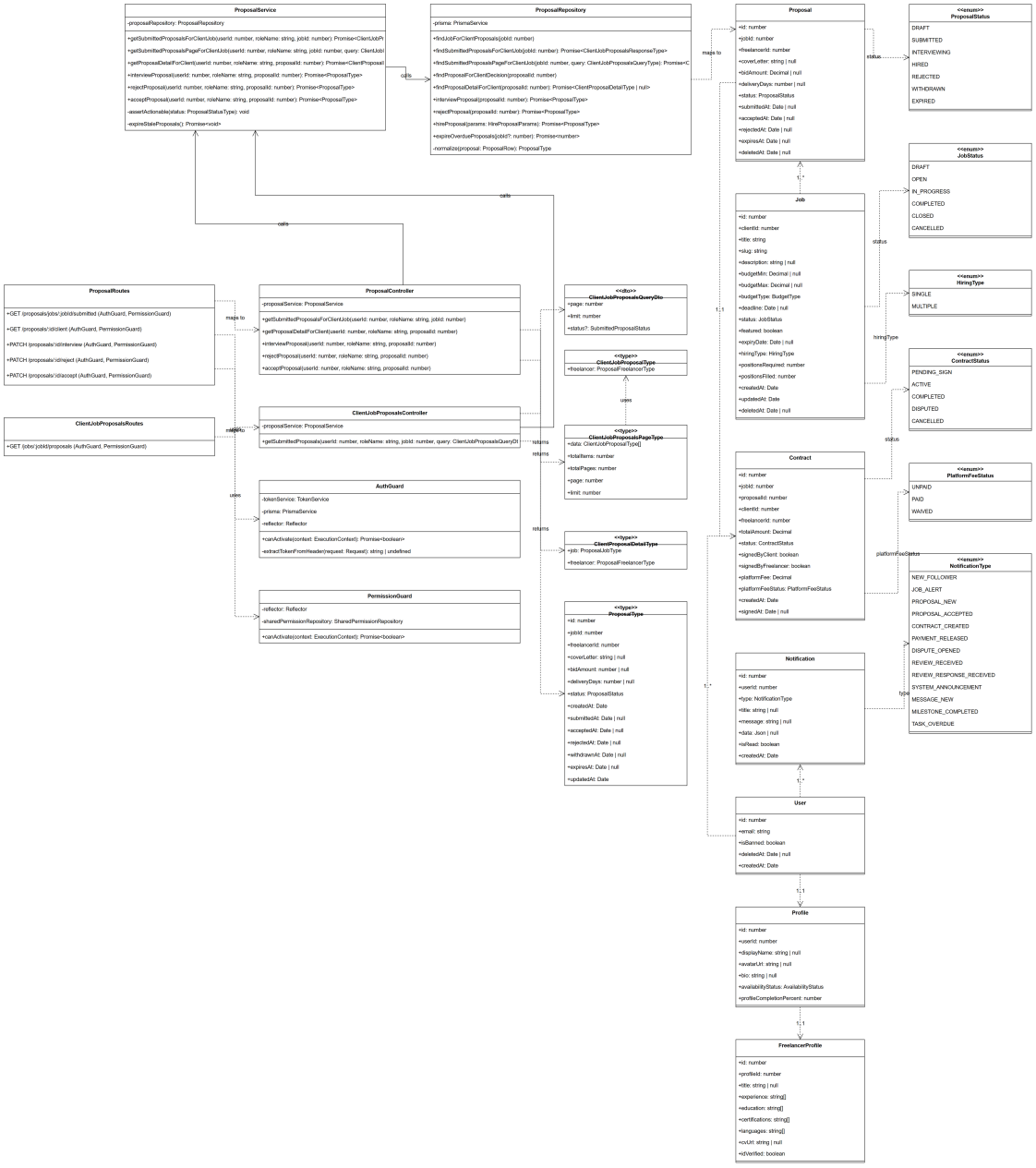

Figure 194: Class diagram of UC-32.08 View Job Proposal List

b. Sequence diagram

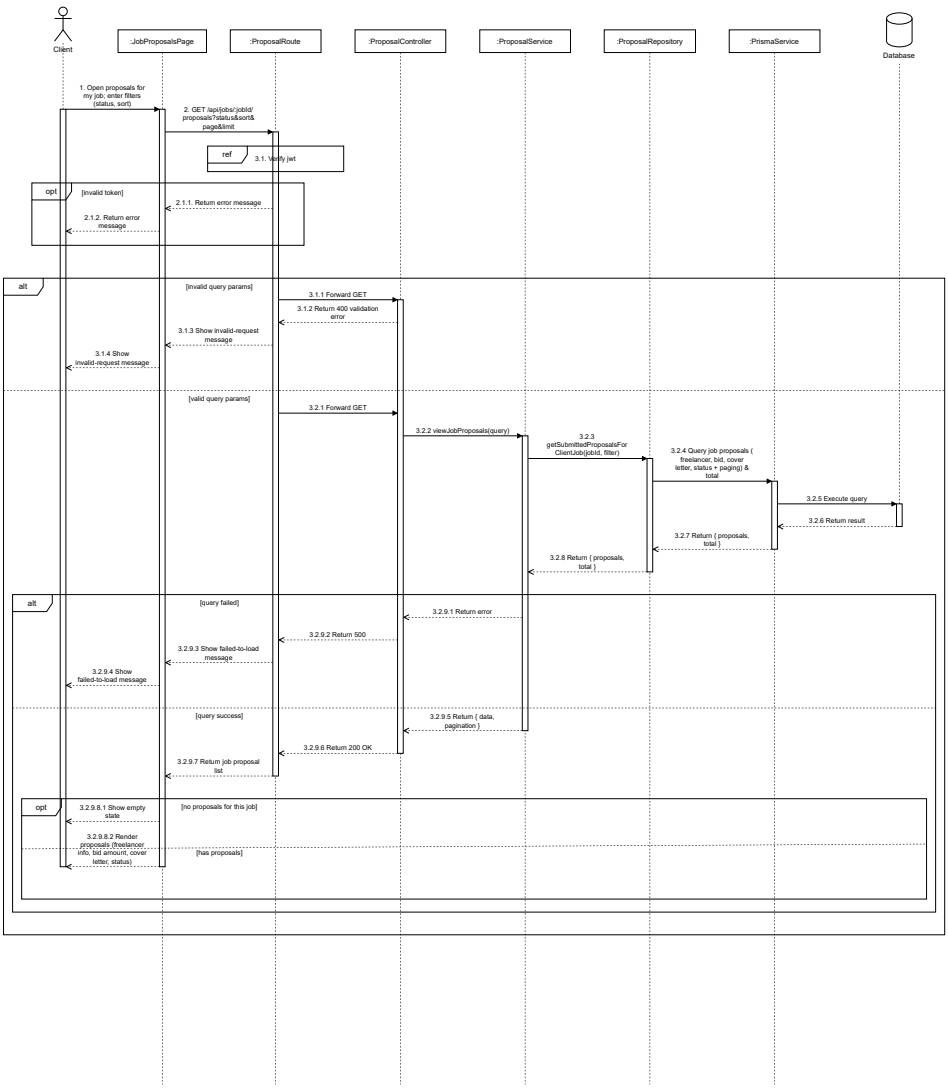

Figure 195: Sequence diagram of UC-32.08 View Job Proposal List

3.17.9 UC-32.09 – View Proposal Detail

a. Class diagram

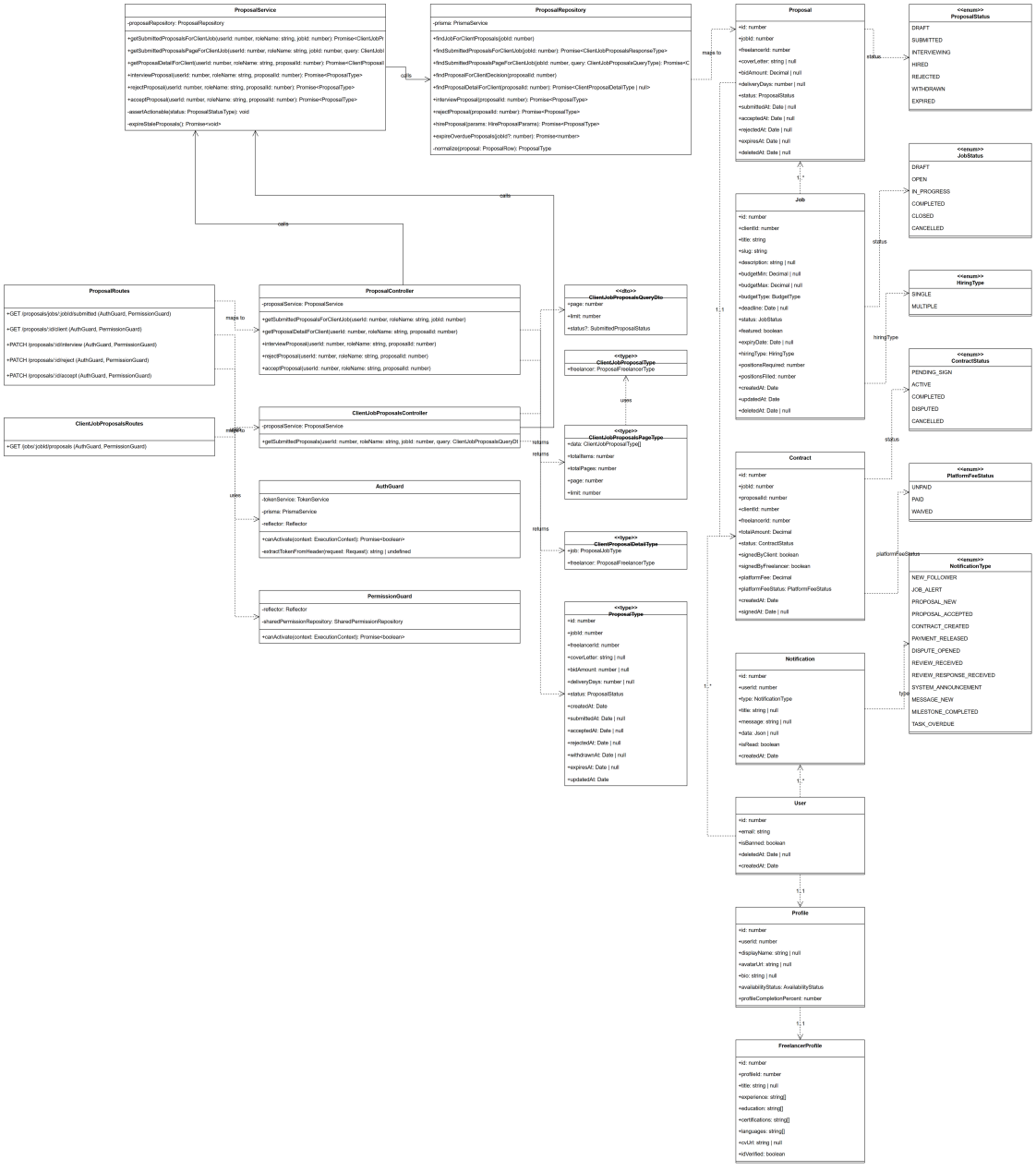

Figure 196: Class diagram of UC-32.09 View Proposal Detail

b. Sequence diagram

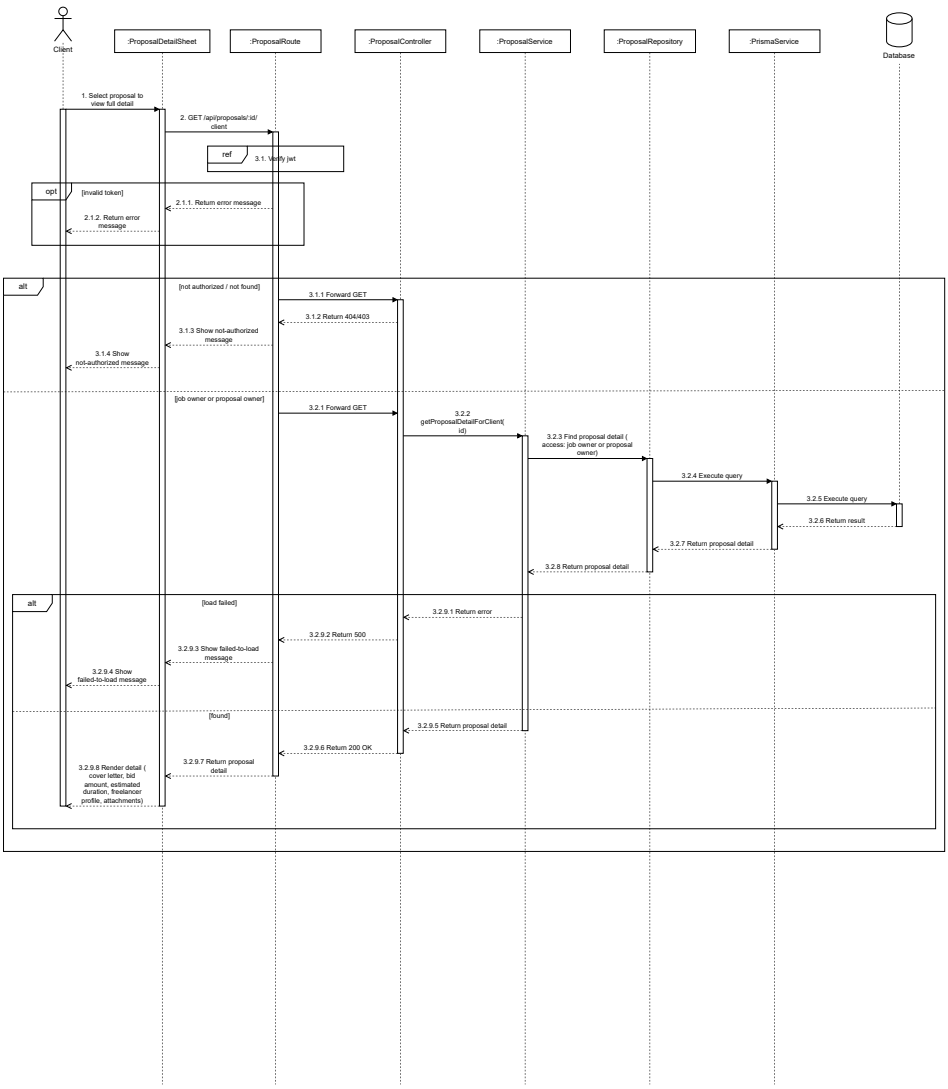

Figure 197: Sequence diagram of UC-32.09 View Proposal Detail

3.18 User Management

User Management provides the administrator with a searchable user catalogue, a consolidated account detail view, and dedicated editors for the Client and Freelancer profiles attached to an account.

3.18.1 UC-28.01 – View User List

a. Class diagram

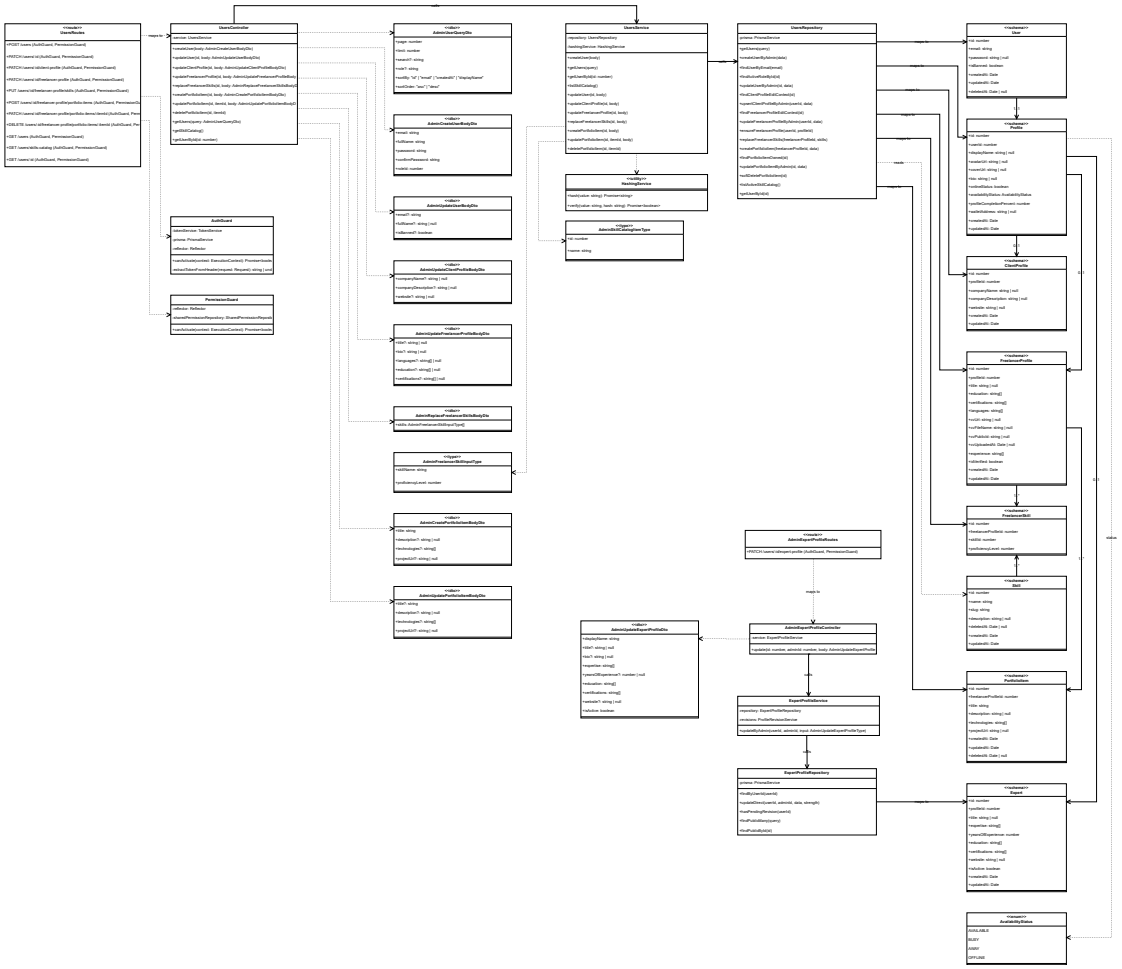

Figure 198: Class diagram of UC-28.01 View User List

b. Sequence diagram

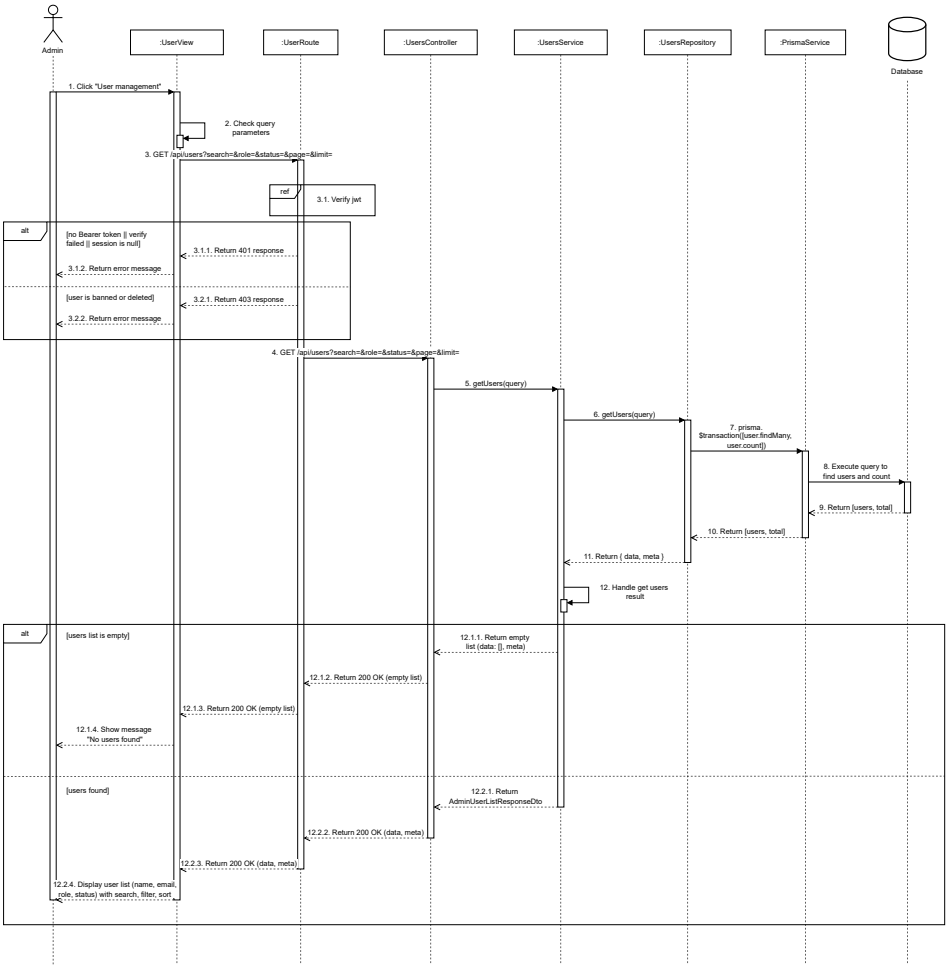

Figure 199: Sequence diagram of UC-28.01 View User List

3.18.2 UC-28.02 – View User Detail

a. Class diagram

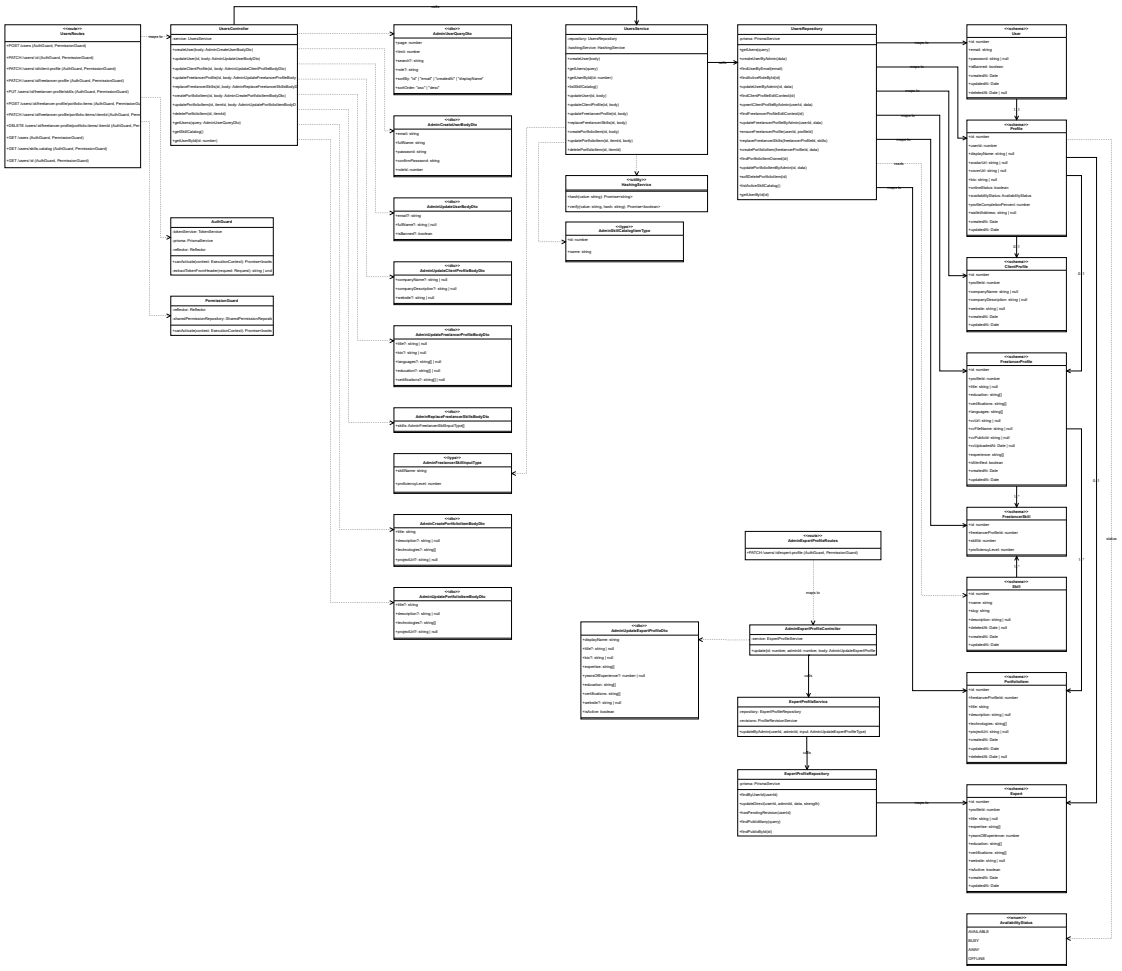

Figure 200: Class diagram of UC-28.02 View User Detail

b. Sequence diagram

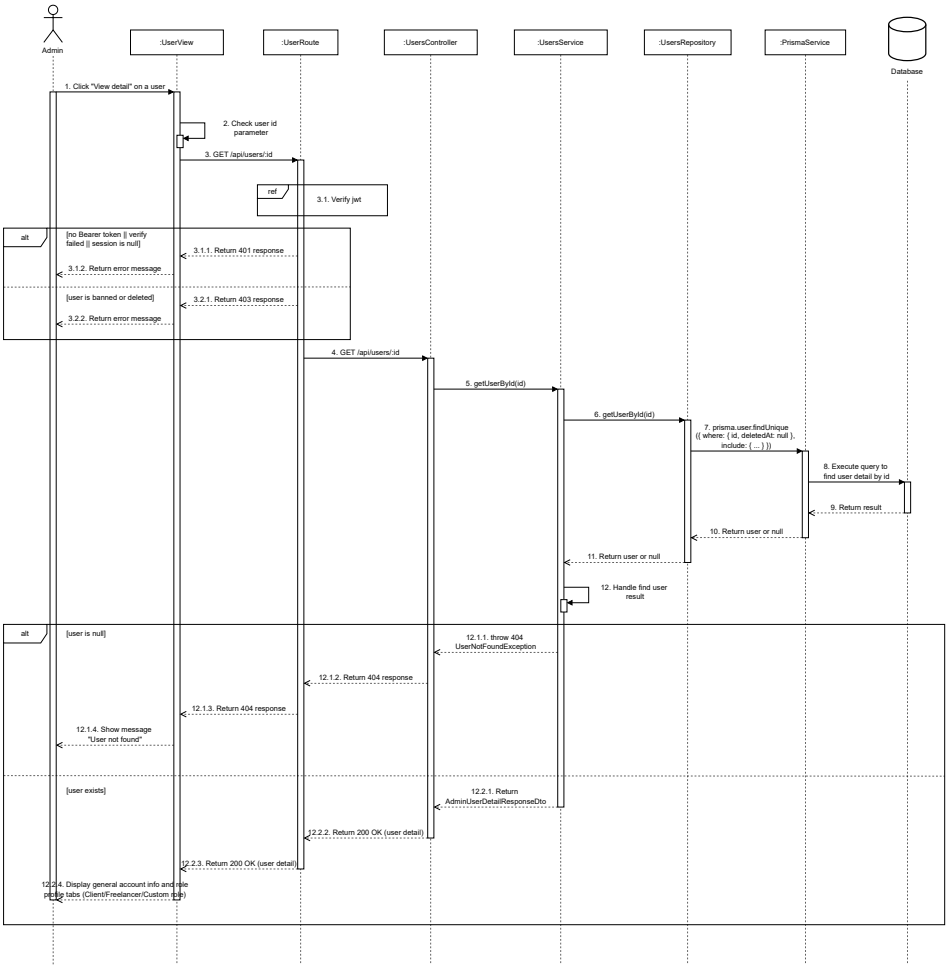

Figure 201: Sequence diagram of UC-28.02 View User Detail

3.18.3 UC-28.03 – Create User

a. Class diagram

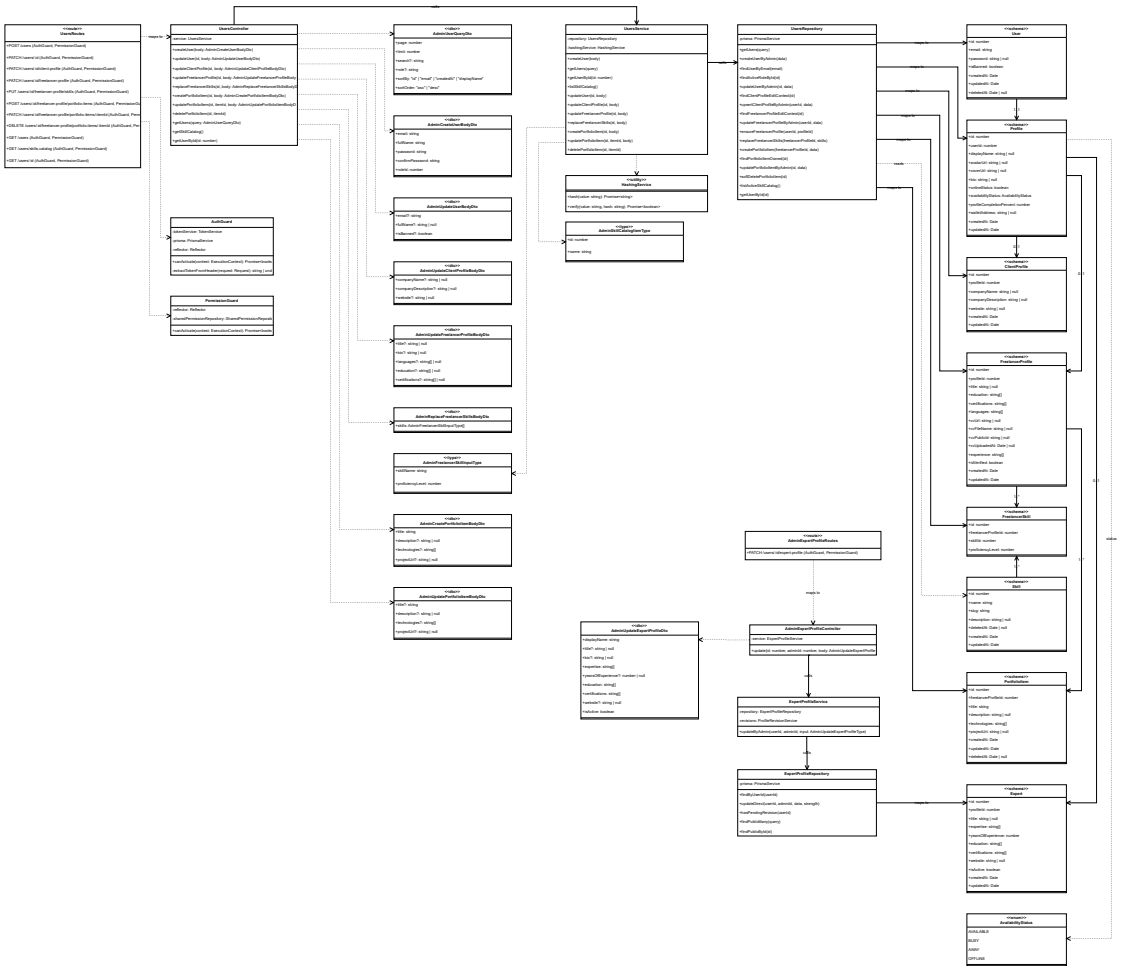

Figure 202: Class diagram of UC-28.03 Create User

b. Sequence diagram

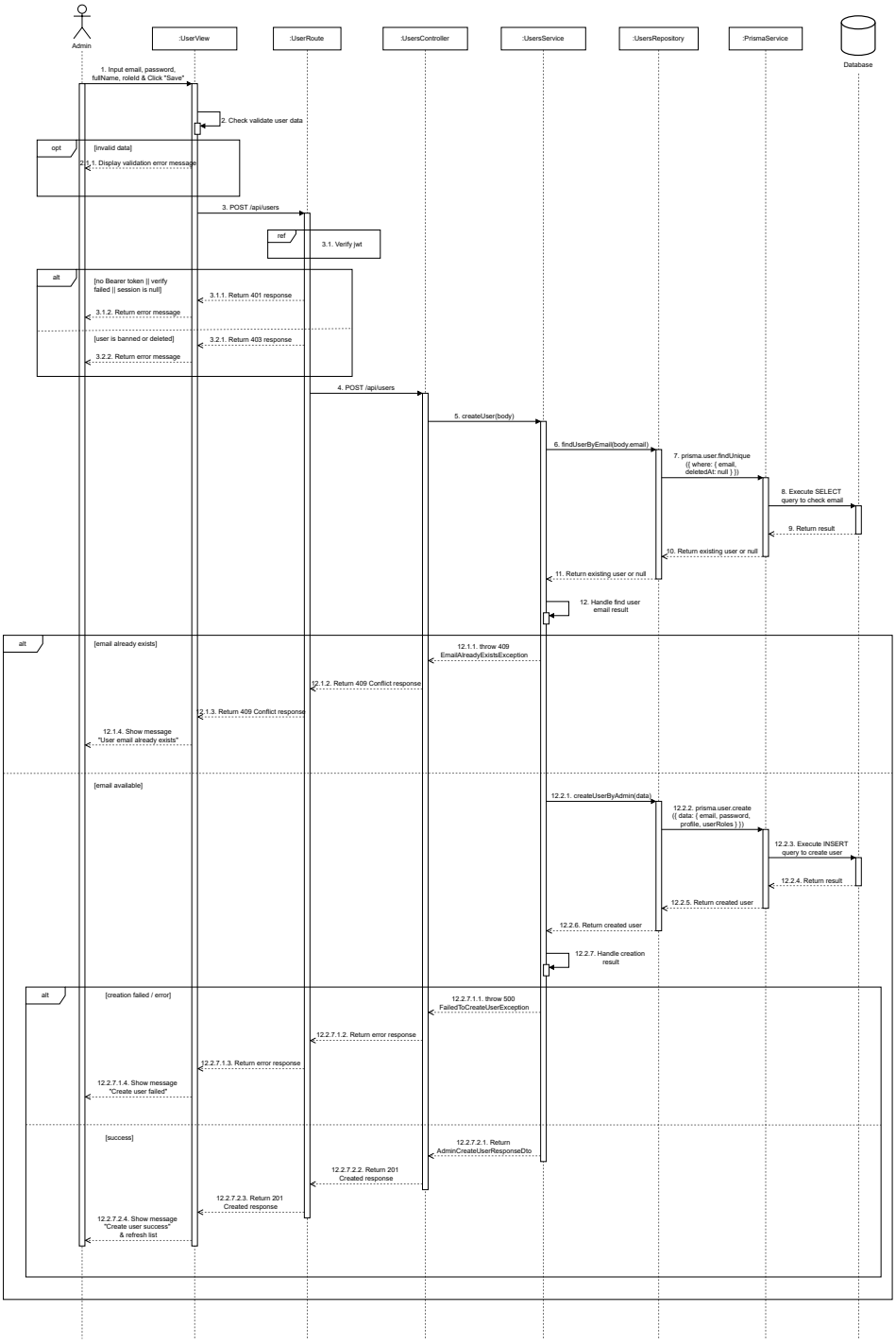

Figure 203: Sequence diagram of UC-28.03 Create User

3.18.4 UC-28.04 – Update User

a. Class diagram

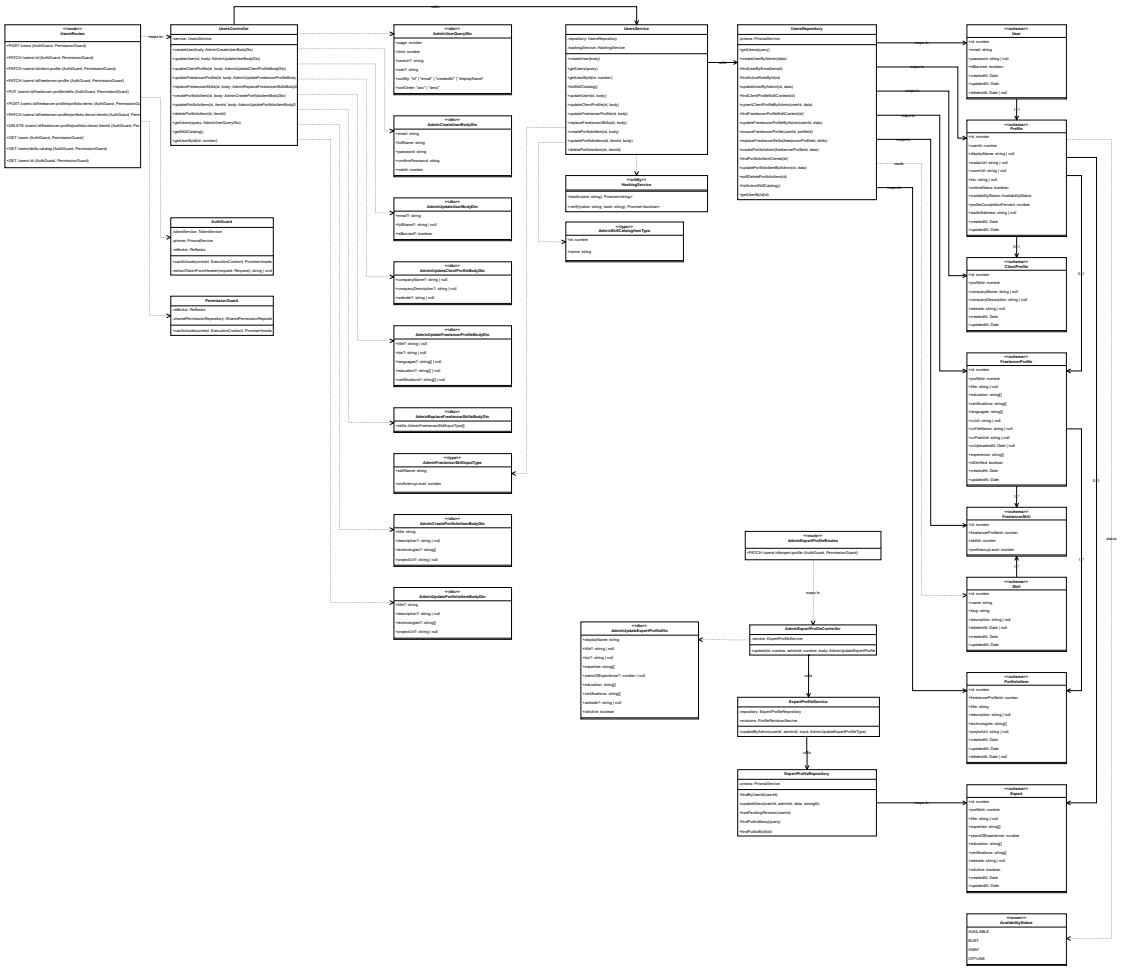

Figure 204: Class diagram of UC-28.04 Update User

b. Sequence diagram

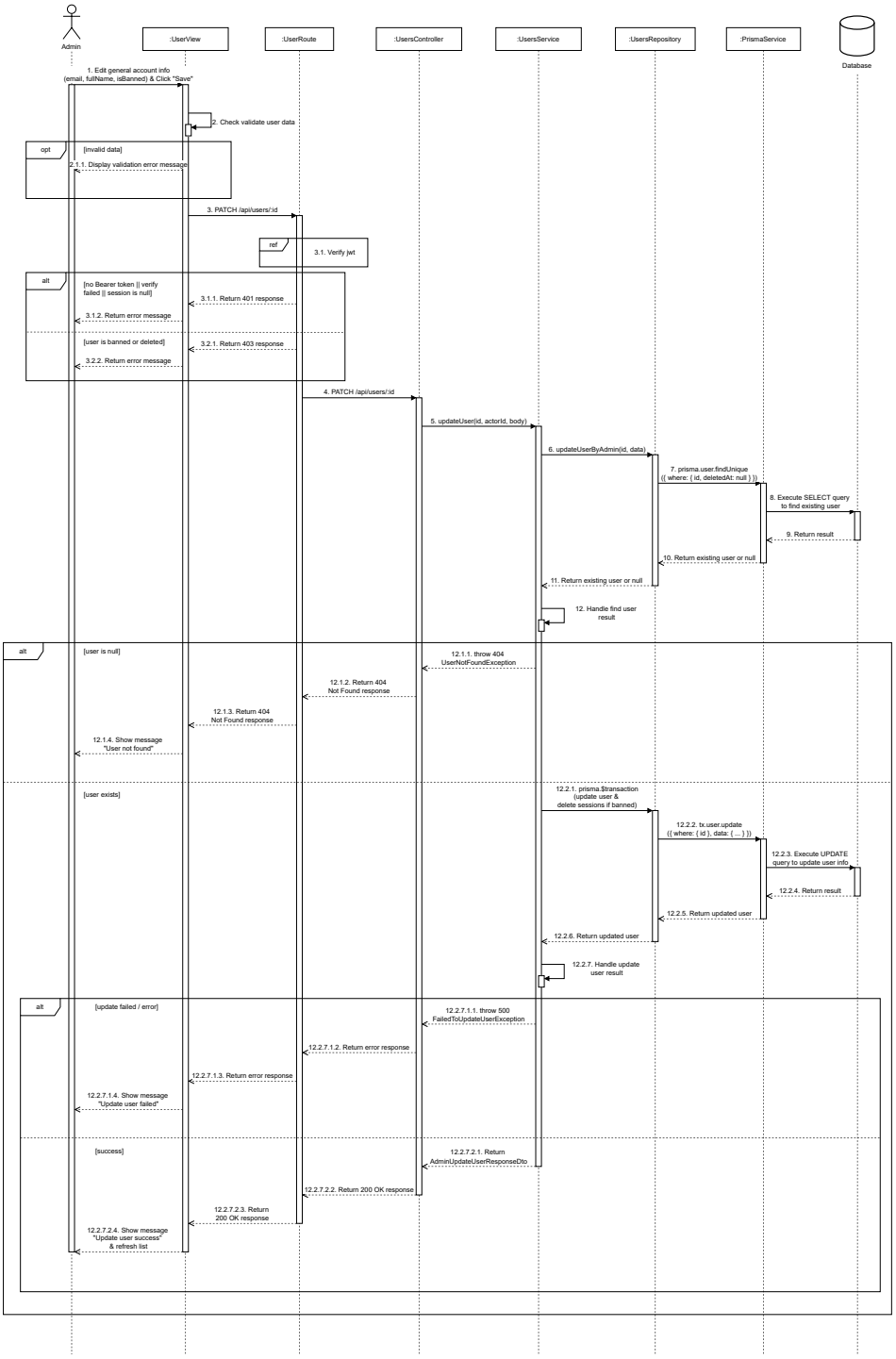

Figure 205: Sequence diagram of UC-28.04 Update User

3.18.5 UC-28.05 – Update Client Profile

a. Class diagram

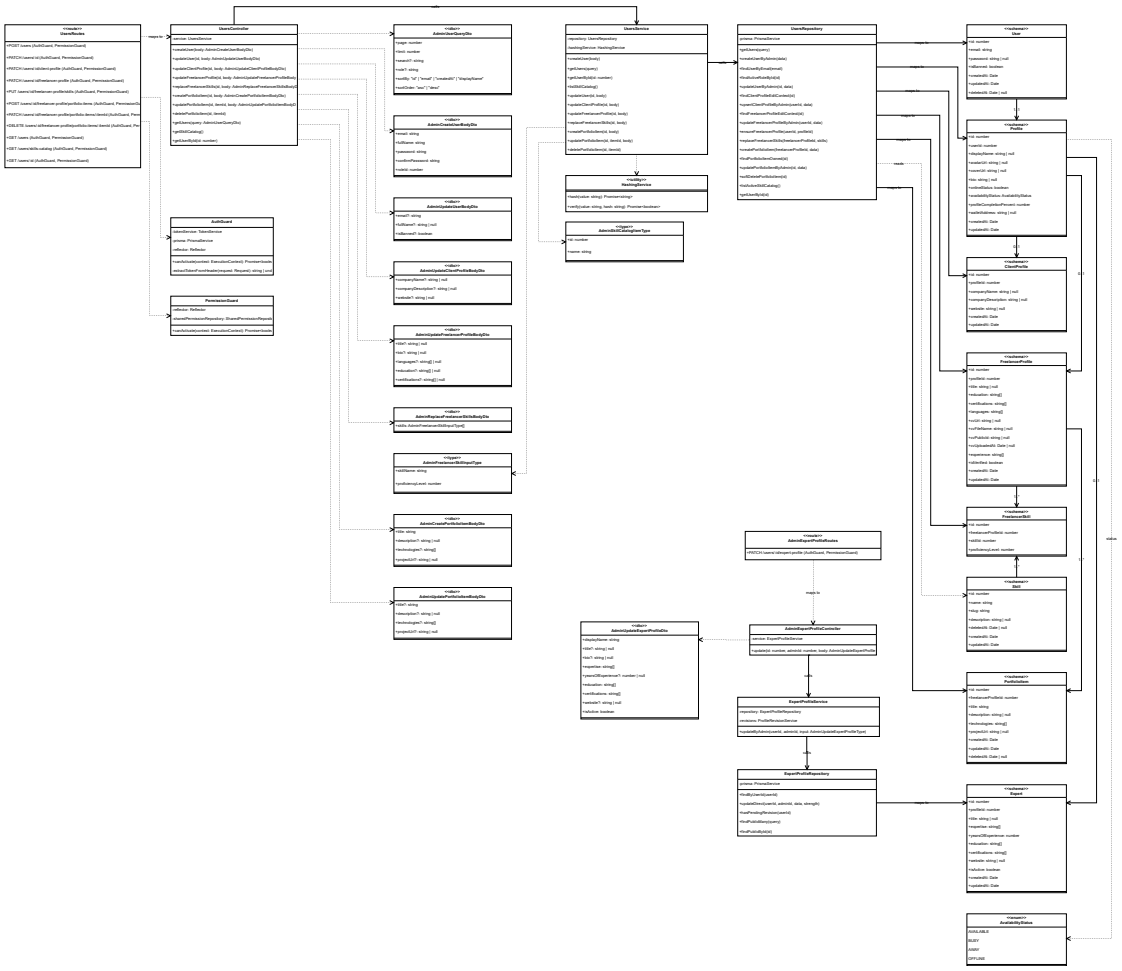

Figure 206: Class diagram of UC-28.05 Update Client Profile

b. Sequence diagram

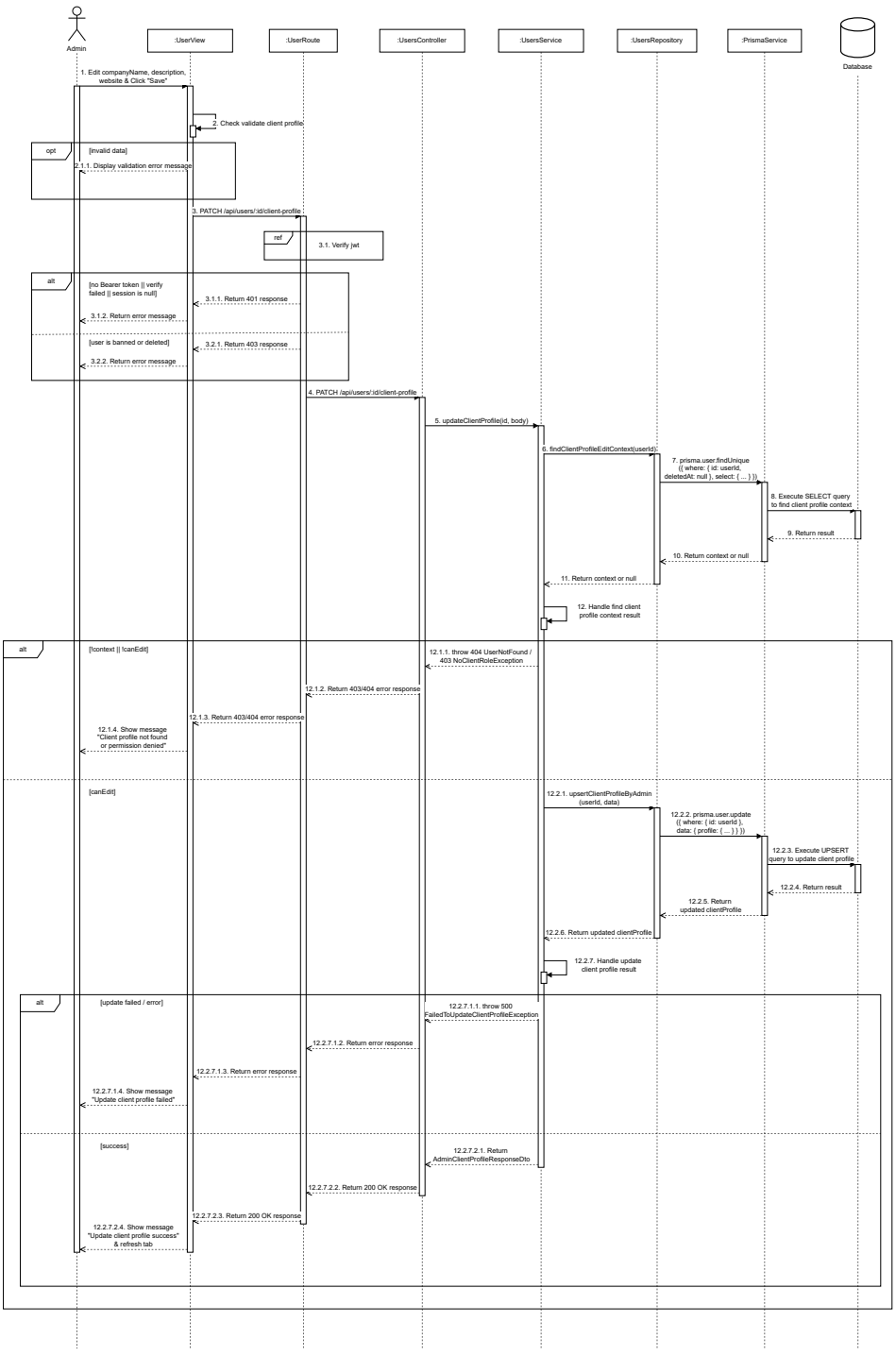

Figure 207: Sequence diagram of UC-28.05 Update Client Profile

3.18.6 UC-28.06 – Update Freelancer Basic Info

a. Class diagram

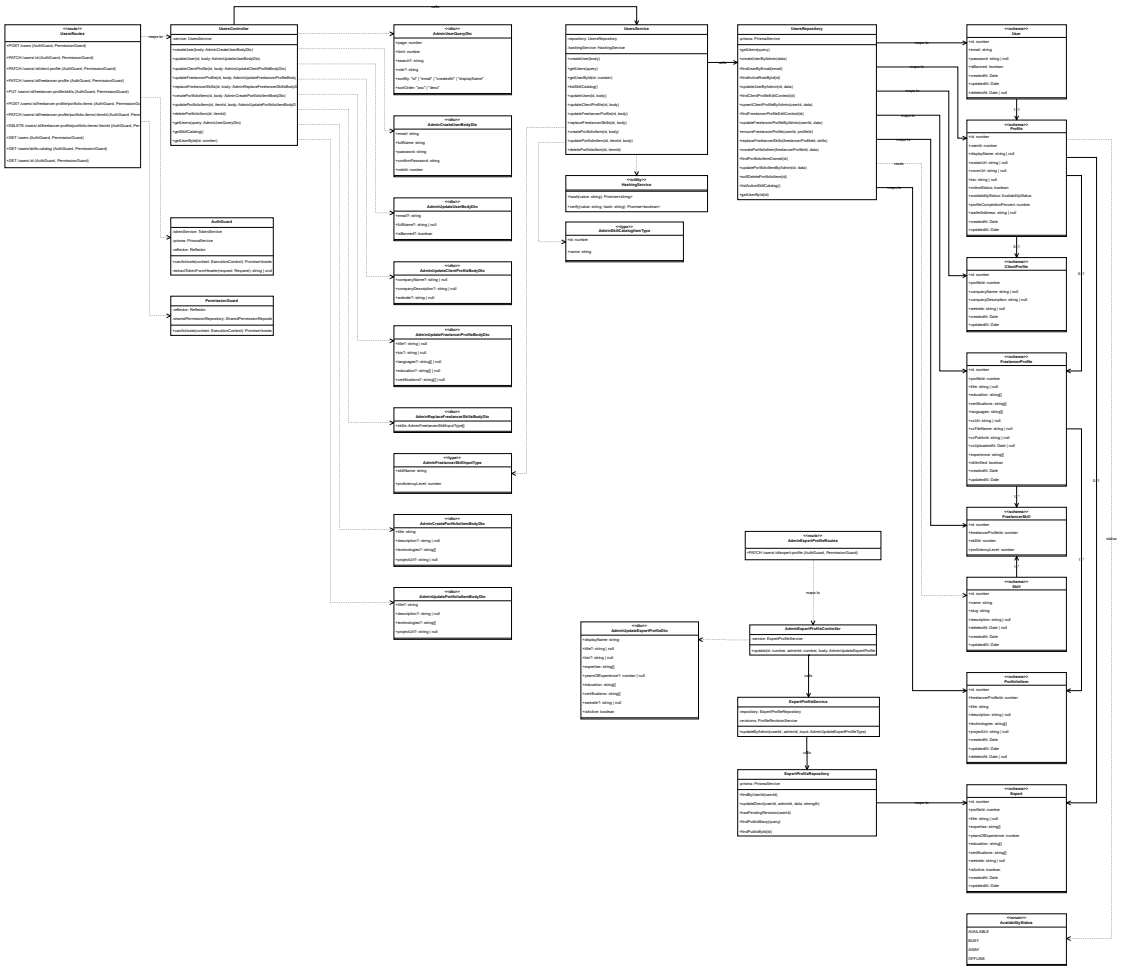

Figure 208: Class diagram of UC-28.06 Update Freelancer Basic Info

b. Sequence diagram

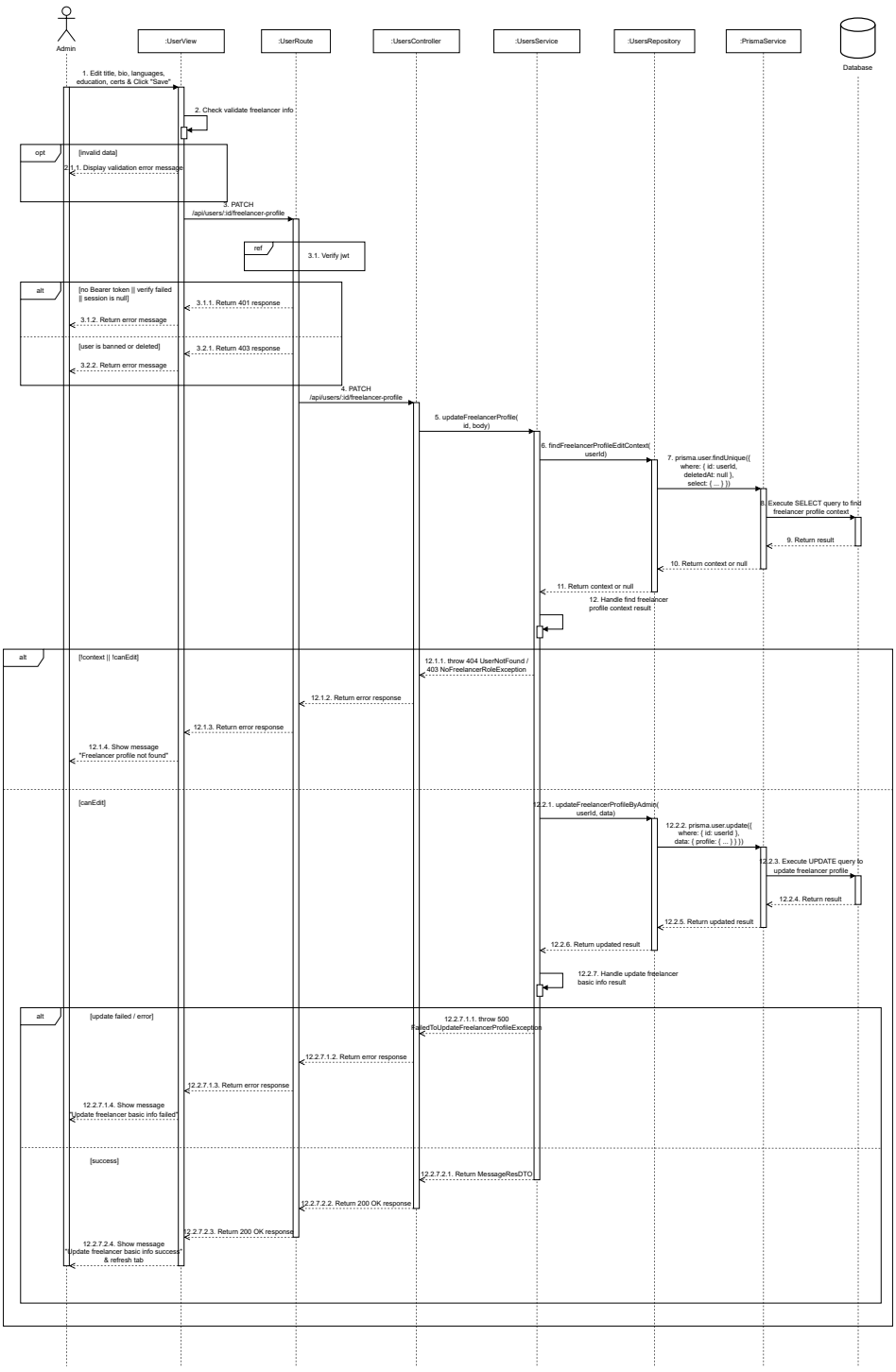

Figure 209: Sequence diagram of UC-28.06 Update Freelancer Basic Info

3.18.7 UC-28.07 – Update Freelancer Skills

a. Class diagram

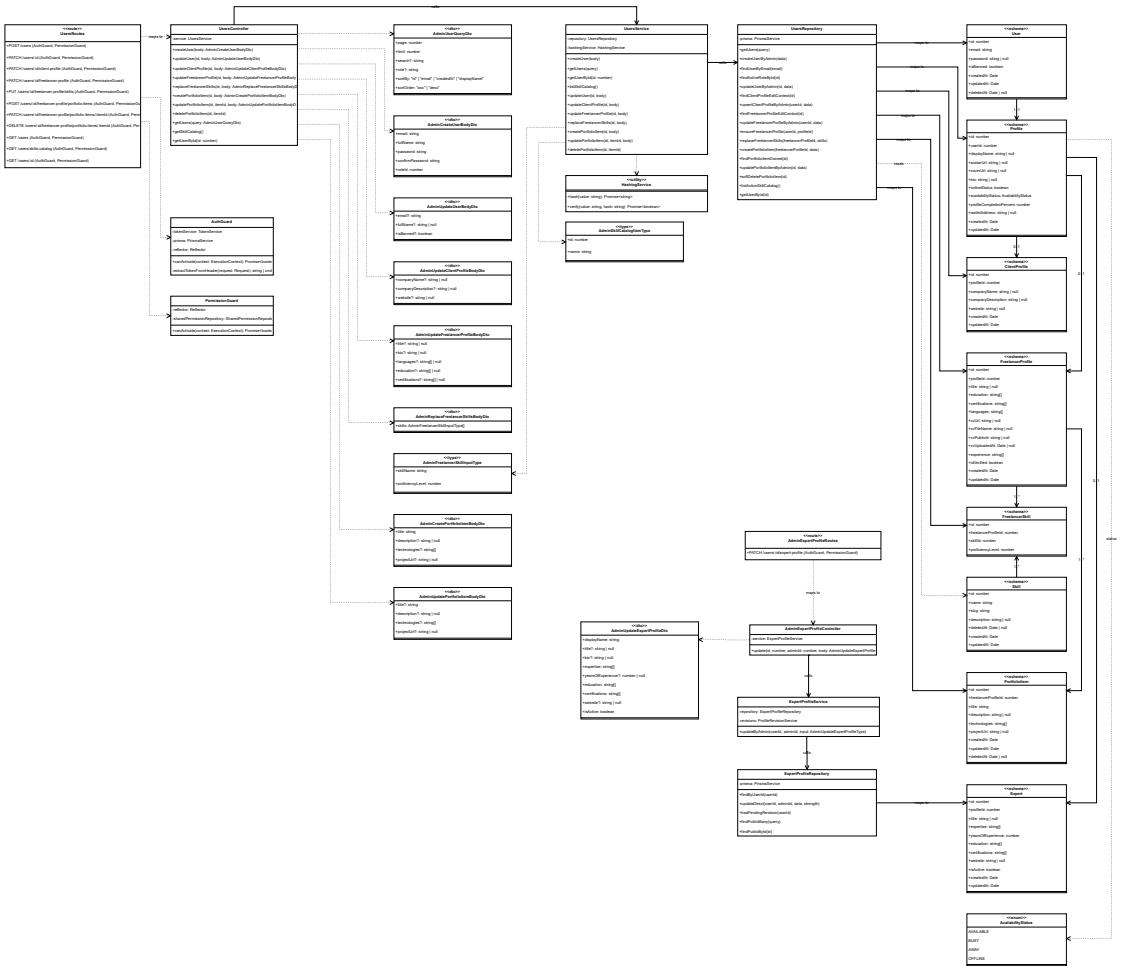

Figure 210: Class diagram of UC-28.07 Update Freelancer Skills

b. Sequence diagram

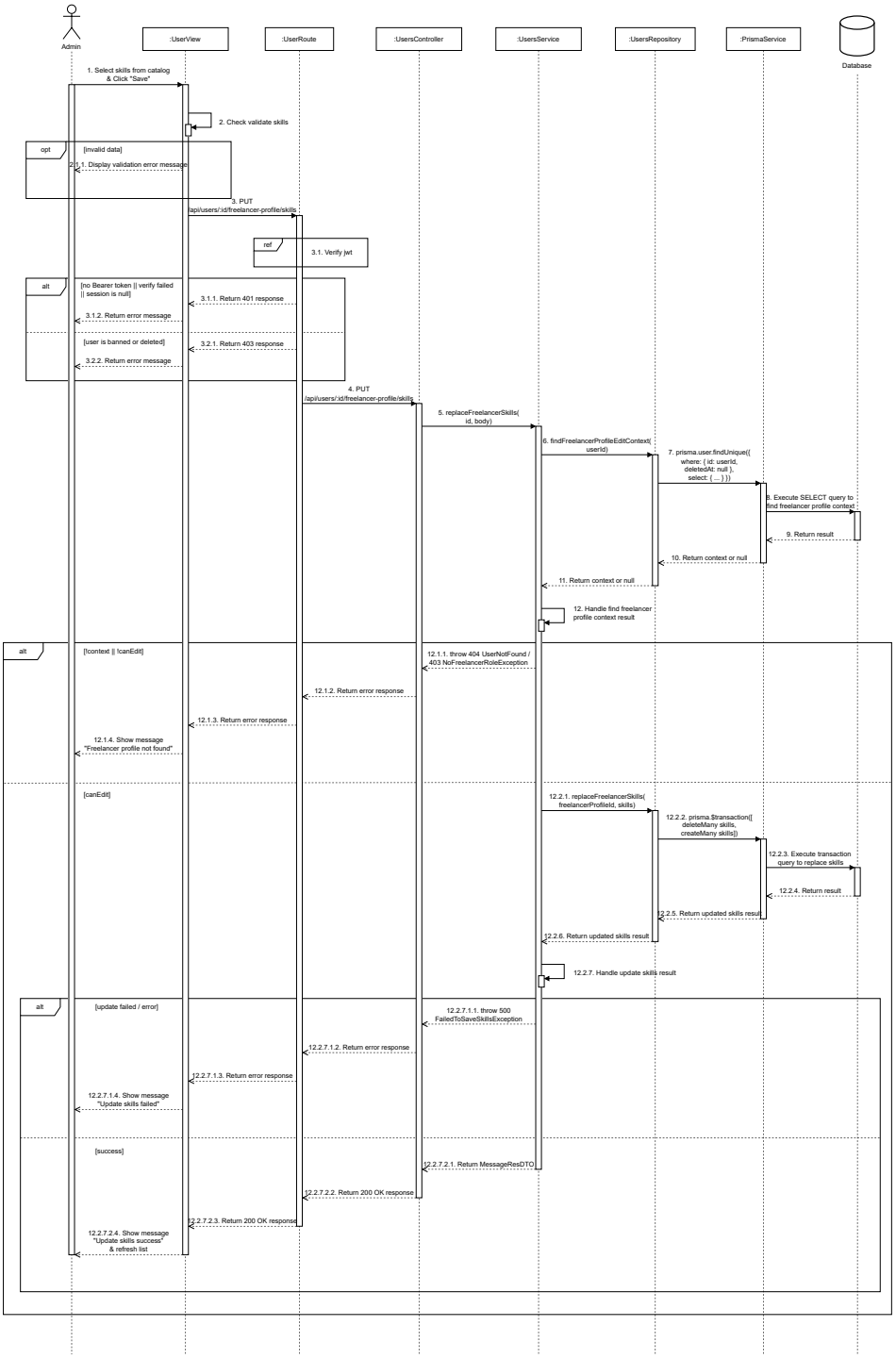

Figure 211: Sequence diagram of UC-28.07 Update Freelancer Skills

3.18.8 UC-28.08 – Update Freelancer Portfolio

a. Class diagram

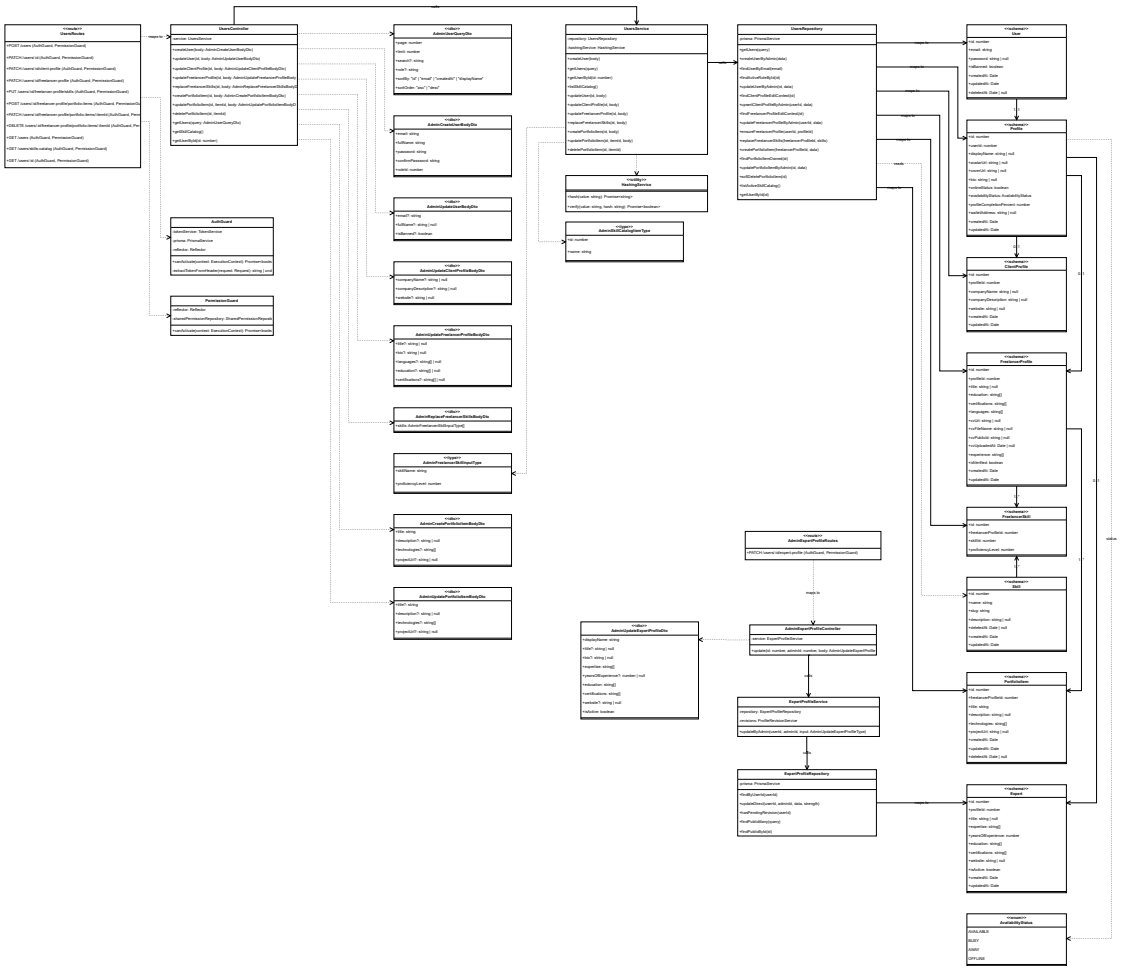

Figure 212: Class diagram of UC-28.08 Update Freelancer Portfolio

b. Sequence diagram

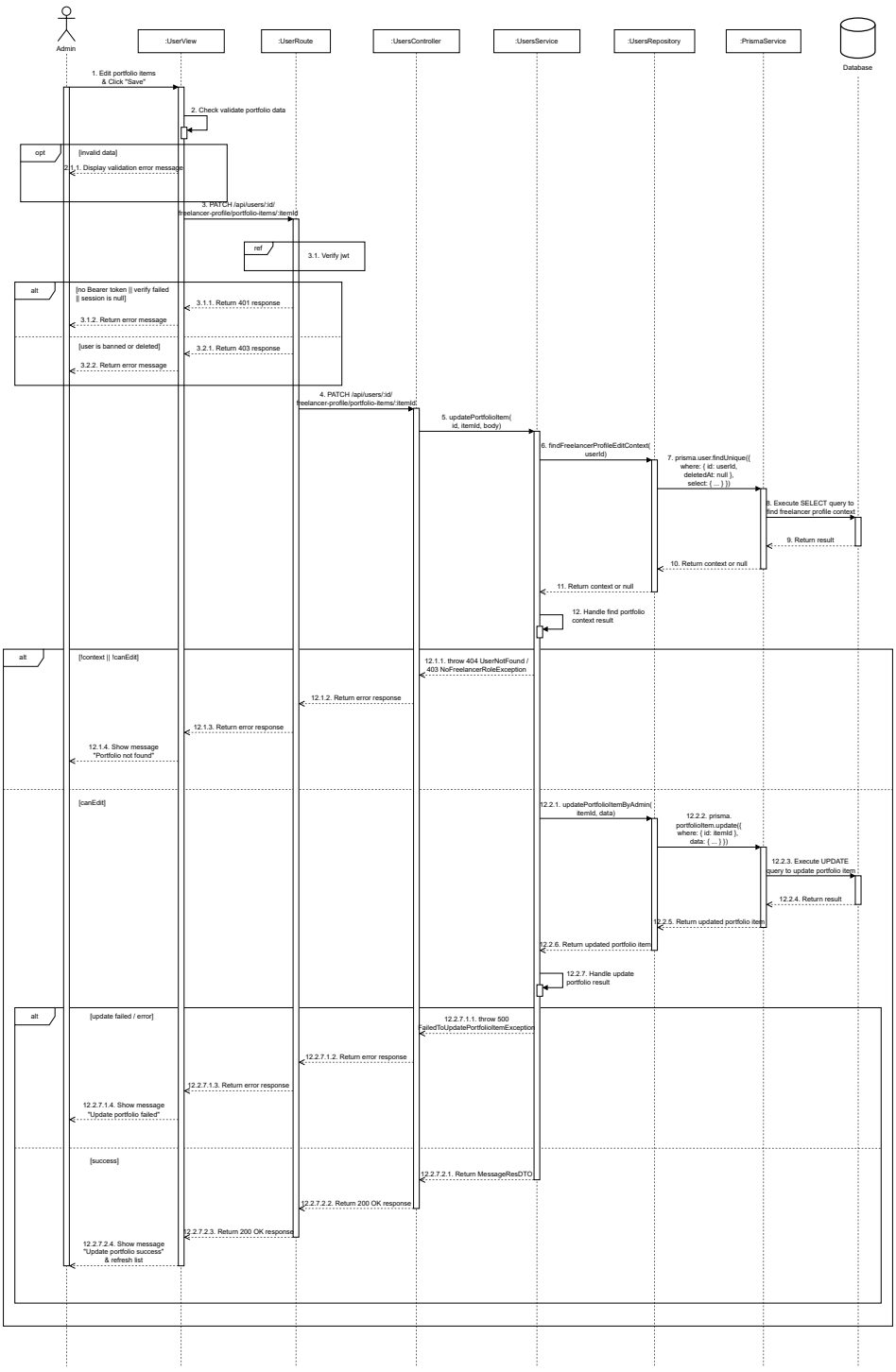

Figure 213: Sequence diagram of UC-28.08 Update Freelancer Portfolio

3.19 Banner Management

Banner Management maintains the advertisement banners shown in configured page positions. The module supports image upload, lifecycle dates, activation state, and hard deletion.

3.19.1 UC-37.01 – View Banner List

a. Class diagram

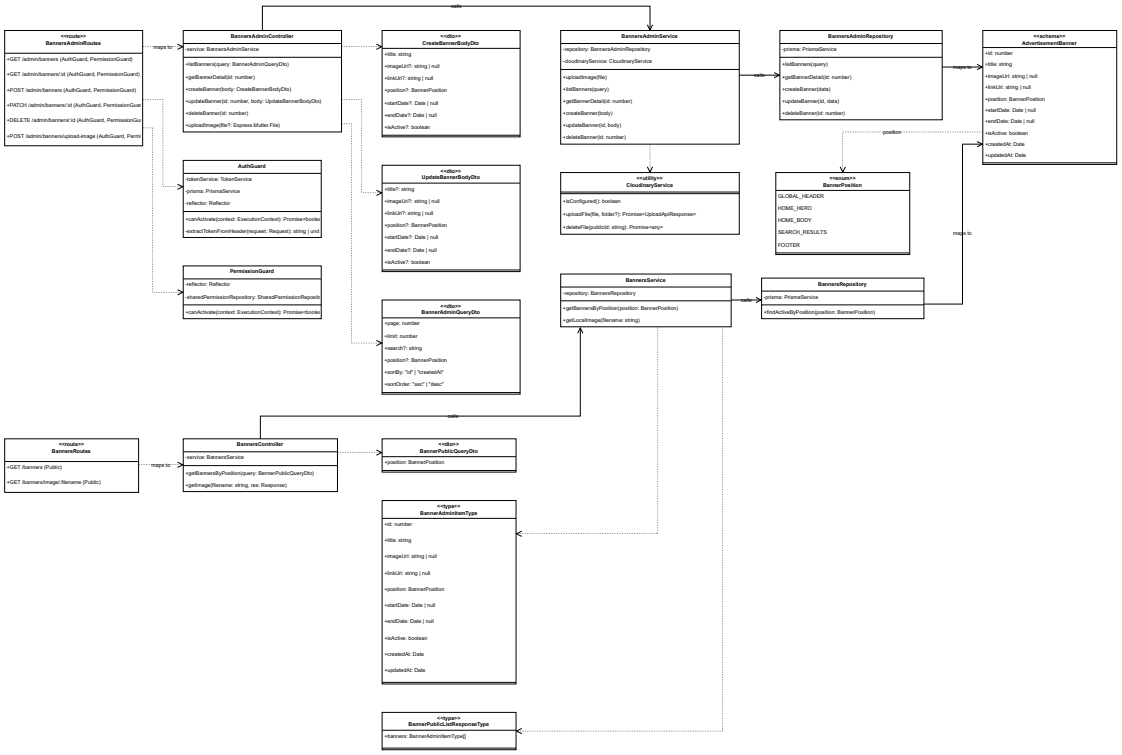

Figure 214: Class diagram of UC-37.01 View Banner List

b. Sequence diagram

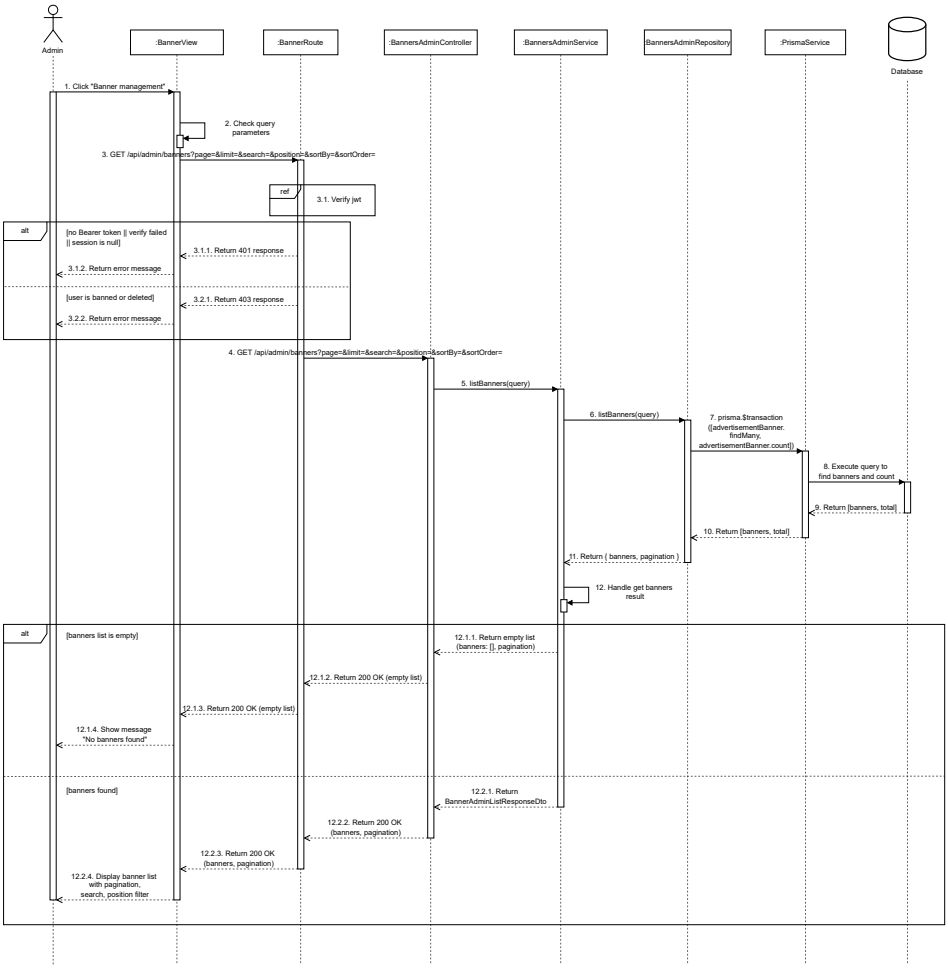

Figure 215: Sequence diagram of UC-37.01 View Banner List

3.19.2 UC-37.02 – View Banner Detail

a. Class diagram

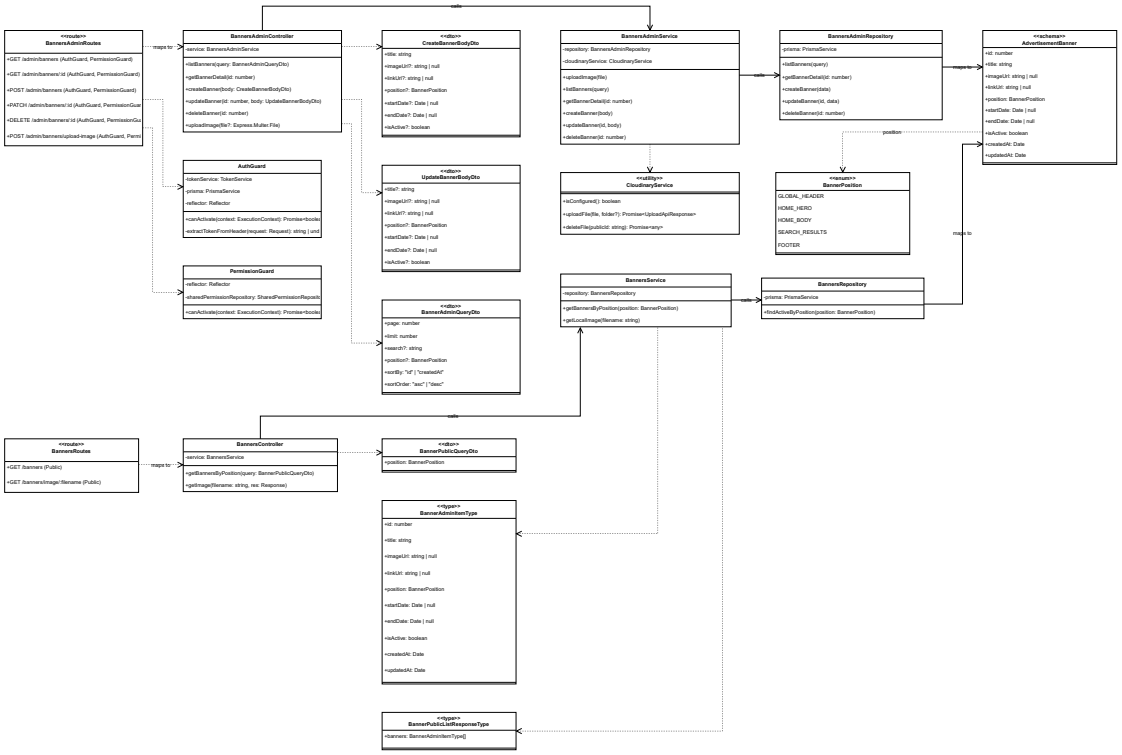

Figure 216: Class diagram of UC-37.02 View Banner Detail

b. Sequence diagram

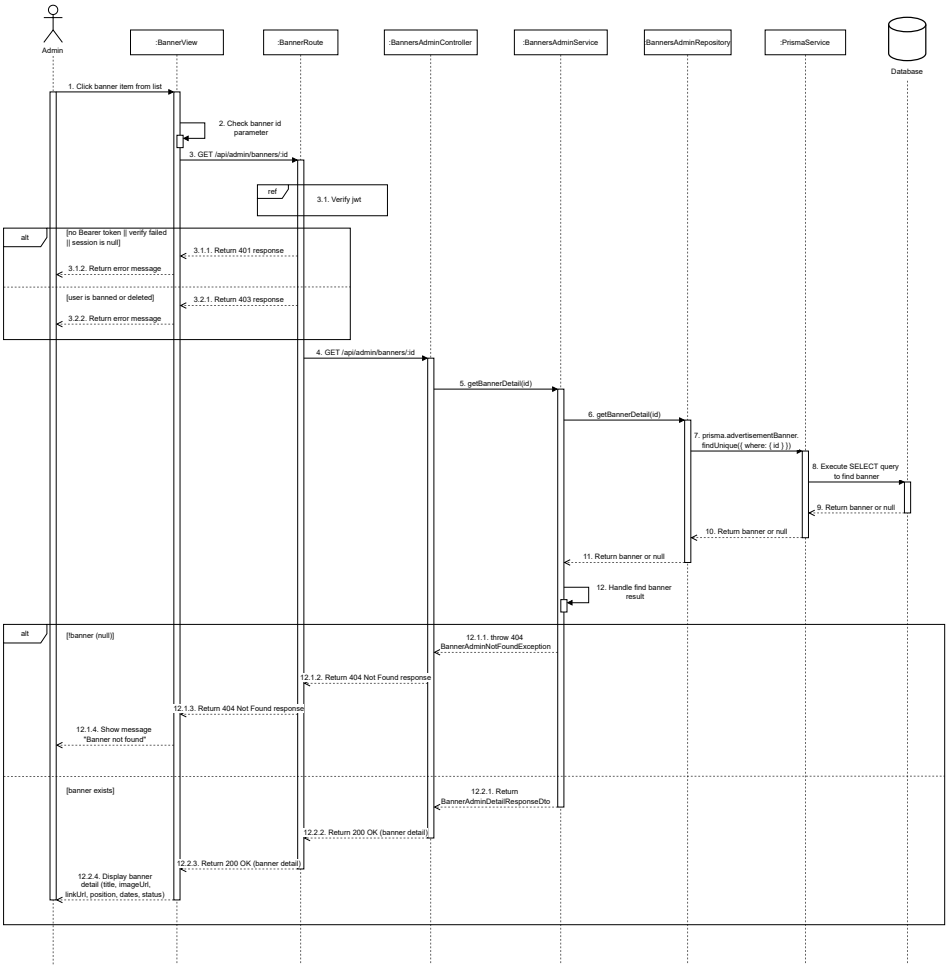

Figure 217: Sequence diagram of UC-37.02 View Banner Detail

3.19.3 UC-37.03 – Create Banner

a. Class diagram

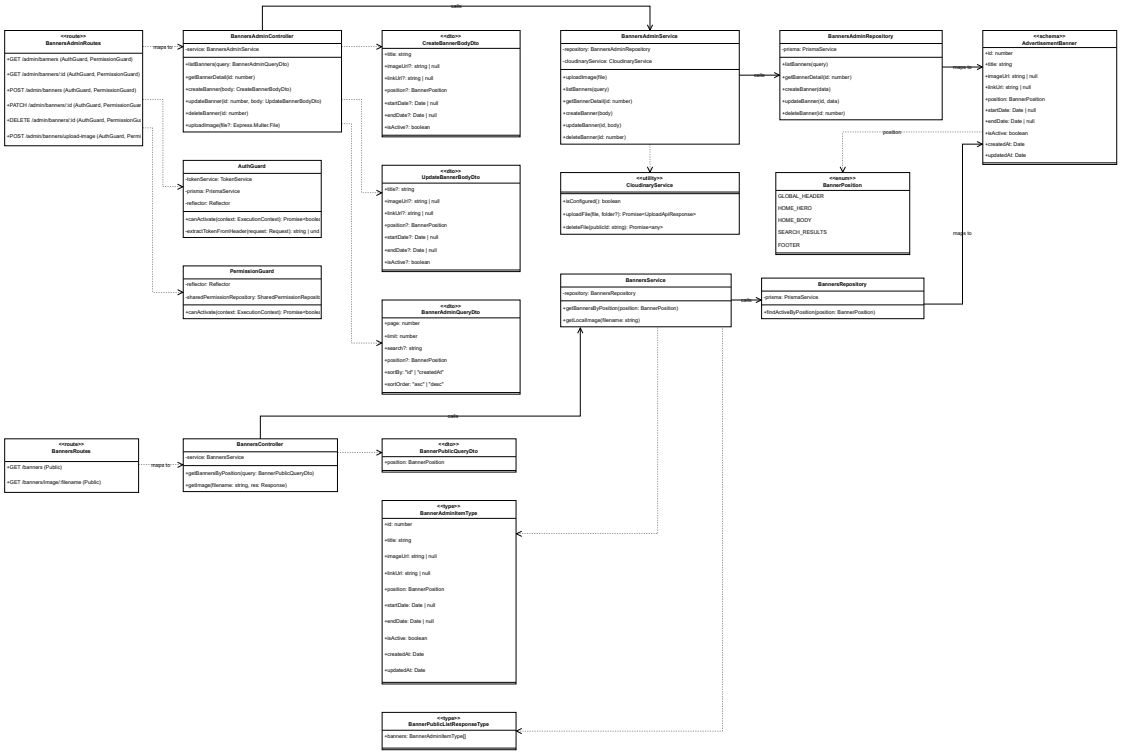

Figure 218: Class diagram of UC-37.03 Create Banner

b. Sequence diagram

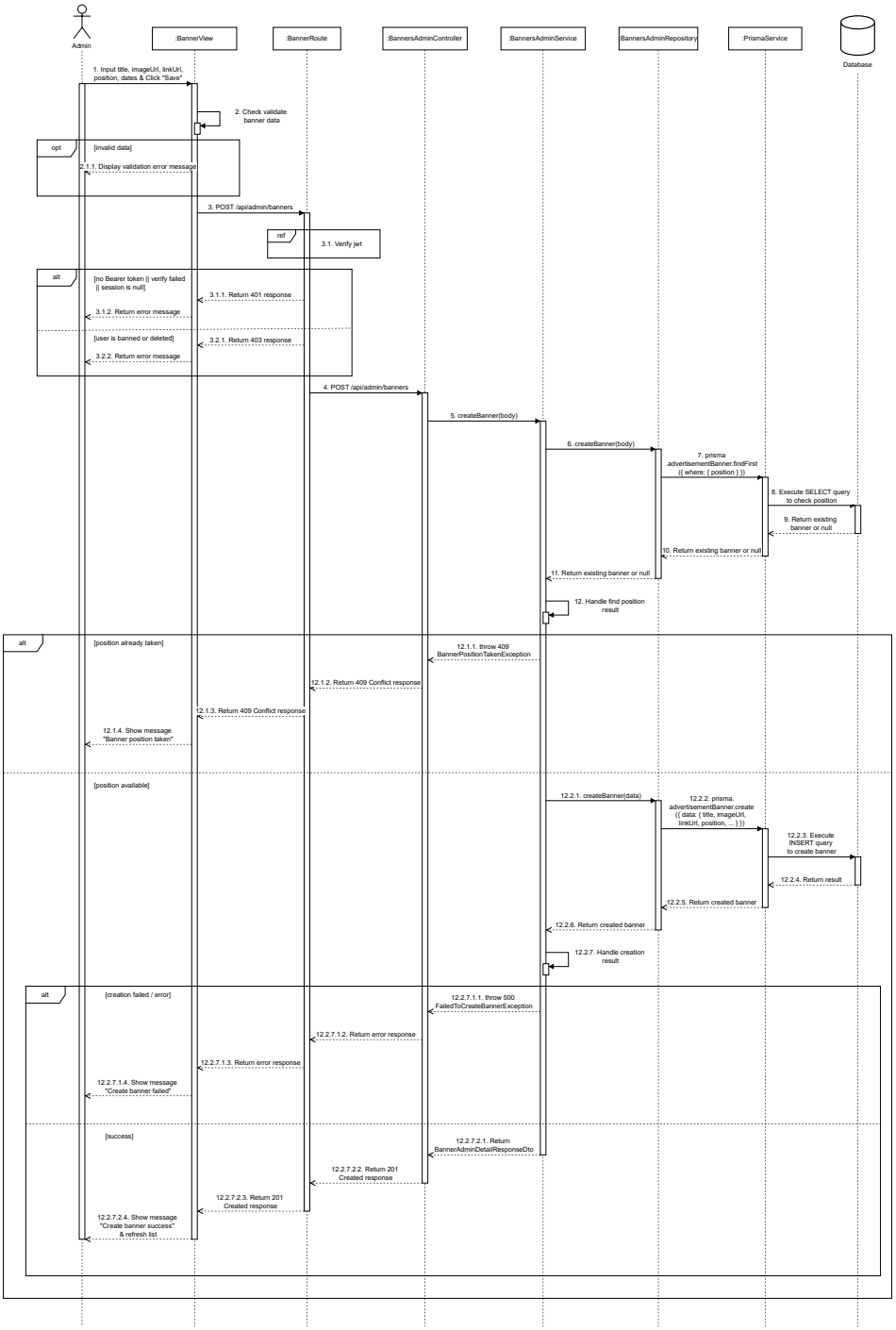

Figure 219: Sequence diagram of UC-37.03 Create Banner

3.19.4 UC-37.04 – Update Banner

a. Class diagram

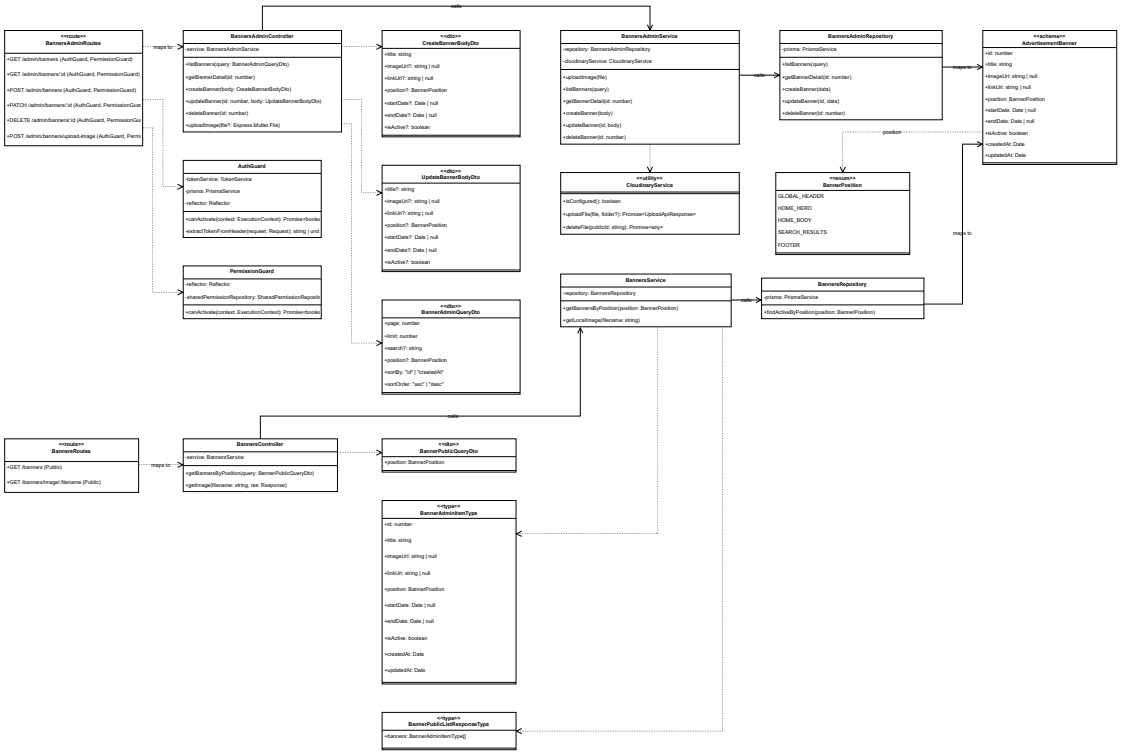

Figure 220: Class diagram of UC-37.04 Update Banner

b. Sequence diagram

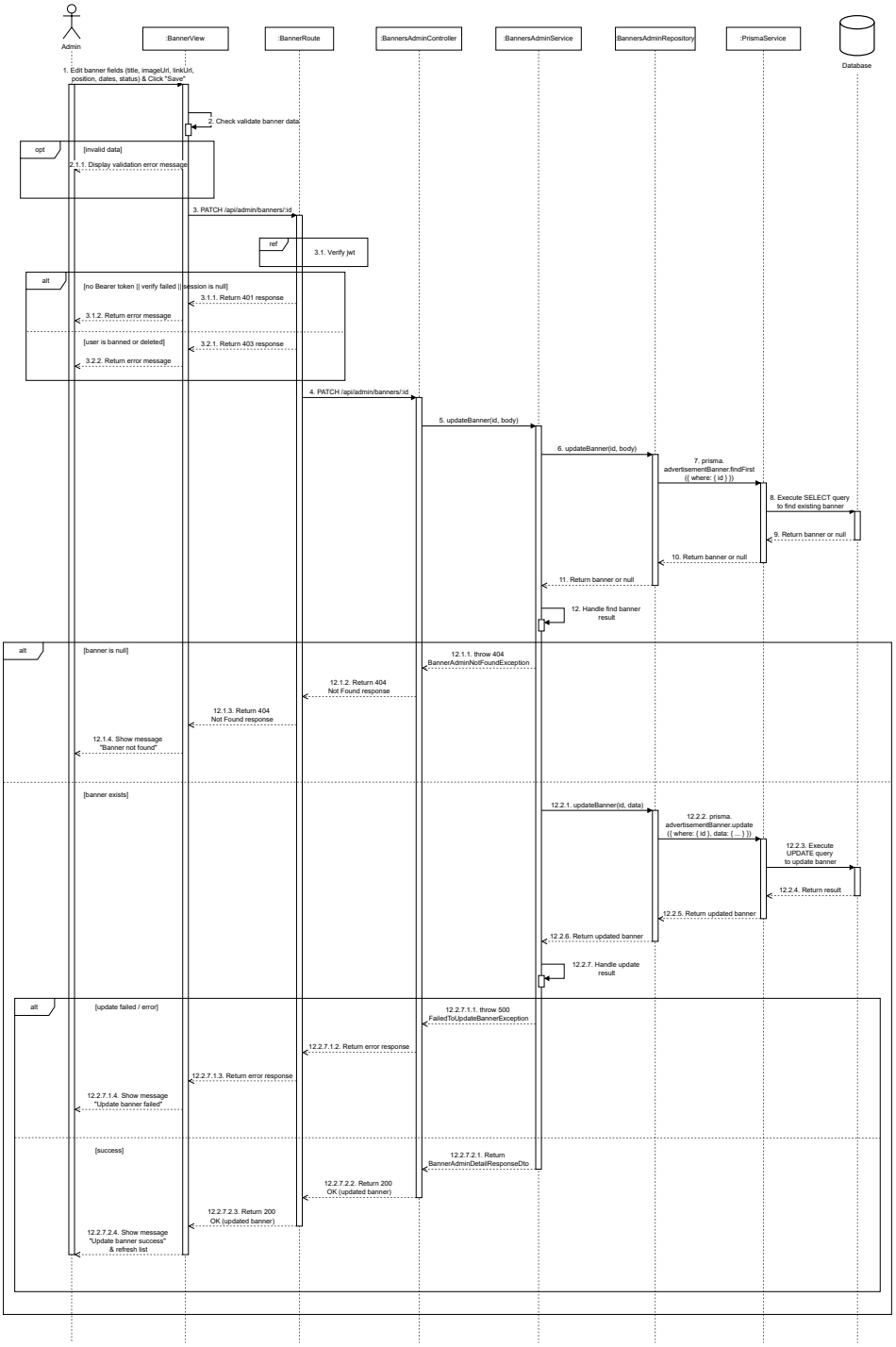

Figure 221: Sequence diagram of UC-37.04 Update Banner

3.19.5 UC-37.05 – Delete Banner

a. Class diagram

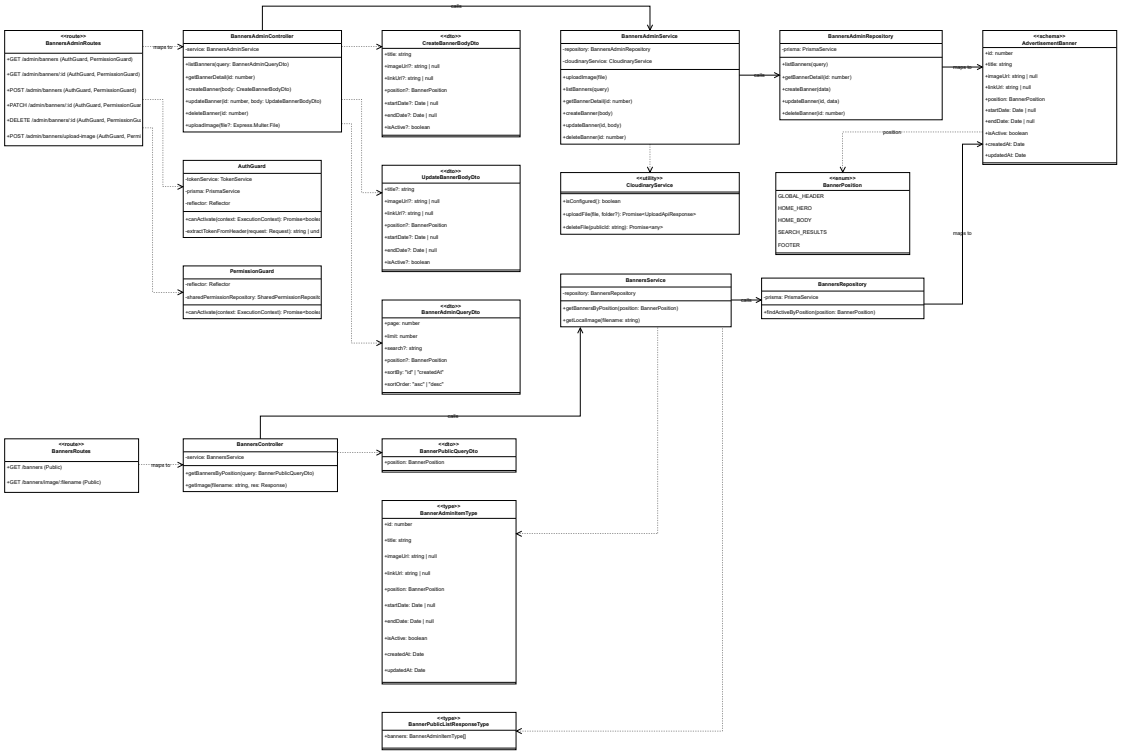

Figure 222: Class diagram of UC-37.05 Delete Banner

b. Sequence diagram

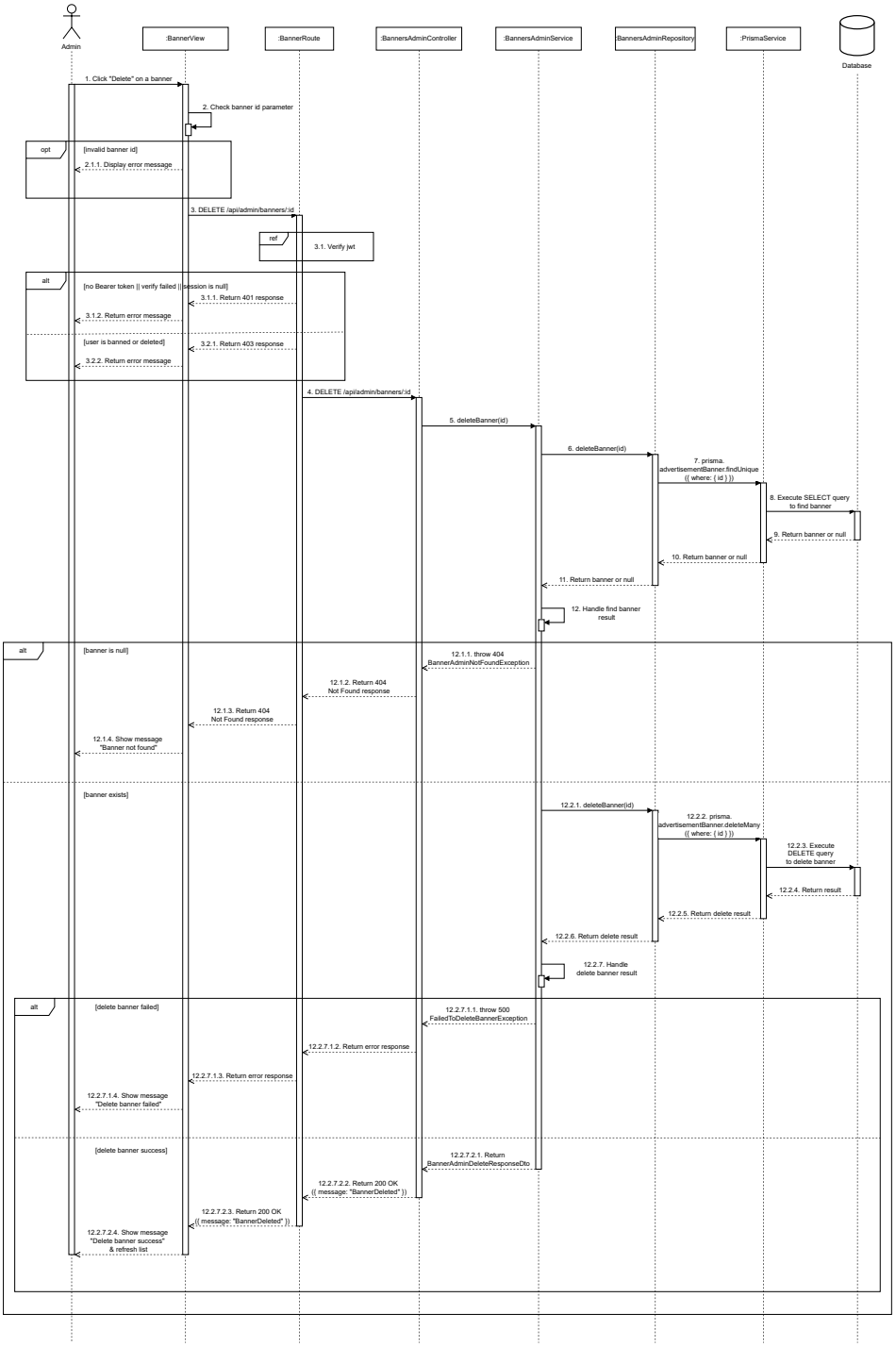

Figure 223: Sequence diagram of UC-37.05 Delete Banner

3.20 Skill Management

Skill Management owns the master skill catalogue used by freelancer profiles and job postings. Skills are soft-deleted so that existing references remain valid, and a deleted skill can be restored.

3.20.1 UC-38.01 – View Skill List

a. Class diagram

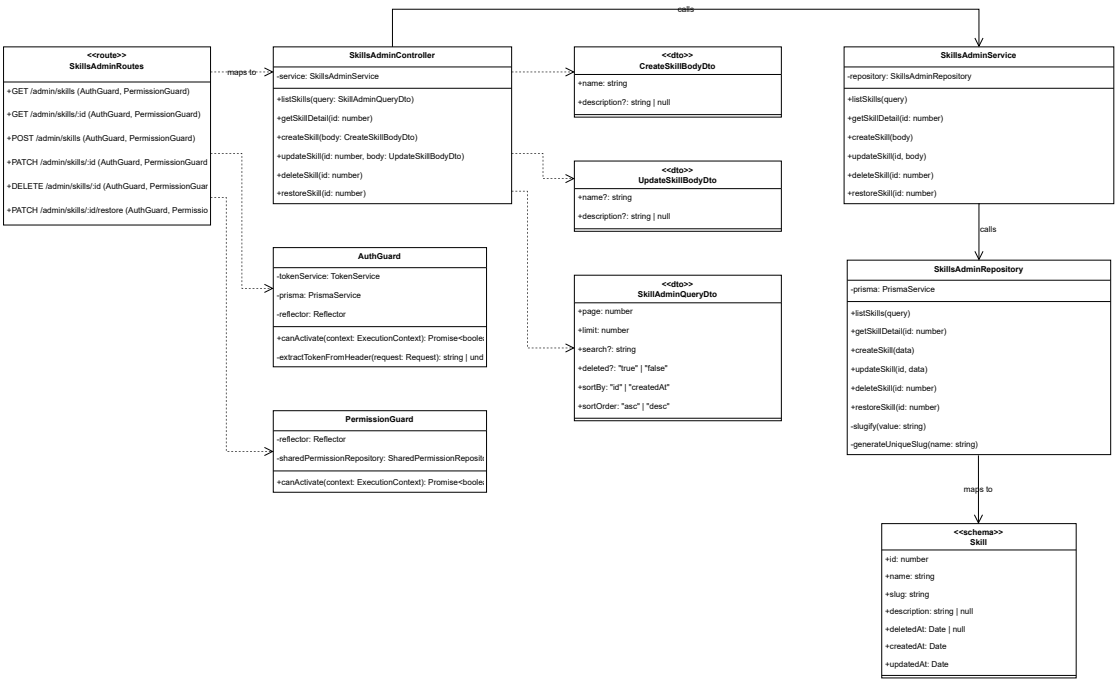

Figure 224: Class diagram of UC-38.01 View Skill List

b. Sequence diagram

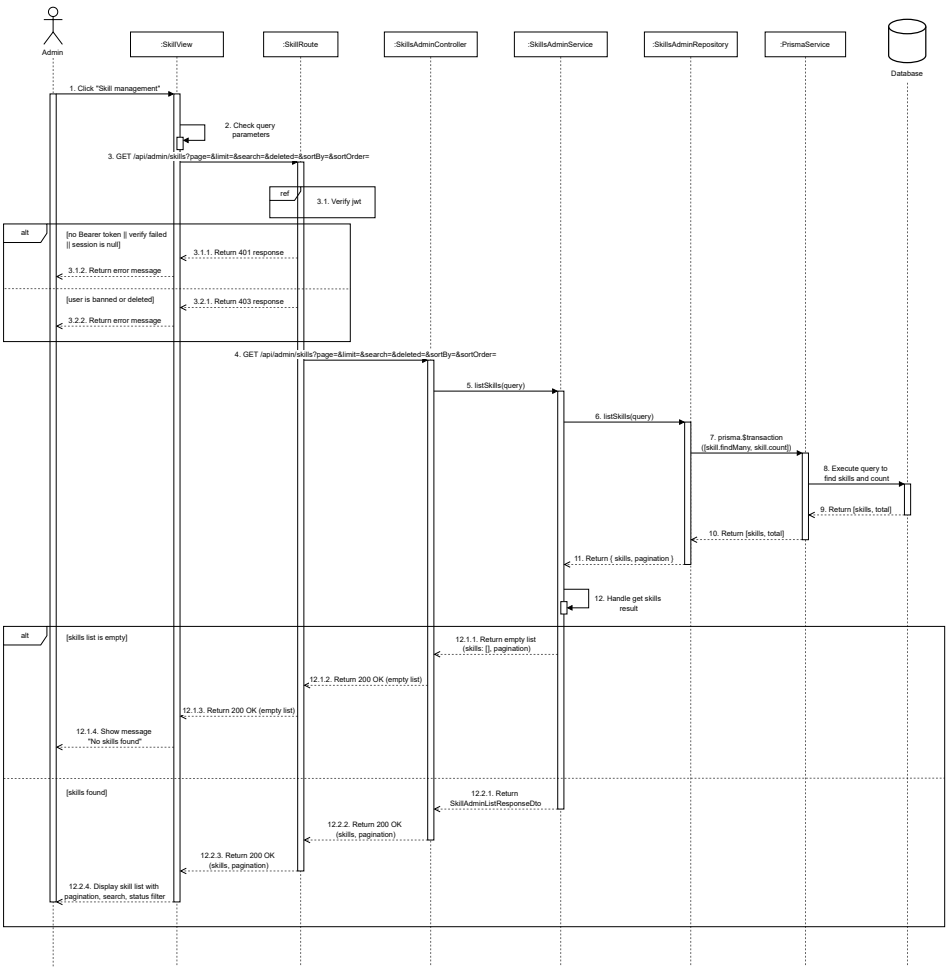

Figure 225: Sequence diagram of UC-38.01 View Skill List

3.20.2 UC-38.02 – View Skill Detail

a. Class diagram

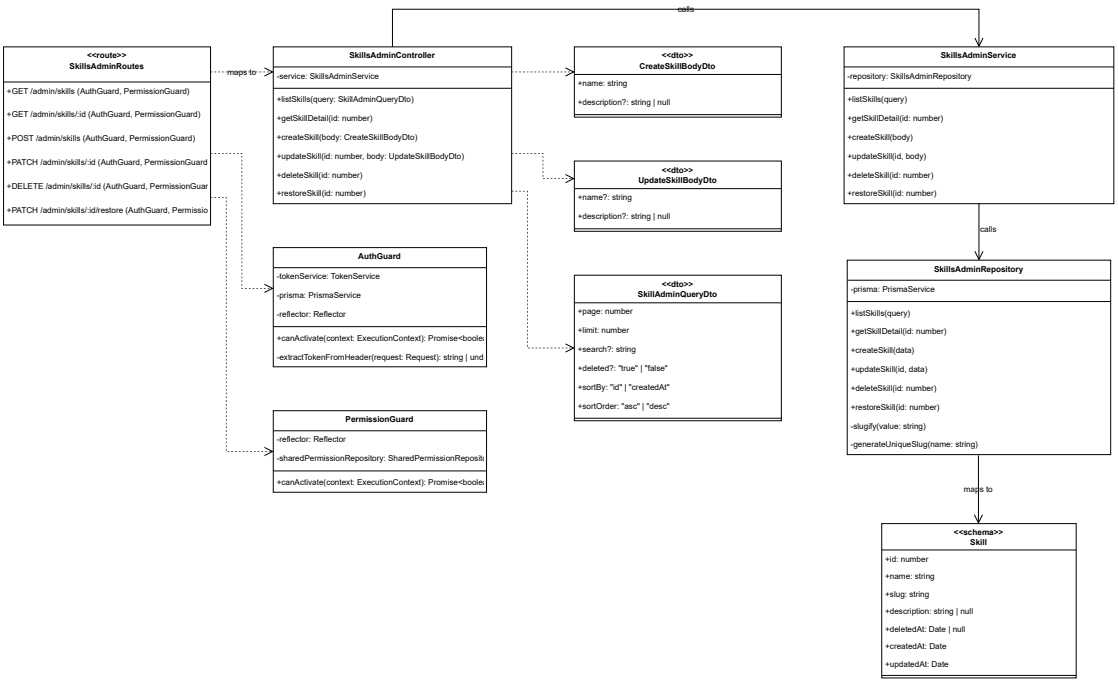

Figure 226: Class diagram of UC-38.02 View Skill Detail

b. Sequence diagram

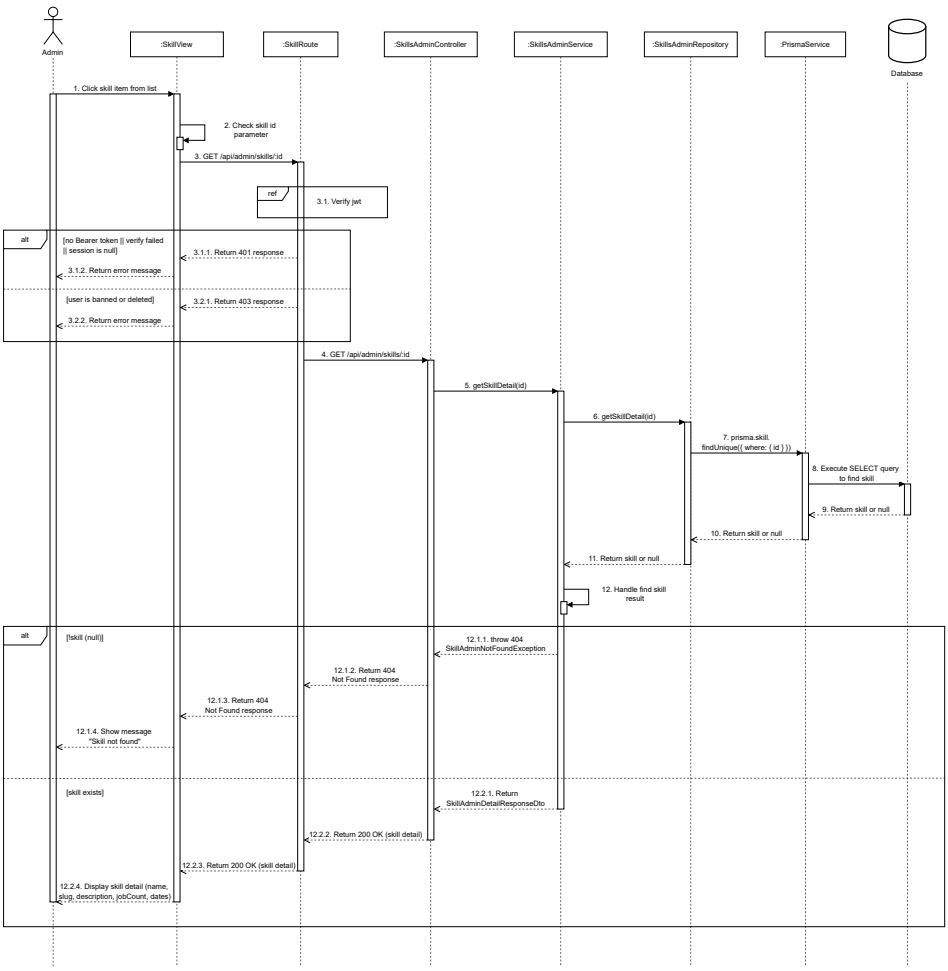

Figure 227: Sequence diagram of UC-38.02 View Skill Detail

3.20.3 UC-38.03 – Create Skill

a. Class diagram

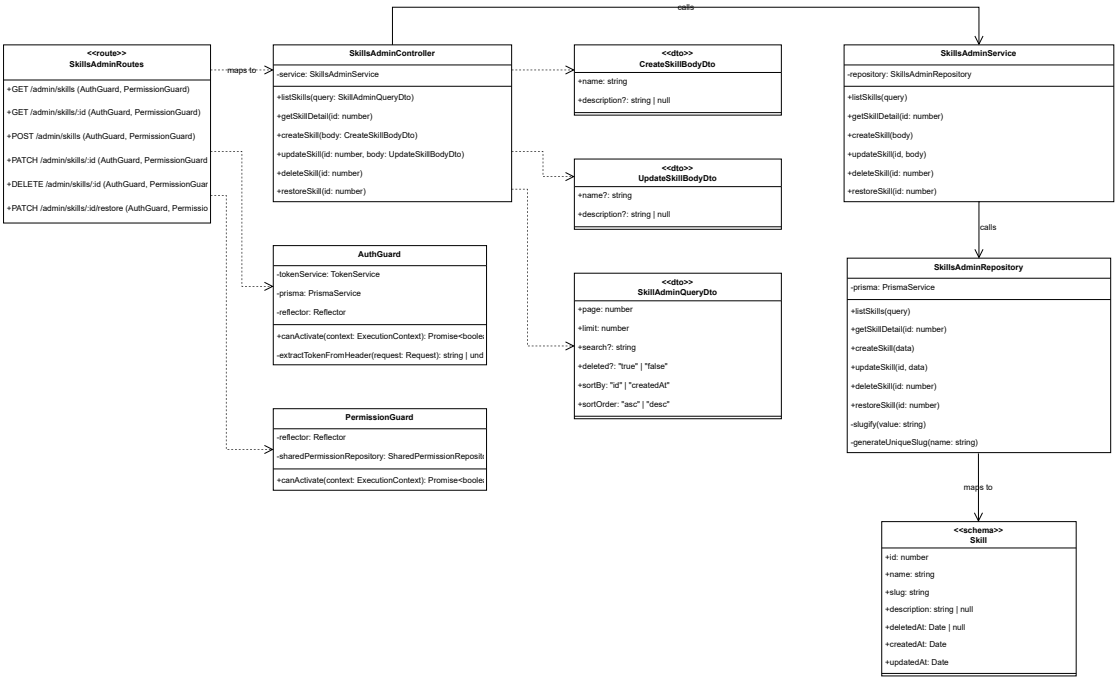

Figure 228: Class diagram of UC-38.03 Create Skill

b. Sequence diagram

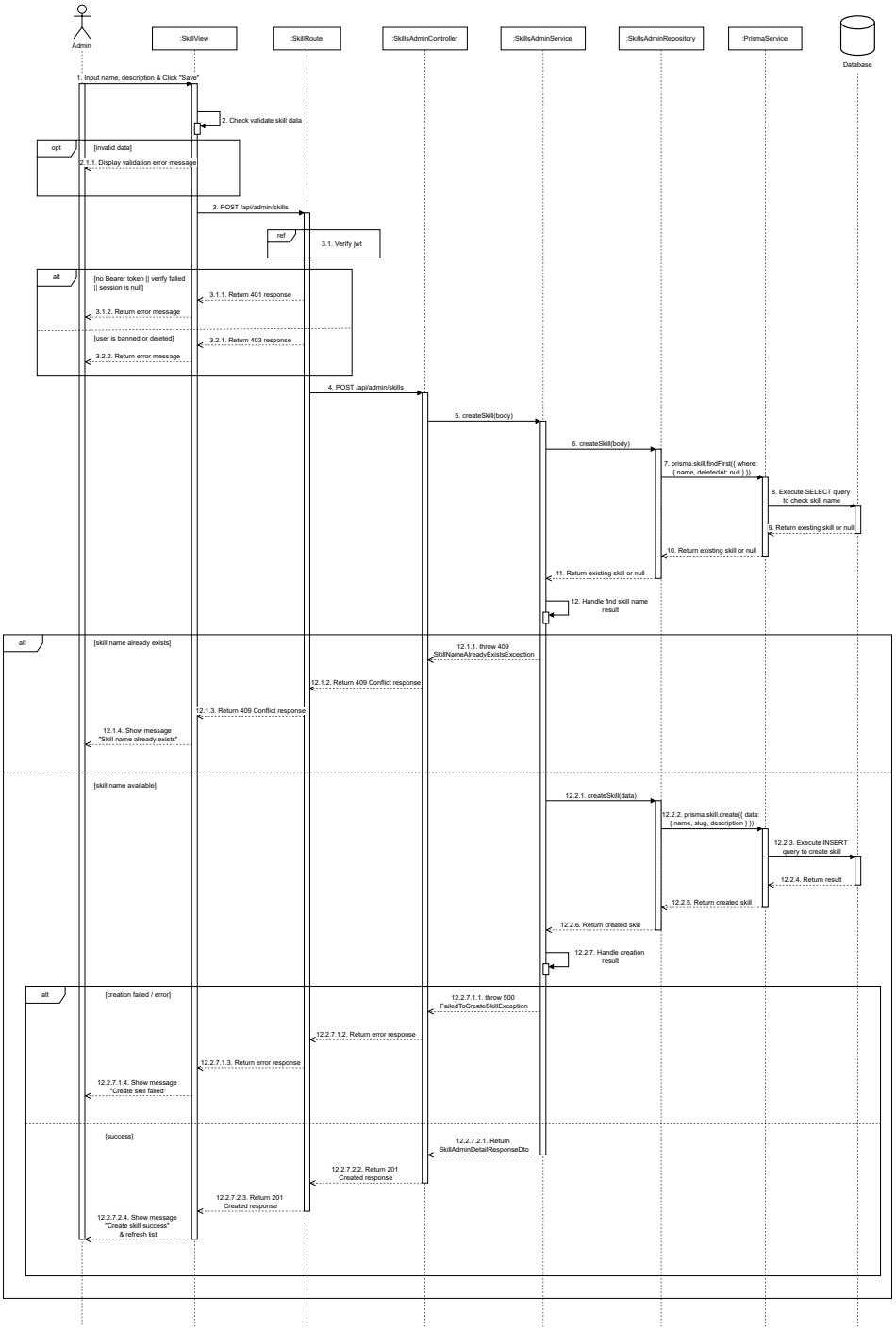

Figure 229: Sequence diagram of UC-38.03 Create Skill

3.20.4 UC-38.04 – Update Skill

a. Class diagram

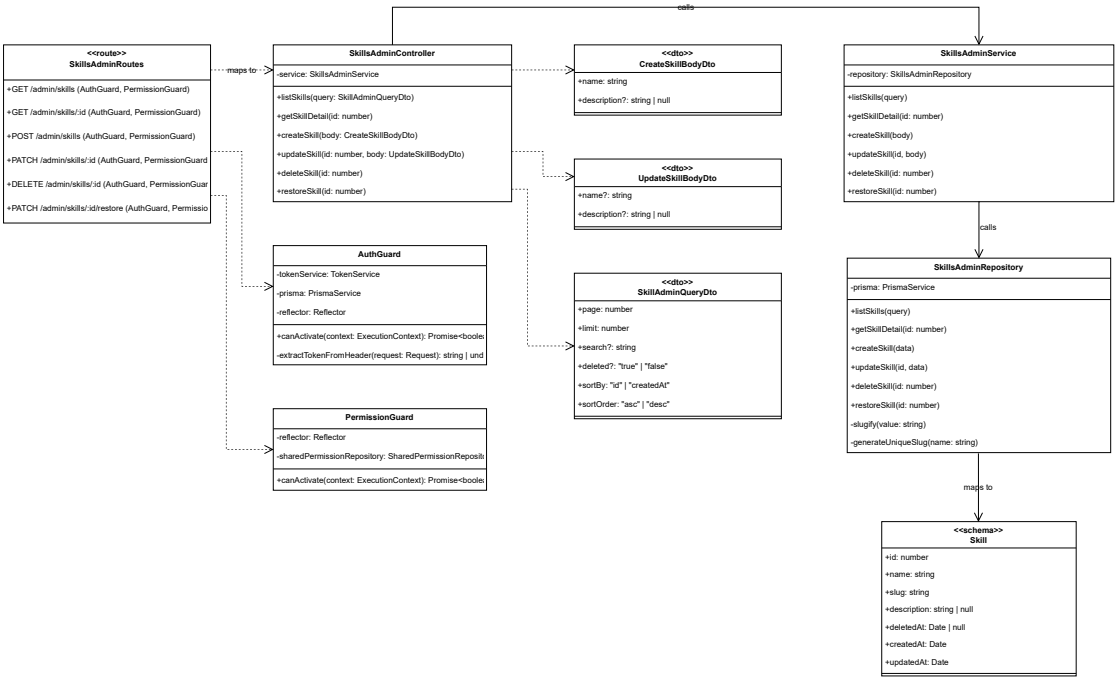

Figure 230: Class diagram of UC-38.04 Update Skill

b. Sequence diagram

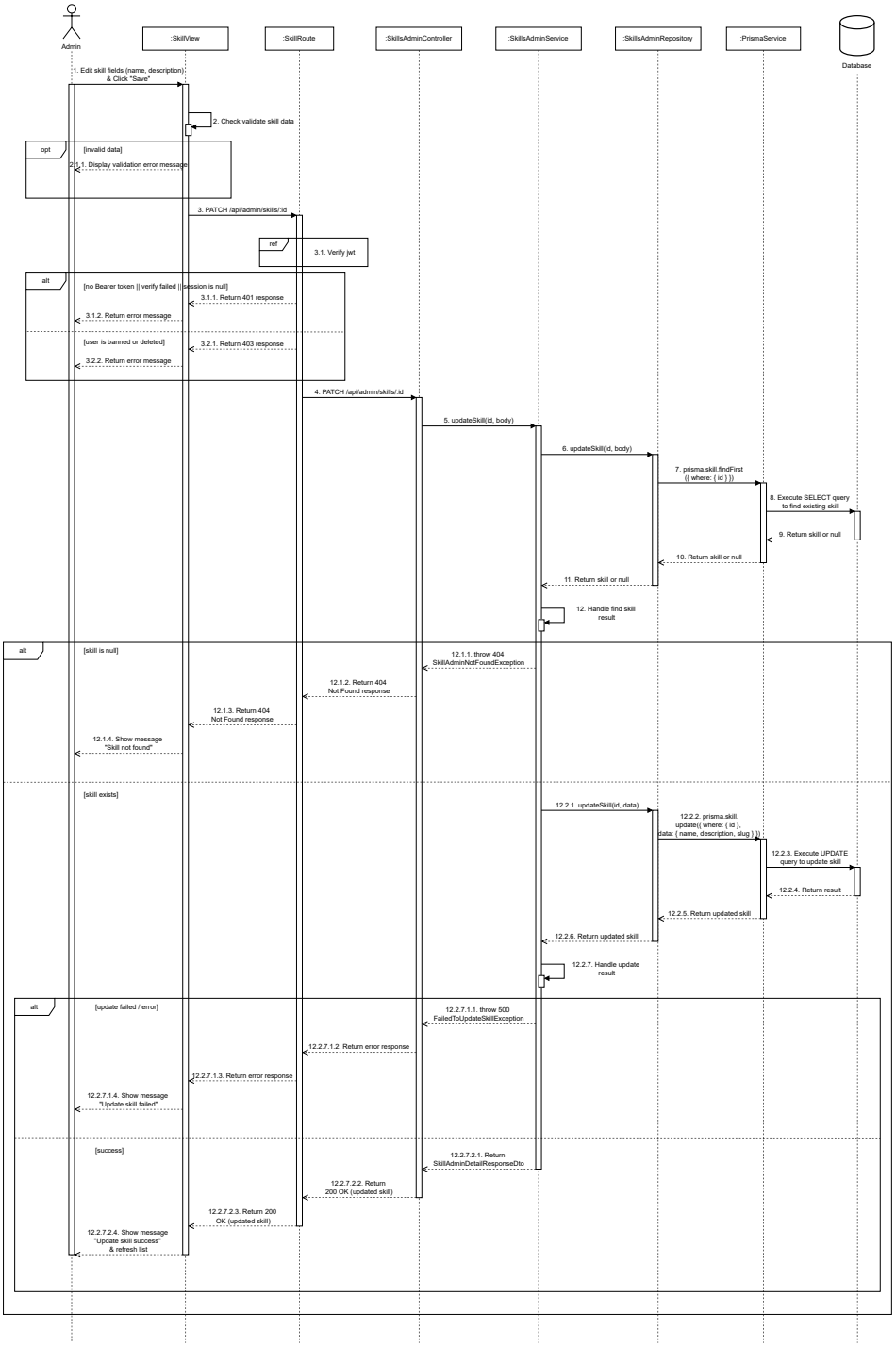

Figure 231: Sequence diagram of UC-38.04 Update Skill

3.20.5 UC-38.05 – Delete Skill

a. Class diagram

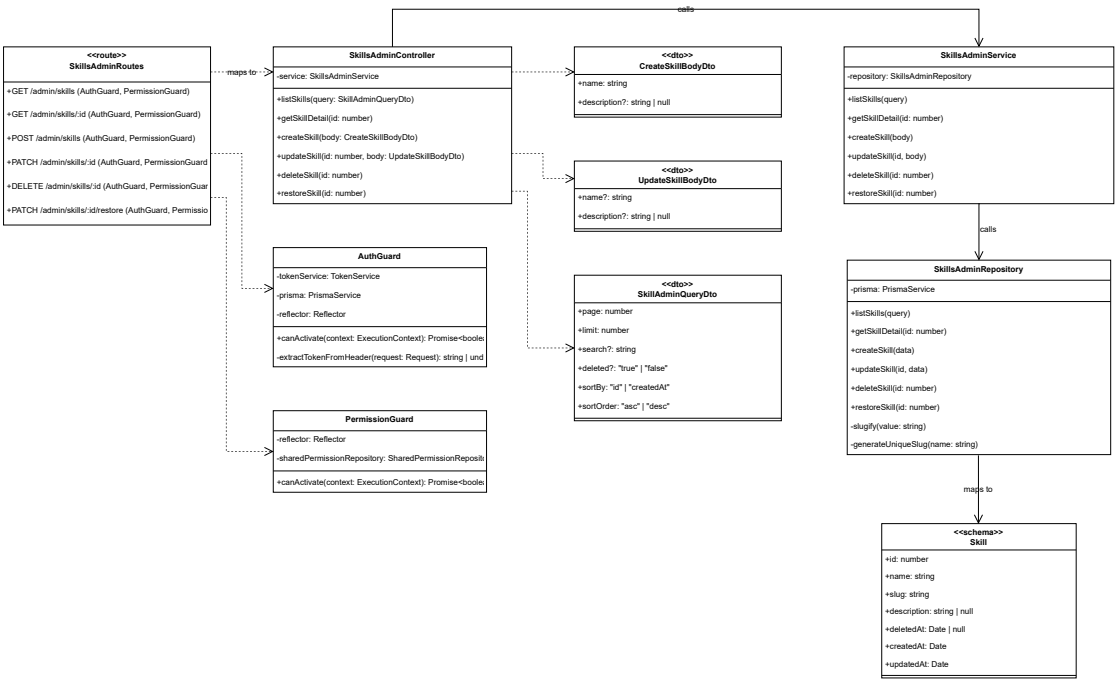

Figure 232: Class diagram of UC-38.05 Delete Skill

b. Sequence diagram

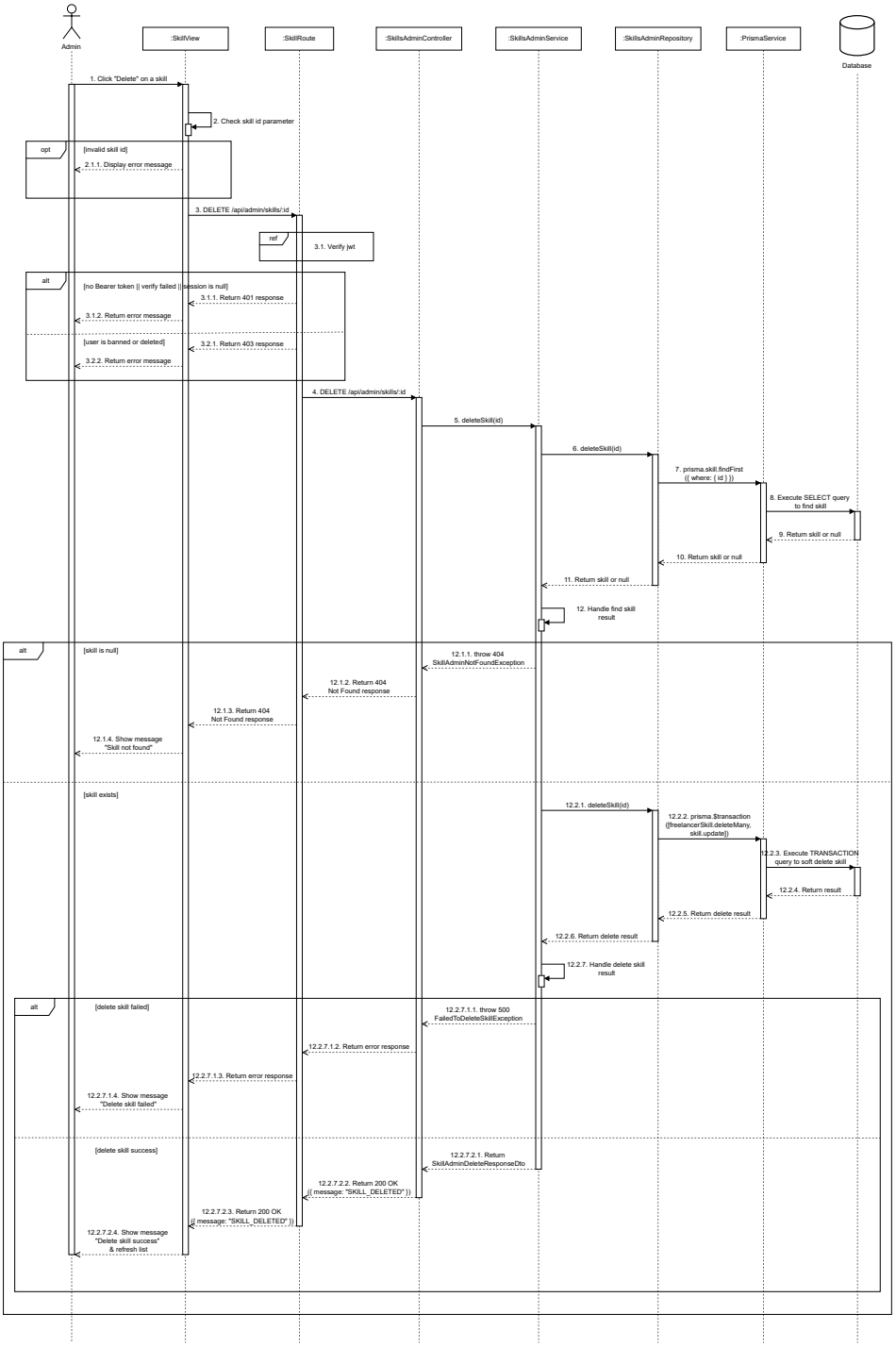

Figure 233: Sequence diagram of UC-38.05 Delete Skill

3.20.6 UC-38.06 – Restore Skill

a. Class diagram

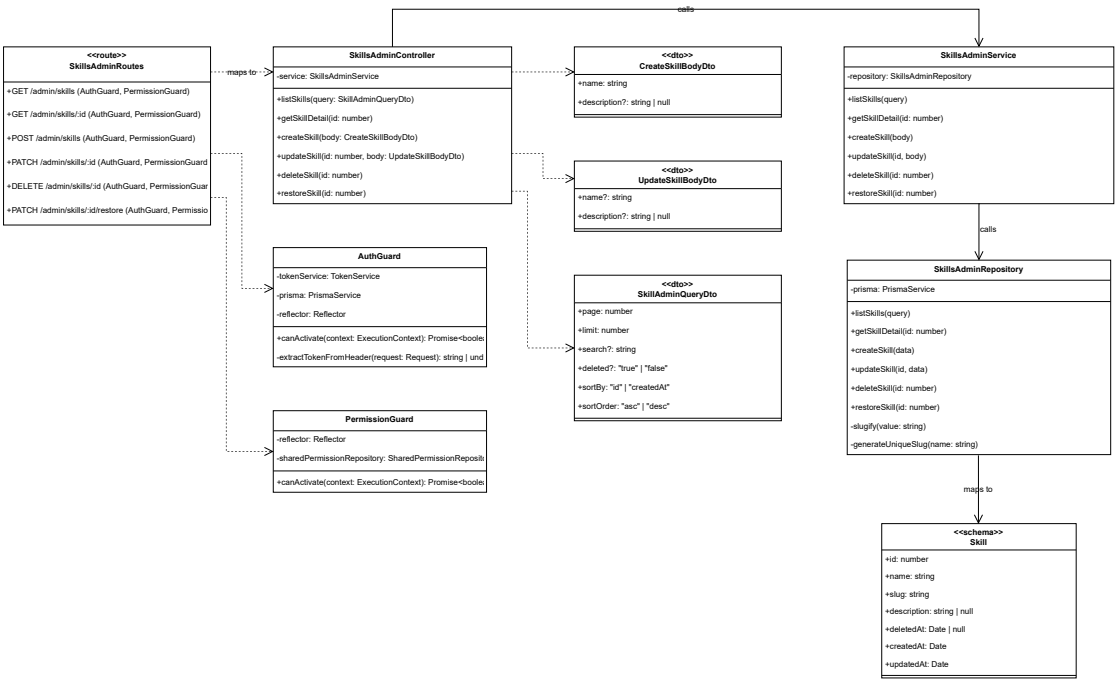

Figure 234: Class diagram of UC-38.06 Restore Skill

b. Sequence diagram

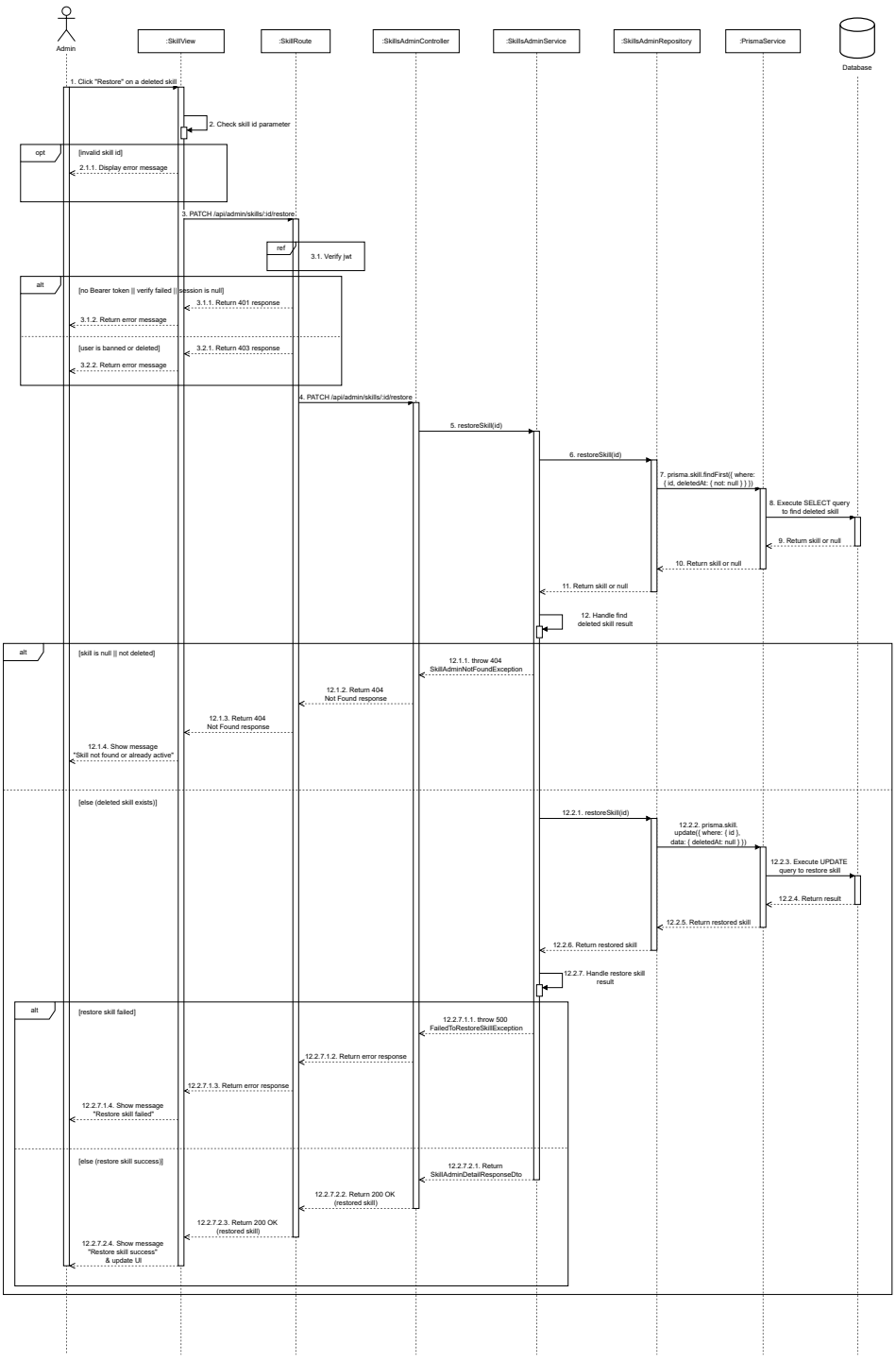

Figure 235: Sequence diagram of UC-38.06 Restore Skill

3.21 Expert Profile Management

The administrator-side Expert Profile editor updates an expert account directly. Any pending self-service revision is superseded, and the account owner is notified of the administrative change.

3.21.1 UC-28.09 – Update Expert Profile

a. Class diagram

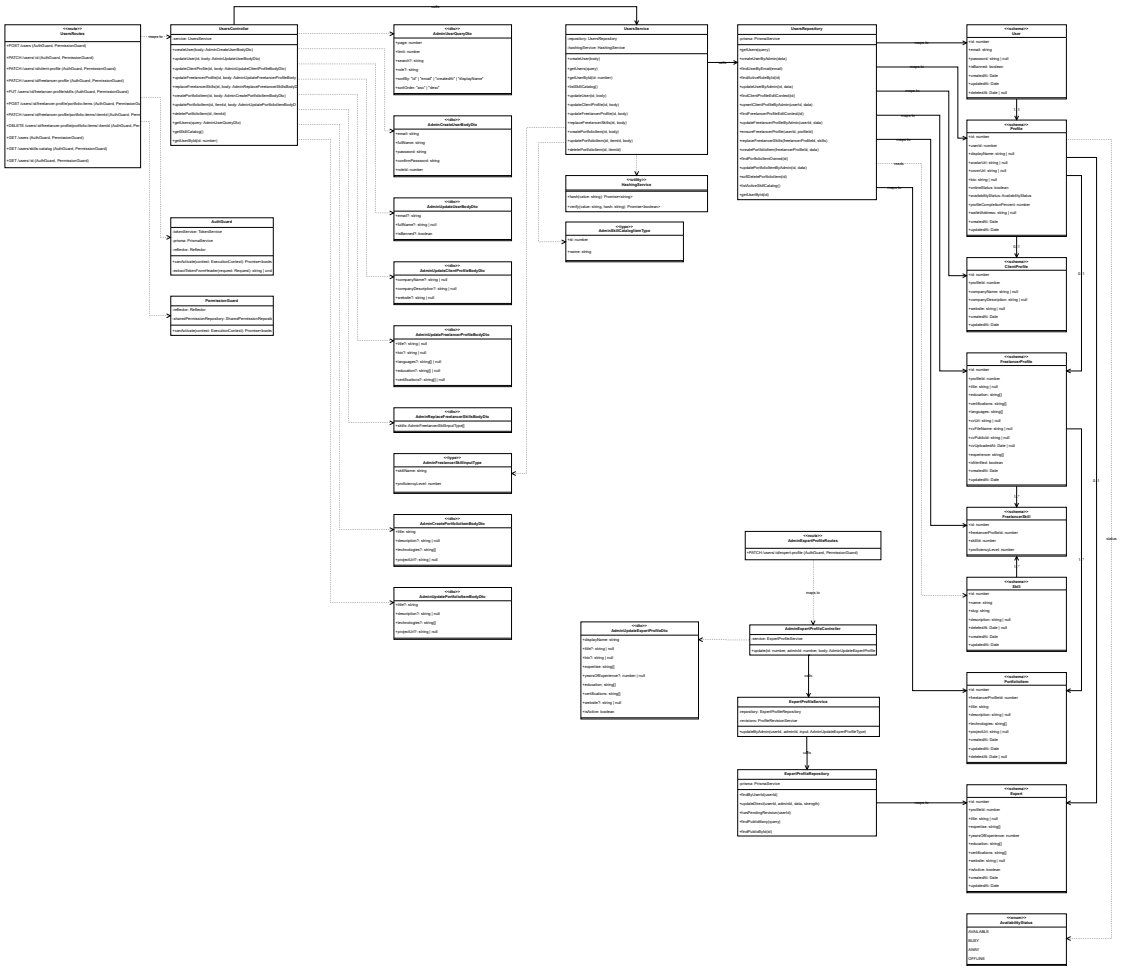

Figure 236: Class diagram of UC-28.09 Update Expert Profile

b. Sequence diagram

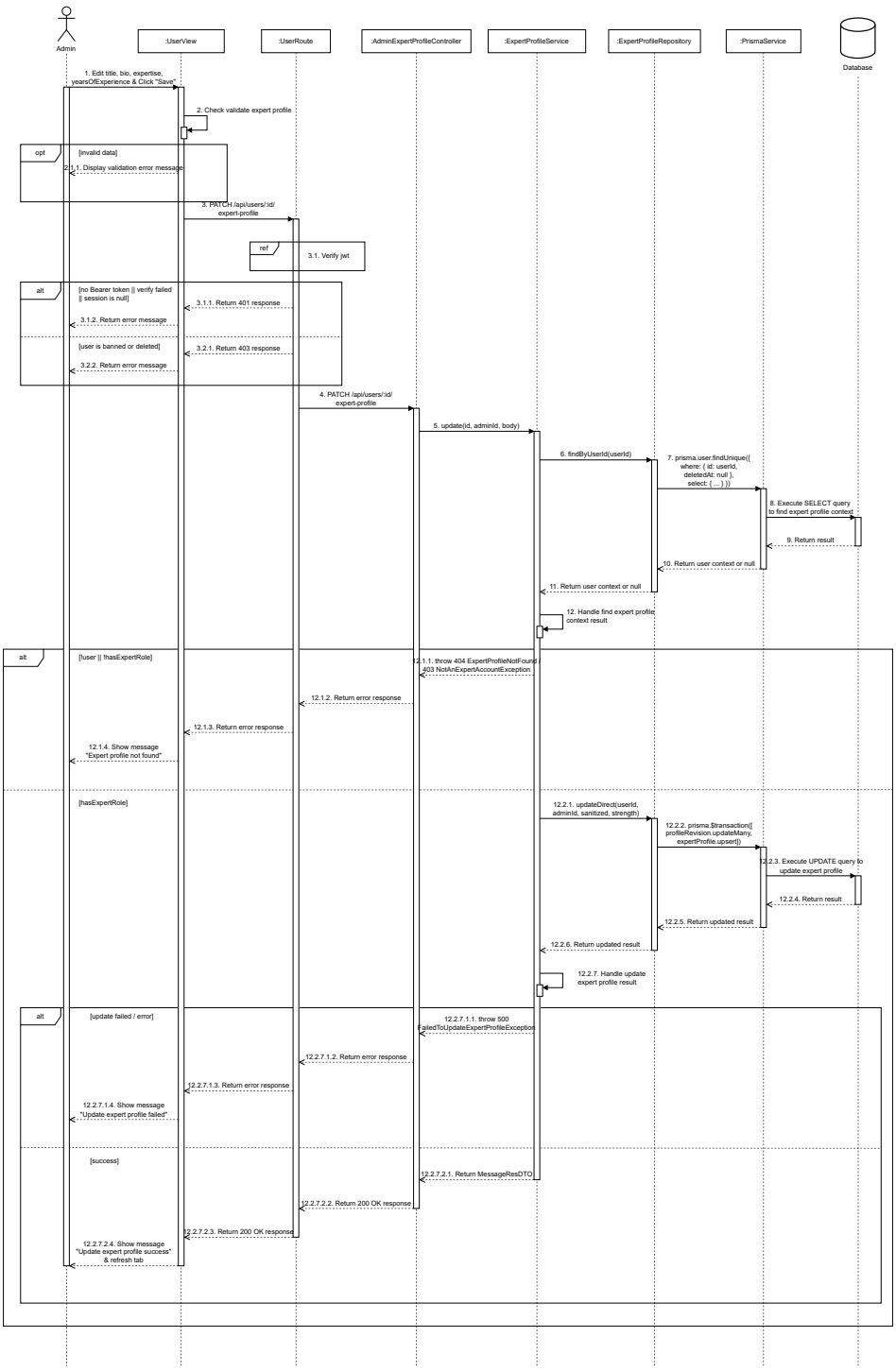

Figure 237: Sequence diagram of UC-28.09 Update Expert Profile

3.22 Job Category Management

Job Category Management maintains the taxonomy used by job postings. A category can be created, updated, soft-deleted, and restored; deletion is blocked while active jobs still reference the category.

3.22.1 UC-46.01 – View Job Category List

a. Class diagram

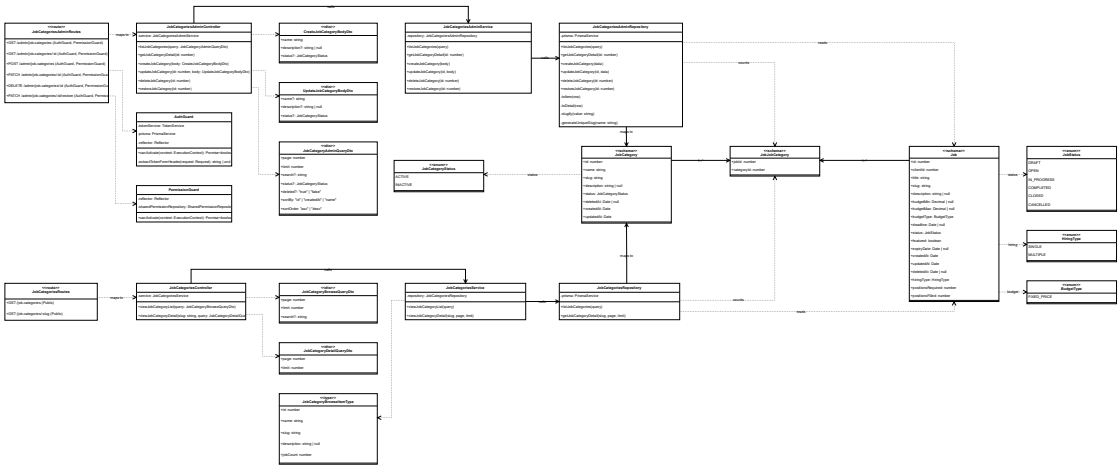

Figure 238: Class diagram of UC-46.01 View Job Category List

b. Sequence diagram

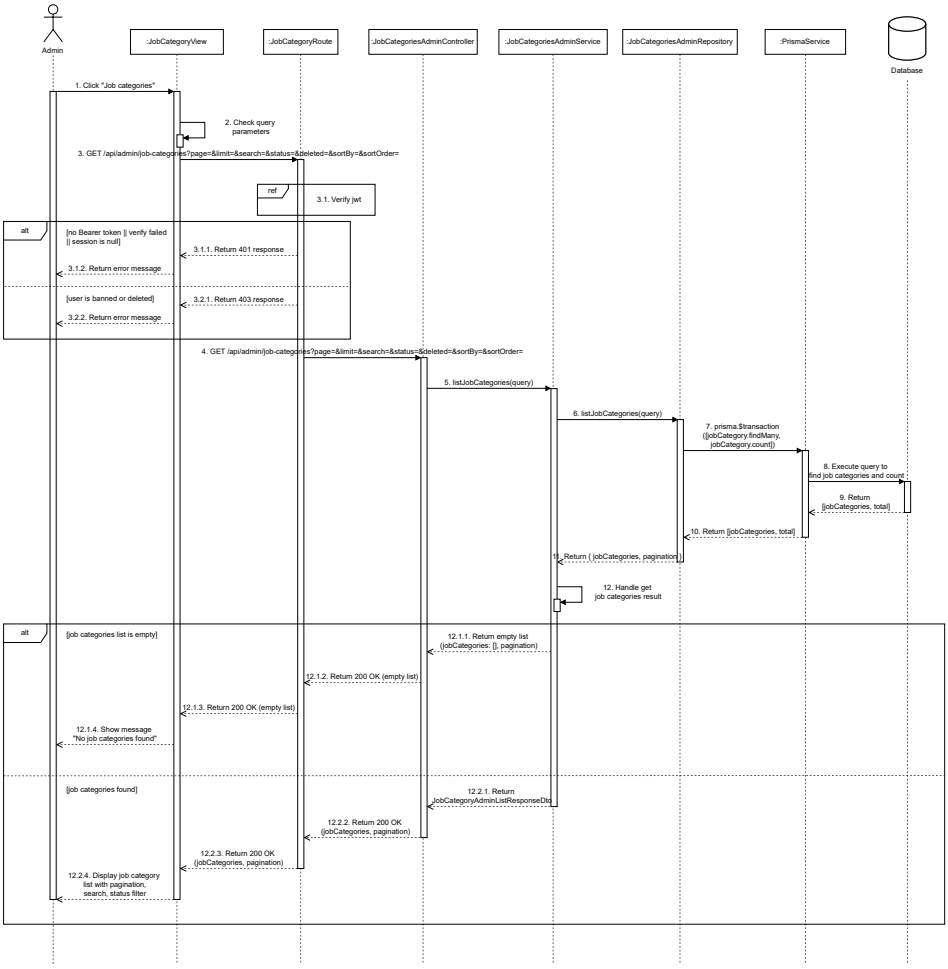

Figure 239: Sequence diagram of UC-46.01 View Job Category List

3.22.2 UC-46.02 – View Job Category Detail

a. Class diagram

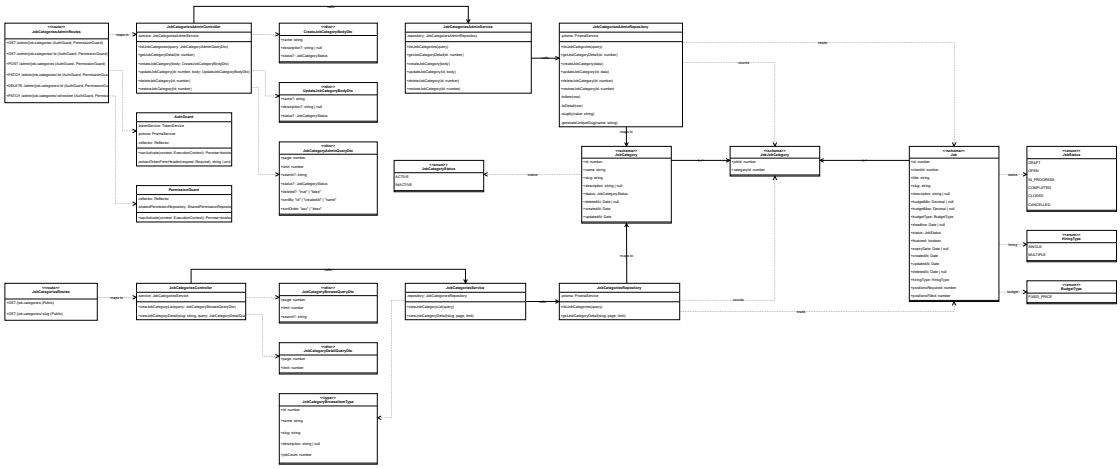

Figure 240: Class diagram of UC-46.02 View Job Category Detail

b. Sequence diagram

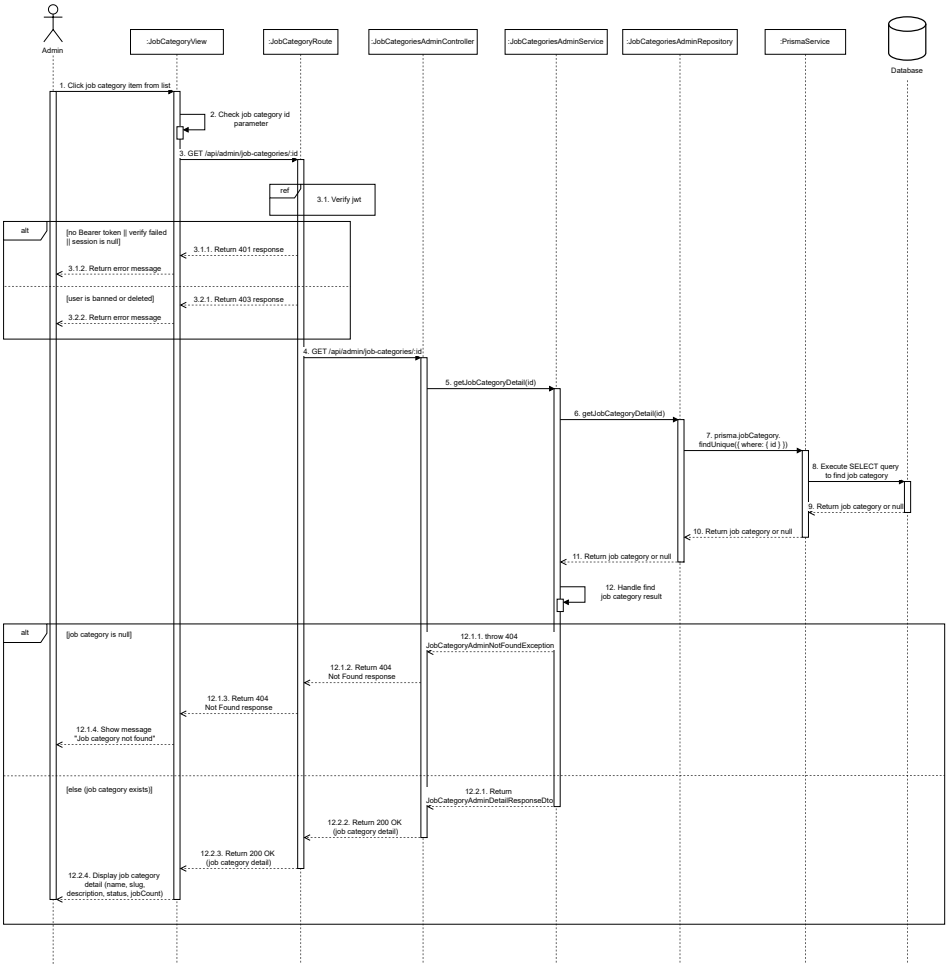

Figure 241: Sequence diagram of UC-46.02 View Job Category Detail

3.22.3 UC-46.03 – Create Job Category

a. Class diagram

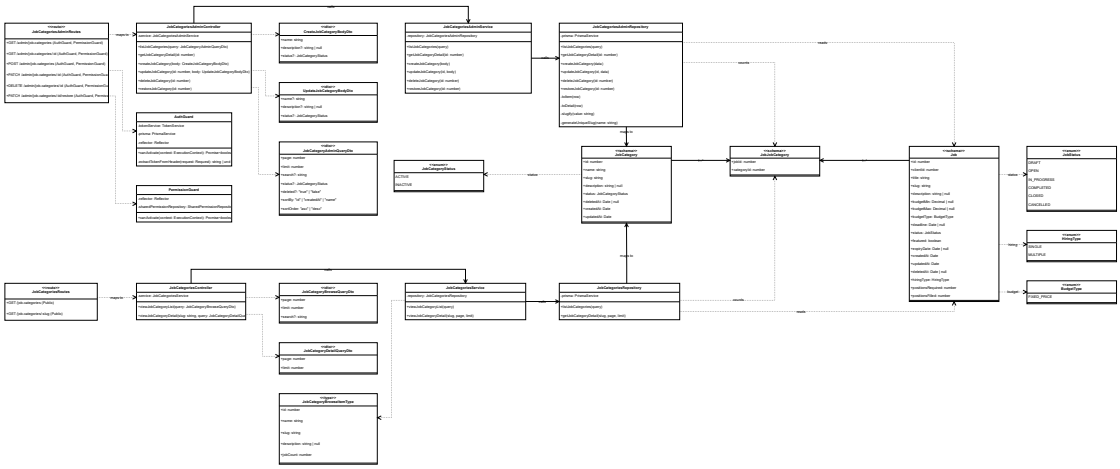

Figure 242: Class diagram of UC-46.03 Create Job Category

b. Sequence diagram

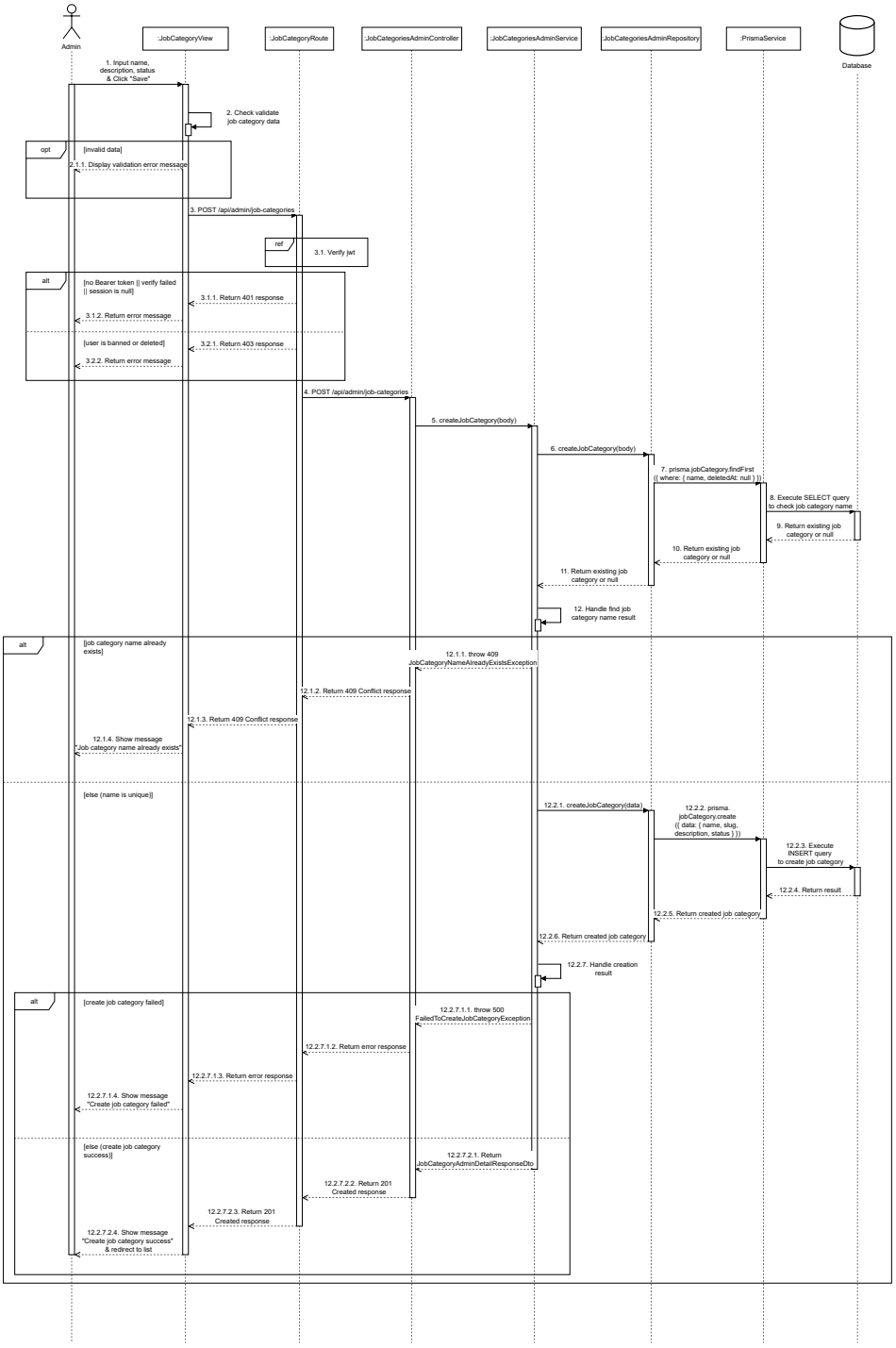

Figure 243: Sequence diagram of UC-46.03 Create Job Category

3.22.4 UC-46.04 – Update Job Category

a. Class diagram

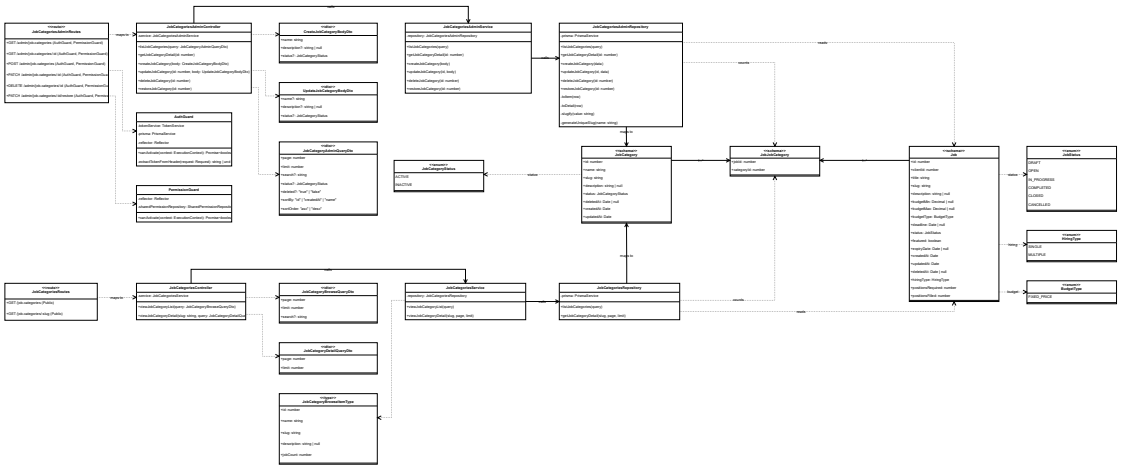

Figure 244: Class diagram of UC-46.04 Update Job Category

b. Sequence diagram

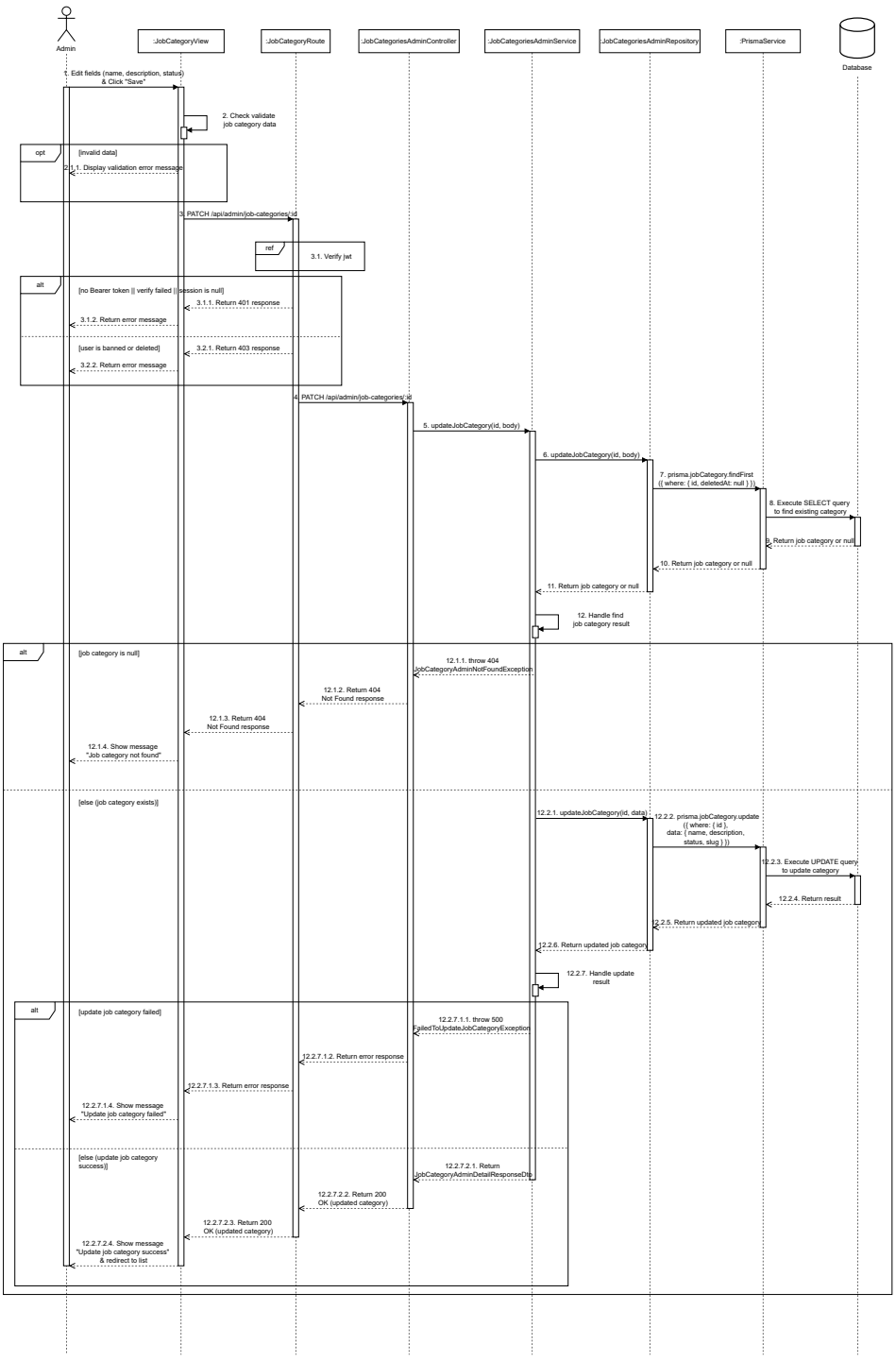

Figure 245: Sequence diagram of UC-46.04 Update Job Category

3.22.5 UC-46.05 – Delete Job Category

a. Class diagram

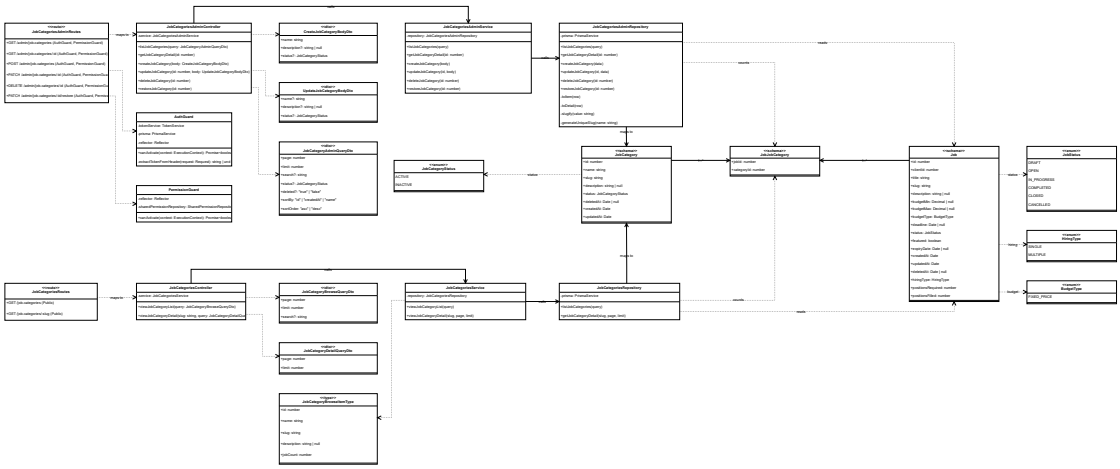

Figure 246: Class diagram of UC-46.05 Delete Job Category

b. Sequence diagram

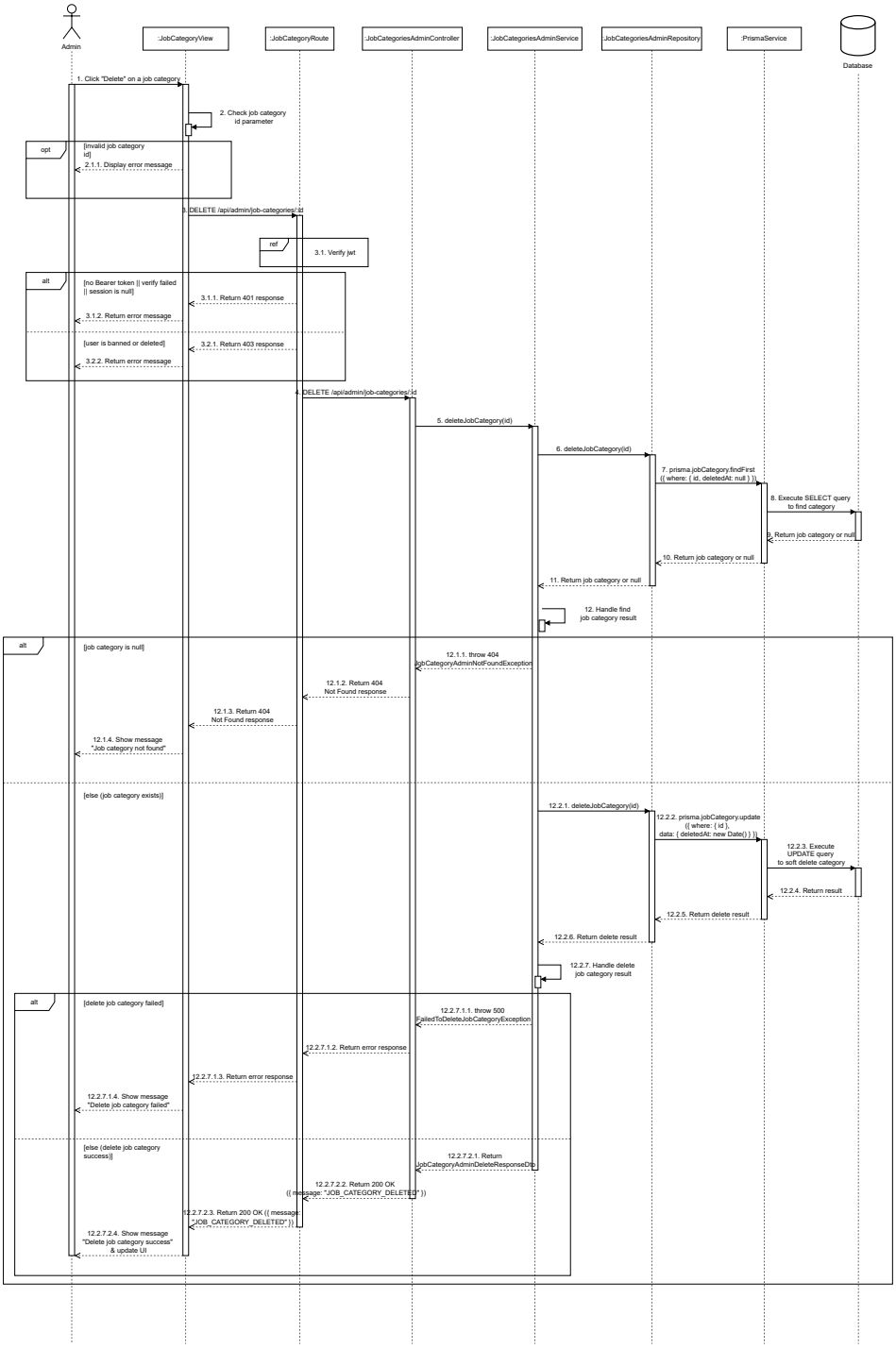

Figure 247: Sequence diagram of UC-46.05 Delete Job Category

3.22.6 UC-46.06 – Restore Job Category

a. Class diagram

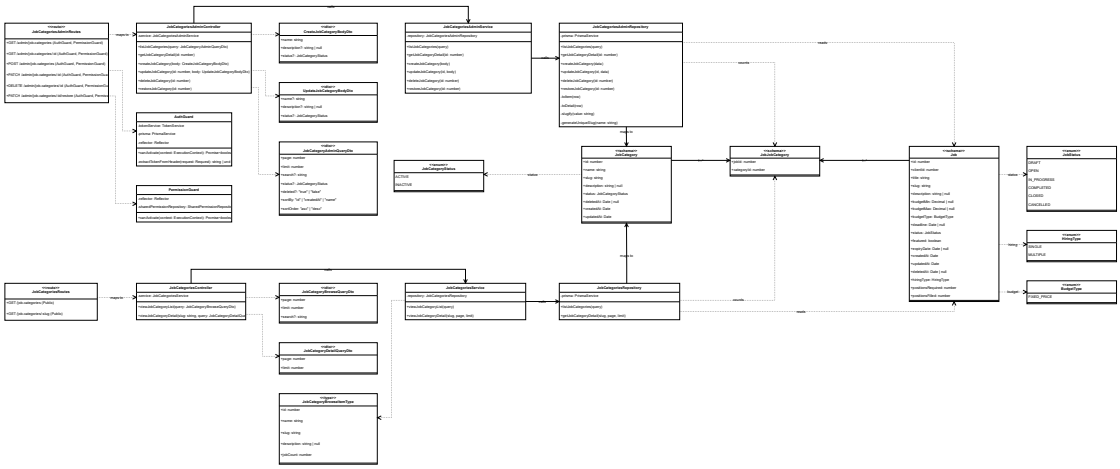

Figure 248: Class diagram of UC-46.06 Restore Job Category

b. Sequence diagram

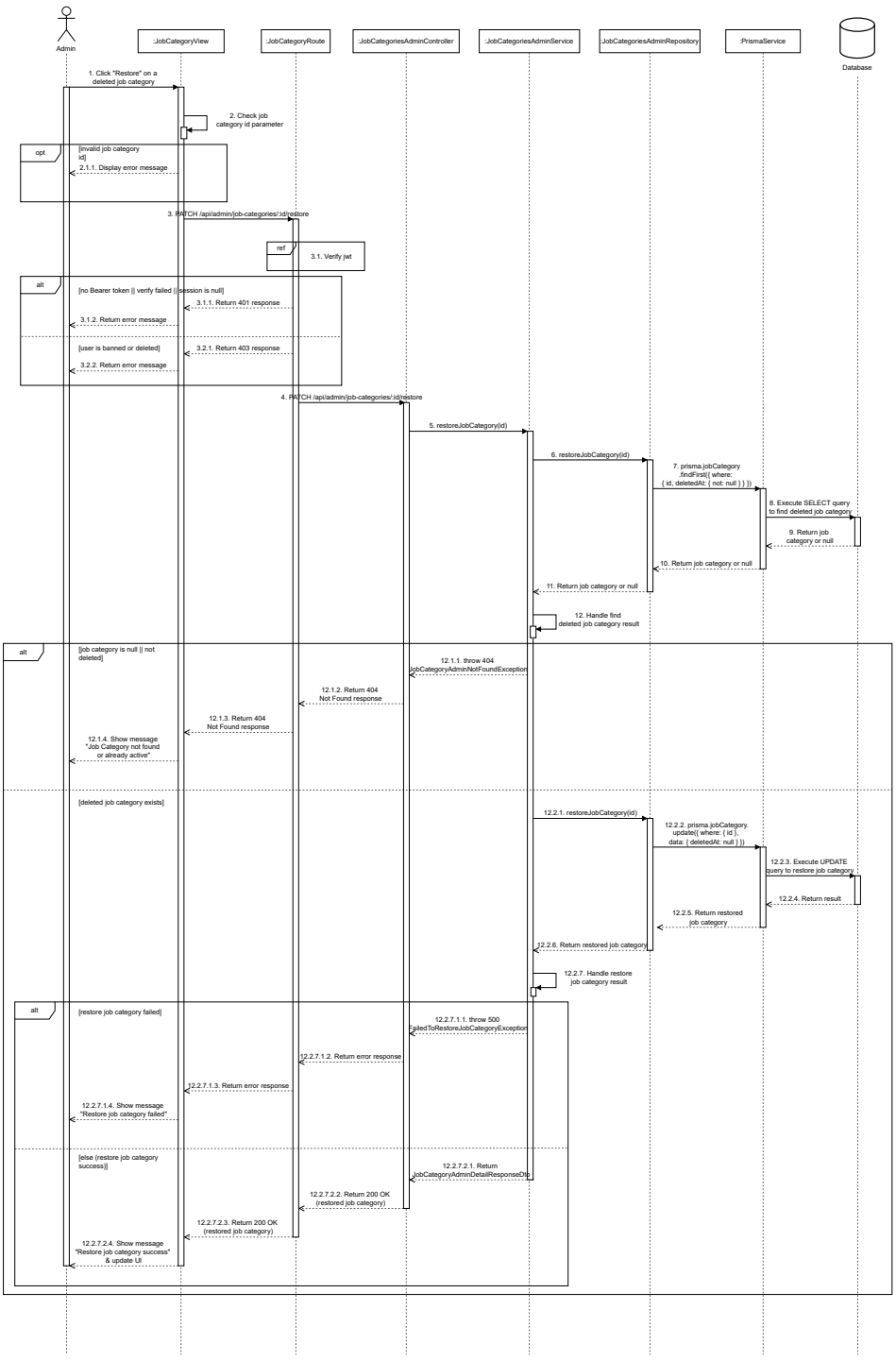

Figure 249: Sequence diagram of UC-46.06 Restore Job Category

4. Reference

[1] OMG, “About the Unified Modeling Language Specification Version 2.5.1,” Object Management Group, Dec. 2017. [Online]. Available: https://www.omg.org/spec/UML/2.5.1/

[2] NestJS, “NestJS Documentation - A progressive Node.js framework,” 2024. [Online]. Available: https://docs.nestjs.com/

[3] Next.js, “Next.js by Vercel - The React Framework for the Web,” 2024. [Online]. Available: https://nextjs.org/docs

[4] Prisma, “Prisma - Next-generation ORM for Node.js and TypeScript,” 2024. [Online]. Available: https://www.prisma.io/docs

[5] PostgreSQL Global Development Group, “PostgreSQL 16 Documentation,” 2024. [Online]. Available: https://www.postgresql.org/docs/16/index.html

[6] Redis, “Redis Documentation and Command Reference,” 2024. [Online]. Available: https://redis.io/docs/

[7] Stripe, “Stripe API Reference - Escrow Payments and Connect Payouts,” 2024. [Online]. Available: https://stripe.com/docs/api

[8] Google Developers, “Google Identity - Using OAuth 2.0 for Web Server Applications,” 2024. [Online]. Available: https://developers.google.com/identity/protocols/oauth2

[9] Tailwind CSS, “Tailwind CSS v4 Documentation,” 2024. [Online]. Available: https://tailwindcss.com/docs